

```
In [1]: import math
from matplotlib.lines import Line2D
import matplotlib.pyplot as plt
from matplotlib.ticker import PercentFormatter
import numpy as np
import pandas as pd
import re
import seaborn as sns
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
# from sklearn.feature_selection import SequentialFeatureSelector # Alternative option
from mlxtend.feature_selection import SequentialFeatureSelector as SFS

pd.set_option("display.max_colwidth", None) # Set maximum column width
pd.set_option("display.max_columns", None) # Set maximum number of columns displayed
pd.set_option("display.width", 300) # Set width for entire output
from IPython.display import display, HTML
display(HTML("<style>.container { width:100% !important; }</style>"))
```

```
In [2]: ##### Notes
# We process the following datasets:
# * ZIP_Locale_Detail.tsv
# * CAASPP entities
# * CAASPP
# * CA_Schl_Table_EthPct.tsv provides LCFF+ indicator
# * UC Admissions "FR Eth by Yr" https://www.universityofcalifornia.edu/about-us/information-center/admissions-source-school
# * UC Admissions "FR GPA by Yr" https://www.universityofcalifornia.edu/about-us/information-center/admissions-source-school
# We only utilize data about California public schools, including charter schools.

# UC Admissions data is indexed by high school name. Each school has a numeric ID, but the IDs differ from the CAASPP IDs.
# Several high schools in the state have the same name. There are even a few duplicate school names in the same county.
# UC Admissions data also includes City and County but not district.
# CAASPP data is indexed by high school name. CAASPP data also includes County, District, and ZIP Code, but not City. We look up City from ZIP using a

testID2Name = {
    "1": "English (ELA)",
    "2": "Math"
}

campuses = ["Berkeley", "UC Davis", "UCI", "UCLA", "UC Merced", "UCR", "UCSB", "UCSC", "UCSD"]
homeCounties = ["Alameda", "Yolo", "Orange", "Los Angeles", "Merced", "Riverside", "Santa Barbara", "Santa Cruz", "San Diego"]
campusesSoCal = ["UCI", "UCLA", "UCR", "UCSB", "UCSD"]
homeCountiesSoCal = ["Orange", "Los Angeles", "Riverside", "Santa Barbara", "San Diego"]

# UC admissions data is censored: if fewer than 3 students enroll, the value is set to 0. So when analyzing Enrollment,
# we have to filter out schools where fewer than 3 students are expected to enroll. This depends on the yield of the campus.
# Here are the yields by campus with minAdms = 10. The first column is for local LCFF+ schools, the second for remote LCFF+ schools:
# Berkeley 65.3% 37.4%
# UC Davis 30.2% 15.6%
# UCI 48.5% 24.1%
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# UCLA      65.5%      63.8%
# UC Merced  22.2%      5.6%
# UCR        25.2%      7.4%
# UCSB       17.6%      7.1%
# UCSC       9.6%       4.9%
# UCSD       48.3%      20.5%
# Set minAdms to about 5 / yield from the first column above.
minAdmsForEnrAnalysis = {
    "Berkeley": 10,
    "UC Davis": 20,
    "UCI": 15,
    "UCLA": 10,
    "UC Merced": 25,
    "UCR": 20,
    "UCSB": 25,
    "UCSC": 30,
    "UCSD": 15
}
```

```
In [3]: ##### Read the ZIP code data to be able to map ZIP code to City.
# The PHYSICAL CITY column indicates the city name. It is blank for a handful records: Pandas converts these to NaN. PHYSICAL STATE is blank for the s
# The LOCALE NAME often indicates the neighborhood name within the city such as "CARMEL MOUNTAIN", "LA JOLLA". It contains other text such as "ANNEX"
# Sometimes the LOCALE indicates a stree location rather than a neighborhood.
filename = "../ZIP_Locale_Detail.tsv"
keepColumns = ["DELIVERY ZIPCODE", "LOCALE NAME", "PHYSICAL CITY", "PHYSICAL STATE"]
dfZIPs = pd.read_csv(filename, delimiter="\t", dtype=str, usecols=keepColumns)
zip2CityDict = {}
zip2City = {}
zip2State = {}
for index, row in dfZIPs.iterrows():
    city = row["PHYSICAL CITY"]
    locale = row["LOCALE NAME"]
    zip = row["DELIVERY ZIPCODE"]
    if city != city: # NaN
        print(zip, locale, city, row["PHYSICAL CITY"])
        continue

# Clean the Locale
locale = re.sub(r" +", " ", locale)
locale = re.sub(r"\bHGT\b", "HEIGHTS", locale)
locale = re.sub(r"^(?:AIRPORT|CPU|DOWNTOWN|FEDERAL(?: BUILDING)?|MAIN OFFICE|VPO) ", "", locale)
locale = re.sub(r"[A-Z]{2}(?: .*)? S&?DC$", "", locale) # E.g. "SANTA ANA CA NORTH GRAND SDC"
locale = re.sub(r"(?: \d+)? CARRIERS(?: \d+)?$", "", locale) # E.g. "NORTHRIDGE CARRIERS 24", "CARMEL VALLEY 21 CARRIERS"
locale = re.sub(r"(?: CARRIERS)?(?: (:ANNEX|ANX|CPLX)(?: .*)?$", "", locale) # E.g. "CARMEL VALLEY CARRIER ANNEX", "MOUNTAIN VIEW CARRIERS ANNEX",
locale = re.sub(r"(?:POSTAL|RETAIL|STAMP) STORE$", "", locale) # E.g. "CARMEL MOUNTAIN POSTAL STORE", "MACYS RETAIL STORE"
locale = re.sub(r"FIN(?:ANCE)?(?: (:STA|STATION|UNIT))?$", "", locale) # E.g. "PARK MANOR FINANCE"
locale = re.sub(r"(?:DPOBU|PRS)$", "", locale) # E.g. "BIG BEAR LAKE DPOBU"

if locale == "" or locale == city:
    pass
elif re.search(r"\b(?:AIRPORT|AVENUE|BANK|BRANCH|CENTER|CTR|DRUGS|HOSPITAL|HWY|MALL|MARKET|MARKETPLACE|MKT|PLAZA|SERVICE|STA|STATION|STORE|STREET|
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elif re.search(city, locale): # E.g. "WASHINGTON LOS ANGELES"
    pass
else:
    if zip not in zip2CityDict:
        zip2CityDict[zip] = {}
        zip2CityDict[zip][locale] = 1

    zip2City[zip] = city
    zip2State[zip] = row["PHYSICAL STATE"]

for zip in zip2CityDict:
    sCity = "/" .join(zip2CityDict[zip].keys())
    zip2City[zip] = "/" .join([sCity, zip2City[zip]])
    if len(zip2CityDict[zip]) > 1 and zip2State[zip] == "CA":
        print(zip2City[zip])

# Add ZIP codes not available in the USPS file
zip2City["90010"] = "LOS ANGELES"
zip2City["90227"] = "unknown"
zip2City["90747"] = "CARSON"
zip2City["91719"] = "unknown"
zip2City["92093"] = "LA JOLLA"
zip2City["92415"] = "SAN BERNARDINO"
zip2City["93740"] = "FRESNO"
zip2City["93741"] = "FRESNO"
zip2City["95057"] = "unknown"
zip2City["95331"] = "unknown"
zip2City["96311"] = "Fleet Post Office AP"
print(len(zip2City))

```

32304 WESTSIDE nan nan
 32305 WESTSIDE nan nan
 32310 WESTSIDE nan nan
 32316 WESTSIDE nan nan
 ALAMEDA/ARCADE/LOS ANGELES
 BARRINGTON/VILLAGE/LOS ANGELES
 LINCOLN/LINCOLN HEIGHTS/LOS ANGELES
 EL SERENO/LINCOLN HEIGHTS/LOS ANGELES
 BICENTENNIAL/MIRACLE MILE/LOS ANGELES
 EAGLE ROCK/HIGHLAND PARK/LOS ANGELES
 LA TIJERA/WESTVERN/LOS ANGELES
 ARCO/BUNKER HILL/FOY/LOS ANGELES
 HUB CITY/WILLOWBROOK/COMPTON
 FIRESTONE/HOLLYDALE/SOUTH GATE
 BELMONT SHORE/LOMA/LONG BEACH
 BIXBY/NORTH/LONG BEACH
 SPRING/VIKING/LONG BEACH
 BRYANT/SPRING/LONG BEACH
 NORTHRIDGE/RANCH/PORTER RANCH
 ROMOLAND/SUN CITY/MENIFEE
 QUAIL VALLEY/SUN CITY/MENIFEE
 PLAYA/SOUTH LAGUNA/LAGUNA BEACH
 BOLSA/GOLDENWEST/WESTMINSTER
 BRISTOL/SOUTH MAIN/SANTA ANA
 DOYLE COLONY/POPLAR/PORTERVILLE
 PINEDALE/WOODWARD PARK/FRESNO
 BARTON/CEDAR/FRESNO
 CHESTNUT/LINDEN/SOUTH SAN FRANCISCO
 GEARY/MENDELL/SAN FRANCISCO
 BAYVIEW/STANTON/BURLINGAME
 MARINA/PRESIDIO/SAN FRANCISCO
 TOWNSEND/ISLAND COVE/SAN FRANCISCO
 MCLAREN/VISITACION/SAN FRANCISCO
 MARINA/MARINA GREEN/SAN FRANCISCO
 CAMBRIDGE/PALO ALTO/EAST PALO ALTO
 MISSION SAN JOSE/WARM SPRINGS/FREMONT
 DIMOND/WEST GRAND/OAKLAND
 GRAND LAKE/WEST GRAND/OAKLAND
 HILLTOP/MIRA VISTA/RICHMOND
 BEAR VALLEY/CAMP CONNELL/ARNOLD
 BROADWAY/FORT SUTTER/SACRAMENTO
 CAMELLIA/FORT SUTTER/SACRAMENTO
 37995

```

In [4]: ##### dfEntities
# Read the CAASPP entities data. This mostly gives us the mapping of county code to county name.
# Read the file. "latin1" (ISO-8859-1) is needed due to Spanish characters in the names of 3 schools.
filename = "CAASPP/sb_ca2024entities_csv.txt"
allEntitiesColumns = ["County Code", "District Code", "School Code", "Type ID", "Filler", "Test Year",
                      "County Name", "District Name", "School Name", "Zip Code"]
keepEntitiesColumns = ["County Code", "District Code", "School Code", "Type ID",
                       "County Name", "District Name", "School Name", "Zip Code"]
dfEntities = pd.read_csv(filename, delimiter="^", dtype=str, encoding="latin1", usecols=keepEntitiesColumns)
  
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dfEntities["School Name"] = dfEntities["School Name"].str.strip() # Trim whitespace

entitiesCounties = dfEntities["County Name"].value_counts() # Count of county names in dataset

code2County = {}
for index, row in dfEntities.iterrows():
    code = row["County Code"]
    name = row["County Name"]
    if code in code2County:
        if code2County[code] != name:
            print("Error")
    else:
        code2County[code] = name

code2City = {}
code2ZIP = {}
for index, row in dfEntities.iterrows():
    code = row["School Code"]
    zip = row["Zip Code"]
    if code == "0000000":
        continue
    if code in code2ZIP:
        if code2ZIP[code] != zip:
            print(code, code2ZIP[code], zip)
    else:
        code2ZIP[code] = zip
        if zip in zip2City:
            code2City[code] = zip2City[zip]
        else:
            print("Unknown ZIP Code:", zip)

```

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In [5]: ##### dfTests 2024
# Read the CAASPP test score data. This includes number of students at the school and test results.
# Fields:
# "Total Students Enrolled" may be "*", in which case "Total Students Tested" = 0, "Total Students Tested with Scores" = 0, and Percentage values are 0.
# Dataframes:
# df: Whole file. Temporary.
# dfTests: Filter out records for grade != 11, etc.
# dfRight = dfTests[keepColumns], then add additional columns.
filename = "CAASPP/sb_ca2024_all_csv_v1.txt"
keepColumns = ["County Code", "District Code", "District Name", "School Code", "School Name", "Type ID", "Test Year", "Test Type",
               "Test ID", "Student Group ID", "Grade",
               "Total Students Enrolled", "Total Students Tested", "Total Students Tested with Scores",
               "Mean Scale Score",
               "Percentage Standard Exceeded", "Count Standard Exceeded",
               "Percentage Standard Met", "Count Standard Met",
               "Percentage Standard Met and Above", "Count Standard Met and Above",
               "Percentage Standard Nearly Met", "Count Standard Nearly Met",
               "Percentage Standard Not Met", "Count Standard Not Met",
               "Overall Total"]

# Read the file. "latin1" (ISO-8859-1) is needed due to Spanish characters in the names of 3 schools.

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df = pd.read_csv(filename, delimiter="^", dtype=str, encoding="latin1", usecols=keepColumns)
df["School Name"] = df["School Name"].str.strip() # Trim whitespace. School Name in the CAASPP and CAASPP Entities data is truncated to 50 characters

dfTests = df[(df["School Code"] != "0000000") & (df["Grade"] == "11") & (df["Student Group ID"] == "1")] # Keep only high schools. Use "&" operator for AND
# df = df.astype({"Total Students Enrolled": "int", "Total Students Tested": "int", "Total Students Tested with Scores": "int"}) # This generates an integer dataframe
df = pd.DataFrame() # Free the memory of the big dataframe
print(dfTests.shape[0])
print(dfTests)
```

5428

pe Test ID	County Code	District Code	District Name	School Code	School Name	Type	ID	Test Year	Test Ty
2206	01	10017	Envision Academy for Arts & Technology	0112607	Envision Academy for Arts & Technology		9	2024	
B	1	1	11	50	48	47	2517.4		
2373	01	10017	Envision Academy for Arts & Technology	0112607	Envision Academy for Arts & Technology		9	2024	
B	2	1	11	50	48	48	2472.5		
4099	01	10017	Alameda County Office of Education	0130401	Alameda County Juvenile Hall/Court		7	2024	
B	1	1	11	13	5	5	*		
4212	01	10017	Alameda County Office of Education	0130401	Alameda County Juvenile Hall/Court		7	2024	
B	2	1	11	13	6	6	*		
4329	01	10017	Alameda County Office of Education	0130419	Alameda County Community		7	2024	
B	1	1	11	15	10	9	*		
...	...	...	...	...	...	...	...	...	...
4050261	58	72769	Wheatland Union High	0123570	Wheatland Community Day High		7	2024	
B	2	1	11	*	*	*	*		
4050294	58	72769	Wheatland Union High	0133751	Edward P. Duplex		7	2024	
B	1	1	11	22	21	21	2469.6		
4050358	58	72769	Wheatland Union High	0133751	Edward P. Duplex		7	2024	
B	2	1	11	22	21	21	2435.6		
4050422	58	72769	Wheatland Union High	5838305	Wheatland Union High		7	2024	
B	1	1	11	240	237	237	2582.1		
4050524	58	72769	Wheatland Union High	5838305	Wheatland Union High		7	2024	
B	2	1	11	240	237	237	2545.9		

	Percentage Standard Exceeded	Count Standard Exceeded	Percentage Standard Met	Count Standard Met	Percentage Standard Not Met	Count Standard Not Met	Overall Total
2206	6.38	3	21.28	10			27.66
13	31.91	15	40.43		19	47	
2373	0.00	0	12.50	6			12.50
6	14.58	7	72.92		35	48	
4099	*	*	*	*			*
*	*	*	*		*	*	
4212	*	*	*	*			*
*	*	*	*		*	*	
4329	*	*	*	*			*
*	*	*	*		*	*	
...	...	...	...	...			...
...	...	...	...		...	...	
4050261	*	*	*	*			*
*	*	*	*		*	*	
4050294	0.00	0	9.52	2			9.52
2	28.57	6	61.90		13	21	
4050358	0.00	0	0.00	0			0.00
0	0.00	0	100.00		21	21	
4050422	20.25	48	33.76	80			54.01
128	24.89	59	21.10		50	237	
4050524	5.91	14	17.30	41			23.21
55	26.16	62	50.63		120	237	

[5428 rows x 26 columns]

```
In [6]: ##### dfTests2025
filename = "CAASPP/sb_ca2025_all_csv_v1.txt"
keepColumns = ["County Code", "District Code", "District Name", "School Code", "School Name", "Type ID", "Test Year", "Test Type",
               "Test ID", "Student Group ID", "Grade",
               "Total Students Enrolled", "Total Students Tested", "Total Students Tested with Scores",
               "Mean Scale Score",
               "Percentage Standard Exceeded", "Count Standard Exceeded",
               "Percentage Standard Met", "Count Standard Met",
               "Percentage Standard Met and Above", "Count Standard Met and Above",
               "Percentage Standard Nearly Met", "Count Standard Nearly Met",
               "Percentage Standard Not Met", "Count Standard Not Met",
               "Overall Total"]

# Read the file. "latin1" (ISO-8859-1) is needed due to Spanish characters in the names of 3 schools.
df = pd.read_csv(filename, delimiter="^", dtype=str, encoding="latin1", usecols=keepColumns)
df["School Name"] = df["School Name"].str.strip() # Trim whitespace. School Name in the CAASPP and CAASPP Entities data is truncated to 50 characters

dfTests2025 = df[(df["School Code"] != "0000000") & (df["Grade"] == "11") & (df["Student Group ID"] == "1")] # Keep only high schools
df = pd.DataFrame() # Free the memory of the big dataframe
print(dfTests2025.shape[0])
```

5412

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In [7]: # CAASPP school names that require special treatment
# * Parentheses are automatically treated as a first name unless added as an exception here.
# * "ALTERNATIVE" (65 schools) and "CONTINUATION" (241 schools) are automatically removed unless added as an exception here.
testParens = {
    # ALTERNATIVE that occurs in UC Schools
    # "not in CAASPP": "SANTA CRUZ ALTERNATIVE", # UCID = 50446
    "VISTA ALTERNATIVE": "VISTA ALTERNATIVE",

    # ALTERNATIVE and CONTINUATION that should not be stripped to avoid duplicates within the same county.
    # (Duplicates are rare and may be automatically handled via lookup.psv, so not critical to avoid them here.)
    "ENTERPRISE ALTERNATIVE": "ENTERPRISE ALTERNATIVE",
    "LIVE OAK ALTERNATIVE": "LIVE OAK ALTERNATIVE",
    "MENDOCINO ALTERNATIVE": "MENDOCINO ALTERNATIVE",
    "POMONA ALTERNATIVE": "POMONA ALTERNATIVE",
    "ROUND VALLEY CONTINUATION": "ROUND VALLEY CONTINUATION",
    "SIERRA VISTA HIGH (ALTERNATIVE)": "SIERRA VISTA ALTERNATIVE HIGH",
    "TEMPLE CITY ALTERNATIVE": "TEMPLE CITY ALTERNATIVE",

    # Parentheses
    "ATASCADERO CHOICES IN EDUCATION ACADEMY (ACE)": "ATASCADERO CHOICES IN EDUCATION ACADEMY (ACE)",
    "BURBANK UNIFIED INDEPENDENT LEARNING ACADEMY (BUIL)": "BURBANK UNIFIED INDEPENDENT LEARNING ACADEMY (BUILA)",
    "EPIC CHARTER (EXCELLENCE PERFORMANCE INNOVATION CI)": "EXCELLENCE PERFORMANCE INNOVATION CITIZENSHIP (EPIC) CHARTER",
    "FIVE KEYS CHARTER (SF SHERIFF'S)": "FIVE KEYS CHARTER (SF SHERIFF'S)",
    "FIVE KEYS INDEPENDENCE HS (SF SHERIFF'S)": "FIVE KEYS INDEPENDENCE HS (SF SHERIFF'S)",
    "HAROLD MCALISTER HIGH (OPPORTUNITY)": "HAROLD MCALISTER HIGH (OPPORTUNITY)",
    "HUMANITIES AND ARTS (HARTS) ACADEMY OF LOS ANGELES": "HUMANITIES AND ARTS (HARTS) ACADEMY OF LOS ANGELES",
    "JUVENILE HALL (ENDEAVOR/VOYAGER SECONDARY)": "ENDEAVOR/VOYAGER SECONDARY (JUVENILE HALL)",
    "KEITH B. BRIGHT HIGH (JUVENILE HALL)": "KEITH B BRIGHT HIGH (JUVENILE HALL)",
    "LAKEPORT ALTERNATIVE (HOME SCHOOL)": "LAKEPORT ALTERNATIVE (HOME SCHOOL)",
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"PACIFIC VALLEY (K-12)": "PACIFIC VALLEY K-12",
"S.F. COUNTY OPPORTUNITY (HILLTOP)": "SAN FRANCISCO COUNTY OPPORTUNITY - HILLTOP",
"SOAR HIGH (STUDENTS ON ACADEMIC RISE)": "STUDENTS ON ACADEMIC RISE (SOAR) HIGH",
"WALNUTWOOD HIGH (INDEPENDENT STUDY)": "WALNUTWOOD HIGH (INDEPENDENT STUDY)",
"WILLIAM J. (PETE) KNIGHT HIGH": "WILLIAM J (PETE) KNIGHT",
"YOSEMITE SCHOOL COMMUNITY DAY (7-12)": "YOSEMITE SCHOOL COMMUNITY DAY 7-12",
}

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In [8]: # Abbreviations are common in UC names, rare in CAASPP names.
# The below abbreviations are unambiguous, so we can expand and then match vs trying to match raw and expanded names.
abbreviations = {
    "ACD": "ACADEMY", # Not used in CAASPP
    "ACAD": "ACADEMY", # Could also mean "ACADEMIES". Not used in CAASPP.
    "ACDM": "ACADEMY",
    "ACDMY": "ACADEMY", # Not used in CAASPP
    "CHAR": "CHARTER", # Not used in CAASPP
    "CHARTR": "CHARTER",
    "CHTR": "CHARTER", # Not used in CAASPP
    "COL": "COLLEG", # Could be "COLLEGE" or "COLLEGIATE". Not used in CAASPP
    "COLG": "COLLEG", # Could be "COLLEGE" or "COLLEGIATE". Not used in CAASPP
    "COLL": "COLLEG", # Could be "COLLEGE" or "COLLEGIATE".
    # "CTR": "?", # Could be "CENTER" or "CHARTER"
    "CTRL": "CENTRAL",
    "HS": "HIGH SCHOOL", # Used 3x in CAASPP
    "INST": "INSTITUTE", # Not used in CAASPP
    "INTERNATIO": "INTERNATIONAL",
    "JR": "JUNIOR",
    "JR.": "JUNIOR", # Only CAASPP
    "JUNIOR-SENIOR": "JUNIOR SENIOR",
    "JUNIOR/SENIOR": "JUNIOR SENIOR",
    "JR/SR": "JUNIOR SENIOR",
    "JR./SR.": "JUNIOR SENIOR",
    "LDRSHP": "LEADERSHIP",
    "LRN": "LEARNING",
    "LRNG": "LEARNING", # Not used in CAASPP
    "L.A.": "LOS ANGELES", # Only CAASPP
    "MATHEMA": "MATH",
    "MATHEMATICS": "MATH",
    "MT": "MOUNT",
    "MT.": "MOUNT",
    "ONE.CHARTER": "ONE CHARTER", # Only CAASPP
    "PREP": "PREPARATORY", # Common use vs strictly abbreviation, in CAASPP.
    "SC": "SCHOOL", # Not used in CAASPP
    "SCH": "SCHOOL", # Used once in CAASPP
    "SCHL": "SCHOOL", # Not used in CAASPP
    "SCI": "SCIENCE", # Not used in CAASPP
    "SF": "SAN FRANCISCO",
    "SIERRA@HOME": "SIERRA AT HOME", # Only CAASPP
    "SR": "SENIOR",
    "SR.": "SENIOR", # Only CAASPP
    "S.F.": "SAN FRANCISCO", # Only CAASPP
    # "TECH": "TECHNOLOGY", # Could also mean "TECHNICAL", in CAASPP
}

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"TECHLGY": "TECHNOLOGY",
"W.E.B.": "W E B", # Only CAASPP
"-SENIOR": "SENIOR", # Only CAASPP
"&": "AND",
"@": "-", # Should be "AT"?
"@MCATEER": "- MCATEER",
}

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In [9]: ##### Formatting
# Hyphens -- Most often, a hyphen surrounded by spaces indicates that a location follows, whereas
# a hyphen not surrounded by whitespace is a hyphenated name, two locations, K-12, etc. This is true in CAASPP and UC data.
# But the use of hyphens and whitespace around hyphens are not consistent. E.g., in CAASPP data:
# ALLIANCE GERTZ-RESSLER RICHARD MERKIN 6-12 COMPLEX|Los Angeles
# ALLIANCE LEICHTMAN-LEVINE FAMILY FOUNDATION ENVIRO|Los Angeles
# ALLIANCE MARINE - INNOVATION AND TECHNOLOGY 6-12 C|Los Angeles
# ALLIANCE OUCHI-O'DONOVAN 6-12 COMPLEX|Los Angeles
# OPTIONS FOR YOUTH - DUARTE|Los Angeles
# OPTIONS FOR YOUTH-ACTON|Los Angeles
# CHAMPS - CHARTER HIGH OF ARTS-MULTIMEDIA AND PERFORMIN
# CALIFORNIA PACIFIC CHARTER - SAN DIEGO|San Diego
# CALIFORNIA PACIFIC CHARTER- LOS ANGELES|Los Angeles
# In UC data:
# ALLIANCE GERTZ-RESSLER HIGH SCHOOL|Los Angeles
# ALLIANCE LEICHTMAN-LEVINE ESHS|Los Angeles
# ALLIANCE MIT 6-12 COMPLEX|Los Angeles
# ALLIANCE OUCHI-O'DONOVAN 6-12|Los Angeles
# CONNECTING WATERS - EAST BAY|Stanislaus
# CONNECTING WATERS-CTRL VALLEY|Stanislaus
# HIGH TECH HIGH - NORTH COUNTY|San Diego
# HIGH TECH HIGH LA|Los Angeles

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In [10]: # testSchoolDetails
testSchoolDetails = {} # Key is ID/SCHOOL/County/CITY/District/ZIP/ExpandedSCHOOL. Value is number of records with this key, typically 2 for Math + E
for index, row in dfTests.iterrows():
    school = row["School Name"].upper()

    # Generate stdSchool, which is tokenized and used to compute expandedSchool.
    if school in testParens:
        stdSchool = testParens[school]
    else:
        stdSchool = re.sub(r", INC$", "", school)
        stdSchool = re.sub(r",", "", stdSchool) # Remove commas
        stdSchool = re.sub(r"([A-Z]): ", r"\1 - ", stdSchool) # Remove colon (CAASPP-specific)
        stdSchool = re.sub(r"(?:ALTERNATIVE|CONTINUATION|\\(ALTERNATIVE\\)|\\(CONTINUATION\\))$", "", stdSchool) # Remove trailing ALTERNATIVE/CONTINUAT
        stdSchool = re.sub(r"^(\.?)* \((.*)\)", r"\2 \1", stdSchool) # Deal with text in parentheses, e.g. "LINCOLN (ABRAHAM) HIGH" -> "ABRAHAM LINCOLN

    # Expand abbreviations
    tokens = stdSchool.split()
    expandedTokens = []
    expanded = False
    for token in tokens:
        if token in abbreviations:

```

```

        expandedTokens.append(abbreviations[token])
        expanded = True
    else:
        noPunctToken = re.sub(r"\.$", "", token) # Remove periods from abbreviations: not used in UC data
        expandedTokens.append(noPunctToken)
    expandedSchool = " ".join(expandedTokens)

    key = "|".join([row["School Code"], school, code2County[row["County Code"]], code2City[row["School Code"]], row["District Name"], code2ZIP[row["Sc
    if row["Total Students Enrolled"] == "*":
        nStudents = 0
    else:
        nStudents = int(row["Total Students Enrolled"])
    if key in testSchoolDetails:
        testSchoolDetails[key][0] += 1
        testSchoolDetails[key][1] += nStudents
    else:
        testSchoolDetails[key] = [1, nStudents]

print(len(testSchoolDetails))

```

2715

```

In [11]: # Create testSchools{} and testStdSchools{} structures that will be used for matching. If testID == "multiple", the match will be skipped.
testID2Details = {}
testRawSchool2Details = {} # Key is school (not expanded and no county)
testSchools = {} # Key is expandedSchool|county
testSchool2Details = {} # Key is expandedSchool|county
testStdSchools = {} # Key is stdSchool|county: drop trailing "SENIOR HIGH SCHOOL" etc. Value is CAASPP school ID or "multiple" if it is not a
testStdSchool2Details = {} # Key is stdSchool|county. Value is \n-joined set of testSchoolDetails keys (id, school, county, city, district, zip, expa
for key in testSchoolDetails:
    (id, school, county, city, district, zip, expandedSchool) = key.split("|")

    if id in testID2Details:
        print("Duplicate ID")
    else:
        testID2Details[id] = key

    if school in testRawSchool2Details:
        testRawSchool2Details[school] = "multiple"
    else:
        testRawSchool2Details[school] = key

    schoolCounty = "|".join([expandedSchool, county])
    if schoolCounty in testSchools:
        testSchools[schoolCounty] = "multiple"
    else:
        testSchools[schoolCounty] = id
        testSchool2Details[schoolCounty] = key

    # Further standardization, starting with expandedSchool
    stdSchool = re.sub(r" SCHOOL$", "", expandedSchool)
    stdSchool = re.sub(r"(?:UNION |(?:JUNIOR )?SENIOR )*(?:HIGH|SECONDARY)$", "", stdSchool)
    stdSchool = re.sub(r"(CHARTER|HIGH|MIDDLE) SCHOOL", r"\1", stdSchool)

```

```

stdSchool = re.sub(r"CENTER|CENTRAL|CHARTER", "CTR", stdSchool)
stdSchool = re.sub(r"COLLEGE|COLLEGIATE", "COLLEG", stdSchool)
stdSchool = re.sub(r"TECHNICAL|TECHNOLOGY", "TECH", stdSchool)
stdSchool = re.sub(r" [1-9K]-12\b", "", stdSchool) # Delete "K-12" etc.
stdSchool = re.sub(r" *- *", " ", stdSchool) # Replace hyphen with a single space
schoolCounty = "|".join([stdSchool, county])
if schoolCounty in testStdSchools:
    testStdSchools[schoolCounty] = "multiple"
    testStdSchool2Details[schoolCounty] = "\n".join([testStdSchool2Details[schoolCounty], key])
else:
    testStdSchools[schoolCounty] = id
    testStdSchool2Details[schoolCounty] = key

print(len(testSchools))
print(len(testStdSchools))

```

2711

2708

In [12]: *# Find duplicate stdSchool names. Especially Look for duplicate stdSchool names that result from normal and ALTERNATIVE/CONTINUATION schools with the*

```

for schoolCounty in sorted(testStdSchools):
    if testStdSchools[schoolCounty] == "multiple":
        print(schoolCounty)
        print(testStdSchool2Details[schoolCounty])
        print()

```


ACADEMY OF INNOVATION|Riverside
0135574|ACADEMY OF INNOVATION|Riverside|NORCO|Corona-Norco Unified|92860|ACADEMY OF INNOVATION
3330503|ACADEMY OF INNOVATION|Riverside|HEMET|Hemet Unified|92543|ACADEMY OF INNOVATION

ALESSANDRO|Riverside
3330065|ALESSANDRO HIGH|Riverside|HEMET|Hemet Unified|92543|ALESSANDRO HIGH
0105940|ALESSANDRO|Riverside|MORENO VALLEY|Moreno Valley Unified|92553|ALESSANDRO

JORDAN|Los Angeles
1934470|JORDAN HIGH|Los Angeles|NORTH/LONG BEACH|Long Beach Unified|90805|JORDAN HIGH
1934454|JORDAN HIGH|Los Angeles|GREENMEAD/LOS ANGELES|Los Angeles Unified|90002|JORDAN HIGH

MONTEREY|Los Angeles
1933332|MONTEREY HIGH (CONTINUATION)|Los Angeles|BURBANK|Burbank Unified|91506|MONTEREY HIGH
1931989|MONTEREY CONTINUATION|Los Angeles|LOS ANGELES|Los Angeles Unified|90022|MONTEREY

SAN ANTONIO|Los Angeles
1931807|SAN ANTONIO HIGH (CONTINUATION)|Los Angeles|LA VERNE|Claremont Unified|91711|SAN ANTONIO HIGH
1930551|SAN ANTONIO CONTINUATION|Los Angeles|SOTO/HUNTINGTON PARK|Los Angeles Unified|90255|SAN ANTONIO

SAVA SACRAMENTO ACADEMIC AND VOCATIONAL ACADEMY|Sacramento
0137281|SAVA - SACRAMENTO ACADEMIC AND VOCATIONAL ACADEMY|Sacramento|LAGUNA/ELK GROVE|SAVA - Sacramento Academic and Vocational|95758|SAVA - SACRAMENTO
ACADEMIC AND VOCATIONAL ACADEMY
0137406|SAVA - SACRAMENTO ACADEMIC AND VOCATIONAL ACADEMY|Sacramento|COLONIAL/SACRAMENTO|SAVA - Sacramento Academic and Vocational|95820|SAVA - SACRAMEN
TO ACADEMIC AND VOCATIONAL ACADEMY
0114272|SAVA - SACRAMENTO ACADEMIC AND VOCATIONAL ACADEMY|Sacramento|NORTH NATOMAS/SACRAMENTO|SAVA - Sacramento Academic and Vocational|95834|SAVA - SAC
RAMENTO ACADEMIC AND VOCATIONAL ACADEMY

```
In [13]: ##### Read the UC admission data
# The Count field indicates what is being counted: "Adm", "App", or "Enr".
directory = "UCAdmissions/2024"
files = {"FR ENR_data.All.csv": "all",
        "FR ENR_data.Berkeley.csv": "Berkeley",
        "FR ENR_data.Davis.csv": "UC Davis",
        "FR ENR_data.Irvine.csv": "UCI",
        "FR ENR_data.LosAngeles.csv": "UCLA",
        "FR ENR_data.Merced.csv": "UC Merced",
        "FR ENR_data.Riverside.csv": "UCR",
        "FR ENR_data.SanDiego.csv": "UCSD",
        "FR ENR_data.SantaBarbara.csv": "UCSB",
        "FR ENR_data.SantaCruz.csv": "UCSC"}

dfs = []
for filename in files.keys():
    dfOneFile = pd.read_csv("/{}/".join([directory, filename]), delimiter=",")
    dfOneFile["Campus"] = files[filename]
    dfs.append(dfOneFile)

dfUC = pd.concat(dfs, ignore_index=True)
dfUC["City"] = dfUC["City"].str.upper() # Convert City to uppercase
print(dfUC.shape[0])
print(dfUC)
```

```

79224
      Calculation1      School      City County/State/ Territory Count uad_uc_ethn_7_cat Pivot Field Values Ca
mpus
0      A B MILLER HIGH SCHOOL50944      A B MILLER HIGH SCHOOL      FONTANA      San Bernardino      Enr      Hispanic/ Latinx      24
all
1      A B MILLER HIGH SCHOOL50944      A B MILLER HIGH SCHOOL      FONTANA      San Bernardino      Adm      Hispanic/ Latinx      56
all
2      A B MILLER HIGH SCHOOL50944      A B MILLER HIGH SCHOOL      FONTANA      San Bernardino      App      Hispanic/ Latinx      66
all
3      A B MILLER HIGH SCHOOL50944      A B MILLER HIGH SCHOOL      FONTANA      San Bernardino      Adm      African American      7
all
4      A B MILLER HIGH SCHOOL50944      A B MILLER HIGH SCHOOL      FONTANA      San Bernardino      App      African American      9
all
...      ...      ...      ...      ...      ...      ...
...
79219 YUCAIPA SENIOR HIGH SCHOOL53820 YUCAIPA SENIOR HIGH SCHOOL      YUCAIPA      San Bernardino      Adm      Hispanic/ Latinx      6
UCSC
79220 YUCAIPA SENIOR HIGH SCHOOL53820 YUCAIPA SENIOR HIGH SCHOOL      YUCAIPA      San Bernardino      App      Hispanic/ Latinx      11
UCSC
79221 YUCAIPA SENIOR HIGH SCHOOL53820 YUCAIPA SENIOR HIGH SCHOOL      YUCAIPA      San Bernardino      Adm      All      16
UCSC
79222 YUCAIPA SENIOR HIGH SCHOOL53820 YUCAIPA SENIOR HIGH SCHOOL      YUCAIPA      San Bernardino      App      All      23
UCSC
79223 YUCCA VALLEY HIGH SCHOOL53818 YUCCA VALLEY HIGH SCHOOL YUCCA VALLEY      San Bernardino      Adm      All      3
UCSC

```

[79224 rows x 8 columns]

```

In [14]: # Compare ucCounties, entitiesCounties
ucCounties = dfUC["County/State/ Territory"].value_counts()
for county in ucCounties.sort_index().index:
    print(county, ucCounties[county], entitiesCounties[county])

```

Alameda 4127 401
Amador 54 17
Butte 277 101
Calaveras 68 28
Colusa 85 25
Contra Costa 2874 306
Del Norte 21 17
El Dorado 332 90
Fresno 1982 382
Glenn 50 34
Humboldt 221 118
Imperial 291 85
Inyo 44 29
Kern 1500 318
Kings 281 82
Lake 74 49
Lassen 16 34
Los Angeles 21068 2290
Madera 276 86
Marin 662 100
Mariposa 22 16
Mendocino 150 74
Merced 671 124
Modoc 14 13
Mono 44 15
Monterey 770 166
Napa 347 46
Nevada 95 45
Orange 6325 677
Placer 1001 158
Plumas 25 16
Riverside 4562 545
Sacramento 2945 401
San Benito 93 41
San Bernardino 3709 618
San Diego 6680 829
San Francisco 1159 127
San Joaquin 1928 263
San Luis Obispo 429 96
San Mateo 1819 210
Santa Barbara 751 150
Santa Clara 4293 460
Santa Cruz 569 99
Shasta 199 119
Sierra 1 7
Siskiyou 57 66
Solano 799 113
Sonoma 861 227
Stanislaus 1040 223
Sutter 227 59
Tehama 35 54
Trinity 4 28

Tulare 820 238
 Tuolumne 42 40
 Ventura 1853 238
 Yolo 458 71
 Yuba 124 48

```
In [15]: ##### Read the UC admission GPA data
directory = "UCAdmissions/2024"
files = {"FR_GPA_by_Year_data.All.csv": "all",
        "FR_GPA_by_Year_data.Berkeley.csv": "Berkeley",
        "FR_GPA_by_Year_data.Davis.csv": "UC Davis",
        "FR_GPA_by_Year_data.Irvine.csv": "UCI",
        "FR_GPA_by_Year_data.LosAngeles.csv": "UCLA",
        "FR_GPA_by_Year_data.Merced.csv": "UC Merced",
        "FR_GPA_by_Year_data.Riverside.csv": "UCR",
        "FR_GPA_by_Year_data.SanDiego.csv": "UCSD",
        "FR_GPA_by_Year_data.SantaBarbara.csv": "UCSB",
        "FR_GPA_by_Year_data.SantaCruz.csv": "UCSC"}

dfs = []
for filename in files.keys():
    dfOneFile = pd.read_csv("/".join([directory, filename]), delimiter="\t")
    dfOneFile["Campus"] = files[filename]
    dfs.append(dfOneFile)
dfGPA = pd.concat(dfs, ignore_index=True)
dfGPA["City"] = dfGPA["City"].str.upper() # Convert City to uppercase
print(dfGPA.shape[0])
print(dfGPA)
```

```
12183
```

	Calculation1	School	City	County/State/Country	App GPA	Adm GPA	Enrl GPA	Campus
0	A B MILLER HIGH SCHOOL050944	A B MILLER HIGH SCHOOL	FONTANA	San Bernardino	3.75	3.83	3.96	all
1	ABLE CHARTER054651	ABLE CHARTER	STOCKTON	San Joaquin	3.77	3.79	NaN	all
2	ABRAHAM LINCOLN HIGH SCHOOL051520	ABRAHAM LINCOLN HIGH SCHOOL	LOS ANGELES	Los Angeles	3.69	3.88	3.94	all
3	ABRAHAM LINCOLN HIGH SCHOOL052910	ABRAHAM LINCOLN HIGH SCHOOL	SAN FRANCISCO	San Francisco	3.82	3.95	4.03	all
4	ABRAHAM LINCOLN HIGH SCHOOL053075	ABRAHAM LINCOLN HIGH SCHOOL	SAN JOSE	Santa Clara	3.94	4.04	4.16	all
...	...	...	...	...	...	...	...	...
12178	YGNACIO VALLEY HIGH SCHOOL050663	YGNACIO VALLEY HIGH SCHOOL	CONCORD	Contra Costa	3.52	3.87	NaN	UCSC
12179	YORBA LINDA HIGH SCHOOL054321	YORBA LINDA HIGH SCHOOL	YORBA LINDA	Orange	3.80	4.02	NaN	UCSC
12180	YOSEMITE HIGH SCHOOL052176	YOSEMITE HIGH SCHOOL	OAKHURST	Madera	3.99	4.07	NaN	UCSC
12181	YUBA CITY HIGH SCHOOL053815	YUBA CITY HIGH SCHOOL	YUBA CITY	Sutter	3.71	4.04	NaN	UCSC
12182	YUCAIPA SENIOR HIGH SCHOOL053820	YUCAIPA SENIOR HIGH SCHOOL	YUCAIPA	San Bernardino	3.86	4.02	NaN	UCSC

[12183 rows x 8 columns]

```
In [16]: # Dictionary of closest UC for each county created using https://www.universityofcalifornia.edu/about-us/information-center/k-12-directory
countyCampus = {
    "all": "all",
    "Alameda": "Berkeley",
    "Amador": "UC Davis",
    "Butte": "UC Davis",
    "Calaveras": "UC Merced",
    "Colusa": "UC Davis",
    "Contra Costa": "Berkeley",
```

```
"Del Norte": "UC Davis",
"El Dorado": "UC Davis",
"Fresno": "UC Merced",
"Glenn": "UC Davis",
"Humboldt": "UC Davis",
"Imperial": "UCSD",
"Inyo": "UC Merced",
"Kern": "UCSB",
"Kings": "UC Merced",
"Lake": "UC Davis",
"Lassen": "UC Davis",
"Los Angeles": "UCLA",
"Madera": "UC Merced",
"Marin": "Berkeley",
"Mariposa": "UC Merced",
"Mendocino": "UC Davis",
"Merced": "UC Merced",
"Modoc": "UC Davis",
"Mono": "UC Merced",
"Monterey": "UCSC",
"Napa": "Berkeley",
"Nevada": "UC Davis",
"Orange": "UCI",
"Placer": "UC Davis",
"Plumas": "UC Davis",
"Riverside": "UCR",
"Sacramento": "UC Davis",
"San Benito": "UCSC",
"San Bernardino": "UCR",
"San Diego": "UCSD",
"San Francisco": "Berkeley",
"San Joaquin": "UC Davis",
"San Luis Obispo": "UCSB",
"San Mateo": "Berkeley",
"Santa Barbara": "UCSB",
"Santa Clara": "UCSC",
"Santa Cruz": "UCSC",
"Shasta": "UC Davis",
"Sierra": "UC Davis",
"Siskiyou": "UC Davis",
"Solano": "UC Davis",
"Sonoma": "Berkeley",
"Stanislaus": "UC Merced",
"Sutter": "UC Davis",
"Tehama": "UC Davis",
"Trinity": "UC Davis",
"Tulare": "UC Merced",
"Tuolumne": "UC Merced",
"Ventura": "UCSB", # Would be UCLA by max count of high schools
"Yolo": "UC Davis",
"Yuba": "UC Davis",
}
```

```

homeCountyCampus = {
    "Alameda": "Berkeley",
    "Yolo": "UC Davis",
    "Orange": "UCI",
    "Los Angeles": "UCLA",
    "Merced": "UC Merced",
    "Riverside": "UCR",
    "Santa Barbara": "UCSB",
    "Santa Cruz": "UCSC",
    "San Diego": "UCSD",
}

areaLA = {}
for county in countyCampus:
    if county in ("Los Angeles", "Riverside", "San Bernardino", "Ventura"):
        areaLA[county] = "Y"
    else:
        areaLA[county] = "N"

```

```

In [17]: # Understand "ALL" in the UC data: it seems to be the count of unique students as opposed to the sum across campuses
for index, row in dfUC.iterrows():
    if (row["School"] == "CANYON CREST ACADEMY" or row["School"] == "GOMPERS PREPARATORY ACADEMY") and row["uad_uc_ethn_7_cat"] == "All" and (row["Cou
        print("|".join(map(str, row.values)))

```

CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|332|all
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|438|all
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|88|all
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|101|all
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|34|Berkeley
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|327|Berkeley
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|3|Berkeley
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|22|Berkeley
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|91|UC Davis
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|346|UC Davis
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|14|UC Davis
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|104|UCI
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|352|UCI
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|27|UCI
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|68|UCI
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|33|UCLA
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|380|UCLA
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|7|UCLA
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|67|UCLA
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|75|UC Merced
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|78|UC Merced
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|10|UC Merced
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|10|UC Merced
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|198|UCR
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|225|UCR
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|50|UCR
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|59|UCR
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|63|UCSD
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|384|UCSD
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|40|UCSD
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|99|UCSD
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|127|UCSB
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|363|UCSB
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|16|UCSB
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|29|UCSB
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|Adm|All|175|UCSC
 CANYON CREST ACADEMY50893|CANYON CREST ACADEMY|SAN DIEGO|San Diego|App|All|275|UCSC
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|Adm|All|23|UCSC
 GOMPERS PREPARATORY ACADEMY54353|GOMPERS PREPARATORY ACADEMY|SAN DIEGO|San Diego|App|All|37|UCSC

```

In [18]: # ucSchoolDetails
# Key is ID|SCHOOL|County|CITY|ExpandedSCHOOL
# Value is [AllApp, AllAdm, AllEnr, Found, FoundID], where:
# AllApp is the number of applications in the All category
# Found is a code indicating whether and how we found this school in the CAASPP data with values:
# 0 => not found.
# 1 => found using initial key match.
# 2 => found after standardization.
# 3 => found by ChatGPT.
ucSchoolDetails = {}
count2i = {"App": 0,
           "Adm": 1,
           "Enr": 2}

```

```

for index, row in dfUC.iterrows():
    schoolID = row["Calculation1"]
    code = re.sub(r"^\.?(\d{5})$", r"\1", schoolID)
    if not re.fullmatch(r"5\d{4}", code): # Five digit code, and the first digit is always 5.
        print(schoolID, code)

    # Expand abbreviations
    school = row["School"]
    stdSchool = re.sub(r"([A-Z])&([A-Z])", r"\1 & \2", school) # R&D -> R & D
    tokens = stdSchool.split()
    expandedTokens = []
    expanded = False
    for token in tokens:
        if token in abbreviations:
            expandedTokens.append(abbreviations[token])
            expanded = True
        else:
            expandedTokens.append(token)
    expandedSchool = " ".join(expandedTokens)

    key = "|".join([code, school, row["County/State/ Territory"], row["City"], expandedSchool])
    if key not in ucSchoolDetails:
        ucSchoolDetails[key] = [0, 0, 0, 0, ""]
    if row["uad_uc_ethn_7_cat"] == "All":
        ucSchoolDetails[key][count2i[row["Count"]]] += row["Pivot Field Values"]

print(len(ucSchoolDetails))

```

1420

```

In [19]: ucSchools = {} # Key is SCHOOL/County
ucSchool2StdSchool = {}
ucStdSchools = {} # Format is same as ucSchools but school name is standardized: drop trailing "SENIOR HIGH SCHOOL" etc.
ucStdSchool2Details = {}
for key in ucSchoolDetails:
    (id, school, county, city, expandedSchool) = key.split("|")

    schoolCounty = "|".join([expandedSchool, county])
    if schoolCounty in ucSchools:
        ucSchools[schoolCounty] += 1
    else:
        ucSchools[schoolCounty] = 1

    # Deal with standardization
    stdSchool = re.sub(r" SCHOOL$", "", expandedSchool)
    stdSchool = re.sub(r" (?:(?:UNION |(?:(?:JUNIOR )?SENIOR )*(?:HIGH|SECONDARY))$", "", stdSchool)
    stdSchool = re.sub(r"(CHARTER|HIGH|MIDDLE) SCHOOL", r"\1", stdSchool)
    stdSchool = re.sub(r"CENTER|CENTRAL|CHARTER", "CTR", stdSchool)
    stdSchool = re.sub(r"COLLEGE|COLLEGIATE", "COLLEG", stdSchool)
    stdSchool = re.sub(r"TECHNICAL|TECHNOLOGY", "TECH", stdSchool)
    stdSchool = re.sub(r"[1-9K]-12\b", "", stdSchool) # Delete "K-12" etc.
    stdSchool = re.sub(r" *- *", " ", stdSchool) # Replace hyphen with a single space
    # stdSchool3 = re.sub(r" INST OF .*\/", " INSTITUTE/", stdSchool)

```



```

ucSchool2StdSchool[expandedSchool] = stdSchool
schoolCounty = "|".join([stdSchool, county])
if schoolCounty in ucStdSchools:
    ucStdSchools[schoolCounty] += 1
    ucStdSchool2Details[schoolCounty] = "\n".join([ucStdSchool2Details[schoolCounty], key])
else:
    ucStdSchools[schoolCounty] = 1
    ucStdSchool2Details[schoolCounty] = key

print(len(ucSchools))
print(len(ucStdSchools))

```

1418

1417

```

In [20]: # Find duplicate school names
for schoolCounty in sorted(ucStdSchools):
    if ucStdSchools[schoolCounty] > 1:
        print(ucStdSchool2Details[schoolCounty])

```

```

50412|SWEETWATER SECONDARY SCHOOL|San Diego|CHULA VISTA|SWEETWATER SECONDARY SCHOOL
52105|SWEETWATER UNION HIGH SCHOOL|San Diego|NATIONAL CITY|SWEETWATER UNION HIGH SCHOOL
54539|WONDERFUL COLLEGE PREP ACADEMY|Kern|DELANO|WONDERFUL COLLEGE PREPARATORY ACADEMY
55584|WONDERFUL COLLEGE PREP ACADEMY|Kern|LOST HILLS|WONDERFUL COLLEGE PREPARATORY ACADEMY
51505|WOODROW WILSON HIGH SCHOOL|Los Angeles|LONG BEACH|WOODROW WILSON HIGH SCHOOL
51840|WOODROW WILSON HIGH SCHOOL|Los Angeles|LOS ANGELES|WOODROW WILSON HIGH SCHOOL

```

```

In [21]: # Try to find each UC school in the set of CAASPP schools, recording which ones we found.
# We will go through ucSchoolDetails. If we find a matching CAASPP school:
# * Store its ID in the value in ucSchoolDetails, else set the value to "".
# * Record that this CAASPP school has been "used".
# Duplicate UC schools and duplicate CAASPP matching schools are both skipped and will be handled through ChatGPT / Lookup.psv
nFound = 0
matchedTestSchools = {} # CAASPP schools that matched a UC school. Key is ??.
tokensNotMatched = {}
usedTestIDs = {}
for key in ucSchoolDetails:
    (id, school, county, city, expandedSchool) = key.split("|")
    schoolCounty = "|".join([expandedSchool, county])
    stdSchool = ucSchool2StdSchool[expandedSchool]
    stdSchoolCounty = "|".join([stdSchool, county])

    # Try various types of heuristic matching & record the results. Then we'll try ChatGPT on the ones that do not unambiguously match.
    # Heuristic matches are based on school name and county only: straight match, match with "HIGH SCHOOL" (etc.) truncated from both.
    if ucStdSchools[stdSchoolCounty] > 1:
        continue # Two schools in UC data with same standardized name, same county
    if schoolCounty in testStdSchools:
        testID = testStdSchools[stdSchoolCounty] # The use of testStdSchools rather than testSchools is intentional: more conservative rejection of t
        if testID == "multiple":
            continue
        if testID in usedTestIDs:
            print("Previously used 1:", testID) # Never happens
            continue

```

```

        ucSchoolDetails[key][3] = 1
        ucSchoolDetails[key][4] = testID
        usedTestIDs[testID] = "|".join([key, "1"])
        nFound += 1
    elif stdSchoolCounty in testStdSchools:
        testID = testStdSchools[stdSchoolCounty]
        if testID == "multiple":
            continue
        if testID in usedTestIDs:
            print("Previously used 2:", testID) # Never happens
            continue
        ucSchoolDetails[key][3] = 2
        ucSchoolDetails[key][4] = testID
        usedTestIDs[testID] = "|".join([key, "2"])
        nFound += 1
    else:
        for token in stdSchool.split():
            if token in tokensNotMatched:
                tokensNotMatched[token] += 1
            else:
                tokensNotMatched[token] = 1

print(nFound)

```

1132

In [22]:

```

# Explore
query = r"^ENTERPRISE|^MONTEREY|ROSELAND|^SAN ANTONIO|SWEETWATER"
print("\nUC Details")
for key in ucSchoolDetails:
    (id, school, county, city, expandedSchool) = key.split("|")
    if re.search(query, school):
        print(key, ucSchoolDetails[key])

print("\nCAASPP")
for schoolCounty in sorted(testSchools):
    if re.search(query, schoolCounty):
        key = testSchool2Details[schoolCounty]
        print(key, testSchoolDetails[key])

```

UC Details

```
52580|ENTERPRISE HIGH SCHOOL|Shasta|REDDING|ENTERPRISE HIGH SCHOOL [114, 52, 22, 2, '4532750']
52050|MONTEREY HIGH SCHOOL|Monterey|MONTEREY|MONTEREY HIGH SCHOOL [566, 241, 56, 2, '2732808']
50902|MONTEREY TRAIL HIGH SCHOOL|Sacramento|ELK GROVE|MONTEREY TRAIL HIGH SCHOOL [828, 331, 111, 2, '0105916']
54850|ROSELAND COLLEGIATE PREP|Sonoma|SANTA ROSA|ROSELAND COLLEGIATE PREPARATORY [111, 49, 13, 0, '']
54044|ROSELAND UNIVERSITY PREP|Sonoma|SANTA ROSA|ROSELAND UNIVERSITY PREPARATORY [163, 68, 16, 0, '']
50412|SWEETWATER SECONDARY SCHOOL|San Diego|CHULA VISTA|SWEETWATER SECONDARY SCHOOL [0, 6, 6, 0, '']
52105|SWEETWATER UNION HIGH SCHOOL|San Diego|NATIONAL CITY|SWEETWATER UNION HIGH SCHOOL [647, 266, 60, 0, '']
```

CAASPP

```
1030469|ENTERPRISE ALTERNATIVE|Fresno|CLOVIS|Clovis Unified|93611|ENTERPRISE ALTERNATIVE [2, 50]
1033422|ENTERPRISE HIGH|Fresno|KERNAN|Kerman Unified|93630|ENTERPRISE HIGH [2, 50]
4532750|ENTERPRISE HIGH|Shasta|REDDING|Shasta Union High|96002|ENTERPRISE HIGH [2, 492]
2730232|MONTEREY COUNTY HOME CHARTER|Monterey|JOHN STEINBECK/SALINAS|Monterey County Office of Education|93901|MONTEREY COUNTY HOME CHARTER [2, 88]
1933332|MONTEREY HIGH (CONTINUATION)|Los Angeles|BURBANK|Burbank Unified|91506|MONTEREY HIGH [2, 87]
2732808|MONTEREY HIGH|Monterey|MONTEREY|Monterey Peninsula Unified|93940|MONTEREY HIGH [2, 680]
2766092|MONTEREY PENINSULA UNIFIED DISTRICT LEVEL PROGRAM|Monterey|MONTEREY|Monterey Peninsula Unified|93940|MONTEREY PENINSULA UNIFIED DISTRICT LEVEL PROGRAM [2, 0]
0112896|MONTEREY PENINSULA UNIFIED SCHOOL DISTRICT COMMUNI|Monterey|SEASIDE|Monterey Peninsula Unified|93955|MONTEREY PENINSULA UNIFIED SCHOOL DISTRICT COMMUNI [1, 0]
0105916|MONTEREY TRAIL HIGH|Sacramento|ELK GROVE|Elk Grove Unified|95624|MONTEREY TRAIL HIGH [2, 1136]
1931989|MONTEREY CONTINUATION|Los Angeles|LOS ANGELES|Los Angeles Unified|90022|MONTEREY [2, 0]
0101923|ROSELAND CHARTER|Sonoma|ROSELAND/SANTA ROSA|Roseland Charter|95407|ROSELAND CHARTER [2, 350]
1931807|SAN ANTONIO HIGH (CONTINUATION)|Los Angeles|LA VERNE|Claremont Unified|91711|SAN ANTONIO HIGH [2, 74]
4935862|SAN ANTONIO HIGH (CONTINUATION)|Sonoma|CASA GRANDE/PETALUMA|Petaluma Joint Union High|94952|SAN ANTONIO HIGH [2, 54]
1930551|SAN ANTONIO CONTINUATION|Los Angeles|SOTO/HUNTINGTON PARK|Los Angeles Unified|90255|SAN ANTONIO [2, 80]
3738226|SWEETWATER HIGH|San Diego|NATIONAL CITY|Sweetwater Union High|91950|SWEETWATER HIGH [2, 1279]
3768411|SWEETWATER UNION HIGH DISTRICT LEVEL PROGRAM|San Diego|CHULA VISTA|Sweetwater Union High|91911|SWEETWATER UNION HIGH DISTRICT LEVEL PROGRAM [2, 622]
```

```
In [23]: # Output Test Schools to CSV files (one per county) to upload to ChatGPT
files = {}
for county in ucCounties.sort_index().index:
    countyNoSpace = re.sub(r" ", "", county)
    filename = "".join(["latestOutput/referenceSchools", countyNoSpace, ".csv"])
    files[county] = open(filename, "w")

for key in testSchoolDetails: # Key is ID/SCHOOL/County/CITY/District/ZIP/ExpandedSCHOOL
    (id, school, county, city, district, zip, expandedSchool) = key.split("|")
    if county not in files:
        continue
    file = files[county]
    file.write("|".join([school, city, district, zip, id]) + "\n")

for county in files:
    files[county].close()
```

```
In [24]: # Output all Test Schools to a single CSV file to upload to ChatGPT to match UC schools not found on the first pass
with open("latestOutput/referenceSchoolsAll.csv", "w") as file:
    for key in testSchoolDetails: # Key is ID/SCHOOL/County/CITY/District/ZIP/ExpandedSCHOOL
        (id, school, county, city, district, zip, expandedSchool) = key.split("|")
        file.write("|".join([school, city, district, zip, id]) + "\n")
```

```
In [25]: # Output UC schools that don't have a match yet to enter in Copilot Bing or ChatGPT UI
for targetCounty in ucCounties.sort_index().index:
    print("\n" + targetCounty)
    for key in ucSchoolDetails:
        (id, school, county, city, expandedSchool) = key.split("|")
        if county == targetCounty and ucSchoolDetails[key][4] == "":
            print("|".join([school, city]))
```

Alameda

ASPIRE GOLDEN STATE PREP ACADEMY|OAKLAND
ASPIRE LIONEL WILSON COLG PREP|OAKLAND
BAY AREA TECHNOLOGY-BAYTECH|OAKLAND
ENVISION ACAD ARTS/TECHNOLOGY|OAKLAND
IMPACT ACAD ARTS/TECHNOLOGY|HAYWARD
LATITUDE 37.8 HS MERRITT COLLEGE|OAKLAND
LEADERSHIP PUBLIC SCH HAYWARD|HAYWARD
LODESTAR LIGHTHOUSE COMM CHARTER SC|OAKLAND
OAKLAND MILITARY INST COL PREP|OAKLAND
SOJOURNER TRUTH ACADEMY|OAKLAND

Amador

Butte

INSPIRE SCHOOL OF ARTS AND SCI|CHICO

Calaveras

Colusa

Contra Costa

ASPIRE RICHMOND CA COLG PREP|RICHMOND
CLAYTON VALLEY HIGH SCHOOL|CONCORD
CONTRA COSTA SCH PRFRMING ARTS|WALNUT CREEK
LEADERSHIP PUBLIC SCH RICHMOND|RICHMOND

Del Norte

DEL NORTE COUNTY HIGH SCHOOL|CRESCENT CITY

El Dorado

CHARTER COMMUNITY SCHOOL|PLACERVILLE
JOHN ADAMS ACADEMY - EL DORADO|EL DORADO HILLS
OCEAN GROVE CHARTER SCHOOL|PLACERVILLE
SKY MOUNTAIN CHARTER SCHOOL|PLACERVILLE

Fresno

CLOVIS ONLINE SCHOOL|CLOVIS
DUNCAN POLYTECHNICAL HS|FRESNO
FARBER SCH OF IND STUDIES & ONLINE|FRESNO
FLOYD B BUCHANAN HIGH SCHOOL|CLOVIS
PHILLIP J PATINO SCH ENTREPREN|FRESNO
THEODORE ROOSEVELT HIGH SCHOOL|FRESNO

Glenn

Humboldt

NORTHCOAST PREP & PERF ARTS A|ARCATA

Imperial

Inyo

Kern

MCFARLAND HIGH SCHOOL|MCFARLAND
ROSAMOND HIGH SCHOOL|ROSAMOND
SHERMAN E BURROUGHS HS|RIDGECREST
WONDERFUL COLLEGE PREP ACADEMY|DELANO
WONDERFUL COLLEGE PREP ACADEMY|LOST HILLS

Kings

Lake

Lassen

Los Angeles

ACADEMIC LEADERSHIP COMMUNITY|LOS ANGELES
ACADEMIES EDUCATION/EMPOWERMNT|CARSON
ACADEMIES OF ANTELOPE VALLEY|LANCASTER
ACADEMY OF SCIENTIFIC EXPLORAT|SAN FERNANDO
ALAIN LEROY LOCKE COLLEGE PREP|LOS ANGELES
ALLIANCE CINDY/BILL SIMON TECH|LOS ANGELES
ALLIANCE COLLINS FAMILY COLRDY|HUNTINGTON PARK
ALLIANCE GERTZ-RESSLER HIGH SC|LOS ANGELES
ALLIANCE JUDY BURTON TECH ACAD|LOS ANGELES
ALLIANCE LEICHTMAN-LEVINE ESHS|LOS ANGELES
ALLIANCE M&E STERN MATH SCI SCH|LOS ANGELES
ALLIANCE MARGARET M BLOOMFIELD|HUNTINGTON PARK
ALLIANCE MIT 6-12 COMPLEX|SUN VALLEY
ALLIANCE OUCHI-O'DONOVAN 6-12|LOS ANGELES
ALLIANCE PATTI & PETER ACADEMY|LOS ANGELES
ALLIANCE PIERA BARBAGLIA HSA|LOS ANGELES
ALLIANCE RENEE & MEYER LUSKIN|LOS ANGELES
ALLIANCE SMIDT TECHNOLOGY HIGH|LOS ANGELES
AMBASSADOR SCHL GLOBAL LDRSHP|LOS ANGELES
ANAHUACALMECAC INTRL UNIV PREP|LOS ANGELES
ANIMO CTY CHAMPIONS CHARTER HS|INGLEWOOD
ANIMO JACKIE ROBINSON CHRT SCH|LOS ANGELES
ANIMO LEADERSHIP CHARTER HS|INGLEWOOD
ANIMO PAT BROWN CHARTER HS|LOS ANGELES
ANIMO S LOS ANGELES CHARTER HS|LOS ANGELES
ANIMO WATTS COLLGE PREP ACDMY|LOS ANGELES
ARTS & ENTERTAINMENT VIRTUAL ACAD|CARSON
ASPIRE OLLIN UNIV PREP ACDMY|HUNTINGTON PARK
BARACK OBAMA GLOBAL PREP ACADEMY|LOS ANGELES
BIRMINGHAM COMMUNITY HS|LAKE BALBOA
BRIO COLLEGE PREP|LOS ANGELES
CA SCH ARTS SAN GABRIEL VALLEY|DUARTE
CALIFORNIA ACAD MATH & SCIENCE|CARSON
CAMINO NUEVO HIGH SCHOOL DALZE|LOS ANGELES
CATCH PREP CHARTER HIGH|LOS ANGELES
CESAR CHAVEZ LA ARTES MAGNET|SAN FERNANDO
CHARTER HS ARTS MEDIA PERFORM|VAN NUYS

CHATSWORTH HIGH SCHOOL|CHATSWORTH
CITY HONORS HIGH SCHOOL|INGLEWOOD
CLARK MAGNET HIGH SCHOOL|LA CRESCENTA
CLEVELAND HIGH SCHOOL|RESEDA
COLLEGE BRIDGE ACADEMY|LOS ANGELES
COLLEGIATE CHARTER HIGH SCHOOL|LOS ANGELES
CRENSHAW HIGH SCHOOL|LOS ANGELES
CRITICAL DESIGN AND GAMING SCH|LOS ANGELES
DANIEL PEARL MAGNET HIGH SCHL|VAN NUYS
DAV STARR JORDAN HIGH SCHOOL|LONG BEACH
DAVID STARR JORDAN HS|LOS ANGELES
DIEGO RIVERA COMM/TECH SCHOOL|LOS ANGELES
DISCOVERY CHARTER PREP SCHOOL|SYLMAR
DOWNTOWN MAGNETS HIGH SCHOOL|LOS ANGELES
DR RICHARD VLADOVIC HARBOR ACA|WILMINGTON
EARLY COLLEGE ACADEMY|LOS ANGELES
EAST COLLEGE PREP|LOS ANGELES
EAST LA PERFORMING ARTS MAGNET|LOS ANGELES
EAST LA RENAISSANCE ACADEMY|LOS ANGELES
EDWARD R ROYBAL LEARNING CNTR|LOS ANGELES
ENGINEERING & TECH ACAD AT THS|LOS ANGELES
ENVIRONMENTAL CHARTER HS|LAWNDALE
ESPERANZA COLLEGE PREP|LOS ANGELES
FELICITAS GONZALO MENDEZ HS|LOS ANGELES
FREMONT ACD ENGINEERING/DESIGN|POMONA
GAHR HIGH SCHOOL|CERRITOS
GIRLS ACADEMIC LEADERSHIP ACAD|LOS ANGELES
GREEN DESIGN STEAM ACADEMY|LOS ANGELES
HIGH TECH HIGH LA|VAN NUYS
HILDA L SOLIS LEARNING ACDMY|LOS ANGELES
HUMANITAS ACADEMY ART AND TECH|LOS ANGELES
HUMANITIES AND ARTS ACADEMY-LA|HARBOR CITY
HUNTINGTON PARK INST APLD MED|HUNTINGTON PARK
ILEAD EXPLORATION HYBRID|ACTON
ILEAD ONLINE CHARTER SCHOOL|ACTON
INTERNATIONAL STUDIES LEARNING|SOUTH GATE
JOHN BURROUGHS HIGH SCHOOL|BURBANK
JOHN GLENN HIGH SCHOOL|NORWALK
JOHN MARSHALL FUNDAMENTAL SCH|PASADENA
JUAN RODRIGUEZ CABRILLO HS|LONG BEACH
KING DREW MAGNET HS MED/SCI|LOS ANGELES
LA CENTER ENRICHED STUDIES|LOS ANGELES
LAKE BALBOA COLLEGE PREP K-12|VAN NUYS
LEADERSHIP&PUBLIC SVC VIRTUAL ACA|LOS ANGELES
LEGACY HIGH SCHOOL VAPA|SOUTH GATE
LENNOX MATH SCIENCE TECH ACAD|INGLEWOOD
LIBRA ACADEMY AT LINDA MARQUEZ HS|HUNTINGTON PARK
LINDA MARQUEZ HS-SOC JUSTICE|HUNTINGTON PARK
LONG BEACH POLYTECHNIC HS|LONG BEACH
LOS ANGELES ACAD ARTS ENTRPRSE|LOS ANGELES
LOS ANGELES SCH GLOBAL STUDIES|LOS ANGELES
MATH SCIENCE TECHNOLOGY MAGNET|LOS ANGELES

MAYWOOD CTR ENRICHED STUDIES|MAYWOOD
MIGUEL CONTRERAS LRNG COMPLEX|LOS ANGELES
MT SAC EARLY COLG ACD W COVINA|WEST COVINA
NARBONNE HIGH SCHOOL|HARBOR CITY
OPPORTUNITIES LRN-WILLIAM HART|VALENCIA
ORTHOPAEDIC HOSP MED MAGNET HS|LOS ANGELES
PERFORMING ARTS COMMUNITY SCHO|LOS ANGELES
PUBLIC SERVICE COMMUNITY SCHO|LOS ANGELES
PUC COMMUNITY CHARTER EARLY COLL HS|SYLMAR
PUC EARLY COLG ACADEMY LEADERS|LOS ANGELES
PUC TRIUMPH CHARTER ACADEMY HS|SYLMAR
RAMON C CORTINES SCH OF VISUAL|LOS ANGELES
RENAISSANCE HIGH SCHOOL ARTS|LONG BEACH
RESEDA HIGH SCHOOL|RESEDA
ROBERT A MILLIKAN HIGH SCHOOL|LONG BEACH
SANTA CLARITA VALLEY INTERNATI|CASTAIC
SATO ACADEMY OF MATH & SCIENCE|LONG BEACH
SCH SOCIAL JUSTICE-CONT LEARN CTR|LOS ANGELES
SCHOOL OF EXTENDED EDU OPTIONS|POMONA
SCHOOL VISUAL ARTS HUMANITIES|LOS ANGELES
SHERMAN OAKS CTR ENRCHED STDS|TARZANA
SOAR HIGH SCHOOL|LANCASTER
SOCIAL JUSTICE HUMANITAS ACAD|SAN FERNANDO
SOCIAL JUSTICE LDRSHP @ TORRES|LOS ANGELES
SONIA SOTOMAYOR ARTS SCI ACAD|LOS ANGELES
SOUTH LA COLLEGE PREP|LOS ANGELES
STEAM LEGACY HIGH SCHOOL CMLPX|SOUTH GATE
STEAM VIRTUAL ACADEMY|LOS ANGELES
STEM ACADEMY OF BOYLE HEIGHTS|LOS ANGELES
STEM ACADEMY OF HOLLYWOOD|HOLLYWOOD
SYLMAR BIOTECH HLTH & ENG MGNT|SYLMAR
SYLMAR HIGH SCHOOL|SYLMAR
TECHNOLOGY PREPARATORY ACADEMY|SAN FERNANDO
TRACY HIGH SCHOOL|CERRITOS
UNIVERSITY HIGH SCHOOL|LOS ANGELES
UNIVERSITY PATHWAYS PUB SVC AC|LOS ANGELES
UP MEDICAL MAGNET ACADEMY|LOS ANGELES
USC HYBRID HIGH COLLEGE PREP|LOS ANGELES
USC MEDIA & ENGINEERING MAGNET SCH|LOS ANGELES
VALLEY ACADEMY ARTS & SCIENCES|GRANADA HILLS
VALLEY OAKS CTR ENRICHED STDS|SUN VALLEY
VAUGHN NEXT CENTURY LEARN CTR|SAN FERNANDO
VIEW PARK PREP ACCEL CHARTER HS|LOS ANGELES
VILLAGE ACADEMY HIGH SCHOOL|POMONA
WARREN HIGH SCHOOL|DOWNEY
WESTBROOK ACADEMY|SOUTH GATE
WESTCHESTER ENRICHED SCIENCE MAGNET|LOS ANGELES
WILLIAM HOWARD TAFT HS|WOODLAND HILLS
WILLIAM J PETE KNIGHT HIGH SCH|PALMDALE
WOODROW WILSON HIGH SCHOOL|LONG BEACH
WOODROW WILSON HIGH SCHOOL|LOS ANGELES
WORKMAN HIGH SCHOOL|CITY OF INDUSTRY

Madera

Marin

Mariposa

Mendocino

Merced

Modoc

Mono

Monterey

KING CITY JOINT UNION HS|KING CITY

Napa

Nevada

SIERRA ACADEMY EXPEDITION LRNG|NEVADA CITY

Orange

ADVANCED LRNG ACDMY EARLY COLG|SANTA ANA
CA CONNECTIONS ACADEMY CAPISTRANO|SAN JUAN CAPISTRANO
CALIFORNIA PACIFIC CHARTER LA|NEWPORT BEACH
EPIC CHARTER SCHOOL - ANAHEIM|ANAHEIM
HECTOR GODINEZ FUNDAMENTAL HS|SANTA ANA
LEGACY COLLEGE PREP|SANTA ANA
ORANGE COUNTY HS OF THE ARTS|SANTA ANA
SAN JOAQUIN HIGH SCHOOL|IRVINE

Placer

JOHN ADAMS ACADEMY|ROSEVILLE

Plumas

Riverside

ACADEMY OF INNOVATION|HEMET
ELITE ACADEMIC ACADEMY-LUCERNE|TEMECULA
HYBRID SCHOOL OF INNOVATION|NORCO
JOHN F KENNEDY MIDDLE COLG HS|NORCO
MARTIN LUTHER KING HIGH SCHOOL|RIVERSIDE
METHOD SCHOOLS|MURRIETA
RIVERSIDE POLY HIGH SCHOOL|RIVERSIDE

Sacramento

ANTELOPE HIGH SCHOOL|ANTELOPE
ASPIRE ALEXANDER TWILIGHT SCND|SACRAMENTO
ENCINA HIGH SCHOOL|SACRAMENTO
HEALTH PROFESSIONS HIGH SCH|SACRAMENTO

MET SACRAMENTO HIGH SCHOOL|SACRAMENTO
NATOMAS CHARTER SCHOOL PERFORM|SACRAMENTO
REX & MARGARET FORTUNE ECHS|ELK GROVE
SACRAMENTO HIGH SCHOOL|SACRAMENTO
SCH ENGINEERING & SCIENCES|SACRAMENTO
UMOJA INTERNATIONAL ACADEMY|SACRAMENTO

San Benito
SAN BENITO COUNTY EVENING HIGH|HOLLISTER

San Bernardino
A B MILLER HIGH SCHOOL|FONTANA
ACADEMY OF CAREERS AND EXPLORA|HELENDAL
COBALT INSTITUTE MATH/SCIENCE|VICTORVILLE
ENCORE JR/SR HIGH SCHOOL|HESPERIA
EXCELSIOR CHARTER SCHOOLS|VICTORVILLE
GORMAN LEARNING CENTER|REDLANDS
GRAND TERRACE HIGH SCHOOL|GRAND TERRACE
OPTIONS FOR YOUTH - ACTON|FONTANA
OPTIONS FOR YOUTH - SAN BERNAR|SAN BERNARDINO
SAGE OAK CHARTER SCHOOL KEPPEL|REDLANDS

San Diego
AUDEO CHARTER SCHOOL|SAN DIEGO
BAYFRONT CHARTER HIGH SCHOOL|CHULA VISTA
CHARTER SCHOOL OF SAN DIEGO|SAN DIEGO
CHULA VISTA LRNG CMNTY CHARTER|CHULA VISTA
COASTAL ACADEMY HS OCEANSIDE|OCEANSIDE
DEL LAGO ACADEMY|ESCONDIDO
GUAJOME PARK ACADEMY|VISTA
HEALTH SCIENCES HIGH/MID COLL|SAN DIEGO
HERBERT HOOVER HIGH SCHOOL|SAN DIEGO
JAMES MADISON HIGH SCHOOL|SAN DIEGO
JULIAN CHARTER SCHOOL LIVE|JULIAN
KEARNY SCH BIOMED SCIENCE TECH|SAN DIEGO
LEARNING CHOICE - CHULA VISTA|CHULA VISTA
LEARNING CHOICE ACADEMY-SAN DIEGO|SAN DIEGO
LITERACY FIRST CHARTER SCHOOLS|ALPINE
OFARRELL CHARTER SCHOOL|SAN DIEGO
PATRICK HENRY HIGH SCHOOL|SAN DIEGO
SAN DIEGO CITY SCHOOLS|SAN DIEGO
SAN DIEGO MET|SAN DIEGO
SAN DIEGO SCHL CREATE/PERFORM|SAN DIEGO
SAN DIEGUITO ACADEMY|ENCINITAS
SCHL ENTREPRENEURSHIP & TECH|SAN DIEGO
SWEETWATER SECONDARY SCHOOL|CHULA VISTA
SWEETWATER UNION HIGH SCHOOL|NATIONAL CITY

San Francisco
ACADEMY-SAN FRAN @ MCATEER|SAN FRANCISCO
GALILEO ACAD OF SCIENCE & TECH|SAN FRANCISCO
PHILLIP & SALA BURTON ACAD HS|SAN FRANCISCO

RUTH ASAWA SCHOOL OF THE ARTS|SAN FRANCISCO
THURGOOD MARSHALL ACADEMIC HS|SAN FRANCISCO

San Joaquin

AMOS ALONZO STAGG HIGH SCHOOL|STOCKTON
ASPIRE BENJAMIN HOLT COLG PREP|STOCKTON
CALIFORNIA CNCTS MONTEREY BAY|RIPON
CALIFORNIA CONNECTIONS ACD RPN|RIPON
MILLENNIUM HIGH SCHOOL|TRACY
TRACY JOINT UNION HIGH SCHOOL|TRACY
VISTA OAKS CHARTER SCHOOL|LODI
WEBER INST OF APPLIED SCI TECH|STOCKTON

San Luis Obispo

San Mateo

SUMMIT PREPARATORY CHARTER HIG|REDWOOD CITY
SUMMIT PUBLIC SCHOOLS SHASTA|DALY CITY

Santa Barbara

SANTA BARBARA SENIOR HIGH SCHO|SANTA BARBARA

Santa Clara

ADRIAN C WILCOX HIGH SCHOOL|SANTA CLARA
DOWNTOWN COLG PREP EL PRIMERO|SAN JOSE
DOWNTOWN COLLEGE PREP ALUM ROC|SAN JOSE
DR TJ OWENS GILROY EARLY COLG|GILROY
HENRY T GUNDERSON HIGH SCHOOL|SAN JOSE
ROBERTO CRUZ LEADERSHIP ACDMY|SAN JOSE
UNIVERSITY PREP ACADEMY|SAN JOSE
W C OVERFELT HIGH SCHOOL|SAN JOSE

Santa Cruz

PACIFIC COLLEGIATE SCHOOL|SANTA CRUZ
SANTA CRUZ ALTERNATIVE|SANTA CRUZ

Shasta

WEST VALLEY HIGH SCHOOL|COTTONWOOD

Sierra

Siskiyou

TULELAKE HIGH SCHOOL|TULELAKE

Solano

BUCKINGHAM COLLEGIATE CHARTER|VACAVILLE
RIO VISTA HIGH SCHOOL|RIO VISTA

Sonoma

ROSELAND COLLEGIATE PREP|SANTA ROSA
ROSELAND UNIVERSITY PREP|SANTA ROSA

Stanislaus
CONNECTING WATERS - EAST BAY|WATERFORD
CONNECTING WATERS-CTRL VALLEY|MODESTO
JOHANSEN HIGH SCHOOL|MODESTO

Sutter

Tehama

Trinity

Tulare
ELEANOR ROOSEVELT COMM LRNG CN|VISALIA
SUMMIT CHARTER COLLEGIATE ACAD|PORTERVILLE
UNIVERSITY PREPARTORY HIGH SCH|VISALIA
VISALIA TECHNICAL EDU CENTER|VISALIA

Tuolumne

Ventura
ARCHITECTURE CONSTRUCT ENG HS|CAMARILLO
CALIFORNIA VIRTUAL ACAD SAN DIEGO|SIMI VALLEY
CALIFORNIA VIRTUAL ACAD SAN MATEO|SIMI VALLEY
CALIFORNIA VIRTUAL ACAD SUTTER|SIMI VALLEY
CALIFORNIA VIRTUAL ACADEMY LA|SIMI VALLEY
CALIFORNIA VIRTUAL ACDM SONOMA|SIMI VALLEY
CALIFORNIA VIRTUAL-SAN JOAQUIN|SIMI VALLEY
COMPASS CHARTER SCHLS OF LA|THOUSAND OAKS
COMPASS CHARTER SCHOOL OF YOLO|THOUSAND OAKS
HIGH SCHOOL AT MOORPARK COLL|MOORPARK

Yolo
DAVIS SCHOOL INDEPENDENT STUDY|DAVIS
DELTA HIGH SCHOOL|CLARKSBURG

Yuba
MARYSVILLE CHARTER ACAD ARTS|MARYSVILLE

Here is the prompt: The attached file contains one row per reference high school. Each row consists of pipe-delimited values schoolName|cityName|districtName|ZIP|ID. Below is a second list of high schools. Each line consists of schoolName|cityName. The names in the two lists do not match exactly due to abbreviations and other variations. For each line below, print the input line followed by "||" as a delimiter followed by the line for the best match school from the reference file. Include all elements from the reference file including districtName, ZIP, and ID. If Copilot Bing indicates "Message exceeds 10240 characters", shorten the list in the query.

In [26]: *# Below, we read mapping of UC school name to CAASPP school name from lookup.psv, which was iteratively created.*

```
# Lines with the following formats contain mapping, others are ignored by this program.
# left || right: mapping found by ChatGPT 1st pass (looking only within one county).
# left ||| right: mapping found by ChatGPT 2nd pass (looking across counties).
# left -- right: manual delete.
# left ++ right: manual add.
# left = "/" delimited (ucSchool, ucCity). This is the ucSchoolDetails key.
# right = "/" delimited (testSchool, testCity, testDistrict, testZIP, testID)

# Error checking includes:
# * Is the format of the line correct?
```

```

# * Does the testID exist?
# * Is the testSchool generated by ChatGPT associated with that testID?
# * Is ucKey a duplicate, meaning that ChatGPT generated the left side of the line incorrectly?
# * Is testID a duplicate, meaning that ChatGPT matched the same CAASPP school to two different UC schools?

```

```

In [27]: # Create Lookup table to get UC key from ChatGPT output
ucSchoolCity2Key = {}
for key in ucSchoolDetails:
    (id, school, county, city, expandedSchool) = key.split("|")
    schoolCity = "|".join([school, city])
    if schoolCity in ucSchoolCity2Key:
        # Check for duplicates: there are none
        print(schoolCity, key, ucSchoolCity2Key[schoolCity])
        continue
    ucSchoolCity2Key[schoolCity] = key

```

```

In [28]: # Read ChatGPT output from Lookup.psv and update ucSchoolDetails, which has the following format:
# Key is ID|SCHOOL|County|CITY|ExpandedSCHOOL
# Value is [ALLApp, ALLAdm, ALLEnr, Found, FoundID]
f = open('lookup.psv', 'rt', encoding='utf8')
for line in f:
    if re.match(r"#", line):
        continue # Skip records that were commented out
    if not re.search(r"\\|\\| ", line): # Only the ChatGPT output lines have this format
        continue
    cleanline = line.rstrip()
    (ucLine, testLine) = cleanline.split(" || ")
    if re.match(r"No ", testLine): # May be "No exact match", "No close match", etc.
        continue
    key = ucSchoolCity2Key[ucLine] # Get the ucSchoolDetails key
    items = testLine.split("|") # Parse testLine to get testID
    if len(items) != 5:
        nItems = len(items)
        print(f"Bad testLine format {nItems} items\n    UC:    {key}\n    Found: {testLine}") # Happens once. Seems like the output was actually ge
        continue
    (testSchool, testCity, testDistrict, testZIP, testID) = items
    testSchool = testSchool.upper() # ChatGPT generated titlecase ~20 times

    if testID not in testID2Details:
        print(f"Bad testLine ID\n    UC:    {key}\n    Found: {testLine}")
        continue
    (idTRef, schoolTRef, countyTRef, cityTRef, districtTRef, zipTRef, expandedSchoolTRef) = testID2Details[testID].split("|")
    if testSchool != schoolTRef:
        actualLine = "|".join([schoolTRef, cityTRef, districtTRef, zipTRef, idTRef])
        if testSchool in testRawSchool2Details:
            if testRawSchool2Details[testSchool] == "multiple":
                sNameLine = "Multiple CAASPP schools with this found school name"
            else:
                (idSName, schoolSName, countySName, citySName, districtSName, zipSName, expandedSchoolSName) = testRawSchool2Details[testSchool].split
                sNameLine = "|".join([schoolSName, citySName, districtSName, zipSName, idSName])
        else:
            sNameLine = "Found school name is hallucinated"

```

```

        print(f"School mismatch\n      UC:      {key}\n      Found: {testLine}\n      FoundN: {sNameLine}\n      Actual: {actualLine}")
        continue

    if ucSchoolDetails[key][3] > 0:
        print(f"Previously found\n      UC:      {key}") # Should not happen because we didn't search using items we previously found.
        continue
    if testID in usedTestIDs:
        print(f"Previously used\n      UC:      {key}\n      PrevUC: {usedTestIDs[testID]}\n      Found: {testLine}")
        continue

    # Found a new item
    ucSchoolDetails[key][3] = 3
    ucSchoolDetails[key][4] = testID
    usedTestIDs[testID] = "|".join([key, "3"])
f.close()

```

Bad testLine format 4 items

UC: 53914|ACADEMY OF INNOVATION|Riverside|HEMET|ACADEMY OF INNOVATION

Found: Academy of Innovation|Hemet Unified|92543|3330503

Previously used

UC: 51443|SAGE OAK CHARTER SCHOOL KEPPEL|San Bernardino|REDLANDS|SAGE OAK CHARTER SCHOOL KEPPEL

PrevUC: 51664|SAGE OAK CHARTER|San Bernardino|REDLANDS|SAGE OAK CHARTER|1

Found: SAGE OAK CHARTER|LUGONIA/REDLANDS|Sage Oak Charter|92373|0136069

Previously used

UC: 54643|CALIFORNIA CONNECTIONS ACD RPN|San Joaquin|RIPON|CALIFORNIA CONNECTIONS ACADEMY RPN

PrevUC: 54952|CALIFORNIA CNCTS MONTEREY BAY|San Joaquin|RIPON|CALIFORNIA CNCTS MONTEREY BAY|3

Found: CALIFORNIA CONNECTIONS ACADEMY NORTHERN CALIFORNIA|STOCKTON|California Connections Academy Northern|95366|0125849

Previously used

UC: 54044|ROSELAND UNIVERSITY PREP|Sonoma|SANTA ROSA|ROSELAND UNIVERSITY PREPARATORY

PrevUC: 54850|ROSELAND COLLEGIATE PREP|Sonoma|SANTA ROSA|ROSELAND COLLEGIATE PREPARATORY|3

Found: ROSELAND CHARTER|ROSELAND/SANTA ROSA|Roseland Charter|95407|0101923

School mismatch

UC: 54057|LOS ANGELES SCH GLOBAL STUDIES|Los Angeles|LOS ANGELES|LOS ANGELES SCHOOL GLOBAL STUDIES

Found: SCHOOL OF GLOBAL STUDIES AT CONTRERAS LEARNING COM|FOY/LOS ANGELES|Los Angeles Unified|90017|0112029

FoundN: Found school name is hallucinated

Actual: SCHOOL OF BUSINESS AND TOURISM AT CONTRERAS LEARNI|FOY/LOS ANGELES|Los Angeles Unified|90017|0112029

School mismatch

UC: 54053|MIGUEL CONTRERAS LRNG COMPLEX|Los Angeles|LOS ANGELES|MIGUEL CONTRERAS LEARNING COMPLEX

Found: MIGUEL CONTRERAS LEARNING COMPLEX GREEN DESIGN STEAM|FOY/LOS ANGELES|Los Angeles Unified|90017|0124503

FoundN: Found school name is hallucinated

Actual: DIEGO RIVERA LEARNING COMPLEX GREEN DESIGN STEAM A|LOS ANGELES|Los Angeles Unified|90001|0124503

School mismatch

UC: 54542|SOCIAL JUSTICE HUMANITAS ACAD|Los Angeles|SAN FERNANDO|SOCIAL JUSTICE HUMANITAS ACADEMY

Found: SOCIAL JUSTICE HUMANITAS ACADEMY AT ESTEBAN TORRES|HAZARD/LOS ANGELES|Los Angeles Unified|90063|0122358

FoundN: Found school name is hallucinated

Actual: HUMANITAS ACADEMY OF ART AND TECHNOLOGY AT ESTEBAN|HAZARD/LOS ANGELES|Los Angeles Unified|90063|0122358

Previously used

UC: 54444|SOCIAL JUSTICE LDRSH @ TORRES|Los Angeles|LOS ANGELES|SOCIAL JUSTICE LEADERSHIP - TORRES

PrevUC: 54695|SCH SOCIAL JUSTICE-CONT LEARN CTR|Los Angeles|LOS ANGELES|SCHOOL SOCIAL JUSTICE-CONT LEARN CTR|3

Found: SOCIAL JUSTICE LEADERSHIP ACADEMY MAGNET AT ESTEBA|HAZARD/LOS ANGELES|Los Angeles Unified|90063|0122366

School mismatch

UC: 51505|WOODROW WILSON HIGH SCHOOL|Los Angeles|LONG BEACH|WOODROW WILSON HIGH SCHOOL

Found: WOODROW WILSON SENIOR HIGH|LONG BEACH|Long Beach Unified|90804|1939875

FoundN: WOODROW WILSON SENIOR HIGH|EL SERENO/LINCOLN HEIGHTS/LOS ANGELES|Los Angeles Unified|90032|1939859

Actual: WILSON HIGH|LONG BEACH|Long Beach Unified|90804|1939875

Previously used

UC: 52842|SAN DIEGO CITY SCHOOLS|San Diego|SAN DIEGO|SAN DIEGO CITY SCHOOLS

PrevUC: 53903|SAN DIEGO HIGH SCHOOL|San Diego|SAN DIEGO|SAN DIEGO HIGH SCHOOL|2

Found: SAN DIEGO HIGH|SAN DIEGO|San Diego Unified|92101|0107219

```
In [29]: # Explore the above issues
query = r"SIERRA VISTA"
desiredZIP = ""
desiredTestID = ""
print("UC")
for key in ucSchoolDetails:
    (ucID, school, county, city, expandedSchool) = key.split("|")
    if re.search(query, school, re.IGNORECASE):
        print(key, ucSchoolDetails[key])
```

```

print("\nCAASPP")
for key in testSchoolDetails:
    (testID, school, county, city, district, zip, expandedSchool) = key.split("|")
    if re.search(query, school, re.IGNORECASE) or zip == desiredZIP or testID == desiredTestID:
        if testID in usedTestIDs:
            print(key, testSchoolDetails[key], " *** ", usedTestIDs[testID])
        else:
            print(key, testSchoolDetails[key])

```

UC

50228|SIERRA VISTA HIGH SCHOOL|Los Angeles|BALDWIN PARK|SIERRA VISTA HIGH SCHOOL [540, 243, 58, 2, '1938166']

CAASPP

1938166|SIERRA VISTA HIGH|Los Angeles|IRWINDALE/BALDWIN PARK|Baldwin Park Unified|91706|SIERRA VISTA HIGH [2, 789] *** 50228|SIERRA VISTA HIGH SCHOOL
|Los Angeles|BALDWIN PARK|SIERRA VISTA HIGH SCHOOL|2
1995653|SIERRA VISTA HIGH (ALTERNATIVE)|Los Angeles|WHITTIER|Whittier Union High|90605|SIERRA VISTA ALTERNATIVE HIGH [2, 220]
0130708|SIERRA VISTA CHARTER HIGH|Tulare|TULARE|Tulare Joint Union High|93274|SIERRA VISTA CHARTER HIGH [2, 126]
5431119|SIERRA VISTA HIGH (CONTINUATION)|Tulare|DINUBA|Dinuba Unified|93618|SIERRA VISTA HIGH [2, 96]

```

In [30]: # Read the 2nd pass ChatGPT output (searching across the whole state rather than by county) and update ucSchoolDetails
f = open('lookup.psv', 'rt', encoding='utf8')
for line in f:
    if not re.search(r"\\|\\| ", line): # Only 2nd pass ChatGPT output lines have this format
        continue
    cleanline = line.rstrip()
    (ucLine, testLine) = cleanline.split(" ||| ")
    if re.match(r"No ", testLine): # May be "No exact match", "No close match", etc.
        continue
    key = ucSchoolCity2Key[ucLine] # Get the ucSchoolDetails key
    items = testLine.split("|") # Parse testLine to get testID
    if len(items) != 5:
        nItems = len(items)
        print(f"Bad testLine format {nItems} items\n      UC:      {key}\n      Found: {testLine}") # Happens once. Seems like the output was actually ge
        continue
    (testSchool, testCity, testDistrict, testZIP, testID) = items
    testSchool = testSchool.upper() # ChatGPT generated titlecase ~20 times

    if testID not in testID2Details:
        print(f"Bad testLine ID\n      UC:      {key}\n      Found: {testLine}")
        continue
    (idTRef, schoolTRef, countyTRef, cityTRef, districtTRef, zipTRef, expandedSchoolTRef) = testID2Details[testID].split("|")
    if testSchool != schoolTRef:
        actualLine = "|".join([schoolTRef, cityTRef, districtTRef, zipTRef, idTRef])
        if testSchool in testRawSchool2Details:
            if testRawSchool2Details[testSchool] == "multiple":
                sNameLine = "Multiple CAASPP schools with this found school name"
            else:
                (idSName, schoolSName, countySName, citySName, districtSName, zipSName, expandedSchoolSName) = testRawSchool2Details[testSchool].split
                sNameLine = "|".join([schoolSName, citySName, districtSName, zipSName, idSName])
        else:
            sNameLine = "Found school name is hallucinated"
        print(f"School mismatch\n      UC:      {key}\n      Found: {testLine}\n      FoundN: {sNameLine}\n      Actual: {actualLine}")

```



```

        continue

    if ucSchoolDetails[key][3] > 0:
        print(f"Previously found\n    UC:      {key}") # Should not happen because we didn't search using items we previously found.
        continue
    if testID in usedTestIDs:
        print(f"Previously used\n    UC:      {key}\n    PrevUC: {usedTestIDs[testID]}\n    Found: {testLine}")
        continue

    # Found a new item
    ucSchoolDetails[key][3] = 4
    ucSchoolDetails[key][4] = testID
    usedTestIDs[testID] = "|".join([key, "4"])
f.close()

```

School mismatch

```

UC:      51799|ELITE ACADEMIC ACADEMY-LUCERNE|Riverside|TEMECULA|ELITE ACADEMIC ACADEMY-LUCERNE
Found:   Elite Academic Academy - Lucerne|TEMECULA|Elite Academic Academy - Lucerne|92590|0136978
FoundN:  ELITE ACADEMIC ACADEMY - LUCERNE|TEMECULA|Elite Academic Academy - Lucerne|92590|0136960
Actual:  ELITE ACADEMIC ACADEMY - MOUNTAIN EMPIRE|TEMECULA|Elite Academic Academy - Mountain Empire|92590|0136978

```

School mismatch

```

UC:      54250|ANTELOPE HIGH SCHOOL|Sacramento|ANTELOPE|ANTELOPE HIGH SCHOOL
Found:   Antelope High|ANTELOPE|Roseville Joint Union High|95843|0113118
FoundN:  ANTELOPE HIGH|ELKHORN/SACRAMENTO|Roseville Joint Union High|95843|0116459
Actual:  DOUGHERTY VALLEY HIGH|SAN RAMON|San Ramon Valley Unified|94582|0113118

```

School mismatch

```

UC:      54846|ILEAD EXPLORATION HYBRID|Los Angeles|ACTON|ILEAD EXPLORATION HYBRID
Found:   iLEAD Hybrid|ACTON|iLEAD Hybrid|93510|0137315
FoundN:  ILEAD HYBRID|ACTON|iLEAD Hybrid|93510|0131987
Actual:  KIPP NAVIGATE COLLEGE PREP|FOOTHILL/SAN JOSE|SBE - KIPP Navigate College Prep|95127|0137315

```

```

In [31]: # Delete incorrect matches
f = open('lookup.psv', 'rt', encoding='utf8')
for line in f:
    if not re.search(r"\-\- ", line):
        continue
    cleanline = line.rstrip()
    (ucLine, testLine) = cleanline.split(" -- ")
    key = ucSchoolCity2Key[ucLine] # Get the ucSchoolDetails key
    items = testLine.split("|") # Parse testLine to get testID
    (testSchool, testCity, testDistrict, testZIP, testID) = items

    if testID not in usedTestIDs:
        print(f"testID not used\n    UC:      {key}\n    Delete: {testLine}")
        continue

    # Delete this item from the "matched" records
    usedWithKey = re.sub(r"\|[1-9]$", "", usedTestIDs[testID])
    if usedWithKey != key:
        print(f"Wrong usedWithKey {key} vs {usedWithKey}")
        continue

    ucSchoolDetails[key][3] = 0

```

```

    ucSchoolDetails[key][4] = ""
    del usedTestIDs[testID]

f.close()

```

```

In [32]: # Add fixed/corrected matches
f = open('lookup.psv', 'rt', encoding='utf8')
for line in f:
    if not re.search(r" \\\+ ", line):
        continue
    cleanline = line.rstrip()
    (ucLine, testLine) = cleanline.split(" ++ ")
    key = ucSchoolCity2Key[ucLine] # Get the ucSchoolDetails key
    items = testLine.split("|") # Parse testLine to get testID
    (testSchool, testCity, testDistrict, testZIP, testID) = items
    testSchool = testSchool.upper() # ChatGPT generated titlecase ~20 times

    if testID not in testID2Details:
        print(f"Bad testLine ID\n    UC:    {key}\n    Found: {testLine}")
        continue
    (idTRef, schoolTRef, countyTRef, cityTRef, districtTRef, zipTRef, expandedSchoolTRef) = testID2Details[testID].split("|")
    if testSchool != schoolTRef:
        actualLine = "|".join([schoolTRef, cityTRef, districtTRef, zipTRef, idTRef])
        if testSchool in testRawSchool2Details:
            if testRawSchool2Details[testSchool] == "multiple":
                sNameLine = "Multiple CAASPP schools with this found school name"
            else:
                (idSName, schoolSName, countySName, citySName, districtSName, zipSName, expandedSchoolSName) = testRawSchool2Details[testSchool].split("|")
                sNameLine = "|".join([schoolSName, citySName, districtSName, zipSName, idSName])
            else:
                sNameLine = "Found school name is hallucinated"
        print(f"School mismatch\n    UC:    {key}\n    Found: {testLine}\n    FoundN: {sNameLine}\n    Actual: {actualLine}")
        continue

    if ucSchoolDetails[key][3] > 0:
        print(f"Previously found\n    UC:    {key}")
        continue
    if testID in usedTestIDs:
        print(f"Previously used\n    UC:    {key}\n    PrevUC: {usedTestIDs[testID]}\n    Found: {testLine}")
        continue

    # Found a new item
    ucSchoolDetails[key][3] = 5
    ucSchoolDetails[key][4] = testID
    usedTestIDs[testID] = "|".join([key, "5"])
f.close()

```

```

In [33]: # How many UC records did we find vs not find?
nCount = 0
yCount = 0
for key in ucSchoolDetails:
    values = ucSchoolDetails[key]

```

```

    if values[3] > 0:
        yCount += 1
    else:
        nCount += 1
print(yCount, nCount)

```

1413 7

```

In [34]: # Which schools from UC records didn't we find?
print("Couldn't find the following schools from UC records:")
for key in ucSchoolDetails:
    values = ucSchoolDetails[key]
    if values[3] > 0:
        continue
    print(key, ucSchoolDetails[key])

```

Couldn't find the following schools from UC records:

```

54053|MIGUEL CONTRERAS LRNG COMPLEX|Los Angeles|LOS ANGELES|MIGUEL CONTRERAS LEARNING COMPLEX [95, 36, 4, 0, '']
54850|ROSELAND COLLEGIATE PREP|Sonoma|SANTA ROSA|ROSELAND COLLEGIATE PREPARATORY [111, 49, 13, 0, '']
52842|SAN DIEGO CITY SCHOOLS|San Diego|SAN DIEGO|SAN DIEGO CITY SCHOOLS [0, 6, 6, 0, '']
54695|SCH SOCIAL JUSTICE-CONT LEARN CTR|Los Angeles|LOS ANGELES|SCHOOL SOCIAL JUSTICE-CONT LEARN CTR [169, 59, 0, 0, '']
54542|SOCIAL JUSTICE HUMANITAS ACAD|Los Angeles|SAN FERNANDO|SOCIAL JUSTICE HUMANITAS ACADEMY [321, 122, 24, 0, '']
54889|UMOJA INTERNATIONAL ACADEMY|Sacramento|SACRAMENTO|UMOJA INTERNATIONAL ACADEMY [73, 40, 3, 0, '']
51687|WESTBROOK ACADEMY|Los Angeles|SOUTH GATE|WESTBROOK ACADEMY [78, 16, 0, 0, '']

```

```

In [35]: # Suspicious if UC nApplications > CAASPP nStudents. ucSchoolDetails sums across 9 campuses. testSchoolDetails sums across 2 test types.
for key in ucSchoolDetails:
    values = ucSchoolDetails[key]
    if values[3] == 0:
        continue # No matching CAASPP record found
    testID = values[4]
    testKey = testID2Details[testID]
    nStudents = testSchoolDetails[testKey][1]
    if values[0] / 9 > nStudents / 2:
        print(key, ucSchoolDetails[key])
        print(testKey, testSchoolDetails[testKey])

```

```

In [36]: # *** Manual Exploration ***
# MIGUEL CONTRERAS LRNG COMPLEX
# query = "contreras/esteba/torres"
# desiredZIP = "90063"
# desiredTestID = ""

# query = "ANTELOPE|ELITE ACADEMIC ACADEMY|envision/ilead/walnutwood"
# desiredZIP = ""
# desiredTestID = ""

# SWEETWATER SECONDARY SCHOOL|San Diego|CHULA VISTA may be https://www.altussouthbay.com/about/Locations 1655 Broadway, Chula Vista, CA 91911, which m
# query = r"sweetwater/altus"
# desiredZIP = ""
# desiredTestID = "3731155"

# Figure out the two STEM ACADEMY UC records and correct in Lookup.psv

```

```

query = r"HOLLYWOOD|STEM ACADEMY|BOYLE HEIGHTS"
desiredZIP = ""
desiredTestID = ""
print("UC")
for key in ucSchoolDetails:
    (ucID, school, county, city, expandedSchool) = key.split("|")
    if re.search(query, school, re.IGNORECASE):
        print(key, ucSchoolDetails[key])

print("\nCAASPP")
for key in testSchoolDetails:
    (testID, school, county, city, district, zip, expandedSchool) = key.split("|")
    if re.search(query, school, re.IGNORECASE) or zip == desiredZIP or testID == desiredTestID:
        if testID in usedTestIDs:
            print(key, testSchoolDetails[key], " *** ", usedTestIDs[testID])
        else:
            print(key, testSchoolDetails[key])

```

UC

```

51615|HOLLYWOOD HIGH SCHOOL|Los Angeles|LOS ANGELES|HOLLYWOOD HIGH SCHOOL [460, 192, 49, 2, '1934033']
52145|NORTH HOLLYWOOD HIGH SCHOOL|Los Angeles|NORTH HOLLYWOOD|NORTH HOLLYWOOD HIGH SCHOOL [1453, 616, 186, 2, '1936350']
51614|ODYSSEY STEM ACADEMY|Los Angeles|LAKEWOOD|ODYSSEY STEM ACADEMY [522, 170, 20, 1, '0136705']
54784|RIVERSIDE STEM ACADEMY|Riverside|RIVERSIDE|RIVERSIDE STEM ACADEMY [287, 120, 44, 1, '0131359']
54780|STEM ACADEMY OF BOYLE HEIGHTS|Los Angeles|LOS ANGELES|STEM ACADEMY OF BOYLE HEIGHTS [54, 27, 0, 5, '0129536']
54624|STEM ACADEMY OF HOLLYWOOD|Los Angeles|HOLLYWOOD|STEM ACADEMY OF HOLLYWOOD [489, 223, 44, 3, '0125989']

```

CAASPP

```

0125989|STEM ACADEMY AT BERNSTEIN HIGH|Los Angeles|HOLLYWOOD/LOS ANGELES|Los Angeles Unified|90028|STEM ACADEMY AT BERNSTEIN HIGH [2, 284] *** 54624|STEM ACADEMY OF HOLLYWOOD|Los Angeles|HOLLYWOOD|STEM ACADEMY OF HOLLYWOOD|3
0129536|BOYLE HEIGHTS SCIENCE, TECHNOLOGY, ENGINEERING AND|Los Angeles|LUGO/LOS ANGELES|Los Angeles Unified|90023|BOYLE HEIGHTS SCIENCE TECHNOLOGY ENGINEERING AND [2, 76] *** 54780|STEM ACADEMY OF BOYLE HEIGHTS|Los Angeles|LOS ANGELES|STEM ACADEMY OF BOYLE HEIGHTS|5
1932821|BOYLE HEIGHTS CONTINUATION|Los Angeles|BOYLE HEIGHTS/LOS ANGELES|Los Angeles Unified|90033|BOYLE HEIGHTS [2, 30]
1934033|HOLLYWOOD SENIOR HIGH|Los Angeles|HOLLYWOOD/LOS ANGELES|Los Angeles Unified|90028|HOLLYWOOD SENIOR HIGH [2, 514] *** 51615|HOLLYWOOD HIGH SCHOOL|Los Angeles|LOS ANGELES|HOLLYWOOD HIGH SCHOOL|2
1936350|NORTH HOLLYWOOD SENIOR HIGH|Los Angeles|CHANDLER/NORTH HOLLYWOOD|Los Angeles Unified|91601|NORTH HOLLYWOOD SENIOR HIGH [2, 1086] *** 52145|NORTH HOLLYWOOD HIGH SCHOOL|Los Angeles|NORTH HOLLYWOOD|NORTH HOLLYWOOD HIGH SCHOOL|2
0136705|ODYSSEY STEM ACADEMY|Los Angeles|LAKEWOOD|Paramount Unified|90712|ODYSSEY STEM ACADEMY [2, 202] *** 51614|ODYSSEY STEM ACADEMY|Los Angeles|LAKEWOOD|ODYSSEY STEM ACADEMY|1
0131359|RIVERSIDE STEM ACADEMY|Riverside|RIVERSIDE|Riverside Unified|92507|RIVERSIDE STEM ACADEMY [2, 108] *** 54784|RIVERSIDE STEM ACADEMY|Riverside|RIVERSIDE|RIVERSIDE STEM ACADEMY|1
0138040|AEROSTEM ACADEMY|Sutter|MARYSVILLE|AeroSTEM Academy|95991|AEROSTEM ACADEMY [2, 8]

```

```

In [37]: # Create a map of UC ID to CAASPP ID
ucID2TestID = {}
for key in ucSchoolDetails:
    (ucID, school, county, city, expandedSchool) = key.split("|")
    testID = ucSchoolDetails[key][4]
    if testID != "":
        ucID2TestID[ucID] = testID
print(len(ucID2TestID))

```

1413

```
In [38]: ##### Create dfLeft, a UC dataframe with acceptance rate by school and campus. Include CAASPP school ID for subsequent join. Do not include UC records
```

#	Calculation1	School	City	County/State/ Territory	Count	uad_uc_ethn_7_cat	Pivot Field Values
# 0	A B MILLER HIGH SCHOOL50944	A B MILLER HIGH SCHOOL	FONTANA	San Bernardino	Enr	Hispanic/ Latinx	24
# 1	A B MILLER HIGH SCHOOL50944	A B MILLER HIGH SCHOOL	FONTANA	San Bernardino	Adm	Hispanic/ Latinx	56
# 2	A B MILLER HIGH SCHOOL50944	A B MILLER HIGH SCHOOL	FONTANA	San Bernardino	App	Hispanic/ Latinx	66
# 3	A B MILLER HIGH SCHOOL50944	A B MILLER HIGH SCHOOL	FONTANA	San Bernardino	Adm	African American	7
# ...	...	...	...	...	...	...	...
# 79222	YUCAIPA SENIOR HIGH SCHOOL53820	YUCAIPA SENIOR HIGH SCHOOL	YUCAIPA	San Bernardino	App	ALL	23
# 79223	YUCCA VALLEY HIGH SCHOOL53818	YUCCA VALLEY HIGH SCHOOL	YUCCA VALLEY	San Bernardino	Adm	ALL	3

```

nRows = 0
for index, row in dfUC.iterrows():
    schoolID = row["Calculation1"]
    ucID = re.sub(r"^\.?(\d{5})$", r"\1", schoolID) # Five digit code, and the first digit is always 5. Already checked that this always succeeds abo
    if ucID not in ucID2TestID:
        continue # Don't have CAASPP data for this school
    testID = ucID2TestID[ucID]
    if row["uad_uc_ethn_7_cat"] != "All":
        continue # Only deal with all ethnicities combined
    school = row["School"]
    city = row["City"]
    county = row["County/State/ Territory"]
    countType = row["Count"]
    countVal = row["Pivot Field Values"]
    campus = row["Campus"]

    if nRows == 0:
        dfLeft = pd.DataFrame({"UC_ID": [ucID], "School": [school], "City": [city], "County": [county], "Campus": [campus], "countyCampus": [countyCam
        nRows += 1
    matchingIndex = dfLeft.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
    if len(matchingIndex) == 0:
        newRow = pd.DataFrame({"UC_ID": [ucID], "School": [school], "City": [city], "County": [county], "Campus": [campus], "countyCampus": [countyCam
        dfLeft = pd.concat([dfLeft, newRow], ignore_index=True)
        nRows += 1
        matchingIndex = dfLeft.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
    iLeftRow = matchingIndex[0]

    dfLeft.at[iLeftRow, countType] += countVal

print(len(dfLeft))

```

12410

```
In [39]: # Merge in the GPA data
dfLeft["App GPA"] = 0.0
dfLeft["Adm GPA"] = 0.0
dfLeft["Enr1 GPA"] = 0.0
for index, row in dfGPA.iterrows():
    schoolID = row["Calculation1"]
    ucID = re.sub(r"^\.?(\d{5})$", r"\1", schoolID) # Five digit code, and the first digit is always 5
    if ucID not in ucID2TestID:
        continue # Don't have CAASPP data for this school
```

```

campus      = row["Campus"]
matchingIndex = dfLeft.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
if len(matchingIndex) == 0:
    continue
iLeftRow = matchingIndex[0]

appGPA      = row["App GPA"]
admGPA      = row["Adm GPA"]
enr1GPA     = row["Enr1 GPA"]
if appGPA == appGPA: # Check for NaN
    dfLeft.at[iLeftRow, "App GPA"] = appGPA
if admGPA == admGPA:
    dfLeft.at[iLeftRow, "Adm GPA"] = admGPA
if enr1GPA == enr1GPA:
    dfLeft.at[iLeftRow, "Enr1 GPA"] = enr1GPA

```

```

In [40]: # If fewer than 5 students applied,      nApp = 0 in the UC data.
# If fewer than 3 students were admitted, nAdm = 0 in the UC data.
# It happens that nApp = 0 and nAdm = 3 or 4.
dfLeft = dfLeft[dfLeft["App"] != 0] # Delete all rows with nApp == 0
print(len(dfLeft))
# dfLeft = dfLeft[dfLeft["Adm"] != 0] # Delete all rows with nAdm == 0
# print(len(dfLeft))
# 12153
# 10609

```

12153

```

In [41]: # Compute the admission rate and enrolment rate
dfLeft["AdmRate"] = dfLeft["Adm"] / dfLeft["App"]
dfLeft["EnrRate"] = dfLeft["Enr"] / dfLeft["Adm"] # Value is NaN when Adm = 0

```

```

In [42]: # Explore
print(dfLeft[dfLeft["Adm"] == 0])
print(dfLeft[dfLeft["Adm"] == 3]) # Smaller values are set to 0 by UC
print(dfLeft[dfLeft["Enr"] == 3]) # Smaller values are set to 0 by UC

```

UC_ID	School	City	County	Campus	countyCampus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr1	GPA
AdmRate	EnrRate														
10	51123	ACADEMY OF CAREERS AND EXPLORA	HELENDAL	San Bernardino	all	UCR	5	0	0	0116723	4.07	0.0	0.0		
0.0	NaN														
11	53914	ACADEMY OF INNOVATION	HEMET	Riverside	all	UCR	5	0	0	3330503	3.68	0.0	0.0		
0.0	NaN														
102	55470	ARTS & ENTERTAINMENT VIRTUAL ACAD	CARSON	Los Angeles	all	UCLA	5	0	0	0141572	3.01	0.0	0.0		
0.0	NaN														
180	54715	CALERO HIGH SCHOOL	SAN JOSE	Santa Clara	all	UCSC	5	0	0	0125427	3.28	0.0	0.0		
0.0	NaN														
286	50635	COLEVILLE HIGH SCHOOL	COLEVILLE	Mono	all	UC Merced	5	0	0	2633006	4.13	0.0	0.0		
0.0	NaN														
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
...	...														
12262	54770	TEACH TECH CHARTER HIGH SCHOOL	LOS ANGELES	Los Angeles	UCSC	UCLA	9	0	0	0129627	3.40	0.0	0.0		
0.0	NaN														
12278	51815	THOMAS JEFFERSON HIGH SCHOOL	LOS ANGELES	Los Angeles	UCSC	UCLA	11	0	0	1934371	3.12	0.0	0.0		
0.0	NaN														
12311	53590	UPPER LAKE HIGH SCHOOL	UPPER LAKE	Lake	UCSC	UC Davis	7	0	0	1737006	3.42	0.0	0.0		
0.0	NaN														
12346	51728	WALLIS ANNENBERG HIGH SCHOOL	LOS ANGELES	Los Angeles	UCSC	UCLA	5	0	0	0100750	3.70	0.0	0.0		
0.0	NaN														
12396	53795	WOODLAKE HIGH SCHOOL	WOODLAKE	Tulare	UCSC	UC Merced	5	0	0	5436282	3.42	0.0	0.0		
0.0	NaN														

[1544 rows x 15 columns]

UC_ID	School	City	County	Campus	countyCampus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr1	GPA	A
dmRate	EnrRate															
117	55034	AUDEO VALLEY CHARTER SCHOOL	MORENO VALLEY	Riverside	all	UCR	5	3	0	0140780	3.66	0.0	0.0	0.0	0.	
600000	0.0															
157	50363	BORREGO SPRINGS HIGH SCHOOL	BORREGO SPRINGS	San Diego	all	UCSD	8	3	0	3730447	3.76	0.0	0.0	0.0	0.	
375000	0.0															
189	52512	CALIFORNIA PACIFIC CHARTER LA	NEWPORT BEACH	Orange	all	UCI	5	3	0	0138800	3.21	0.0	0.0	0.0	0.	
600000	0.0															
195	54278	CALIFORNIA VIRTUAL ACDM SONOMA	SIMI VALLEY	Ventura	all	UCSB	5	3	0	0107284	3.48	0.0	0.0	0.0	0.	
600000	0.0															
199	55003	CAMBRIDGE VIRTUAL ACADEMY	ANAHEIM	Orange	all	UCI	5	3	0	0140707	3.95	0.0	0.0	0.0	0.	
600000	0.0															
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
...	...															
12257	53485	TAFT UNION HIGH SCHOOL	TAFT	Kern	UCSC	UCSB	7	3	0	1535905	3.49	0.0	0.0	0.0	0.	
428571	0.0															
12264	54601	TECHNOLOGY PREPARATORY ACADEMY	SAN FERNANDO	Los Angeles	UCSC	UCLA	6	3	0	0124404	3.72	0.0	0.0	0.0	0.	
500000	0.0															
12287	53315	TRACY INDEPENDENT STUDY CHARTR	TRACY	San Joaquin	UCSC	UC Davis	5	3	0	0139949	3.57	0.0	0.0	0.0	0.	
600000	0.0															
12352	53704	WATERFORD HIGH SCHOOL	WATERFORD	Stanislaus	UCSC	UC Merced	12	3	0	5030259	3.30	0.0	0.0	0.0	0.	
250000	0.0															
12390	53790	WINTERS HIGH SCHOOL	WINTERS	Yolo	UCSC	UC Davis	8	3	0	5738505	3.56	0.0	0.0	0.0	0.	
375000	0.0															

[671 rows x 15 columns]

	UC_ID	School	City	County	Campus	countyCampus	App	Adm	Enr	CAASPP_ID	App GPA	Adm GPA	Enr1 GPA	A
dmRate	EnrRate													
1	54651	ABLE CHARTER	STOCKTON	San Joaquin	all	UC Davis	14	10	3	0126755	3.77	3.79	0.0	0.
714286	0.300000													
6	54247	ACADEMIA AVANCE CHARTER SCHOOL	LOS ANGELES	Los Angeles	all	UCLA	12	10	3	0109926	3.58	3.67	0.0	0.
833333	0.300000													
51	54527	ALLIANCE TENNENBAUM FAMILY TECH HS	LOS ANGELES	Los Angeles	all	UCLA	8	8	3	0121293	3.88	3.88	0.0	1.
000000	0.375000													
106	54260	ASPIRE GOLDEN STATE PREP ACADEMY	OAKLAND	Alameda	all	Berkeley	29	14	3	0118224	3.32	3.53	0.0	0.
482759	0.214286													
116	53199	AUDEO CHARTER SCHOOL	SAN DIEGO	San Diego	all	UCSD	5	4	3	3731395	3.82	0.00	0.0	0.
800000	0.750000													
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
...	...													
12360	53521	WEST HIGH SCHOOL	TORRANCE	Los Angeles	UCSC	UCLA	82	57	3	1939602	3.92	4.11	0.0	0.
695122	0.052632													
12389	53787	WINDSOR HIGH SCHOOL	WINDSOR	Sonoma	UCSC	Berkeley	25	16	3	4930228	3.73	4.05	0.0	0.
640000	0.187500													
12391	51705	WISH ACADEMY HIGH SCHOOL	LOS ANGELES	Los Angeles	UCSC	UCLA	13	10	3	0135632	3.88	4.08	0.0	0.
769231	0.300000													
12395	52598	WOODCREEK HIGH SCHOOL	ROSEVILLE	Placer	UCSC	UC Davis	28	17	3	3130176	3.82	4.04	0.0	0.
607143	0.176471													
12407	53815	YUBA CITY HIGH SCHOOL	YUBA CITY	Sutter	UCSC	UC Davis	40	26	3	5139001	3.71	4.04	0.0	0.
650000	0.115385													

[1034 rows x 15 columns]

```
In [43]: ##### Create dfLeft2025 with 2025 UC Admissions data released 3/13/2026.
# Format is same as UC admission data for California private schools (below), different from the initial format for 2024 UC Admissions.
directory = "UCAdmissions/2025"
files = {"FR ENR_data.All.csv": "all",
        "FR ENR_data.Berkeley.csv": "Berkeley",
        "FR ENR_data.Davis.csv": "UC Davis",
        "FR ENR_data.Irvine.csv": "UCI",
        "FR ENR_data.LosAngeles.csv": "UCLA",
        "FR ENR_data.Merced.csv": "UC Merced",
        "FR ENR_data.Riverside.csv": "UCR",
        "FR ENR_data.SanDiego.csv": "UCSD",
        "FR ENR_data.SantaBarbara.csv": "UCSB",
        "FR ENR_data.SantaCruz.csv": "UCSC"}

dfs = []
for filename in files.keys():
    dfOneFile = pd.read_csv("/".join([directory, filename]), delimiter="\t", encoding="utf-16")
    dfOneFile["Campus"] = files[filename]
    dfs.append(dfOneFile)
dfPublic2025 = pd.concat(dfs, ignore_index=True)
dfPublic2025["City"] = dfPublic2025["City"].str.upper() # Convert City to uppercase
# print(dfPublic2025.shape[0])
# print(dfPublic2025)

nRows = 0
for index, row in dfPublic2025.iterrows():
```



```

schoolID = row["Calculation1"]
ucID = re.sub(r"^\.?(\d{5})$", r"\1", schoolID) # Five digit code
if ucID not in ucID2TestID:
    continue # Don't have CAASPP data for this school
testID = ucID2TestID[ucID]
school = row["School"]
city = row["City"]
county = row["County/State/ Country"]
countType = row["Count"]
countVal = row["All"] # Only deal with all ethnicities combined
if countVal != countVal or countVal == 0:
    # NaN
    if countType == "App":
        continue
    else:
        countVal = 0
campus = row["Campus"]

if nRows == 0:
    dfLeft2025 = pd.DataFrame({"UC_ID": [ucID], "School": [school], "City": [city], "County": [county], "Campus": [campus], "countyCampus": [count
    nRows += 1
matchingIndex = dfLeft2025.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
if len(matchingIndex) == 0:
    newRow = pd.DataFrame({"UC_ID": [ucID], "School": [school], "City": [city], "County": [county], "Campus": [campus], "countyCampus": [countyCam
    dfLeft2025 = pd.concat([dfLeft2025, newRow], ignore_index=True) # Slow way to do this. Should create list of records and merge them into a df
    nRows += 1
    matchingIndex = dfLeft2025.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
iLeftRow = matchingIndex[0]

dfLeft2025.at[iLeftRow, countType] += countVal

print(len(dfLeft2025))
# dfLeft2024 with GPAs: [12153 rows x 15 columns]

```

13747

```

In [44]: # Read the UC admission GPA data (2025)
directory = "UCAdmissions/2025"
files = {"FR GPA by Year.All.csv": "all",
        "FR GPA by Year.Berkeley.csv": "Berkeley",
        "FR GPA by Year.Davis.csv": "UC Davis",
        "FR GPA by Year.Irvine.csv": "UCI",
        "FR GPA by Year.LosAngeles.csv": "UCLA",
        "FR GPA by Year.Merced.csv": "UC Merced",
        "FR GPA by Year.Riverside.csv": "UCR",
        "FR GPA by Year.SanDiego.csv": "UCSD",
        "FR GPA by Year.SantaBarbara.csv": "UCSB",
        "FR GPA by Year.SantaCruz.csv": "UCSC"}

dfs = []
for filename in files.keys():
    dfOneFile = pd.read_csv("/".join([directory, filename]), delimiter="\t", encoding="utf-16")
    dfOneFile["Campus"] = files[filename]
    dfs.append(dfOneFile)

```

```
dfGPA2025 = pd.concat(dfs, ignore_index=True)
dfGPA2025["City"] = dfGPA2025["City"].str.upper() # Convert City to uppercase
print(dfGPA2025.shape[0])
print(dfGPA2025)
```

12427

	Calculation1	School	City	County/State/	Country	App GPA	Adm GPA	Enrl GPA	Campus
0	A B MILLER HIGH SCHOOL050944	A B MILLER HIGH SCHOOL	FONTANA	San Bernardino		3.69	3.76	3.88	all
1	ABLE CHARTER054651	ABLE CHARTER	STOCKTON	San Joaquin		3.77	3.86	NaN	all
2	ABRAHAM LINCOLN HIGH SCHOOL051520	ABRAHAM LINCOLN HIGH SCHOOL	LOS ANGELES	Los Angeles		3.68	3.80	3.93	all
3	ABRAHAM LINCOLN HIGH SCHOOL052910	ABRAHAM LINCOLN HIGH SCHOOL	SAN FRANCISCO	San Francisco		3.90	4.01	4.10	all
4	ABRAHAM LINCOLN HIGH SCHOOL053075	ABRAHAM LINCOLN HIGH SCHOOL	SAN JOSE	Santa Clara		3.95	4.00	4.13	all
...	...	...	...	...		...	...	...	...
12422	YORBA LINDA HIGH SCHOOL054321	YORBA LINDA HIGH SCHOOL	YORBA LINDA	Orange		3.81	4.05	NaN	UCSC
12423	YOSEMITE HIGH SCHOOL052176	YOSEMITE HIGH SCHOOL	OAKHURST	Madera		3.84	3.98	NaN	UCSC
12424	YREKA HIGH SCHOOL053810	YREKA HIGH SCHOOL	YREKA	Siskiyou		3.68	3.95	NaN	UCSC
12425	YUBA CITY HIGH SCHOOL053815	YUBA CITY HIGH SCHOOL	YUBA CITY	Sutter		3.74	3.92	NaN	UCSC
12426	YUCAIPA SENIOR HIGH SCHOOL053820	YUCAIPA SENIOR HIGH SCHOOL	YUCAIPA	San Bernardino		3.82	3.99	NaN	UCSC

[12427 rows x 8 columns]

```
In [45]: # Merge in the GPA data
dfLeft2025["App GPA"] = 0.0
dfLeft2025["Adm GPA"] = 0.0
dfLeft2025["Enrl GPA"] = 0.0
for index, row in dfGPA2025.iterrows():
    schoolID = row["Calculation1"]
    ucID = re.sub(r"^\.*?(\d{5})$", r"\1", schoolID) # Five digit code, and the first digit is always 5
    if ucID not in ucID2TestID:
        continue # Don't have CAASPP data for this school

    campus = row["Campus"]
    matchingIndex = dfLeft2025.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
    if len(matchingIndex) == 0:
        continue
    iLeftRow = matchingIndex[0]

    appGPA = row["App GPA"]
    admGPA = row["Adm GPA"]
    enr1GPA = row["Enrl GPA"]
    if appGPA == appGPA: # Check for NaN
        dfLeft2025.at[iLeftRow, "App GPA"] = appGPA
    if admGPA == admGPA:
        dfLeft2025.at[iLeftRow, "Adm GPA"] = admGPA
    if enr1GPA == enr1GPA:
        dfLeft2025.at[iLeftRow, "Enrl GPA"] = enr1GPA
```

```
In [46]: # Compute the admission rate and enrolment rate for each record
dfLeft2025["AdmRate"] = dfLeft2025["Adm"] / dfLeft2025["App"]
dfLeft2025["EnrRate"] = dfLeft2025["Enr"] / dfLeft2025["Adm"] # Value is NaN when Adm = 0
```

```
In [47]: print(dfLeft)
print()
print(dfLeft2025)
```

[illegible]

```
[12153 rows x 15 columns]
```

[illegible]

[13747 rows x 15 columns]

```
In [48]: ##### PRIVATE SCHOOLS
# Read the UC admission data for California private schools
directory = "UCAdmissions/2024"
files = {"FR ENR_private.All.csv": "all",
        "FR ENR_private.Berkeley.csv": "Berkeley",
        "FR ENR_private.Davis.csv": "UC Davis",
        "FR ENR_private.Irvine.csv": "UCI",
        "FR ENR_private.LosAngeles.csv": "UCLA",
        "FR ENR_private.Merced.csv": "UC Merced",
        "FR ENR_private.Riverside.csv": "UCR",
        "FR ENR_private.SanDiego.csv": "UCSD",
        "FR ENR_private.SantaBarbara.csv": "UCSB",
        "FR ENR_private.SantaCruz.csv": "UCSC"}

dfs = []
for filename in files.keys():
    dfOneFile = pd.read_csv("/".join([directory, filename]), delimiter="\t", encoding="utf-16")
    dfOneFile["Campus"] = files[filename]
    dfs.append(dfOneFile)
dfPrivate = pd.concat(dfs, ignore_index=True)
dfPrivate["City"] = dfPrivate["City"].str.upper() # Convert City to uppercase
print(dfPrivate.shape[0])
print(dfPrivate)
```

12765

			Calculation1			School		City	County/State/ Territory	Count	All	African American	American I
ndian	Hispanic/	Latinx	Pacific Islander	Asian	White	Domestic Unknown	Int'l Campus						
0			A1 COLLEGE PREP55050			A1 COLLEGE PREP	LOS ANGELES		Los Angeles	App	NaN		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN all						
1			A1 COLLEGE PREP55050			A1 COLLEGE PREP	LOS ANGELES		Los Angeles	Adm	NaN		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN all						
2			A1 COLLEGE PREP55050			A1 COLLEGE PREP	LOS ANGELES		Los Angeles	Enr	NaN		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN all						
3			ACACIAWOOD SCHOOL50381			ACACIAWOOD SCHOOL	ANAHEIM		Orange	App	NaN		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN all						
4			ACACIAWOOD SCHOOL50381			ACACIAWOOD SCHOOL	ANAHEIM		Orange	Adm	3.0		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN all						
...			...			...	...		...	...	...		...
...			...			...	...						
12760	XAVIER COLLEGE PREPARATORY HS54103			XAVIER COLLEGE PREPARATORY HS		PALM DESERT			Riverside	Adm	18.0		NaN
NaN		4.0	NaN	3.0	7.0	NaN	NaN UCSC						
12761	XAVIER COLLEGE PREPARATORY HS54103			XAVIER COLLEGE PREPARATORY HS		PALM DESERT			Riverside	Enr	NaN		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN UCSC						
12762			YORK SCHOOL52053			YORK SCHOOL	MONTEREY		Monterey	App	19.0		NaN
NaN		5.0	NaN	NaN	7.0	NaN	NaN UCSC						
12763			YORK SCHOOL52053			YORK SCHOOL	MONTEREY		Monterey	Adm	7.0		NaN
NaN		NaN	NaN	NaN	3.0	NaN	NaN UCSC						
12764			YORK SCHOOL52053			YORK SCHOOL	MONTEREY		Monterey	Enr	NaN		NaN
NaN		NaN	NaN	NaN	NaN	NaN	NaN UCSC						

[12765 rows x 15 columns]

```

In [49]: # Create a UC dataframe with acceptance rate by school and campus for ***private schools***
nRows = 0
for index, row in dfPrivate.iterrows():
    schoolID = row["Calculation1"]
    ucID = re.sub(r"^\.?(\d{5})$", r"\1", schoolID) # Five digit code
    testID = ""
    school = row["School"]
    city = row["City"]
    county = row["County/State/ Territory"]
    countType = row["Count"]
    countVal = row["All"] # Only deal with all ethnicities combined
    campus = row["Campus"]

    if nRows == 0:
        dfLeftPrivate = pd.DataFrame({"UC_ID": [ucID], "School": [school], "City": [city], "County": [county], "Campus": [campus], "countyCampus": [co
        nRows += 1
    matchingIndex = dfLeftPrivate.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
    if len(matchingIndex) == 0:
        newRow = pd.DataFrame({"UC_ID": [ucID], "School": [school], "City": [city], "County": [county], "Campus": [campus], "countyCampus": [countyCam
        dfLeftPrivate = pd.concat([dfLeftPrivate, newRow], ignore_index=True)
        nRows += 1
    matchingIndex = dfLeftPrivate.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index
    iLeftRow = matchingIndex[0]

    dfLeftPrivate.at[iLeftRow, countType] += countVal

print(len(dfLeftPrivate))

# If too few students applied, App = NaN in the UC data. Delete those rows.
dfLeftPrivate = dfLeftPrivate[dfLeftPrivate["App"] == dfLeftPrivate["App"]]
print(len(dfLeftPrivate))

# If only 1 or 2 students were admitted, Adm is set to NaN in the UC data. Delete those rows.
dfLeftPrivate = dfLeftPrivate[dfLeftPrivate["Adm"] == dfLeftPrivate["Adm"]]
print(len(dfLeftPrivate))

# Compute the admission rate and enrolment rate for each record
dfLeftPrivate["AdmRate"] = dfLeftPrivate["Adm"] / dfLeftPrivate["App"]
dfLeftPrivate["EnrRate"] = dfLeftPrivate["Enr"] / dfLeftPrivate["Adm"]

4255
2573
2070

```

```

In [50]: # Classify private schools as religious vs other
dfLeftPrivate["Religious"] = False
dfLeftPrivate.loc[dfLeftPrivate["School"].str.contains(r"\b(?:CATHOLIC|CHRISTIAN|SAINT|OUR LADY)\b|^(?:ROCK ACADEMY)$", regex=True), "Religious"] = Tr

```

```

In [51]: # Compute admission statistics for private high schools: all vs local vs secular vs local secular
selectCampus = "UCSD"
dfLeftPrivateCampus = dfLeftPrivate[(dfLeftPrivate["Campus"] == selectCampus)]
dfLeftPrivateLocalCampus = dfLeftPrivate[(dfLeftPrivate["Campus"] == selectCampus) & (dfLeftPrivate["countyCampus"] == dfLeftPrivate["Campus"])]

```

```

dfLeftPrivateCampusNonRelig = dfLeftPrivateCampus[dfLeftPrivateCampus["Religious"] == False]
dfLeftPrivateLocalCampusNonRelig = dfLeftPrivateLocalCampus[dfLeftPrivateLocalCampus["Religious"] == False]

print(dfLeftPrivateLocalCampusNonRelig)
print()

nApp = dfLeftPrivateCampus["App"].astype(int).sum()
nAdm = dfLeftPrivateCampus["Adm"].astype(int).sum()
admRate = nAdm / nApp
print(f"{selectCampus} private schools {nApp} {nAdm} {admRate:.1%}")

nApp = dfLeftPrivateLocalCampus["App"].astype(int).sum()
nAdm = dfLeftPrivateLocalCampus["Adm"].astype(int).sum()
admRate = nAdm / nApp
print(f"{selectCampus} local private schools {nApp} {nAdm} {admRate:.1%}")

nApp = dfLeftPrivateCampusNonRelig["App"].astype(int).sum()
nAdm = dfLeftPrivateCampusNonRelig["Adm"].astype(int).sum()
admRate = nAdm / nApp
print(f"{selectCampus} secular private schools {nApp} {nAdm} {admRate:.1%}")

nApp = dfLeftPrivateLocalCampusNonRelig["App"].astype(int).sum()
nAdm = dfLeftPrivateLocalCampusNonRelig["Adm"].astype(int).sum()
admRate = nAdm / nApp
print(f"{selectCampus} local secular private schools {nApp} {nAdm} {admRate:.1%}")

```

	UC_ID	School	City	County	Campus	countyCampus	App	Adm	Enr	CAASPP_ID	AdmRate	EnrRate	Religious
3007	50470	ARMY AND NAVY ACADEMY	CARLSBAD	San Diego	UCSD	UCSD	20.0	6.0	4.0		0.300000	0.666667	False
3030	51330	BISHOPS SCHOOL	LA JOLLA	San Diego	UCSD	UCSD	68.0	15.0	NaN		0.220588	NaN	False
3049	54785	CAMBRIDGE SCHOOL	SAN DIEGO	San Diego	UCSD	UCSD	12.0	8.0	NaN		0.666667	NaN	False
3115	52843	FRANCIS W PARKER SCHOOL	SAN DIEGO	San Diego	UCSD	UCSD	58.0	8.0	NaN		0.137931	NaN	False
3159	53036	HIGH BLUFF ACADEMY	RANCHO SANTA FE	San Diego	UCSD	UCSD	13.0	3.0	NaN		0.230769	NaN	False
3189	51333	LA JOLLA COUNTRY DAY SCHOOL	LA JOLLA	San Diego	UCSD	UCSD	51.0	11.0	3.0		0.215686	0.272727	False
3269	50449	PACIFIC RIDGE SCHOOL	CARLSBAD	San Diego	UCSD	UCSD	51.0	11.0	NaN		0.215686	NaN	False

```

UCSD private schools 10014 2146 21.4%
UCSD local private schools 712 152 21.3%
UCSD secular private schools 7343 1548 21.1%
UCSD local secular private schools 273 62 22.7%

```

```

In [52]: ##### Back to CAASPP
# Explore
dfExplore = dfTests[dfTests["Percentage Standard Met and Above"] != "*"]
print(dfExplore[(dfExplore["Test ID"] == "2") & (dfExplore["Percentage Standard Met and Above"].astype(float) >= 82)])

```

County Code		District Code		District Name		School Code		School Name		Type	ID	Test	Year	Test							
Type	Test	ID	Student	Group	ID	Grade	Total	Students	Enrolled	Total	Students	Tested	Total	Students	Tested	with Scores	Mean	Scale	Score	\	
35675			01		61176				Fremont Unified	0135244						Mission San Jose High				7	2024
B	2			1	11			418		410						410		2759.3			
763410			19		64212				ABC Unified	1931880						Whitney (Gretchen) High				7	2024
B	2			1	11			176		176						176		2803.1			
952958			19		64725				Long Beach Unified	1995539						California Academy of Mathematics and Science				7	2024
B	2			1	11			170		170						170		2773.5			
982948			19		64733				Renaissance Arts Academy	0101683						Renaissance Arts Academy				9	2024
B	2			1	11			35		35						35		2776.4			
985924			19		64733				Los Angeles Unified	0102921						Dr. Richard A. Vladovic Harbor Teacher Preparation				7	2024
B	2			1	11			113		113						113		2728.5			
1063230			19		64733				Los Angeles Unified	0133736						Science Academy STEM Magnet				7	2024
B	2			1	11			43		43						43		2810.7			
1369324			19		64964				San Marino Unified	1937754						San Marino High				7	2024
B	2			1	11			203		194						194		2746.1			
1741155			29		66357				Nevada Joint Union High	0112367						William & Marian Ghidotti High				7	2024
B	2			1	11			36		34						34		2711.0			
1775438			30		66431				Anaheim Union High	3030616						Oxford Academy				7	2024
B	2			1	11			210		207						207		2756.5			
2888317			37		68346				San Dieguito Union High	0106328						Canyon Crest Academy				7	2024
B	2			1	11			553		544						544		2748.3			
3367766			43		69468				Fremont Union High	4334421						Lynbrook High				7	2024
B	2			1	11			404		398						398		2792.6			
3367948			43		69468				Fremont Union High	4334462						Monta Vista High				7	2024
B	2			1	11			438		424						424		2767.6			
3369460			43		69484				Gilroy Unified	0114496						Dr. TJ Owens Gilroy Early College Academy				7	2024
B	2			1	11			70		70						70		2718.7			
3382544			43		69534				Los Gatos-Saratoga Union High	4337762						Saratoga High				7	2024
B	2			1	11			293		283						283		2741.3			

Percentage Standard Exceeded	Count Standard Exceeded	Percentage Standard Met	Count Standard Met	Percentage Standard Met and Above	Count Standard Met and Above	Percentage Standard Nearly Met	Count Standard Nearly Met	Percentage Standard Not Met	Count Standard Not Met	Overall Total
35675	69.27	284	16.10	66	85.37					
350	10.73	44	3.90	16	410					
763410	88.64	156	10.80	19	99.43					
175	0.00	0	0.57	1	176					
952958	75.88	129	17.65	30	93.53					
159	5.29	9	1.18	2	170					
982948	80.00	28	17.14	6	97.14					
34	0.00	0	2.86	1	35					
985924	58.41	66	31.86	36	90.27					
102	7.96	9	1.77	2	113					
1063230	95.35	41	4.65	2	100.00					
43	0.00	0	0.00	0	43					
1369324	67.01	130	20.10	39	87.11					
169	8.25	16	4.64	9	194					
1741155	38.24	13	44.12	15	82.35					
28	14.71	5	2.94	1	34					
1775438	69.08	143	21.26	44	90.34					
187	7.25	15	2.42	5	207					
2888317	63.60	346	18.93	103	82.54					

449	11.58	63	5.88	32	544	
3367766	81.91	326	10.30	41		92.21
367	4.77	19	3.02	12	398	
3367948	72.17	306	17.45	74		89.62
380	6.13	26	4.25	18	424	
3369460	50.00	35	32.86	23		82.86
58	12.86	9	4.29	3	70	
3382544	64.31	182	19.79	56		84.10
238	8.13	23	7.77	22	283	

```
In [53]: print(dfTests[dfTests["School Name"].str.contains(r"Middle College|Canyon Crest|Torrey Pines", regex=True)])
```

County Code		District Code		District Name		School Code		School Name		Type	ID	Test	Year	Test										
Type	Test	ID	Student	Group	ID	Grade	Total	Students	Enrolled	Total	Students	Tested	Total	Students	Tested	with	Scores	Mean	Scale	Score	\	Test	Year	Test
277980			07		61796				West Contra Costa Unified		0730291						Middle College	High			7		2024	
B	1			1	11			72			72						72		2689.6					
278066			07		61796				West Contra Costa Unified		0730291						Middle College	High			7		2024	
B	2			1	11			72			72						72		2638.0					
388051			10		62166				Fresno Unified		0108555				Design Science	Middle College	High				7		2024	
B	1			1	11			55			55						55		2668.7					
388127			10		62166				Fresno Unified		0108555				Design Science	Middle College	High				7		2024	
B	2			1	11			55			55						55		2642.2					
432610			10		62265				Kings Canyon Joint Unified		0126292				Reedley	Middle College	High				10		2024	
B	1			1	11			56			56						56		2701.7					
432676			10		62265				Kings Canyon Joint Unified		0126292				Reedley	Middle College	High				10		2024	
B	2			1	11			56			56						56		2616.5					
719814			16		63982				Lemoore Middle College	High	0110205				Lemoore	Middle College	High				9		2024	
B	1			1	11			58			57						57		2680.9					
719904			16		63982				Lemoore Middle College	High	0110205				Lemoore	Middle College	High				9		2024	
B	2			1	11			58			58						58		2621.4					
985708			19		64733				Los Angeles Unified		0102913						Middle College	High			7		2024	
B	1			1	11			78			78						78		2666.4					
985772			19		64733				Los Angeles Unified		0102913						Middle College	High			7		2024	
B	2			1	11			78			78						78		2635.7					
1750030			30		10306				Unity Middle College	High	0133959				Unity	Middle College	High				9		2024	
B	1			1	11			12			12						12		2537.2					
1750086			30		10306				Unity Middle College	High	0133959				Unity	Middle College	High				9		2024	
B	2			1	11			12			12						12		2463.6					
1923914			30		66670				Santa Ana Unified		3030608						Middle College	High			7		2024	
B	1			1	11			93			93						93		2707.7					
1923992			30		66670				Santa Ana Unified		3030608						Middle College	High			7		2024	
B	2			1	11			93			93						93		2650.8					
2083588			33		66993				Beaumont Unified		0141366				Beaumont	Middle College	High				7		2024	
B	1			1	11			21			21						21		2628.5					
2083652			33		66993				Beaumont Unified		0141366				Beaumont	Middle College	High				7		2024	
B	2			1	11			21			21						21		2618.3					
2211426			33		67249				San Jacinto Unified		0142117				San Jacinto	Middle College	High				7		2024	
B	1			1	11			10			10						10		*					
2211478			33		67249				San Jacinto Unified		0142117				San Jacinto	Middle College	High				7		2024	
B	2			1	11			10			10						10		*					
2590486			36		67876				San Bernardino City Unified		3631090						Middle College	High			7		2024	
B	1			1	11			71			70						70		2692.6					
2590562			36		67876				San Bernardino City Unified		3631090						Middle College	High			7		2024	
B	2			1	11			71			70						70		2666.6					
2753221			37		68130				Grossmont Union High		3731536				Grossmont	Middle College	High				7		2024	
B	1			1	11			23			22						22		2691.7					
2753275			37		68130				Grossmont Union High		3731536				Grossmont	Middle College	High				7		2024	
B	2			1	11			23			21						21		2680.6					
2795011			37		68296				Poway Unified		0141390				Poway to Palomar	Middle College	High				7		2024	
B	1			1	11			42			38						38		2649.6					
2795079			37		68296				Poway Unified		0141390				Poway to Palomar	Middle College	High				7		2024	
B	2			1	11			42			38						38		2603.3					
2820000			37		68338				Health Sciences High and Middle College		0114462				Health Sciences High and Middle College						9		2024	
B	1			1	11			155			153						153		2571.7					

2820092		37	68338	Health Sciences High and Middle College	0114462	Health Sciences High and Middle College		9	2024
B	2		11	155	153	153	2512.1		
2821514		37	68338	San Diego Unified	0118315	East Village Middle College High		7	2024
B	1		11	29	28	28	2657.9		
2821594		37	68338	San Diego Unified	0118315	East Village Middle College High		7	2024
B	2		11	29	28	28	2594.9		
2888223		37	68346	San Dieguito Union High	0106328	Canyon Crest Academy		7	2024
B	1		11	553	545	545	2715.6		
2888317		37	68346	San Dieguito Union High	0106328	Canyon Crest Academy		7	2024
B	2		11	553	544	544	2748.3		
2888697		37	68346	San Dieguito Union High	3730033	Torrey Pines High		7	2024
B	1		11	640	622	621	2670.5		
2888799		37	68346	San Dieguito Union High	3730033	Torrey Pines High		7	2024
B	2		11	639	621	620	2650.7		
3046939		39	68585	Lodi Unified	3930443	Middle College High		7	2024
B	1		11	78	71	71	2720.3		
3047023		39	68585	Lodi Unified	3930443	Middle College High		7	2024
B	2		11	78	71	71	2702.1		
3456862		43	73387	Milpitas Unified	0140673	Milpitas Middle College High		7	2024
B	1		11	19	19	19	2627.1		
3456934		43	73387	Milpitas Unified	0140673	Milpitas Middle College High		7	2024
B	2		11	19	18	18	2631.1		
3950228		56	72546	Oxnard Union High	0136804	Oxnard Middle College High		7	2024
B	1		11	48	48	48	2690.9		
3950296		56	72546	Oxnard Union High	0136804	Oxnard Middle College High		7	2024
B	2		11	48	48	48	2633.7		
4018838		57	72694	Washington Unified	0135939	Washington Middle College High		10	2024
B	1		11	21	21	21	2678.6		
4018906		57	72694	Washington Unified	0135939	Washington Middle College High		10	2024
B	2		11	21	21	21	2651.0		

	Percentage Standard Exceeded	Count Standard Exceeded	Percentage Standard Met	Count Standard Met	Percentage Standard Not Met	Count Standard Not Met	Overall Total
277980	55.56	40	36.11	26			91.67
66	8.33	6	0.00		0	72	
278066	20.83	15	31.94	23			52.78
38	27.78	20	19.44		14	72	
388051	50.91	28	29.09	16			80.00
44	14.55	8	5.45		3	55	
388127	27.27	15	34.55	19			61.82
34	20.00	11	18.18		10	55	
432610	62.50	35	32.14	18			94.64
53	5.36	3	0.00		0	56	
432676	8.93	5	46.43	26			55.36
31	19.64	11	25.00		14	56	
719814	52.63	30	35.09	20			87.72
50	7.02	4	5.26		3	57	
719904	13.79	8	39.66	23			53.45
31	24.14	14	22.41		13	58	
985708	48.72	38	37.18	29			85.90
67	10.26	8	3.85		3	78	
985772	12.82	10	42.31	33			55.13

43	34.62	27	10.26	8	78	
1750030	8.33	1	16.67	2		25.00
3	41.67	5	33.33	4	12	
1750086	0.00	0	8.33	1		8.33
1	0.00	0	91.67	11	12	
1923914	63.44	59	31.18	29		94.62
88	4.30	4	1.08	1	93	
1923992	21.51	20	34.41	32		55.91
52	34.41	32	9.68	9	93	
2083588	28.57	6	38.10	8		66.67
14	23.81	5	9.52	2	21	
2083652	19.05	4	38.10	8		57.14
12	9.52	2	33.33	7	21	
2211426	*	*	*	*		*
*	*	*	*	*	*	
2211478	*	*	*	*		*
*	*	*	*	*	*	
2590486	58.57	41	31.43	22		90.00
63	8.57	6	1.43	1	70	
2590562	31.43	22	34.29	24		65.71
46	24.29	17	10.00	7	70	
2753221	59.09	13	22.73	5		81.82
18	9.09	2	9.09	2	22	
2753275	42.86	9	28.57	6		71.43
15	19.05	4	9.52	2	21	
2795011	42.11	16	28.95	11		71.05
27	18.42	7	10.53	4	38	
2795079	13.16	5	28.95	11		42.11
16	23.68	9	34.21	13	38	
2820000	19.61	30	28.10	43		47.71
73	26.14	40	26.14	40	153	
2820092	5.88	9	11.11	17		16.99
26	18.30	28	64.71	99	153	
2821514	53.57	15	25.00	7		78.57
22	7.14	2	14.29	4	28	
2821594	17.86	5	25.00	7		42.86
12	21.43	6	35.71	10	28	
2888223	67.16	366	24.77	135		91.93
501	5.69	31	2.39	13	545	
2888317	63.60	346	18.93	103		82.54
449	11.58	63	5.88	32	544	
2888697	48.63	302	33.01	205		81.64
507	12.72	79	5.64	35	621	
2888799	31.61	196	28.23	175		59.84
371	18.71	116	21.45	133	620	
3046939	69.01	49	28.17	20		97.18
69	1.41	1	1.41	1	71	
3047023	39.44	28	33.80	24		73.24
52	19.72	14	7.04	5	71	
3456862	31.58	6	36.84	7		68.42
13	21.05	4	10.53	2	19	
3456934	33.33	6	16.67	3		50.00

9	16.67	3	33.33	6	18	
3950228	52.08	25	39.58	19		91.67
44	8.33	4	0.00	0	48	
3950296	22.92	11	25.00	12		47.92
23	33.33	16	18.75	9	48	
4018838	52.38	11	33.33	7		85.71
18	14.29	3	0.00	0	21	
4018906	14.29	3	42.86	9		57.14
12	28.57	6	14.29	3	21	

```
In [54]: # Test for conditions that shouldn't exist in the data
print(dfTests[dfTests["Test Year"] != "2024"])
print(dfTests[dfTests["Test Type"] != "B"])
print(dfTests[dfTests["Student Group ID"] != "1"])
print(dfTests[dfTests["Grade"] != "11"])
```

Empty DataFrame

Columns: [County Code, District Code, District Name, School Code, School Name, Type ID, Test Year, Test Type, Test ID, Student Group ID, Grade, Total Students Enrolled, Total Students Tested, Total Students Tested with Scores, Mean Scale Score, Percentage Standard Exceeded, Count Standard Exceeded, Percentage Standard Met, Count Standard Met, Percentage Standard Met and Above, Count Standard Met and Above, Percentage Standard Nearly Met, Count Standard Nearly Met, Percentage Standard Not Met, Count Standard Not Met, Overall Total]

Index: []

Empty DataFrame

Columns: [County Code, District Code, District Name, School Code, School Name, Type ID, Test Year, Test Type, Test ID, Student Group ID, Grade, Total Students Enrolled, Total Students Tested, Total Students Tested with Scores, Mean Scale Score, Percentage Standard Exceeded, Count Standard Exceeded, Percentage Standard Met, Count Standard Met, Percentage Standard Met and Above, Count Standard Met and Above, Percentage Standard Nearly Met, Count Standard Nearly Met, Percentage Standard Not Met, Count Standard Not Met, Overall Total]

Index: []

Empty DataFrame

Columns: [County Code, District Code, District Name, School Code, School Name, Type ID, Test Year, Test Type, Test ID, Student Group ID, Grade, Total Students Enrolled, Total Students Tested, Total Students Tested with Scores, Mean Scale Score, Percentage Standard Exceeded, Count Standard Exceeded, Percentage Standard Met, Count Standard Met, Percentage Standard Met and Above, Count Standard Met and Above, Percentage Standard Nearly Met, Count Standard Nearly Met, Percentage Standard Not Met, Count Standard Not Met, Overall Total]

Index: []

Empty DataFrame

Columns: [County Code, District Code, District Name, School Code, School Name, Type ID, Test Year, Test Type, Test ID, Student Group ID, Grade, Total Students Enrolled, Total Students Tested, Total Students Tested with Scores, Mean Scale Score, Percentage Standard Exceeded, Count Standard Exceeded, Percentage Standard Met, Count Standard Met, Percentage Standard Met and Above, Count Standard Met and Above, Percentage Standard Nearly Met, Count Standard Nearly Met, Percentage Standard Not Met, Count Standard Not Met, Overall Total]

Index: []

```
In [55]: # 2024: Create a CAASPP dataframe with pass rate by school. Filter out:
# * Rows with "Total Students Tested with Scores" = "" or "Total Students Tested with Scores" < 50
# * Select English Language Arts/Literacy ("Test ID" = 1) or Mathematics ("Test ID" = 2)
keepColumns = ["County Code", "District Code", "District Name", "School Code", "School Name", "Type ID", "Test ID",
               "Total Students Enrolled", "Total Students Tested", "Total Students Tested with Scores", "Mean Scale Score",
               "Count Standard Exceeded", "Count Standard Met", "Count Standard Met and Above", "Count Standard Nearly Met", "Count Standard Not Met"]
dfRight = dfTests[keepColumns] # 5428 rows
dfRight = dfRight[dfRight["Total Students Tested with Scores"] != ""] # 5016 rows remain
dfRight = dfRight[dfRight["Total Students Tested with Scores"].astype(int) >= 50] # 3105 rows remain
print(len(dfRight))

# Add columns for PassRate, County name, countyCampus
```

```

dfRight["PassRate"] = dfRight["Count Standard Met and Above"].astype(int) / dfRight["Total Students Tested with Scores"].astype(int)
dfRight["CAASPP County Name"] = dfRight["County Code"].map(code2County)
dfRight["CAASPP County Campus"] = dfRight["CAASPP County Name"].map(countyCampus)

```

3105

```

In [56]: # 2025: Create a CAASPP dataframe with pass rate by school. Filter out:
# * Rows with "Total Students Tested with Scores" = "*" or "Total Students Tested with Scores" < 50
# * Select English Language Arts/Literacy ("Test ID" = 1) or Mathematics ("Test ID" = 2)
keepColumns = ["County Code", "District Code", "District Name", "School Code", "School Name", "Type ID", "Test ID",
               "Total Students Enrolled", "Total Students Tested", "Total Students Tested with Scores", "Mean Scale Score",
               "Count Standard Exceeded", "Count Standard Met", "Count Standard Met and Above", "Count Standard Nearly Met", "Count Standard Not Met"]

dfRight2025 = dfTests2025[keepColumns]
dfRight2025 = dfRight2025[dfRight2025["Total Students Tested with Scores"] != "*"]
dfRight2025 = dfRight2025[dfRight2025["Total Students Tested with Scores"].astype(int) >= 50]
print(len(dfRight2025))

# Add columns for PassRate, County name, countyCampus
dfRight2025["PassRate"] = dfRight2025["Count Standard Met and Above"].astype(int) / dfRight2025["Total Students Tested with Scores"].astype(int)
dfRight2025["CAASPP County Name"] = dfRight2025["County Code"].map(code2County)
dfRight2025["CAASPP County Campus"] = dfRight2025["CAASPP County Name"].map(countyCampus)

```

3107

```

In [57]: ##### Read the Schools Table that includes a LCFF+ indicator. The "CDS Code" in this table concatenates the "County Code", "District Code", and "School
filename = "CA_Schl_Table_EthPct.tsv"
dfSchoolsTable = pd.read_csv(filename, delimiter="\t", dtype=str, encoding="latin1")
# dfSchoolsTable = dfSchoolsTable[dfSchoolsTable["Highest Grade"] == "12"] # What is highest grade of "San Jacinto Leadership Academy - Magnet 1", "N

for index, row in dfSchoolsTable.iterrows():
    school = row["School Name"]
    truncSchool = school[:50].strip()
    district = row["District"]
    if district != district: # NaN
        continue
    key = "|".join([truncSchool, district])
    dfSchoolsTable.at[index, "key"] = key
    dfSchoolsTable.at[index, "Trunc School Name"] = truncSchool
print(dfSchoolsTable)

```

	School and City and Zip Code	County	District Highest Grade	CDS Code	LCFF Plus School t U
C Feeder High School t Closest UC Campus	SAPEP Programs Serving School	School Name	Street \		
0	21st Century Learning Institute,Beaumont,92223-1927	Riverside	Beaumont Unified	12	33669930129882
N	UCR (none)	21st Century Learning Institute	939 East 10th Street		
1	'O Me-nok Learning Center,Klamath,95548-0065	Del Norte	Del Norte County Unified	6	08618206005417
N	UCD (none)	'O Me-nok Learning Center	300 Minot Creek Road		
2	A. E. Arnold Elementary,Cypress,90630-2724	Orange	Cypress Elementary	6	30664806027767
N	UCI (none)	A. E. Arnold Elementary	9281 Denni Street		
3	A. G. Cook Elementary,Garden Grove,92844-2819	Orange	Garden Grove Unified	6	30665226028211
N	UCI (none)	A. G. Cook Elementary	9802 Woodbury Avenue		
4	A. H. Morgan Elementary,Rialto,92376-3665	San Bernardino	Rialto Unified	5	36678506036669
N	UCR (none)	A. H. Morgan Elementary	1571 North Sycamore Avenue		
...	...	...	...	...	...
...	...	...	...	...	...
10393	Zehnder Ranch Elementary,Elk Grove,95757-6262	Sacramento	Elk Grove Unified	6	34673140135400
N	UCD (none)	Zehnder Ranch Elementary	9880 Denali Circle		
10394	Zela Davis,Hawthorne,90250-6228	Los Angeles	Hawthorne	5	19645926014039
N	UCLA P-20 Regional Intersegmental Alliances		Zela Davis	13435	South Yukon Avenue
10395	Zephyr Lane Elementary,Bakersfield,93307-3180	Kern	Fairfax Elementary	6	15634610129130
N	UCSB (none)	Zephyr Lane Elementary	6327 Zephyr Lane		
10396	Zoe Barnum High,Eureka,95501-4074	Humboldt	Eureka City Schools	12	12755151232057
N	UCD (none)	Zoe Barnum High	216 West Harris Street		
10397	Zupanic Virtual Academy,Rialto,92376-6926	San Bernardino	Rialto Unified	12	36678503630530
Y	UCR (none)	Zupanic Virtual Academy	266 West Randall Avenue		

	City	Zip Code	ATP Code	Students Served	Masked Enrollment	Total % African American	% Amerian Indian	% Chicanx/Latinx	% Asian & Pac. Is
1. % White	% Other/Unknown	Geodesic Distance	Calpads	Upp	key	Trunc	School Name		
0	Beaumont	92223-1927	NaN	NaN	668	8.1%	NaN	55.4%	3.
9%	26.9%	5.7%	21	1	21st Century Learning Institute Beaumont Unified	21st Century Learning Institute			
1	Klamath	95548-0065	NaN	NaN	190	NaN	58.9%	14.7%	N
aN	14.7%	11.6%	240	1	'O Me-nok Learning Center Del Norte County Unified	'O Me-nok Learning Center			
2	Cypress	90630-2724	NaN	NaN	1,270	5.5%	NaN	35.0%	32.
4%	22.2%	4.6%	18	1	A. E. Arnold Elementary Cypress Elementary	A. E. Arnold Elementary			
3	Garden Grove	92844-2819	NaN	NaN	694	NaN	NaN	20.2%	71.
2%	4.6%	4.0%	11	1	A. G. Cook Elementary Garden Grove Unified	A. G. Cook Elementary			
4	Rialto	92376-3665	NaN	NaN	1,018	12.4%	NaN	82.3%	1.
0%	1.8%	2.2%	11	1	A. H. Morgan Elementary Rialto Unified	A. H. Morgan Elementary			
...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...
10393	Elk Grove	95757-6262	NaN	NaN	2,070	6.5%	NaN	15.7%	56.
6%	11.9%	9.3%	22	0	Zehnder Ranch Elementary Elk Grove Unified	Zehnder Ranch Elementary			
10394	Hawthorne	90250-6228	NaN	NaN	1,798	30.3%	NaN	61.6%	3.
4%	1.3%	3.1%	13	1	Zela Davis Hawthorne	Zela Davis			
10395	Bakersfield	93307-3180	NaN	NaN	1,514	0.9%	NaN	88.9%	2.
6%	5.8%	1.6%	79	1	Zephyr Lane Elementary Fairfax Elementary	Zephyr Lane Elementary			
10396	Eureka	95501-4074	NaN	NaN	150	NaN	9.3%	20.0%	4.
0%	50.7%	16.0%	200	1	Zoe Barnum High Eureka City Schools	Zoe Barnum High			
10397	Rialto	92376-6926	053967	NaN	1,196	10.5%	NaN	83.1%	1.
0%	2.8%	2.2%	8	1	Zupanic Virtual Academy Rialto Unified	Zupanic Virtual Academy			

[10398 rows x 26 columns]

```

In [58]: # 2024: For each CAASPP school, find matching Schools Table records and add the LCFF Plus indicator to dfRight
dfRight["LCFF Plus School t"] = "?" # Add a new column with default value for all rows
nNone = 0
nOne = 0
nMore = 0
for index, row in dfRight.iterrows():
    cdsCode = "".join([row["County Code"], row["District Code"], row["School Code"]])
    school = row["School Name"]
    district = row["District Name"]
    dfResult = dfSchoolsTable[dfSchoolsTable["CDS Code"] == cdsCode]
    if len(dfResult) == 1: # Exactly one match
        nOne += 1
        dfRight.at[index, "LCFF Plus School t"] = dfResult.iloc[0]["LCFF Plus School t"]
        # if dfResult.iloc[0]["Highest Grade"] != "12":
        #     print(dfResult.iloc[0])
        # The following schools have "Highest Grade" = 10:
        # * San Jacinto Leadership Academy - Magnet
        # * Norton Science and Language Academy
    elif len(dfResult) > 1: # More than one match
        nMore += 1
    else: # No school/district match
        print("NO MATCH:", school, row["Test ID"])
        nNone += 1

print(nNone, nOne, nMore)

```

```

NO MATCH: Los Angeles Unified District Level Program 1
NO MATCH: Los Angeles Unified District Level Program 2
NO MATCH: Sweetwater Union High District Level Program 1
NO MATCH: Sweetwater Union High District Level Program 2
4 3101 0

```

```

In [59]: # 2025: For each CAASPP school, find matching Schools Table records and add the LCFF Plus indicator to dfRight2025
dfRight2025["LCFF Plus School t"] = "?" # Add a new column with default value for all rows
nNone = 0
nOne = 0
nMore = 0
for index, row in dfRight2025.iterrows():
    cdsCode = "".join([row["County Code"], row["District Code"], row["School Code"]])
    school = row["School Name"]
    district = row["District Name"]
    result = dfSchoolsTable[dfSchoolsTable["CDS Code"] == cdsCode]
    if len(result) == 1: # Exactly one match
        nOne += 1
        dfRight2025.at[index, "LCFF Plus School t"] = result.iloc[0]["LCFF Plus School t"]
    elif len(result) > 1: # More than one match
        nMore += 1
    else: # No school/district match
        print("NO MATCH:", school, row["Test ID"])
        nNone += 1

print(nNone, nOne, nMore)

```


NO MATCH: Los Angeles Unified District Level Program 1
 NO MATCH: Los Angeles Unified District Level Program 2
 NO MATCH: Sweetwater Union High District Level Program 1
 NO MATCH: Sweetwater Union High District Level Program 2
 4 3103 0

```
In [60]: print(dfRight[dfRight["School Name"].str.contains(r"Canyon Crest|Torrey Pines|Gompers|Preuss", regex=True)])
```

County Code	District Code	District Name	School Code	School Name	Type	ID	Test ID	Total Students	Enrolled	Total S
2821922	37	68338 Gompers Preparatory Academy	0119610	Gompers Preparatory Academy		9	1	158		
157		157 2603.9	39	55				94		
2822143	37	68338 Gompers Preparatory Academy	0119610	Gompers Preparatory Academy		9	2	157		
156		156 2502.8	3	11				14		
2833602	37	68338 Preuss School UCSD	3731189	Preuss School UCSD		9	1	117		
116		116 2652.5	52	38				90		
2833810	37	68338 Preuss School UCSD	3731189	Preuss School UCSD		9	2	117		
117		117 2617.7	26	28				54		
2888223	37	68346 San Dieguito Union High	0106328	Canyon Crest Academy		7	1	553		
545		545 2715.6	366	135				501		
2888317	37	68346 San Dieguito Union High	0106328	Canyon Crest Academy		7	2	553		
544		544 2748.3	346	103				449		
2888697	37	68346 San Dieguito Union High	3730033	Torrey Pines High		7	1	640		
622		621 2670.5	302	205				507		
2888799	37	68346 San Dieguito Union High	3730033	Torrey Pines High		7	2	639		
621		620 2650.7	196	175				371		

Count	Standard	Nearly Met	Count	Standard	Not Met	PassRate	CAASPP	County	Name	CAASPP	County	Campus	LCFF	Plus	School	t
2821922		40			23	0.598726		San	Diego			UCSD				Y
2822143		41			101	0.089744		San	Diego			UCSD				Y
2833602		19			7	0.775862		San	Diego			UCSD				Y
2833810		29			34	0.461538		San	Diego			UCSD				Y
2888223		31			13	0.919266		San	Diego			UCSD				N
2888317		63			32	0.825368		San	Diego			UCSD				N
2888697		79			35	0.816425		San	Diego			UCSD				N
2888799		116			133	0.598387		San	Diego			UCSD				N

```
In [61]: print(dfRight2025[dfRight2025["School Name"].str.contains(r"Canyon Crest|Torrey Pines|Gompers|Preuss", regex=True)])
```

County Code	District Code	District Name	School Code	School Name	Type	ID	Test ID	Total Students Enrolled	Total S
2810944	37	68338 Gompers Preparatory Academy	0119610	Gompers Preparatory Academy		9	1	177	
176		176 2621.2	43	77				120	
2811152	37	68338 Gompers Preparatory Academy	0119610	Gompers Preparatory Academy		9	2	177	
175		175 2543.5	6	25				31	
2822873	37	68338 Preuss School UCSD	3731189	Preuss School UCSD		9	1	115	
114		113 2666.6	50	43				93	
2823091	37	68338 Preuss School UCSD	3731189	Preuss School UCSD		9	2	115	
114		114 2626.5	28	28				56	
2878175	37	68346 San Dieguito Union High	0106328	Canyon Crest Academy		7	1	572	
561		561 2724.8	404	117				521	
2878267	37	68346 San Dieguito Union High	0106328	Canyon Crest Academy		7	2	572	
559		559 2756.7	370	113				483	
2878653	37	68346 San Dieguito Union High	3730033	Torrey Pines High		7	1	605	
582		581 2669.4	289	175				464	
2878751	37	68346 San Dieguito Union High	3730033	Torrey Pines High		7	2	605	
584		584 2648.0	202	123				325	

Count	Standard	Nearly Met	Count	Standard	Not Met	PassRate	CAASPP	County Name	CAASPP	County	Campus	LCFF	Plus	School	t
2810944		40			16	0.681818		San Diego			UCSD			Y	
2811152		63			81	0.177143		San Diego			UCSD			Y	
2822873		17			3	0.823009		San Diego			UCSD			Y	
2823091		23			35	0.491228		San Diego			UCSD			Y	
2878175		27			13	0.928699		San Diego			UCSD			N	
2878267		38			38	0.864043		San Diego			UCSD			N	
2878653		73			44	0.798623		San Diego			UCSD			N	
2878751		117			142	0.556507		San Diego			UCSD			N	

```
In [62]: # For each CAASPP school, find matching Schools Table records based on name and district
# nOne = 0
# nNone = 0
# nMore1 = 0
# nMore2 = 0
# nBadDistrict = 0
# for index, row in dfRight.iterrows():
#     if row["Test ID"] != "2":
#         continue
#     school = row["School Name"]
#     district = row["District Name"]
#     key = "|".join([school, district])
#     result1 = dfSchoolsTable[dfSchoolsTable["key"] == key] # Match school and district names
#     result2 = dfSchoolsTable[dfSchoolsTable["Trunc School Name"] == school] # Match school name only
#     if len(result1) == 1: # Exactly one school/district match
#         nOne += 1
#     elif len(result1) > 1: # More than one school/district match
#         nMore1 += 1
#     else: # No school/district match
#         if len(result2) == 1: # Exactly one school match => district name must differ
#             nBadDistrict += 1
#         # if nNone + nBadDistrict < 10:
#         #     print(district, "|", result2.iloc[0]["District"])
#     elif len(result2) > 1: # More than one school match
```

```

#         nMore2 += 1
#         matchingDistricts = " | ".join(result2["District"])
#         matchingCodes = " | ".join(result2["CDS Code"])
#         print("BAD DISTRICT:", school, "||", district, "||", matchingDistricts, "||", row["County Code"], row["District Code"], row["School Code"])
#     else: # No school match
#         nNone += 1
#         print("NO MATCH:", school)

# print()
# print(nOne, nBadDistrict, "nNone:", nNone, "nMore1:", nMore1, "nMore2:", nMore2)

```

```

In [63]: # UC San Diego partners with 13 high schools across San Diego County through its TRiO Outreach Programs.
# These programs are designed for first-generation and/or low-income students who are motivated to attend college.
# Students receive year-round support including tutoring, academic advising, career exposure, college application help, and summer programs.
# UCSD's outreach goal is to help students successfully matriculate into higher education, including UC campuses, CSU campuses, community colleges, and
trioSchoolNames = {
    "Chula Vista High": 1,
    "Clairemont High": 1,
    "El Cajon Valley High": 1,
    "Hoover High": 1,
    "Lincoln High": 1,
    "Mar Vista High": 1,
    "Mission Bay High": 1,
    "Monte Vista High": 1,
    "Mount Miguel High": 1,
    "Mountain Empire High": 1,
    "San Diego High": 1,
    "Southwest High": 1,
    "Sweetwater Union High": 1,
}

for index, row in dfRight.iterrows():
    if row["Test ID"] != "2" or row["County Code"] != "37":
        continue
    cdsCode = "".join([row["County Code"], row["District Code"], row["School Code"]])
    schoolName = row["School Name"]
    if schoolName in trioSchoolNames:
        print(f'"{cdsCode}": "TRiO",    # {schoolName}')

"37681303731692": "TRiO",    # El Cajon Valley High
"37681303734548": "TRiO",    # Monte Vista High
"37681303734761": "TRiO",    # Mount Miguel High
"37682133734878": "TRiO",    # Mountain Empire High
"37683380107219": "TRiO",    # San Diego High
"37683380114025": "TRiO",    # Lincoln High
"37683383731213": "TRiO",    # Clairemont High
"37683383732997": "TRiO",    # Hoover High
"37683383734431": "TRiO",    # Mission Bay High

```

```

In [64]: # Still need to find:
# Chula Vista High School
# Mar Vista High School
# Southwest High School

```

```
# Sweetwater Union High School
query = r"Chula Vista|Mar Vista|Southwest|Sweetwater"
for index, row in dfRight.iterrows():
    if row["Test ID"] != "2" or row["County Code"] != "37":
        continue
    cdsCode = "".join([row["County Code"], row["District Code"], row["School Code"]])
    schoolName = row["School Name"]
    if re.search(query, schoolName, re.IGNORECASE):
        print(f'"{cdsCode}": "TRiO",      # {schoolName}')
```

```
"37680236115778": "TRiO",      # Chula Vista Learning Community Charter
"37684113730124": "TRiO",      # Southwest Senior High
"37684113731064": "TRiO",      # Chula Vista Senior High
"37684113733953": "TRiO",      # Mar Vista Senior High
"37684113738226": "TRiO",      # Sweetwater High
"37684113768411": "TRiO",      # Sweetwater Union High District Level Program
"37764710114678": "TRiO",      # High Tech High Chula Vista
```

```
In [65]: # Dictionaries for category column to highlight specific schools in plots
highlight = {
    "19647251995539": "CAMS",      # CALIFORNIA ACAD MATH & SCIENCE
    "37683460106328": "CCA",      # CANYON CREST ACADEMY
    # "37682960118935": "Norte",   # Del Norte High
    # "37683463730033": "Torrey",  # Torrey Pines High
    "37683380119610": "Partner",  # GOMPERS PREPARATORY ACADEMY
    "37683383731189": "Partner",  # PREUSS SCHOOL UCSD
    "37683380107086": "KEARNY",   # Kearny School of Biomedical Science and Technology
    "37681303731692": "TRiOy",    # El Cajon Valley High
    "37681303734548": "TRiOy",    # Monte Vista High
    "37681303734761": "TRiOy",    # Mount Miguel High
    "37682133734878": "TRiOn",    # Mountain Empire High
    "37683380107219": "TRiOn",    # San Diego High
    "37683380114025": "TRiOy",    # Lincoln High
    "37683383731213": "TRiOn",    # Clairemont High
    "37683383732997": "TRiOy",    # Hoover High
    "37683383734431": "TRiOn",    # Mission Bay High
    "37684113730124": "TRiOy",    # Southwest Senior High
    "37684113731064": "TRiOy",    # Chula Vista Senior High
    "37684113733953": "TRiOy",    # Mar Vista Senior High
    "37684113738226": "TRiOy",    # Sweetwater High
    "": "", #
}

customPalette = {
    "CAMS": "purple",
    "CCA": "black",
    "Partner": "green",
    "KEARNY": "yellow",
    "TRiOn": "darkred",
    "TRiOy": "red",
    " LCFF+": "grey",
    " ": "#1f77b4",
}
```

```

categoryText = {
    "CAMS": "California Academy of Math & Science (not LCFF+)",
    "CCA": "Canyon Crest Academy (not LCFF+)",
    "Partner": "UCSD Partners Gompers & Preuss (LCFF+)",
    "KEARNY": "Kearny School of Biomedical Science & Tech (LCFF+)",
    "TRiOn": "UCSD TRiO Outreach Program (not LCFF+)",
    "TRiOy": "UCSD TRiO Outreach Program (LCFF+)",
    " LCFF+": "Other LCFF+",
    " ": "Other not LCFF+",
}

categoryTextNarrow = {
    "CAMS": "CA Acad Ma&Sci\n(not LCFF+)",
    "CCA": "CCA\n(not LCFF+)",
    "Partner": "UCSD Partners\n(LCFF+)",
    "KEARNY": "Kearny Biomed\n(LCFF+)",
    "TRiOy": "UCSD TRiO\nLCFF+",
    "TRiOn": "UCSD TRiO\nnot LCFF+",
    " LCFF+": "Other\nLCFF+",
    " ": "Other\nnot LCFF+",
}

```

```

In [66]: # Add a new column 'Category'
dfRight["Category"] = " " # Initialize with " " for all rows. " " sorts before strings beginning with letters.
for index, row in dfRight.iterrows():
    cdsCode = "".join([row["County Code"], row["District Code"], row["School Code"]])
    if cdsCode in highlight:
        dfRight.at[index, "Category"] = highlight[cdsCode]
        if row["Test ID"] == "2":
            print(cdsCode, row["School Name"], row["LCFF Plus School t"])
    elif row["LCFF Plus School t"] == "Y":
        dfRight.at[index, "Category"] = " LCFF+"

```

```

19647251995539 California Academy of Mathematics and Science N
37681303731692 El Cajon Valley High Y
37681303734548 Monte Vista High Y
37681303734761 Mount Miguel High Y
37682133734878 Mountain Empire High N
37683380107086 Kearny School of Biomedical Science and Technology Y
37683380107219 San Diego High N
37683380114025 Lincoln High Y
37683380119610 Gompers Preparatory Academy Y
37683383731189 Preuss School UCSD Y
37683383731213 Clairemont High N
37683383732997 Hoover High Y
37683383734431 Mission Bay High N
37683460106328 Canyon Crest Academy N
37684113730124 Southwest Senior High Y
37684113731064 Chula Vista Senior High Y
37684113733953 Mar Vista Senior High Y
37684113738226 Sweetwater High Y

```

```
In [67]: print(dfRight)
```

County Code	District Code	District Name	School Code	School Name	Type	ID	Test	ID	Total	Students	Enro							
lled	Total	Students	Tested	Total	Students	Tested	with	Scores	Mean	Scale	Score	Count	Standard	Exceeded	Count	Standard	Met	\
11469	01		61119		Alameda	Unified		0130229		Alameda	High	7		1				
467		440		440				2647.1				181		157				
11569	01		61119		Alameda	Unified		0130229		Alameda	High	7		2				
466		438		438				2623.7				131		109				
11791	01		61119		Alameda	Community	Learning	Center		0130609		Alameda	Community	Learning	Center		9	1
53		51		51				2556.0				8		10				
11999	01		61119		Alameda	Community	Learning	Center		0130609		Alameda	Community	Learning	Center		9	2
53		50		50				2515.0				3		10				
12220	01		61119		Alameda	Unified		0132142		Encinal	Junior/Senior	High	7		1			
239		218		218				2627.4				69		77				
...	...		...					...				...		...				
...		...		...				...				...		...				
4039195	58		72736		Marysville	Joint	Unified		5830096	South	Lindhurst	Continuation	High	7		2		
140		128		128				2439.4				0		1				
4039649	58		72736		Marysville	Joint	Unified		5835202		Marysville	High	7		1			
231		223		223				2542.9				29		56				
4039751	58		72736		Marysville	Joint	Unified		5835202		Marysville	High	7		2			
231		227		227				2501.7				5		17				
4050422	58		72769		Wheatland	Union	High		5838305		Wheatland	Union	High	7		1		
240		237		237				2582.1				48		80				
4050524	58		72769		Wheatland	Union	High		5838305		Wheatland	Union	High	7		2		
240		237		237				2545.9				14		41				

Count	Standard	Met	and Above	Count	Standard	Nearly	Met	Count	Standard	Not	Met	PassRate	CAASPP	County	Name	CAASPP	County	Campus	LCFF	Plus	School
l t	Category																				
11469			338		52		50	0.768182	Alameda		Berkeley										
N																					
11569			240		72		126	0.547945	Alameda		Berkeley										
N																					
11791			18		20		13	0.352941	Alameda		Berkeley										
N																					
11999			13		8		29	0.260000	Alameda		Berkeley										
N																					
12220			146		39		33	0.669725	Alameda		Berkeley										
N																					
...			...		...		...	...													
...	...																				
4039195			1		10		117	0.007812	Yuba		UC Davis										
Y LCFF+																					
4039649			85		65		73	0.381166	Yuba		UC Davis										
N																					
4039751			22		58		147	0.096916	Yuba		UC Davis										
N																					
4050422			128		59		50	0.540084	Yuba		UC Davis										
Y LCFF+																					
4050524			55		62		120	0.232068	Yuba		UC Davis										
Y LCFF+																					

[3105 rows x 21 columns]

```
In [68]: # Overall CAASPP stats, not grouped by school types
dfRightByYear = {
    "2024": dfRight,
    "2025": dfRight2025
}
for year in ("2024", "2025"):
    df = dfRightByYear[year]
    for selectTestID in ("2", "1"):
        filteredRight = df[df["Test ID"] == selectTestID]
        nStudents = filteredRight["Total Students Enrolled"].astype(int).sum()
        nStudentsMet = filteredRight["Count Standard Met and Above"].astype(int).sum()
        nStudentsNotMet = filteredRight["Count Standard Nearly Met"].astype(int).sum() + filteredRight["Count Standard Not Met"].astype(int).sum()
        rateStudentsMet = nStudentsMet / nStudents
        print(f"{year} {testID2Name[selectTestID]:20} {nStudents:6} {nStudentsMet:6} {nStudentsNotMet:6} {rateStudentsMet:5.1%}")

2024 Math          438903 119026 296830 27.1%
2024 English (ELA) 439205 236153 180330 53.8%
2025 Math          433498 129025 284181 29.8%
2025 English (ELA) 433722 239913 173986 55.3%
```

```
In [69]: # Compute CAASPP statistics independent of UC applications (2024). Same info is buried in statsCAASPP() output below.
filteredRight = dfRight[dfRight["Test ID"] == "2"]
nStudentsAll = filteredRight["Total Students Enrolled"].astype(int).sum()
filteredRight = filteredRight[filteredRight["PassRate"] <= 0.25]
nStudents25 = filteredRight["Total Students Enrolled"].astype(int).sum()
rate = nStudents25 / nStudentsAll
print(f"CAASPP Pass <= 25% Schools {rate:.1%} {nStudents25} {nStudentsAll}")
```

CAASPP Pass <= 25% Schools 52.4% 229828 438903

```
In [70]: def statsCAASPP(df, year):
    # Fraction of schools < 25%
    # Fraction of students in schools < 25%
    # Fraction of LCFF+ schools
    # Fraction of students in LCFF+ schools
    for selectTestID in ("2", "1"):
        filteredRight = df[df["Test ID"] == selectTestID]

        nSchools = len(filteredRight)
        nStudents = filteredRight["Total Students Enrolled"].astype(int).sum()

        dfLowCAASPP = filteredRight[filteredRight["PassRate"] <= 0.25]
        nSchoolsLowCAASPP = len(dfLowCAASPP)
        nStudentsLowCAASPP = dfLowCAASPP["Total Students Enrolled"].astype(int).sum()
        rateSchoolsLowCAASPP = nSchoolsLowCAASPP / nSchools
        rateStudentsLowCAASPP = nStudentsLowCAASPP / nStudents

        dfHighCAASPP = filteredRight[filteredRight["PassRate"] >= 0.75]
        nSchoolsHighCAASPP = len(dfHighCAASPP)
        nStudentsHighCAASPP = dfHighCAASPP["Total Students Enrolled"].astype(int).sum()
        rateSchoolsHighCAASPP = nSchoolsHighCAASPP / nSchools
        rateStudentsHighCAASPP = nStudentsHighCAASPP / nStudents
```



```

dfLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "Y"]
nSchoolsLCFF = len(dfLCFF)
nStudentsLCFF = dfLCFF["Total Students Enrolled"].astype(int).sum()
rateSchoolsLCFF = nSchoolsLCFF / nSchools
rateStudentsLCFF = nStudentsLCFF / nStudents
topSchoolLCFF = dfLCFF.loc[dfLCFF["PassRate"].idxmax(), "School Name"]

dfLCFFLowCAASPP = dfLCFF[dfLCFF["PassRate"] <= 0.25]
nSchoolsLCFFLowCAASPP = len(dfLCFFLowCAASPP)
nStudentsLCFFLowCAASPP = dfLCFFLowCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsLCFFLowCAASPP = nSchoolsLCFFLowCAASPP / nSchoolsLCFF
rateStudentsLCFFLowCAASPP = nStudentsLCFFLowCAASPP / nStudentsLCFF

dfLCFFHighCAASPP = dfLCFF[dfLCFF["PassRate"] >= 0.75]
nSchoolsLCFFHighCAASPP = len(dfLCFFHighCAASPP)
nStudentsLCFFHighCAASPP = dfLCFFHighCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsLCFFHighCAASPP = nSchoolsLCFFHighCAASPP / nSchoolsLCFF
rateStudentsLCFFHighCAASPP = nStudentsLCFFHighCAASPP / nStudentsLCFF

dfNotLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "N"]
nSchoolsNotLCFF = len(dfNotLCFF)
nStudentsNotLCFF = dfNotLCFF["Total Students Enrolled"].astype(int).sum()
rateSchoolsNotLCFF = nSchoolsNotLCFF / nSchools
rateStudentsNotLCFF = nStudentsNotLCFF / nStudents

dfNotLCFFLowCAASPP = dfNotLCFF[dfNotLCFF["PassRate"] <= 0.25]
nSchoolsNotLCFFLowCAASPP = len(dfNotLCFFLowCAASPP)
nStudentsNotLCFFLowCAASPP = dfNotLCFFLowCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsNotLCFFLowCAASPP = nSchoolsNotLCFFLowCAASPP / nSchoolsNotLCFF
rateStudentsNotLCFFLowCAASPP = nStudentsNotLCFFLowCAASPP / nStudentsNotLCFF

dfNotLCFFHighCAASPP = dfNotLCFF[dfNotLCFF["PassRate"] >= 0.75]
nSchoolsNotLCFFHighCAASPP = len(dfNotLCFFHighCAASPP)
nStudentsNotLCFFHighCAASPP = dfNotLCFFHighCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsNotLCFFHighCAASPP = nSchoolsNotLCFFHighCAASPP / nSchoolsNotLCFF
rateStudentsNotLCFFHighCAASPP = nStudentsNotLCFFHighCAASPP / nStudentsNotLCFF

ratioSchoolsFailLCFF = rateSchoolsLowCAASPP / rateSchoolsLCFF
ratioStudentsFailLCFF = rateStudentsLowCAASPP / rateStudentsLCFF

print(f"{year} {testID2Name[selectTestID]:20} CAASPP LCFF Not LCFF")
print("      n nLow nHigh rLow rHigh nLCFF nNot rLCFF rNot Low/LCFF rLow rHigh rLow rHigh topSchoolLCFF")
print(f"All      Counties nSchools: {nSchools:6} {nSchoolsLowCAASPP:6} {nSchoolsHighCAASPP:6} {rateSchoolsLowCAASPP:5.1%} {rateSchoolsHighCAASPP:5.1%}")
print(f"All      Counties nStudents: {nStudents:6} {nStudentsLowCAASPP:6} {nStudentsHighCAASPP:6} {rateStudentsLowCAASPP:5.1%} {rateStudentsHighCAASPP:5.1%}")

for selectCampus in campuses:
    filteredRight = df[df["Test ID"] == selectTestID]
    filteredRight = filteredRight[filteredRight["CAASPP County Campus"] == selectCampus]

    nSchools = len(filteredRight)

```

```

nStudents = filteredRight["Total Students Enrolled"].astype(int).sum()

dfLowCAASPP = filteredRight[filteredRight["PassRate"] <= 0.25]
nSchoolsLowCAASPP = len(dfLowCAASPP)
nStudentsLowCAASPP = dfLowCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsLowCAASPP = nSchoolsLowCAASPP / nSchools
rateStudentsLowCAASPP = nStudentsLowCAASPP / nStudents

dfHighCAASPP = filteredRight[filteredRight["PassRate"] >= 0.75]
nSchoolsHighCAASPP = len(dfHighCAASPP)
nStudentsHighCAASPP = dfHighCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsHighCAASPP = nSchoolsHighCAASPP / nSchools
rateStudentsHighCAASPP = nStudentsHighCAASPP / nStudents

dfLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "Y"]
nSchoolsLCFF = len(dfLCFF)
nStudentsLCFF = dfLCFF["Total Students Enrolled"].astype(int).sum()
rateSchoolsLCFF = nSchoolsLCFF / nSchools
rateStudentsLCFF = nStudentsLCFF / nStudents
topSchoolLCFF = dfLCFF.loc[dfLCFF["PassRate"].idxmax(), "School Name"]

dfLCFFLowCAASPP = dfLCFF[dfLCFF["PassRate"] <= 0.25]
nSchoolsLCFFLowCAASPP = len(dfLCFFLowCAASPP)
nStudentsLCFFLowCAASPP = dfLCFFLowCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsLCFFLowCAASPP = nSchoolsLCFFLowCAASPP / nSchoolsLCFF
rateStudentsLCFFLowCAASPP = nStudentsLCFFLowCAASPP / nStudentsLCFF

dfLCFFHighCAASPP = dfLCFF[dfLCFF["PassRate"] >= 0.75]
nSchoolsLCFFHighCAASPP = len(dfLCFFHighCAASPP)
nStudentsLCFFHighCAASPP = dfLCFFHighCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsLCFFHighCAASPP = nSchoolsLCFFHighCAASPP / nSchoolsLCFF
rateStudentsLCFFHighCAASPP = nStudentsLCFFHighCAASPP / nStudentsLCFF

dfNotLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "N"]
nSchoolsNotLCFF = len(dfNotLCFF)
nStudentsNotLCFF = dfNotLCFF["Total Students Enrolled"].astype(int).sum()
rateSchoolsNotLCFF = nSchoolsNotLCFF / nSchools
rateStudentsNotLCFF = nStudentsNotLCFF / nStudents

dfNotLCFFLowCAASPP = dfNotLCFF[dfNotLCFF["PassRate"] <= 0.25]
nSchoolsNotLCFFLowCAASPP = len(dfNotLCFFLowCAASPP)
nStudentsNotLCFFLowCAASPP = dfNotLCFFLowCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsNotLCFFLowCAASPP = nSchoolsNotLCFFLowCAASPP / nSchoolsNotLCFF
rateStudentsNotLCFFLowCAASPP = nStudentsNotLCFFLowCAASPP / nStudentsNotLCFF

dfNotLCFFHighCAASPP = dfNotLCFF[dfNotLCFF["PassRate"] >= 0.75]
nSchoolsNotLCFFHighCAASPP = len(dfNotLCFFHighCAASPP)
nStudentsNotLCFFHighCAASPP = dfNotLCFFHighCAASPP["Total Students Enrolled"].astype(int).sum()
rateSchoolsNotLCFFHighCAASPP = nSchoolsNotLCFFHighCAASPP / nSchoolsNotLCFF
rateStudentsNotLCFFHighCAASPP = nStudentsNotLCFFHighCAASPP / nStudentsNotLCFF

```

```
ratioSchoolsFailLCFF = rateSchoolsLowCAASPP / rateSchoolsLCFF
ratioStudentsFailLCFF = rateStudentsLowCAASPP / rateStudentsLCFF

print(f"{selectCampus:9} Counties nSchools: {nSchools:6} {nSchoolsLowCAASPP:6} {nSchoolsHighCAASPP:6} {rateSchoolsLowCAASPP:5.1%} {rateSc
print(f"{selectCampus:9} Counties nStudents: {nStudents:6} {nStudentsLowCAASPP:6} {nStudentsHighCAASPP:6} {rateStudentsLowCAASPP:5.1%} {ra
print("")
```

```
In [71]: # Get CAASPP statistics independent of UC applications for 2024 and 2025
statsCAASPP(dfRight, 2024)
statsCAASPP(dfRight2025, 2025)
```

2024 Math			CAASPP					LCFF					LCFF		Not LCFF		topSchool	LCFF
			n	nLow	nHigh	rLow	rHigh	nLCFF	nNot	rLCFF	rNot	Low/LCFF	rLow	rHigh	rLow	rHigh		
All	Counties	nSchools:	1551	955	20	61.6%	1.3%	727	821	46.9%	52.9%	1.31	86.7%	0.0%	39.2%	2.4%	Middle College High	
All	Counties	nStudents:	438903	229828	6232	52.4%	1.4%	171356	267042	39.0%	60.8%	1.34	85.4%	0.0%	39.2%	2.4%		
Berkeley	Counties	nSchools:	176	88	3	50.0%	1.7%	51	125	29.0%	71.0%	1.73					AIMS College Prep High	
Berkeley	Counties	nStudents:	47380	16848	1451	35.6%	3.1%	7723	39657	16.3%	83.7%	2.18						
UC Davis	Counties	nSchools:	226	145	0	64.2%	0.0%	53	173	23.5%	76.5%	2.74					Willows High	
UC Davis	Counties	nStudents:	55094	30496	0	55.4%	0.0%	10473	44621	19.0%	81.0%	2.91						
UCI	Counties	nSchools:	92	37	1	40.2%	1.1%	41	50	44.6%	54.3%	0.90					La Quinta High	
UCI	Counties	nStudents:	36516	10704	210	29.3%	0.6%	12867	23586	35.2%	64.6%	0.83						
UCLA	Counties	nSchools:	383	228	5	59.5%	1.3%	247	135	64.5%	35.2%	0.92					Downtown Business High	
UCLA	Counties	nStudents:	99878	49115	1016	49.2%	1.0%	50101	49647	50.2%	49.7%	0.98						
UC Merced	Counties	nSchools:	152	120	1	78.9%	0.7%	92	60	60.5%	39.5%	1.30					Reedley High	
UC Merced	Counties	nStudents:	39858	30096	123	75.5%	0.3%	23406	16452	58.7%	41.3%	1.29						
UCR	Counties	nSchools:	199	153	1	76.9%	0.5%	120	79	60.3%	39.7%	1.28					Middle College High	
UCR	Counties	nStudents:	62393	42722	54	68.5%	0.1%	33015	29378	52.9%	47.1%	1.29						
UCSB	Counties	nSchools:	98	69	0	70.4%	0.0%	43	55	43.9%	56.1%	1.60					Cesar E. Chavez High	
UCSB	Counties	nStudents:	30940	21302	0	68.8%	0.0%	13952	16988	45.1%	54.9%	1.53						
UCSC	Counties	nSchools:	95	50	7	52.6%	7.4%	33	62	34.7%	65.3%	1.52					KIPP San Jose Collegiate	
UCSC	Counties	nStudents:	28227	11498	2188	40.7%	7.8%	7022	21205	24.9%	75.1%	1.64						
UCSD	Counties	nSchools:	130	65	2	50.0%	1.5%	47	82	36.2%	63.1%	1.38					Preuss School UCSD	
UCSD	Counties	nStudents:	38617	17047	1190	44.1%	3.1%	12797	25508	33.1%	66.1%	1.33						
2024 English (ELA)			CAASPP					LCFF					LCFF		Not LCFF		topSchool	LCFF
			n	nLow	nHigh	rLow	rHigh	nLCFF	nNot	rLCFF	rNot	Low/LCFF	rLow	rHigh	rLow	rHigh		
All	Counties	nSchools:	1554	203	211	13.1%	13.6%	725	826	46.7%	53.2%	0.28	23.3%	2.8%	4.0%	23.1%	Middle College High	
All	Counties	nStudents:	439205	25569	70808	5.8%	16.1%	171143	267559	39.0%	60.9%	0.15	13.0%	2.0%	4.0%	23.1%		
Berkeley	Counties	nSchools:	176	22	36	12.5%	20.5%	50	126	28.4%	71.6%	0.44					Dozier-Libbey Medical High	
Berkeley	Counties	nStudents:	47391	2839	13192	6.0%	27.8%	7632	39759	16.1%	83.9%	0.37						
UC Davis	Counties	nSchools:	229	30	16	13.1%	7.0%	55	174	24.0%	76.0%	0.55					ABLE Charter	
UC Davis	Counties	nStudents:	55293	2907	3831	5.3%	6.9%	10618	44675	19.2%	80.8%	0.27						
UCI	Counties	nSchools:	92	13	23	14.1%	25.0%	40	51	43.5%	55.4%	0.33					La Quinta High	
UCI	Counties	nStudents:	36514	2157	10472	5.9%	28.7%	12796	23655	35.0%	64.8%	0.17						
UCLA	Counties	nSchools:	383	32	58	8.4%	15.1%	247	135	64.5%	35.2%	0.13					Middle College High	
UCLA	Counties	nStudents:	99888	4660	17861	4.7%	17.9%	50106	49652	50.2%	49.7%	0.09						
UC Merced	Counties	nSchools:	153	28	13	18.3%	8.5%	92	61	60.1%	39.9%	0.30					Reedley High	
UC Merced	Counties	nStudents:	39918	3537	3999	8.9%	10.0%	23376	16542	58.6%	41.4%	0.15						
UCR	Counties	nSchools:	199	40	14	20.1%	7.0%	120	79	60.3%	39.7%	0.33					Middle College High	
UCR	Counties	nStudents:	62401	5420	4803	8.7%	7.7%	33017	29384	52.9%	47.1%	0.16						
UCSB	Counties	nSchools:	98	16	7	16.3%	7.1%	43	55	43.9%	56.1%	0.37					Wonderful College Prep Academy	
UCSB	Counties	nStudents:	30945	1762	1292	5.7%	4.2%	13938	17007	45.0%	55.0%	0.13						
UCSC	Counties	nSchools:	93	8	21	8.6%	22.6%	31	62	33.3%	66.7%	0.26					KIPP San Jose Collegiate	
UCSC	Counties	nStudents:	28075	649	6991	2.3%	24.9%	6872	21203	24.5%	75.5%	0.09						
UCSD	Counties	nSchools:	131	14	23	10.7%	17.6%	47	83	35.9%	63.4%	0.30					Preuss School UCSD	
UCSD	Counties	nStudents:	38780	1638	8367	4.2%	21.6%	12788	25682	33.0%	66.2%	0.13						
2025 Math			CAASPP					LCFF					LCFF		Not LCFF		topSchool	LCFF
			n	nLow	nHigh	rLow	rHigh	nLCFF	nNot	rLCFF	rNot	Low/LCFF	rLow	rHigh	rLow	rHigh		
All	Counties	nSchools:	1552	840	28	54.1%	1.8%	711	838	45.8%	54.0%	1.18	78.6%	0.0%	33.2%	3.3%	La Quinta High	
All	Counties	nStudents:	433498	198845	7959	45.9%	1.8%	167444	265619	38.6%	61.3%	1.19	79.0%	0.0%	33.2%	3.3%		
Berkeley	Counties	nSchools:	171	77	4	45.0%	2.3%	45	126	26.3%	73.7%	1.71					John Henry High	
Berkeley	Counties	nStudents:	46431	14152	1634	30.5%	3.5%	6960	39471	15.0%	85.0%	2.03						

UC Davis	Counties	nSchools:	233	133	0	57.1%	0.0%	53	180	22.7%	77.3%	2.51		Anderson High
UC Davis	Counties	nStudents:	55271	26239	0	47.5%	0.0%	10584	44687	19.1%	80.9%	2.48		
UCI	Counties	nSchools:	94	36	2	38.3%	2.1%	43	50	45.7%	53.2%	0.84		La Quinta High
UCI	Counties	nStudents:	36050	9608	761	26.7%	2.1%	12531	23464	34.8%	65.1%	0.77		
UCLA	Counties	nSchools:	386	180	9	46.6%	2.3%	245	140	63.5%	36.3%	0.73		Middle College High
UCLA	Counties	nStudents:	98071	38795	1800	39.6%	1.8%	48626	49337	49.6%	50.3%	0.80		
UC Merced	Counties	nSchools:	152	114	1	75.0%	0.7%	89	63	58.6%	41.4%	1.28		Corcoran High
UC Merced	Counties	nStudents:	39830	28433	120	71.4%	0.3%	23131	16699	58.1%	41.9%	1.23		
UCR	Counties	nSchools:	194	132	2	68.0%	1.0%	116	78	59.8%	40.2%	1.14		Middle College High
UCR	Counties	nStudents:	61528	36130	236	58.7%	0.4%	32222	29306	52.4%	47.6%	1.12		
UCSB	Counties	nSchools:	98	59	1	60.2%	1.0%	42	56	42.9%	57.1%	1.40		California Virtual Academy at Maricopa
UCSB	Counties	nStudents:	31261	19165	368	61.3%	1.2%	14430	16831	46.2%	53.8%	1.33		
UCSC	Counties	nSchools:	92	44	8	47.8%	8.7%	30	62	32.6%	67.4%	1.47		KIPP San Jose Collegiate
UCSC	Counties	nStudents:	27426	10551	2468	38.5%	9.0%	6613	20813	24.1%	75.9%	1.60		
UCSD	Counties	nSchools:	132	65	1	49.2%	0.8%	48	83	36.4%	62.9%	1.35		Preuss School UCSD
UCSD	Counties	nStudents:	37630	15772	572	41.9%	1.5%	12347	25011	32.8%	66.5%	1.28		

2025 English (ELA)			CAASPP						LCFF				LCFF				Not LCFF				
			n	nLow	nHigh	rLow	rHigh	nLCFF	nNot	rLCFF	rNot	Low/LCFF	rLow	rHigh	rLow	rHigh	topSchool	LCFF			
All	Counties	nSchools:	1555	197	232	12.7%	14.9%	714	838	45.9%	53.9%	0.28	23.1%	4.1%	3.6%	24.2%	Middle College High				
All	Counties	nStudents:	433722	24623	76067	5.7%	17.5%	167646	265643	38.7%	61.2%	0.15	12.9%	2.7%	3.6%	24.2%					
Berkeley	Counties	nSchools:	172	22	30	12.8%	17.4%	46	126	26.7%	73.3%	0.48					Dozier-Libbey Medical High				
Berkeley	Counties	nStudents:	46511	2761	11445	5.9%	24.6%	7045	39466	15.1%	84.9%	0.39									
UC Davis	Counties	nSchools:	235	25	28	10.6%	11.9%	54	181	23.0%	77.0%	0.46					Anderson High				
UC Davis	Counties	nStudents:	55394	2804	6330	5.1%	11.4%	10644	44750	19.2%	80.8%	0.26									
UCI	Counties	nSchools:	94	15	22	16.0%	23.4%	43	50	45.7%	53.2%	0.35					La Quinta High				
UCI	Counties	nStudents:	36052	2221	10649	6.2%	29.5%	12521	23476	34.7%	65.1%	0.18									
UCLA	Counties	nSchools:	385	29	67	7.5%	17.4%	245	139	63.6%	36.1%	0.12					Middle College High				
UCLA	Counties	nStudents:	98015	3959	19122	4.0%	19.5%	48629	49278	49.6%	50.3%	0.08									
UC Merced	Counties	nSchools:	152	24	12	15.8%	7.9%	90	62	59.2%	40.8%	0.27					Summit Charter Academy				
UC Merced	Counties	nStudents:	39859	2725	3536	6.8%	8.9%	23199	16660	58.2%	41.8%	0.12									
UCR	Counties	nSchools:	194	44	17	22.7%	8.8%	116	78	59.8%	40.2%	0.38					San Jacinto Leadership Academy				
- Magnet																					
UCR	Counties	nStudents:	61526	5799	7003	9.4%	11.4%	32221	29305	52.4%	47.6%	0.18									
UCSB	Counties	nSchools:	99	14	8	14.1%	8.1%	42	57	42.4%	57.6%	0.33					Wonderful College Prep Academy				
UCSB	Counties	nStudents:	31314	1774	1775	5.7%	5.7%	14427	16887	46.1%	53.9%	0.12									
UCSC	Counties	nSchools:	91	10	21	11.0%	23.1%	29	62	31.9%	68.1%	0.34					KIPP San Jose Collegiate				
UCSC	Counties	nStudents:	27367	860	7516	3.1%	27.5%	6553	20814	23.9%	76.1%	0.13									
UCSD	Counties	nSchools:	133	14	27	10.5%	20.3%	49	83	36.8%	62.4%	0.29					Preuss School UCSD				
UCSD	Counties	nStudents:	37684	1720	8691	4.6%	23.1%	12407	25007	32.9%	66.4%	0.14									

```
In [72]: # Get LCFF+ % of schools and % of students by region (EDGE)
selectTestID = "2"
filteredRight = dfRight[dfRight["Test ID"] == selectTestID]

nSchools = len(filteredRight)
nStudents = filteredRight["Total Students Enrolled"].astype(int).sum()

dfLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "Y"]
nSchoolsLCFF = len(dfLCFF)
nStudentsLCFF = dfLCFF["Total Students Enrolled"].astype(int).sum()
```

```

rateSchoolsLCFF = nSchoolsLCFF / nSchools
rateStudentsLCFF = nStudentsLCFF / nStudents
topSchoolLCFF = dfLCFF.loc[dfLCFF["PassRate"].idxmax(), "School Name"]
selectCampus = "All"
print("Campus      Schools  Students")
print(f"{selectCampus:9} {rateSchoolsLCFF:5.0%}    {rateStudentsLCFF:5.0%}")

for selectCampus in campuses:
    filteredRight = dfRight[dfRight["Test ID"] == selectTestID]
    filteredRight = filteredRight[filteredRight["CAASPP County Campus"] == selectCampus]

    nSchools = len(filteredRight)
    nStudents = filteredRight["Total Students Enrolled"].astype(int).sum()

    dfLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "Y"]
    nSchoolsLCFF = len(dfLCFF)
    nStudentsLCFF = dfLCFF["Total Students Enrolled"].astype(int).sum()
    rateSchoolsLCFF = nSchoolsLCFF / nSchools
    rateStudentsLCFF = nStudentsLCFF / nStudents

    print(f"{selectCampus:9} {rateSchoolsLCFF:5.0%}    {rateStudentsLCFF:5.0%}")

```

Campus	Schools	Students
All	47%	39%
Berkeley	29%	16%
UC Davis	23%	19%
UCI	45%	35%
UCLA	64%	50%
UC Merced	61%	59%
UCR	60%	53%
UCSB	44%	45%
UCSC	35%	25%
UCSD	36%	33%

```

In [73]: def createPlot(ax, dfLeft, dfRight, selectCampus, selectCounties, xAxis="PassRate", yAxis="AdmRate", filterMetrics=[], selectTest="Math", selectMinApp
    # selectCampus = "UCSD"
    # selectCounties = "San Diego"
    # selectTest = "Math" # "English", "ELA", or "Math"
    # xAxis = "PassRate" # "PassRate", "App GPA"
    # yAxis = "AdmRate" # "AdmRate", "Adm GPA", "App GPA", "EnrRate", "SeniorAdmRate"
    # selectClassSize = 100 # Records with <50 were already filtered in dfRight
    # title = "auto" # "" for no title, "auto", or a string for the title

    if selectTest in ("English", "ELA"):
        selectTestID = "1"
    else:
        selectTestID = "2" # Math

    # Filter the dataframes
    filteredLeft = dfLeft[dfLeft["Campus"] == selectCampus]
    filteredLeft = filteredLeft[filteredLeft["App"] >= selectMinApps]
    if selectCounties == "Local":
        filteredLeft = filteredLeft[filteredLeft["countyCampus"] == selectCampus]

```

```

elif selectCounties != "All":
    filteredLeft = filteredLeft[filteredLeft["County"] == selectCounties]
if xAxis == "App GPA" or yAxis == "App GPA":
    filteredLeft = filteredLeft[filteredLeft["App GPA"] >= 3.0]
if yAxis == "Adm GPA":
    filteredLeft = filteredLeft[filteredLeft["Adm GPA"] >= 3.0]
elif yAxis == "EnrRate":
    filteredLeft = filteredLeft[filteredLeft["Adm"] >= minAdmsForEnrAnalysis[selectCampus]]

filteredRight = dfRight[dfRight["Test ID"] == selectTestID]
filteredRight = filteredRight[filteredRight["Total Students Tested with Scores"].astype(int) >= selectClassSize]

# Join
joined = pd.merge(filteredLeft, filteredRight, left_on="CAASPP_ID", right_on="School Code", how="inner")

joined["SeniorAdmRate"] = joined["Adm"].astype(int) / joined["Total Students Enrolled"].astype(int) # Fraction of seniors admitted
joined["SeniorAppRate"] = joined["App"].astype(int) / joined["Total Students Enrolled"].astype(int) # Fraction of seniors who applied

# Filter out specified rows
for spec in filterMetrics:
    filterMetric = spec[0]
    filterOp = spec[1]
    filterVal = spec[2]
    if filterOp == "<=":
        joined = joined[joined[filterMetric] <= filterVal]
    elif filterOp == ">=":
        joined = joined[joined[filterMetric] >= filterVal]
    elif filterOp == "==":
        joined = joined[joined[filterMetric] == filterVal]

dfSorted = joined.sort_values(by="Category")

# Scatterplot with bubble sizes and weighted linear fit.
# Seaborn's scatterplot returns a Matplotlib Axes object (ax), which we customize.
sns.scatterplot(
    x=xAxis,
    y=yAxis,
    size="SeniorAppRate", # Bubble size is application rate
    sizes=(20, 300), # min/max point sizes
    hue="Category",
    palette=customPalette,
    data=dfSorted,
    legend=False,
    alpha=0.6,
    ax=ax
)

# Format axes as percentages
if re.search(r"Rate", xAxis):
    ax.xaxis.set_major_formatter(PercentFormatter(xmax=1, decimals=0)) # denominator=1 means 1.0 → 100%
if re.search(r"Rate", yAxis):
    ax.yaxis.set_major_formatter(PercentFormatter(xmax=1, decimals=0))

```

```

# Set axis labels
if compact == "Y":
    axisFontSize=11
    labelpad = 1
else:
    axisFontSize=12
    labelpad = 2
if xAxis == "PassRate":
    ax.set_xlabel(f"CAASPP {selectTest} Pass Rate", fontsize=axisFontSize, labelpad=labelpad)
elif xAxis == "App GPA":
    ax.set_xlabel(f"GPA of Applicants to {selectCampus}", fontsize=axisFontSize, labelpad=labelpad)
elif xAxis == "SeniorAppRate":
    ax.set_xlabel(f"Fraction of Seniors who Apply to {selectCampus}", fontsize=axisFontSize, labelpad=labelpad)
if yAxis == "AdmRate":
    if compact == "Y":
        ax.set_ylabel(f"{selectCampus} Adm. Rate", fontsize=axisFontSize, labelpad=labelpad)
    else:
        ax.set_ylabel(f"{selectCampus} Admission Rate", fontsize=axisFontSize, labelpad=labelpad)
elif yAxis == "Adm GPA":
    ax.set_ylabel(f"GPA of Admitted Applicants to {selectCampus}", fontsize=axisFontSize, labelpad=labelpad)
elif yAxis == "App GPA":
    ax.set_ylabel(f"GPA of Applicants to {selectCampus}", fontsize=axisFontSize, labelpad=labelpad)
elif yAxis == "EnrRate":
    ax.set_ylabel(f"{selectCampus} Enrollment Rate", fontsize=axisFontSize, labelpad=labelpad)
elif yAxis == "SeniorAdmRate":
    ax.set_ylabel(f"{selectCampus} nAdmitted/nSeniors", fontsize=axisFontSize, labelpad=labelpad)

# Set plot title
if title == "auto":
    if selectCounties in ("All", "Local"):
        ax.set_title(f"High Schools in {selectCounties} Counties")
    else:
        ax.set_title(f"High Schools in {selectCounties} County")
elif title != "":
    ax.set_title(title)

# Compute the weighted regression separately and then plot the line.
X = joined[xAxis]
y = joined[yAxis]
if yAxis == "Adm GPA":
    weights = joined["Adm"] # Weight is number of students admitted
elif yAxis == "EnrRate":
    weights = joined["Adm"] # Weight is number of students admitted
elif yAxis == "SeniorAdmRate":
    weights = joined["SeniorAppRate"] # Weight is fraction of seniors who applied. Is this best???
    # weights = joined["Total Students Enrolled"].astype(float) # Weight is number of students
else:
    weights = joined["App"].astype(float) # Weight is number of applications
X_const = sm.add_constant(X) # Add constant for intercept

# Weighted Least squares regression

```



```

modelWLSq = sm.WLS(y, X_const, weights=weights)
results = modelWLSq.fit()

# Get fitted line
x_vals = np.linspace(X.min(), X.max(), 100)
y_vals = results.predict(sm.add_constant(x_vals))

# Add the regression line to the plot
# If the p-value is small (commonly < 0.05), we reject the null hypothesis that the slope = 0 => there is evidence of a trend between X and Y.
slopeRange = results.conf_int().loc[xAxis]
angle0 = math.degrees(math.atan(slopeRange[0]))
angle1 = math.degrees(math.atan(slopeRange[1]))
angleDiff = abs(angle1 - angle0)
# if results.rsquared > 0.01 or results.pvalues[xAxis] < 0.03:
if angleDiff < 30:
    sns.lineplot(x=x_vals, y=y_vals, color="blue", ax=ax)

if legend == "Y":
    ax.legend([], [], frameon=False) # Remove Seaborn's automatic legend

# Build custom legend handles
legendElements = []
for category in categoryText:
    legendElements.append(Line2D([0], [0], marker='o', color='w', label=categoryText[category], markerfacecolor=customPalette[category], markeredgewidth=2))

# Add custom legend
ax.legend(handles=legendElements, bbox_to_anchor=(1.05, 1), loc="upper left")

return

```

```

In [74]: def createPlots(dfLeft, dfRight, selectCounties, xAxis="PassRate", yAxis="AdmRate", filterMetrics=[], selectTest="Math", selectMinApps=20, selectClassSize=10):
# Create a 3x3 grid of subplots. axes is a 2D numpy array of Axes objects axes[row, col]. We'll pass each to createPlot().
fig, axes = plt.subplots(3, 3, figsize=(12, 12))

for iRow in range(3):
    for iCol in range(3):
        iCampus = 3 * iRow + iCol
        campus = campuses[iCampus]
        createPlot(axes[iCol, iRow], dfLeft, dfRight, campus, selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize)

fig.suptitle(title, fontsize=16)
plt.tight_layout() # Adjust layout so plots don't overlap
plt.show()

```

```

In [75]: def createPlots10(dfLeft, dfRight, selectCounties, xAxis="PassRate", yAxis="AdmRate", filterMetrics=[], selectTest="Math", selectMinApps=20, selectClassSize=10):
# Create a 3x3 grid of subplots
fig, axes = plt.subplots(5, 2, figsize=(10, 12))

# axes is a 2D numpy array of Axes objects axes[row, col]. Pass each ax object to createPlot().
createPlot(axes[0, 0], dfLeft, dfRight, "Berkeley", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[0, 1], dfLeft, dfRight, "UC Davis", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[1, 0], dfLeft, dfRight, "UCI", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",

```

```

createPlot(axes[1, 1], dfLeft, dfRight, "UCLA", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[2, 0], dfLeft, dfRight, "UC Merced", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[2, 1], dfLeft, dfRight, "UCR", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[3, 0], dfLeft, dfRight, "UCSB", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[3, 1], dfLeft, dfRight, "UCSC", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
createPlot(axes[4, 0], dfLeft, dfRight, "UCSD", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, "",
legendAxis = axes[4, 1]
legendAxis.axis("off")
legendAxis.legend([], [], frameon=False) # Remove Seaborn's automatic Legend
legendElements = [] # Build custom Legend handles
for category in categoryText:
    legendElements.append(Line2D([0], [0], marker='o', color='w', label=categoryText[category], markerfacecolor=customPalette[category], markersize=10))
legendAxis.legend(handles=legendElements, bbox_to_anchor=(-0.1, 1), loc="upper left") # Add custom Legend

fig.suptitle(title, fontsize=16)
plt.tight_layout() # Adjust layout so plots don't overlap
plt.show()

```

```

In [76]: # Compare UCSD and UCI in a pair of plots
def createPlots2(dfLeft, dfRight, selectCounties, xAxis="PassRate", yAxis="AdmRate", filterMetrics=[], selectTest="Math", selectMinApps=20, selectClassSize=100):
    # Create a 1x2 grid of subplots
    fig, axes = plt.subplots(1, 2, figsize=(width, height))

    if campusTitles == "Y":
        title1 = "UCI"
        title2 = "UCSD"
    else:
        title1 = ""
        title2 = ""

    createPlot(axes[0], dfLeft, dfRight, "UCI", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, title1)
    createPlot(axes[1], dfLeft, dfRight, "UCSD", selectCounties, xAxis, yAxis, filterMetrics, selectTest, selectMinApps, selectClassSize, title2, legendElements)

    fig.suptitle(title, fontsize=16, y=titleGap) # titleGap specifies gap between title and plots. Use 0.93-0.96.
    plt.tight_layout() # Adjust layout so plots don't overlap
    plt.show()

```

```

In [77]: def computeStats(dfLeft, dfRight, selectCampus, selectCounties, metrics=["App", "Adm"], filterMetrics=[["PassRate", ">=", 0.0], ["PassRate", "<=", 0.2]], selectTest="Math", selectMinApps=20, selectClassSize=100):
    intMetrics = ["Total Students Enrolled"]
    if selectTest in ("English", "ELA"):
        selectTestID = "1"
    else:
        selectTestID = "2" # Math

    # Filter the dataframes
    filteredLeft = dfLeft[dfLeft["Campus"] == selectCampus]
    filteredLeft = filteredLeft[filteredLeft["App"] >= selectMinApps]
    filteredLeft = filteredLeft[filteredLeft["Adm"] >= selectMinAdms]
    if selectCounties == "Local":
        filteredLeft = filteredLeft[filteredLeft["countyCampus"] == selectCampus]
    elif selectCounties != "All":
        filteredLeft = filteredLeft[filteredLeft["County"] == selectCounties]

```

```

if "App GPA" in metrics:
    filteredLeft = filteredLeft[filteredLeft["App GPA"] >= 3.0]
if "Adm GPA" in metrics:
    filteredLeft = filteredLeft[filteredLeft["Adm GPA"] >= 3.0]

filteredRight = dfRight[dfRight["Test ID"] == selectTestID]
filteredRight = filteredRight[filteredRight["Total Students Tested with Scores"].astype(int) >= selectClassSize]

# Join
joined = pd.merge(filteredLeft, filteredRight, left_on="CAASPP_ID", right_on="School Code", how="inner")

# Calculate additional metrics
joined["SeniorAdmRate"] = joined["Adm"].astype(int) / joined["Total Students Enrolled"].astype(int) # Fraction of seniors admitted
joined["SeniorAppRate"] = joined["App"].astype(int) / joined["Total Students Enrolled"].astype(int) # Fraction of seniors who applied

# Filter out specified rows
for spec in filterMetrics:
    filterMetric = spec[0]
    filterOp = spec[1]
    filterVal = spec[2]
    if filterOp == "<=":
        joined = joined[joined[filterMetric] <= filterVal]
    elif filterOp == "<":
        joined = joined[joined[filterMetric] < filterVal]
    elif filterOp == ">=":
        joined = joined[joined[filterMetric] >= filterVal]
    elif filterOp == ">":
        joined = joined[joined[filterMetric] > filterVal]
    elif filterOp == "==":
        joined = joined[joined[filterMetric] == filterVal]
    elif filterOp == "!=":
        joined = joined[joined[filterMetric] != filterVal]
    else:
        print(f"{filterOp} is not supported by computeStats().")

# Compute the stats
results = {}
for metric in metrics:
    if metric in intMetrics:
        results[metric] = joined[metric].astype(int).sum()
    else:
        results[metric] = joined[metric].sum()

return results

```

In [78]: *# Explore overall UC application stats vs. UCSD vs. filter out schools with few applications to avoid small counts set to 0 issue*
Compare "all" to sum across campuses. The latter is smaller due to small counts set to 0 issue.

```

print("selectMinApps=0")
print("campus", "App", "Adm", "Enr")
campus = "all"
result = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=0)
print(f'{campus:20} {result["App"]:6} {result["Adm"]:5} {result["Enr"]:5}')

```

```

sumEnr = 0
for campus in campuses:
    result = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=0)
    sumEnr += result["Enr"]
    print(f'{campus:20} {result["App"]:6} {result["Adm"]:5} {result["Enr"]:5}')
print(sumEnr)
print()

print("selectMinApps=20")
print("campus", "App", "Adm", "Enr")
campus = "all"
result = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=20)
print(f'{campus:20} {result["App"]:6} {result["Adm"]:5} {result["Enr"]:5}')
sumEnr = 0
for campus in campuses:
    result = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=20)
    sumEnr += result["Enr"]
    print(f'{campus:20} {result["App"]:6} {result["Adm"]:5} {result["Enr"]:5}')
print(sumEnr)

```

```

selectMinApps=0
campus App Adm Enr
all          115596 82315 37381
Berkeley     60715  8399  3741
UC Davis     58361 21159  4108
UCI          76709 16857  4139
UCLA         78383  6836  3882
UC Merced    23032 21440  1283
UCR          43780 32935  4424
UCSB         63346 20927  3004
UCSC         49306 30725  3048
UCSD         75631 20223  4762
32391

```

```

selectMinApps=20
campus App Adm Enr
all          113577 80882 36858
Berkeley     56499  8031  3706
UC Davis     53396 19108  3916
UCI          73419 16418  4126
UCLA         75525  6780  3871
UC Merced    16076 15057  1110
UCR          39603 30015  4351
UCSB         59023 19045  2876
UCSC         44208 27716  2946
UCSD         72003 19158  4679
31581

```

```

In [79]: print("
          # Denominator: All schools
          results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[])
          appAll = results["App"]

```

```

admAll = results["Adm"]
enrAll = results["Enr"]

# CAASPP Pass Low vs high
results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25]])
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rate = float(adm) / float(app)
appRate = app / appAll
admRate = adm / admAll
enrRate = enr / enrAll
print(f"CAASPP Pass <= 25%      {campus:9} {rate:5.1%} {adm:5} {app:5}      {appRate:5.1%} {admRate:5.1%} {enrRate:5.1%}")
rate0 = rate

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.75]])
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
if app > 0:
    rate = float(adm) / float(app)
else:
    rate = 0
rateRatio = rate0 / rate
print(f"CAASPP Pass >= 75%      {campus:9} {rate:5.1%} {adm:5} {app:5} {rateRatio:.2f}")

# LCFF+ Y vs N
results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"]], selectMin
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rate = float(adm) / float(app)
appRate = app / appAll
admRate = adm / admAll
enrRate = enr / enrAll
print(f"LCFF+ Y      {campus:9} {rate:5.1%} {adm:5} {app:5}      {appRate:5.1%} {admRate:5.1%} {enrRate:5.1%}")
rate0 = rate

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "N"]], selectMin
adm = results["Adm"]
app = results["App"]
enr = results["Enr"]
rate = float(adm) / float(app)
rateRatio = rate0 / rate
print(f"LCFF+ N      {campus:9} {rate:5.1%} {adm:5} {app:5} {rateRatio:.2f}")

# LCFF+ = "Y"
results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rate = float(adm) / float(app)

```

```

appRate = app / appAll
admRate = adm / admAll
enrRate = enr / enrAll
print(f"    LCFF+, Pass <= 25% {campus:9} {rate:5.1%} {adm:5} {app:5}      {appRate:5.1%} {admRate:5.1%} {enrRate:5.1%}")
rate0 = rate

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.50], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
if app > 0:
    rate = float(adm) / float(app)
    rateRatio = rate0 / rate
else:
    rate = 0
    rateRatio = 0
print(f"    LCFF+, Pass >= 50% {campus:9} {rate:5.1%} {adm:5} {app:5} {rateRatio:.2f}")

# LCFF+ = "N"
results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rate = float(adm) / float(app)
appRate = app / appAll
admRate = adm / admAll
enrRate = enr / enrAll
print(f"Not LCFF+, Pass <= 25% {campus:9} {rate:5.1%} {adm:5} {app:5}      {appRate:5.1%} {admRate:5.1%} {enrRate:5.1%}")
rate0 = rate

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.75], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
if app > 0:
    rate = float(adm) / float(app)
else:
    rate = 0
rateRatio = rate0 / rate
print(f"Not LCFF+, Pass >= 75% {campus:9} {rate:5.1%} {adm:5} {app:5} {rateRatio:.2f}")
print()

```

		Acceptance Rate	App%	Adm%	Enr%
CAASPP Pass <= 25%	Berkeley	14.1% 1949 13831	24.5%	24.3%	18.1%
CAASPP Pass >= 75%	Berkeley	14.3% 538 3775	0.99		
LCFF+ Y	Berkeley	14.3% 1884 13135	23.2%	23.5%	14.2%
LCFF+ N	Berkeley	13.9% 6484 46718	1.03		
LCFF+, Pass <= 25%	Berkeley	14.4% 1166 8086	14.3%	14.5%	10.0%
LCFF+, Pass >= 50%	Berkeley	10.5% 29 277	1.38		
Not LCFF+, Pass <= 25%	Berkeley	13.6% 783 5745	10.2%	9.7%	8.1%
Not LCFF+, Pass >= 75%	Berkeley	14.3% 538 3775	0.96		
CAASPP Pass <= 25%	UC Davis	42.4% 5242 12368	23.2%	27.4%	28.8%
CAASPP Pass >= 75%	UC Davis	23.5% 882 3747	1.80		
LCFF+ Y	UC Davis	42.6% 4779 11206	21.0%	25.0%	16.5%
LCFF+ N	UC Davis	34.7% 16046 46222	1.23		
LCFF+, Pass <= 25%	UC Davis	43.2% 2830 6551	12.3%	14.8%	11.7%
LCFF+, Pass >= 50%	UC Davis	44.6% 78 175	0.97		
Not LCFF+, Pass <= 25%	UC Davis	41.5% 2412 5817	10.9%	12.6%	17.2%
Not LCFF+, Pass >= 75%	UC Davis	23.5% 882 3747	1.76		
CAASPP Pass <= 25%	UCI	21.7% 4511 20789	28.3%	27.5%	24.0%
CAASPP Pass >= 75%	UCI	25.5% 1039 4078	0.85		
LCFF+ Y	UCI	22.4% 4810 21491	29.3%	29.3%	28.2%
LCFF+ N	UCI	22.1% 11994 54316	1.01		
LCFF+, Pass <= 25%	UCI	21.9% 3121 14279	19.4%	19.0%	18.1%
LCFF+, Pass >= 50%	UCI	29.6% 141 476	0.74		
Not LCFF+, Pass <= 25%	UCI	21.4% 1390 6510	8.9%	8.5%	6.0%
Not LCFF+, Pass >= 75%	UCI	25.5% 1039 4078	0.84		
CAASPP Pass <= 25%	UCLA	8.6% 1859 21666	28.7%	27.4%	25.5%
CAASPP Pass >= 75%	UCLA	8.5% 350 4109	1.01		
LCFF+ Y	UCLA	8.6% 1819 21085	27.9%	26.8%	25.4%
LCFF+ N	UCLA	8.9% 5014 56520	0.97		
LCFF+, Pass <= 25%	UCLA	8.4% 1220 14483	19.2%	18.0%	16.0%
LCFF+, Pass >= 50%	UCLA	10.1% 40 398	0.84		
Not LCFF+, Pass <= 25%	UCLA	8.9% 639 7183	9.5%	9.4%	9.5%
Not LCFF+, Pass >= 75%	UCLA	8.5% 350 4109	1.04		
CAASPP Pass <= 25%	UC Merced	91.4% 4923 5388	33.5%	32.7%	51.4%
CAASPP Pass >= 75%	UC Merced	96.8% 1293 1336	0.94		
LCFF+ Y	UC Merced	91.3% 6740 7386	45.9%	44.8%	45.9%
LCFF+ N	UC Merced	94.3% 13045 13832	0.97		
LCFF+, Pass <= 25%	UC Merced	91.5% 3464 3786	23.6%	23.0%	36.4%
LCFF+, Pass >= 50%	UC Merced	96.2% 50 52	0.95		
Not LCFF+, Pass <= 25%	UC Merced	91.1% 1459 1602	10.0%	9.7%	15.0%
Not LCFF+, Pass >= 75%	UC Merced	96.8% 1293 1336	0.94		
CAASPP Pass <= 25%	UCR	67.8% 8175 12060	30.5%	27.2%	29.4%
CAASPP Pass >= 75%	UCR	82.0% 2086 2545	0.83		
LCFF+ Y	UCR	67.5% 9705 14384	36.3%	32.3%	27.7%
LCFF+ N	UCR	79.6% 22349 28089	0.85		
LCFF+, Pass <= 25%	UCR	66.7% 6232 9339	23.6%	20.8%	21.3%
LCFF+, Pass >= 50%	UCR	72.9% 186 255	0.91		

Not LCFF+, Pass <= 25% UCR	71.4%	1943	2721	6.9%	6.5%	8.1%
Not LCFF+, Pass >= 75% UCR	82.0%	2086	2545	0.87		
CAASPP Pass <= 25% UCSB	37.8%	4689	12407	21.0%	24.6%	19.5%
CAASPP Pass >= 75% UCSB	29.9%	1172	3915	1.26		
LCFF+ Y UCSB	38.1%	4628	12146	20.6%	24.3%	11.8%
LCFF+ N UCSB	31.6%	15900	50303	1.21		
LCFF+, Pass <= 25% UCSB	38.2%	2656	6953	11.8%	13.9%	7.3%
LCFF+, Pass >= 50% UCSB	31.9%	74	232	1.20		
Not LCFF+, Pass <= 25% UCSB	37.3%	2033	5454	9.2%	10.7%	12.2%
Not LCFF+, Pass >= 75% UCSB	29.9%	1172	3915	1.25		
CAASPP Pass <= 25% UCSC	59.4%	5999	10099	22.8%	21.6%	16.0%
CAASPP Pass >= 75% UCSC	62.7%	1946	3106	0.95		
LCFF+ Y UCSC	54.8%	5702	10400	23.5%	20.6%	8.6%
LCFF+ N UCSC	64.5%	24424	37884	0.85		
LCFF+, Pass <= 25% UCSC	55.4%	3212	5794	13.1%	11.6%	5.7%
LCFF+, Pass >= 50% UCSC	61.6%	77	125	0.90		
Not LCFF+, Pass <= 25% UCSC	64.7%	2787	4305	9.7%	10.1%	10.3%
Not LCFF+, Pass >= 75% UCSC	62.7%	1946	3106	1.03		
CAASPP Pass <= 25% UCSD	36.8%	6695	18196	25.3%	34.9%	31.8%
CAASPP Pass >= 75% UCSD	16.8%	701	4177	2.19		
LCFF+ Y UCSD	41.7%	7381	17692	24.6%	38.5%	35.1%
LCFF+ N UCSD	22.2%	12675	57012	1.88		
LCFF+, Pass <= 25% UCSD	41.3%	4723	11426	15.9%	24.7%	22.9%
LCFF+, Pass >= 50% UCSD	46.5%	171	368	0.89		
Not LCFF+, Pass <= 25% UCSD	29.1%	1972	6770	9.4%	10.3%	8.9%
Not LCFF+, Pass >= 75% UCSD	16.8%	701	4177	1.74		

```
In [80]: # Campus LowAccept% HighAccept% Ratio NotLCFF+LowAccept% NotLCFF+HighAccept% Ratio
print("Public Schools Non-LCFF+ Schools LCFF+ Schools Private")
print("Campus All LowMath HighMath Ratio LowMath HighMath Ratio LowMath HighMath Ratio All")
for campus in campuses:
    # Denominators
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[])
    app = results["App"]
    adm = results["Adm"]
    rateAll = float(adm) / float(app)

    # CAASPP Pass Low vs high
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25]])
    app = results["App"]
    adm = results["Adm"]
    rateLow = float(adm) / float(app)

    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.75]])
    app = results["App"]
    adm = results["Adm"]
    rateHigh = float(adm) / float(app)
    rateRatio = rateHigh / rateLow
```



```

# LCFF+ = "N"
results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
rateLowNotLCFF = float(adm) / float(app)

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.75], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
rateHighNotLCFF = float(adm) / float(app)
rateRatioNotLCFF = rateHighNotLCFF / rateLowNotLCFF

# LCFF+ = "Y"
results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
rateLowLCFF = float(adm) / float(app)

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.50], ["LCFF Plus School
app = results["App"]
adm = results["Adm"]
rateHighLCFF = float(adm) / float(app)
rateRatioLCFF = rateHighLCFF / rateLowLCFF

dfLeftPrivateCampus = dfLeftPrivate[dfLeftPrivate["Campus"] == campus]
app = dfLeftPrivateCampus["App"].astype(int).sum()
adm = dfLeftPrivateCampus["Adm"].astype(int).sum()
ratePrivate = adm / app

print(f"{campus:9}   {rateAll:5.1%}   {rateLow:5.1%}   {rateHigh:5.1%}   {rateRatio:.2f}   {rateLowNotLCFF:5.1%}   {rateHighNotLCFF:5.1%}   {rateRa

```

	Public Schools				Non-LCFF+ Schools			LCFF+ Schools			Private
Campus	All	LowMath	HighMath	Ratio	LowMath	HighMath	Ratio	LowMath	HighMath	Ratio	All
Berkeley	14.2%	14.1%	14.3%	1.01	13.6%	14.3%	1.05	14.4%	10.5%	0.73	18.2%
UC Davis	35.8%	42.4%	23.5%	0.56	41.5%	23.5%	0.57	43.2%	44.6%	1.03	36.9%
UCI	22.4%	21.7%	25.5%	1.17	21.4%	25.5%	1.19	21.9%	29.6%	1.36	23.1%
UCLA	9.0%	8.6%	8.5%	0.99	8.9%	8.5%	0.96	8.4%	10.1%	1.19	11.2%
UC Merced	93.7%	91.4%	96.8%	1.06	91.1%	96.8%	1.06	91.5%	96.2%	1.05	92.5%
UCR	75.8%	67.8%	82.0%	1.21	71.4%	82.0%	1.15	66.7%	72.9%	1.09	76.1%
UCSB	32.3%	37.8%	29.9%	0.79	37.3%	29.9%	0.80	38.2%	31.9%	0.84	29.0%
UCSC	62.7%	59.4%	62.7%	1.05	64.7%	62.7%	0.97	55.4%	61.6%	1.11	64.7%
UCSD	26.6%	36.8%	16.8%	0.46	29.1%	16.8%	0.58	41.3%	46.5%	1.12	21.4%

```

In [81]: # What fraction of admitted students enroll?
print(" Enrollment Rate by Math CAASPP Group")
print("Campus      All      Low      Mid      High")
for campus in campuses:
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinAdms=minAdmsForEnrAnalysis[campus
    app = results["App"]
    adm = results["Adm"]
    enr = results["Enr"]
    rateAll = float(enr) / float(adm)

```

```

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25]], selectMinAdms=minA
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rateLow = float(enr) / float(adm)

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">", 0.25], ["PassRate", "<", 0.
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rateMed = float(enr) / float(adm)

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.75]], selectMinAdms=minA
app = results["App"]
adm = results["Adm"]
enr = results["Enr"]
rateHigh = float(enr) / float(adm)

print(f"{campus:9} {rateAll:5.1%} {rateLow:5.1%} {rateMed:5.1%} {rateHigh:5.1%}")

```

Enrollment Rate by Math CAASPP Group

Campus	All	Low	Mid	High
Berkeley	53.3%	46.4%	53.4%	61.7%
UC Davis	21.9%	26.8%	21.4%	14.7%
UCI	28.5%	28.5%	28.7%	26.8%
UCLA	60.5%	63.4%	61.1%	50.5%
UC Merced	7.6%	13.5%	5.5%	3.1%
UCR	15.1%	17.2%	14.8%	10.7%
UCSB	16.9%	16.5%	17.6%	11.5%
UCSC	12.2%	13.5%	12.6%	7.5%
UCSD	26.9%	25.7%	27.7%	22.9%

```

In [82]: # Check composition of UC Campus student body: what fraction of applicants and enrolled students from each school perf type.
print("
          Composition by Math CAASPP Group")
print("
      nApplications    %Apps    nEnrolled    %Enr")
print("Campus    All    Low    High    High    All    Low    High    Low    Mid    High")
for campus in campuses:
    # Denominators
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=0)
    appAll = results["App"]
    adm = results["Adm"]
    enrAll = results["Enr"]

    # CAASPP Pass Low to High
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", "<=", 0.25]], selectMinApps=0)
    appLow = results["App"]
    adm = results["Adm"]
    enrLow = results["Enr"]

    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">", 0.25], ["PassRate", "<", 0.
    appMid = results["App"]
    adm = results["Adm"]
    enrMid = results["Enr"]

```

```

results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["PassRate", ">=", 0.75]], selectMinApps=0)
appHigh = results["App"]
adm = results["Adm"]
enrHigh = results["Enr"]

appHighRate = float(appHigh) / float(appAll)
enrLowRate = float(enrLow) / float(enrAll)
enrMidRate = float(enrMid) / float(enrAll)
enrHighRate = float(enrHigh) / float(enrAll)

print(f"{campus:9}    {appAll:5} {appLow:5} {appHigh:4}    {appHighRate:5.1%}    {enrAll:4} {enrLow:4} {enrHigh:4}    {enrLowRate:5.1%} {enrMidRate:5.1%} {enrHighRate:5.1%}")

```

Campus	nApplications			%Apps	nEnrolled			%Enr		
	All	Low	High		All	Low	High	Low	Mid	High
Berkeley	60715	16884	3775	6.2%	3741	700	330	18.7%	72.5%	8.82%
UC Davis	58361	15990	3764	6.4%	4108	1270	131	30.9%	65.9%	3.19%
UCI	76709	23285	4078	5.3%	4139	1002	269	24.2%	69.3%	6.50%
UCLA	78383	23766	4109	5.2%	3882	991	173	25.5%	70.0%	4.46%
UC Merced	23032	9287	1407	6.1%	1283	676	44	52.7%	43.9%	3.43%
UCR	43780	14750	2585	5.9%	4424	1332	223	30.1%	64.9%	5.04%
UCSB	63346	15649	3915	6.2%	3004	666	136	22.2%	73.3%	4.53%
UCSC	49306	13858	3123	6.3%	3048	531	147	17.4%	77.8%	4.82%
UCSD	75631	21008	4177	5.5%	4762	1542	151	32.4%	64.4%	3.17%

```

In [83]: # Check for adverse selection of admitted applicants: is the enrollment rate higher for admitted students from local LCFF+ schools than from remote LCFF+ schools?
print("Applicants from LCFF+ schools")
print("Campus    enrRateLocal enrRateRemote")
for campus in campuses:
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"], ["CAASPP C", ">=", 0.75]], selectMinApps=0)
    app = results["App"]
    adm = results["Adm"]
    enr = results["Enr"]
    enrRateLocal = float(enr) / float(adm)

    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"], ["CAASPP C", "<=", 0.75]], selectMinApps=0)
    app = results["App"]
    adm = results["Adm"]
    enr = results["Enr"]
    enrRateRemote = float(enr) / float(adm)

    print(f"{campus:9}    {enrRateLocal:5.1%}    {enrRateRemote:5.1%}")

```

Campus	enrRateLocal	enrRateRemote
Berkeley	65.3%	37.4%
UC Davis	38.8%	17.4%
UCI	49.0%	26.4%
UCLA	65.5%	63.8%
UC Merced	22.9%	5.8%
UCR	25.3%	8.3%
UCSB	24.1%	10.6%
UCSC	9.8%	12.2%
UCSD	48.7%	21.8%

```
In [84]: dfRight["LA Area"] = dfRight["CAASPP County Name"].map(areaLA)
```

```
In [85]: # Check for adverse selection of admitted applicants (EDGE)
sRatioLCFF = 0
sRatioNotLCFF = 0

print("                non-LCFF+ schools                LCFF+ Schools")
print("Campus    enrRateLocal enrRateRemote ratio    enrRateLocal enrRateRemote ratio")
for campus in campuses:
    results1 = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "N"], ["CAASPP
    results2 = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "N"], ["CAASPP
    results3 = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "Y"], ["CAASPP
    results4 = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "Y"], ["CAASPP

    adm = results1["Adm"]
    enr = results1["Enr"]
    enrRateLocalNotLCFF = float(enr) / float(adm)
    adm = results2["Adm"]
    enr = results2["Enr"]
    enrRateRemoteNotLCFF = float(enr) / float(adm)
    ratioNotLCFF = enrRateRemoteNotLCFF / enrRateLocalNotLCFF
    sRatioNotLCFF += ratioNotLCFF

    adm = results3["Adm"]
    enr = results3["Enr"]
    enrRateLocalLCFF = float(enr) / float(adm)
    adm = results4["Adm"]
    enr = results4["Enr"]
    enrRateRemoteLCFF = float(enr) / float(adm)
    ratioLCFF = enrRateRemoteLCFF / enrRateLocalLCFF
    sRatioLCFF += ratioLCFF

    print(f"{campus:9}    {enrRateLocalNotLCFF:5.0%}    {enrRateRemoteNotLCFF:5.0%}    {ratioNotLCFF:0.2}    {enrRateLocalLCFF:5.0%}

ratioNotLCFF = sRatioNotLCFF / 9
ratioLCFF = sRatioLCFF / 9
print(f"average    {ratioNotLCFF:0.2}    {ratioLCFF:0.2}")

# Different look at UCLA local area
# campus = "UCLA"
```

```

# results1 = computeStats(dfLeft, dfRight, campus, "ALL", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "N"], ["LA Area",
# results2 = computeStats(dfLeft, dfRight, campus, "ALL", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "N"], ["LA Area",
# results3 = computeStats(dfLeft, dfRight, campus, "ALL", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"], ["LA Area",
# results4 = computeStats(dfLeft, dfRight, campus, "ALL", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"], ["LA Area",

# adm = results1["Adm"]
# enr = results1["Enr"]
# enrRateLocalNotLCFF = float(enr) / float(adm)
# adm = results2["Adm"]
# enr = results2["Enr"]
# enrRateRemoteNotLCFF = float(enr) / float(adm)
# ratioNotLCFF = enrRateRemoteNotLCFF / enrRateLocalNotLCFF

# adm = results3["Adm"]
# enr = results3["Enr"]
# enrRateLocalLCFF = float(enr) / float(adm)
# adm = results4["Adm"]
# enr = results4["Enr"]
# enrRateRemoteLCFF = float(enr) / float(adm)
# ratioLCFF = enrRateRemoteLCFF / enrRateLocalLCFF

# campus = "UCLA*"
# print(f"{campus:9}   {enrRateLocalNotLCFF:5.0%}           {enrRateRemoteNotLCFF:5.0%}           {ratioNotLCFF:0.2}           {enrRateLocalLCFF:5.0%}           {e

```

	non-LCFF+ schools			LCFF+ Schools		
Campus	enrRateLocal	enrRateRemote	ratio	enrRateLocal	enrRateRemote	ratio
Berkeley	64%	51%	0.79	65%	37%	0.57
UC Davis	39%	19%	0.49	39%	17%	0.45
UCI	39%	25%	0.64	49%	26%	0.54
UCLA	66%	57%	0.86	65%	64%	0.97
UC Merced	18%	5%	0.28	23%	6%	0.25
UCR	28%	13%	0.46	25%	8%	0.33
UCSB	21%	17%	0.81	24%	11%	0.44
UCSC	13%	12%	0.92	10%	12%	1.2
UCSD	40%	25%	0.63	49%	22%	0.45
average			0.65			0.58

```

In [86]: def printCounts(row):
          print(f"{row[0]:11} {row[1]:13} {row[2]:13} {row[3]:13} {row[4]:13} {row[5]:13} {row[6]:13} {row[7]:13} {row[8]:13} {row[9]:13}")

          def printRates(row):
              print(f"{row[0]:11} {row[1]:13.0%} {row[2]:13.0%} {row[3]:13.0%} {row[4]:13.0%} {row[5]:13.0%} {row[6]:13.0%} {row[7]:13.0%} {row[8]:13.0%} {row[9]:13.0%}")

          def printCountsSoCal(row):
              print(f"{row[0]:11} {row[1]:13} {row[2]:13} {row[3]:13} {row[4]:13} {row[5]:13}")

          def printRatesSoCal(row):
              print(f"{row[0]:11} {row[1]:13.0%} {row[2]:13.0%} {row[3]:13.0%} {row[4]:13.0%} {row[5]:13.0%} {row[6]:13.2%}")

```

```

In [87]: # Campus x County applications, admissions, admission rate, enrolment rate by LCFF+ N/Y
          header = ["Campus"]
          for county in homeCountiesSoCal:
              header.append(county)

```

```

header.append("rLocal/Remote")

for lcff in ("N", "Y"):
    print(f"*** LCFF+ = {lcff} ***")
    row = header
    print(f"{row[0]:11} {row[1]:>13} {row[2]:>13} {row[3]:>13} {row[4]:>13} {row[5]:>13} {row[6]:>13}")

    rows = []
    for campus in campusesSoCal:
        apps = [campus + " apps"]
        adms = [" adms"]
        admRates = [" admRate"]
        enrRates = [" enrRate"]
        localAdmRate = 0.0
        localEnrRate = 0.0
        remoteApps = 0
        remoteAdms1 = 0
        remoteAdms2 = 0
        remoteEnrs = 0
        for county in homeCountiesSoCal:
            results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", lcff], [
            app = results["App"]
            apps.append(app)

            adm = results["Adm"]
            adms.append(adm)

            if app > 0:
                admRate = float(adm) / float(app)
            else:
                admRate = 0
            admRates.append(admRate)

            if countyCampus[county] == campus:
                localAdmRate = admRate
            else:
                remoteApps += app
                remoteAdms1 += adm

            results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", lcff], [
            adm = results["Adm"]
            enr = results["Enr"]
            if adm > 0:
                enrRate = float(enr) / float(adm)
            else:
                enrRate = 0
            enrRates.append(enrRate)

            if countyCampus[county] == campus:
                localEnrRate = enrRate
            else:
                remoteAdms2 += adm

```

```
remoteEnrs += enr

remoteAdmRate = float(remoteAdms1) / float(remoteApps)
admRates.append(localAdmRate / remoteAdmRate)
remoteEnrRate = float(remoteEnrs) / float(remoteAdms2)
enrRates.append(localEnrRate / remoteEnrRate)

printCountsSoCal(apps)
printCountsSoCal(adms)
printRatesSoCal(admRates)
printRatesSoCal(enrRates)
```

```

*** LCFF+ = N ***
Campus      Orange  Los Angeles  Riverside Santa Barbara  San Diego rLocal/Remote
UCI apps    6922    12631      2584      397      5121
  adms      2090    2612      631      115      1241
  admRate   30%     21%      24%      29%      24%      1.4
  enrRate   39%     27%      25%      13%      21%      1.5
UCLA apps   6659    13980      2129      552      5402
  adms      635     1342      203      58      501
  admRate   10%     10%      10%      11%      9%      1.0
  enrRate   60%     66%      56%      61%      59%      1.1
UCR apps    3705    6496      2436      90      2180
  adms      3040    5164      1851      69      1746
  admRate   82%     79%      76%      77%      80%      0.95
  enrRate   17%     15%      29%      0%      11%      2.0
UCSB apps   5676    11163      1522      514      4848
  adms      1757    3589      498      184      1581
  admRate   31%     32%      33%      36%      33%      1.1
  enrRate   12%     18%      18%      19%      22%      1.1
UCSD apps   6703    12443      2616      469      6370
  adms      1444    3064      726      105      1627
  admRate   22%     25%      28%      22%      26%      1.1
  enrRate   27%     25%      39%      9%      40%      1.5

*** LCFF+ = Y ***
Campus      Orange  Los Angeles  Riverside Santa Barbara  San Diego rLocal/Remote
UCI apps    2409    9368      2211      184      1410
  adms      658     1949      534      34      304
  admRate   27%     21%      24%      18%      22%      1.3
  enrRate   49%     30%      30%      29%      12%      1.7
UCLA apps   1873    9681      1739      177      1319
  adms      173     936      155      10      107
  admRate   9%      10%      9%      6%      8%      1.1
  enrRate   72%     65%      63%      0%      40%      1.0
UCR apps    1251    6131      2371      74      782
  adms      892     4086      1690      30      515
  admRate   71%     67%      71%      41%      66%      1.1
  enrRate   8%      9%      27%      0%      4%      3.2
UCSB apps   840     4750      958      198      607
  adms      316     1571      354      53      239
  admRate   38%     33%      37%      27%      39%      0.77
  enrRate   0%      13%      0%      30%      0%      3.0
UCSD apps   1739    6300      1991      147      1994
  adms      778     2685      827      57      770
  admRate   45%     43%      42%      39%      39%      0.9
  enrRate   20%     23%      26%      9%      49%      2.1

```

```

In [88]: ##### Get data for adverse selection plots. Some of this was already computed above, but here we pull it all together.
selectTestID = "2"
# Initialize dict-of-lists that we will convert to dfAdverse dataframe for the plots. key = column name, value = list of values.
dAdverse = {
    "campus": [],
    "county": [],
}

```



```
In [89]: # LCFF+ % of students by county
dAdverse["pct Students from LCFF+ Schools in County"] = []
for county in homeCounties:
    campus = countyCampus[county]
    filteredRight = dfRight[dfRight["Test ID"] == selectTestID]
    filteredRight = filteredRight[filteredRight["CAASPP County Name"] == county]

    nStudents = filteredRight["Total Students Enrolled"].astype(int).sum() # number of 11th graders

    dfLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "Y"]
    nStudentsLCFF = dfLCFF["Total Students Enrolled"].astype(int).sum()
    rateStudentsLCFF = nStudentsLCFF / nStudents

    dAdverse["campus"].append(campus)
    dAdverse["county"].append(county)
    dAdverse["pct Students from LCFF+ Schools in County"].append(rateStudentsLCFF)
    print(f"{campus:9} {county:13} {nStudents:>5} {nStudentsLCFF:>5} {rateStudentsLCFF:5.0%}")
```

Berkeley	Alameda	15240	3028	20%
UC Davis	Yolo	1990	348	17%
UCI	Orange	36516	12867	35%
UCLA	Los Angeles	99878	50101	50%
UC Merced	Merced	4450	3030	68%
UCR	Riverside	32572	17057	52%
UCSB	Santa Barbara	4866	1732	36%
UCSC	Santa Cruz	2937	986	34%
UCSD	San Diego	35924	10814	30%

```
In [90]: # % of student body that is LCFF+
dAdverse["pct Campus Enrolled Students from LCFF+ Schools"] = []
for campus in campuses:
    # Denominator: ALL schools
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=0)
    appAll = results["App"]
    admAll = results["Adm"]
    enrAll = results["Enr"]

    # LCFF+
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"]], selectMinApps=0)
    adm = results["Adm"]
    app = results["App"]
    enr = results["Enr"]
    rate = float(enr) / float(enrAll)

    dAdverse["pct Campus Enrolled Students from LCFF+ Schools"].append(rate)
    print(f"{campus:9} {enrAll:>5} {enr:>5} {rate:5.0%}")
```

Berkeley	3741	527	14%
UC Davis	4108	652	16%
UCI	4139	1162	28%
UCLA	3882	982	25%
UC Merced	1283	527	41%
UCR	4424	1212	27%
UCSB	3004	345	11%
UCSC	3048	254	8%
UCSD	4762	1647	35%

```
In [91]: # Admission rate LCFF+ / non-LCFF+
dAdverse["Admission rate LCFF+/non-LCFF+"] = []
for campus in campuses:
    # non-LCFF+
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "N"]])
    adm = results["Adm"]
    app = results["App"]
    enr = results["Enr"]
    rateN = float(adm) / float(app)

    # LCFF+
    results = computeStats(dfLeft, dfRight, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "==", "Y"]])
    adm = results["Adm"]
    app = results["App"]
    enr = results["Enr"]
    rateY = float(adm) / float(app)

    ratio = rateY / rateN
    dAdverse["Admission rate LCFF+/non-LCFF+"].append(ratio)
    print(f"{campus:9} {ratio:5.2}")
```

Berkeley	1.1
UC Davis	1.2
UCI	1.0
UCLA	1.0
UC Merced	0.97
UCR	0.85
UCSB	1.2
UCSC	0.85
UCSD	1.9

```
In [92]: # % of student body from LCFF+ schools / % of students in county from LCFF+ schools
dAdverse["% of student body from LCFF+ schools / % of students in county from LCFF+ schools"] = []
i = 0
for campus in campuses:
    ratio = dAdverse["pct Campus Enrolled Students from LCFF+ Schools"][i] / dAdverse["pct Students from LCFF+ Schools in County"][i]
    dAdverse["% of student body from LCFF+ schools / % of students in county from LCFF+ schools"].append(ratio)
    print(f"{campus:9} {ratio:5.2}")
    i += 1
```

Berkeley	0.71
UC Davis	0.91
UCI	0.8
UCLA	0.5
UC Merced	0.6
UCR	0.52
UCSB	0.32
UCSC	0.25
UCSD	1.1

```
In [93]: dfAdverse = pd.DataFrame(dAdverse)
```

```
In [94]: # Plot % of student body that is LCFF+ (Y-axis) vs. %LCFF+ students in county (X-axis). Label = campus. Bubble size = Admission rate LCFF+/non-LCFF+.
# Seaborn's scatterplot returns a Matplotlib Axes object (ax), which we customize.
xAxis = "pct Students from LCFF+ Schools in County"
yAxis = "pct Campus Enrolled Students from LCFF+ Schools"
fig, ax = plt.subplots()

sns.scatterplot(
    data=dfAdverse,
    x=xAxis,
    y=yAxis,
    size="Admission rate LCFF+/non-LCFF+",
    alpha=0.7,      # transparency
    legend=False,
    ax=ax
)

for _, row in dfAdverse.iterrows():
    ax.text(
        row[xAxis] + 0.005,
        row[yAxis] + 0.005,
        str(row["campus"]),      # column containing labels
        fontsize=9,
        ha="left",
        va="center"
    )

ax.set_title("2024")

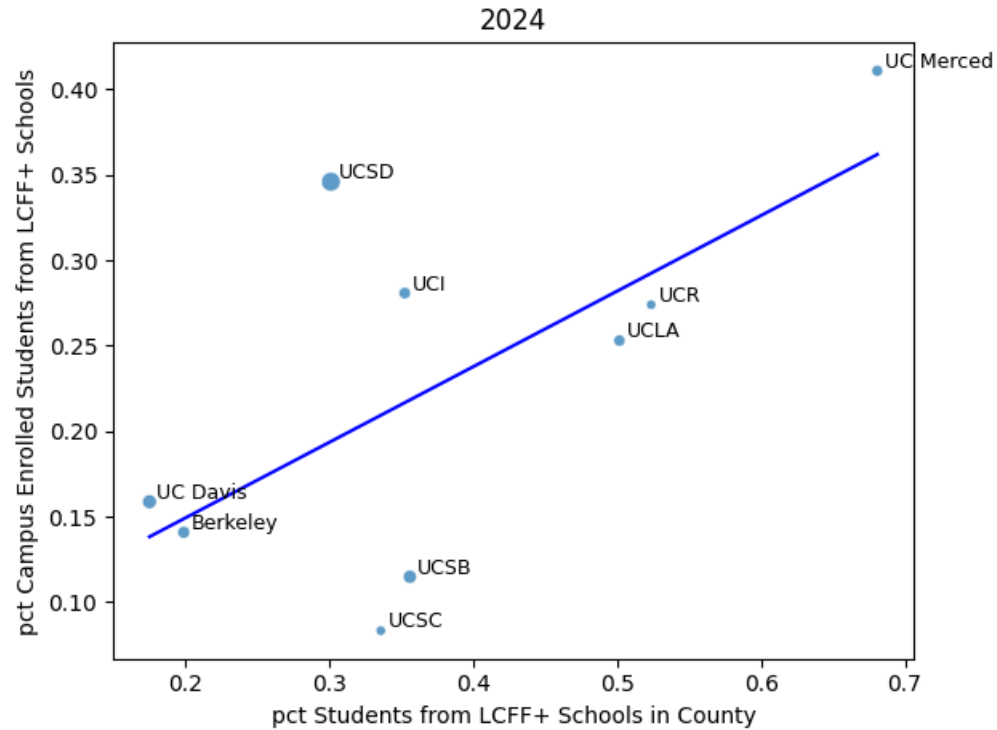
# Compute the weighted regression separately and then plot the line.
X = dfAdverse[xAxis]
y = dfAdverse[yAxis]
X_const = sm.add_constant(X) # Add constant for intercept

# Weighted least squares regression
modelWLSq = sm.WLS(y, X_const)
results = modelWLSq.fit()

# Get fitted line
x_vals = np.linspace(X.min(), X.max(), 100)
y_vals = results.predict(sm.add_constant(x_vals))
```

```
# Add the regression line to the plot
sns.lineplot(x=x_vals, y=y_vals, color="blue", ax=ax)

plt.tight_layout()
plt.show()
```



```
In [95]: # Plot admission rate LCFF+ / non-LCFF+ (Y-axis) vs. % of student body that is LCFF+ / % LCFF+ students in county (X-axis). Bubble size = ??? . Label =
xAxis = "% of student body from LCFF+ schools / % of students in county from LCFF+ schools"
yAxis = "Admission rate LCFF+/non-LCFF+"
fig, ax = plt.subplots()

sns.scatterplot(
    data=dfAdverse,
    x=xAxis,
    y=yAxis,
    # size="Admission rate LCFF+/non-LCFF+",
    s=40,          # point size
    alpha=0.7,     # transparency
    legend=False,
    ax=ax
)

for _, row in dfAdverse.iterrows():
```

```

    ax.text(
        row[xAxis] + 0.01,
        row[yAxis] + 0.01,
        str(row["campus"]), # column containing labels
        fontsize=9,
        ha="left",
        va="center"
    )

ax.set_title("2024")
ax.set(xlabel="% enrolled from LCFF+ schools / % students in county from LCFF+ schools",
       ylabel="Admission rate LCFF+/non-LCFF+",
       xlim=(0.2, 1.2),
       ylim=(0.7, 1.92))

# Compute a weighted regression separately and then plot the best fit curve
X = dfAdverse[xAxis]
y = dfAdverse[yAxis]

# *** LINE ***
X_const = sm.add_constant(X) # Add constant for intercept

# Weighted least squares regression
modelWLSq = sm.WLS(y, X_const)
results = modelWLSq.fit()

# Get fitted line
x_vals = np.linspace(X.min(), X.max(), 100)
y_vals = results.predict(sm.add_constant(x_vals))
r2 = results.rsquared

# Add the regression line to the plot
sns.lineplot(x=x_vals, y=y_vals, color="blue", ax=ax)

# *** QUADRATIC ***
# X_poly = pd.DataFrame({
#     "const": 1,
#     "x": X,
#     "x2": X**2
# })

# modelWLSq = sm.WLS(y, X_poly)
# results = modelWLSq.fit()

# x_vals = np.linspace(X.min(), X.max(), 100)
# X_pred = pd.DataFrame({
#     "const": 1,
#     "x": x_vals,
#     "x2": x_vals**2
# })
# y_vals2 = results.predict(X_pred)

```

```

# *** EXPONENTIAL *** ensures monotonicity
# y_log = np.log(y)
# weights = y
# X_const = sm.add_constant(X)

# modelWLSq = sm.WLS(y_log, X_const, weights=weights)
# results = modelWLSq.fit()

# x_vals = np.linspace(X.min(), X.max(), 100)
# X_pred = sm.add_constant(x_vals)

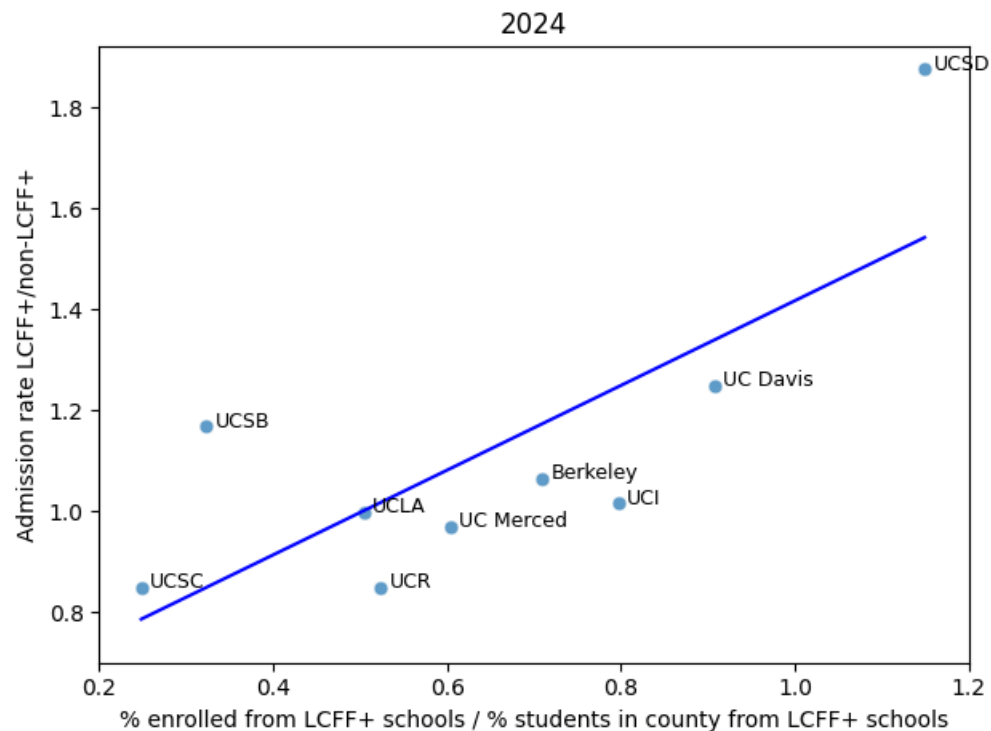
# y_vals3 = np.exp(results.predict(X_pred)) # predict log(y), then exponentiate

# # Add the regression line to the plot
# sns.lineplot(x=x_vals, y=y_vals3, color="green", ax=ax)

plt.tight_layout()
plt.show()

print(f"R squared for linear fit: {r2:.3f}")

```



R squared for linear fit: 0.579

```

In [151... ##### 2025 adverse selection plots
selectTestID = "2"

```

```

# Initialize dict-of-lists that we will convert to dfAdverse dataframe for the plots. key = column name, value = list of values.
dAdverse = {
    "campus": [],
    "county": [],
}

# LCFF+ % of students by county
print("County % students LCFF+")
dAdverse["pct Students from LCFF+ Schools in County"] = []
for county in homeCounties:
    campus = countyCampus[county]
    filteredRight = dfRight2025[dfRight2025["Test ID"] == selectTestID]
    filteredRight = filteredRight[filteredRight["CAASPP County Name"] == county]

    nStudents = filteredRight["Total Students Enrolled"].astype(int).sum() # number of 11th graders

    dfLCFF = filteredRight[filteredRight["LCFF Plus School t"] == "Y"]
    nStudentsLCFF = dfLCFF["Total Students Enrolled"].astype(int).sum()
    rateStudentsLCFF = nStudentsLCFF / nStudents

    dAdverse["campus"].append(campus)
    dAdverse["county"].append(county)
    dAdverse["pct Students from LCFF+ Schools in County"].append(rateStudentsLCFF)
    print(f"{campus:9} {county:13} {nStudents:>5} {nStudentsLCFF:>5} {rateStudentsLCFF:5.0%}")

# % of student body that is LCFF+
print("Campus admAll admLCFF+ enrAll enrLCFF+ %_enr_LCFF+")
dAdverse["pct Campus Enrolled Students from LCFF+ Schools"] = []
for campus in campuses:
    # Denominator: All schools
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[], selectMinApps=0)
    appAll = results["App"]
    admAll = results["Adm"]
    enrAll = results["Enr"]

    # LCFF+
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "Y"]], s
    adm = results["Adm"]
    app = results["App"]
    enr = results["Enr"]
    rate = float(enr) / float(enrAll)

    dAdverse["pct Campus Enrolled Students from LCFF+ Schools"].append(rate)
    print(f"{campus:9} {admAll:>5} {adm:>5} {enrAll:>5} {enr:>5} {rate:5.0%}")

# Admission rate LCFF+ / non-LCFF+
print("Admission rate LCFF+ / non-LCFF+")
dAdverse["Admission rate LCFF+/non-LCFF+"] = []
for campus in campuses:
    # non-LCFF+
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "N"]])
    adm = results["Adm"]

```

```

app = results["App"]
enr = results["Enr"]
rateN = float(adm) / float(app)

# LCFF+
results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"], filterMetrics=[["LCFF Plus School t", "=", "Y"]])
adm = results["Adm"]
app = results["App"]
enr = results["Enr"]
rateY = float(adm) / float(app)

ratio = rateY / rateN
dAdverse["Admission rate LCFF+/non-LCFF+"].append(ratio)
print(f"{campus:9} {ratio:5.2}")

# % of student body from LCFF+ schools / % of students in county from LCFF+ schools
print("Campus % enrolled LCFF+ / County % students LCFF+")
dAdverse["% of student body from LCFF+ schools / % of students in county from LCFF+ schools"] = []
i = 0
for campus in campuses:
    ratio = dAdverse["pct Campus Enrolled Students from LCFF+ Schools"][i] / dAdverse["pct Students from LCFF+ Schools in County"][i]
    dAdverse["% of student body from LCFF+ schools / % of students in county from LCFF+ schools"].append(ratio)
    print(f"{campus:9} {ratio:5.2}")
    i += 1

# Adverse selection
print("Enrolment rates")
print("Campus      EnrNL      EnrNR      EnrYL      EnrYR")
for county in homeCounties:
    campus = countyCampus[county]
    # non-LCFF+
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"],
                           filterMetrics=[["LCFF Plus School t", "=", "N"], ["CAASPP County Name", "=", county]],
                           selectMinAdms=minAdmsForEnrAnalysis[campus])

    admNL = results["Adm"]
    appNL = results["App"]
    enrNL = results["Enr"]
    rateNL = float(enrNL) / float(admNL)
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"],
                           filterMetrics=[["LCFF Plus School t", "=", "N"], ["CAASPP County Campus", "!=", campus]],
                           selectMinAdms=minAdmsForEnrAnalysis[campus])

    admNR = results["Adm"]
    appNR = results["App"]
    enrNR = results["Enr"]
    rateNR = float(enrNR) / float(admNR)

    # LCFF+
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"],
                           filterMetrics=[["LCFF Plus School t", "=", "Y"], ["CAASPP County Name", "=", county]],
                           selectMinAdms=minAdmsForEnrAnalysis[campus])

    admYL = results["Adm"]
    appYL = results["App"]

```



```

    enrYL = results["Enr"]
    rateYL = float(enrYL) / float(admYL)
    results = computeStats(dfLeft2025, dfRight2025, campus, "All", metrics=["App", "Adm", "Enr"],
                           filterMetrics=[["LCFF Plus School t", "==", "Y"], ["CAASPP County Campus", "!=", campus]],
                           selectMinAdms=minAdmsForEnrAnalysis[campus])

    admYR = results["Adm"]
    appYR = results["App"]
    enrYR = results["Enr"]
    rateYR = float(enrYR) / float(admYR)

    print(f"{campus:9} {rateNL:.4}    {rateNR:.4}    {rateYL:.4}    {rateYR:.4}")

dfAdverse2025 = pd.DataFrame(dAdverse)

# Plot % of student body that is LCFF+ (Y-axis) vs. %LCFF+ students in county (X-axis). Label = campus. Bubble size = Admission rate LCFF+/non-LCFF+.
# Seaborn's scatterplot returns a Matplotlib Axes object (ax), which we customize.
xAxis = "pct Students from LCFF+ Schools in County"
yAxis = "pct Campus Enrolled Students from LCFF+ Schools"
fig, ax = plt.subplots()

sns.scatterplot(
    data=dfAdverse2025,
    x=xAxis,
    y=yAxis,
    size="Admission rate LCFF+/non-LCFF+",
    alpha=0.7,      # transparency
    legend=False,
    ax=ax
)

for _, row in dfAdverse2025.iterrows():
    ax.text(
        row[xAxis] + 0.005,
        row[yAxis] + 0.005,
        str(row["campus"]),      # column containing labels
        fontsize=9,
        ha="left",
        va="center"
    )

ax.set_title("2025")

# Compute the weighted regression separately and then plot the line.
X = dfAdverse2025[xAxis]
y = dfAdverse2025[yAxis]
X_const = sm.add_constant(X) # Add constant for intercept

# Weighted least squares regression
modelWLSq = sm.WLS(y, X_const)
results = modelWLSq.fit()

# Get fitted line

```

```

x_vals = np.linspace(X.min(), X.max(), 100)
y_vals = results.predict(sm.add_constant(x_vals))

# Add the regression line to the plot
sns.lineplot(x=x_vals, y=y_vals, color="blue", ax=ax)

plt.tight_layout()
plt.show()

# Plot admission rate LCFF+ / non-LCFF+ (Y-axis) vs. % of student body that is LCFF+ / % LCFF+ students in county (X-axis). Bubble size = ????. Label =
xAxis = "% of student body from LCFF+ schools / % of students in county from LCFF+ schools"
yAxis = "Admission rate LCFF+/non-LCFF+"
fig, ax = plt.subplots()

sns.scatterplot(
    data=dfAdverse2025,
    x=xAxis,
    y=yAxis,
    # size="Admission rate LCFF+/non-LCFF+",
    s=40,          # point size
    alpha=0.7,     # transparency
    legend=False,
    ax=ax
)

for _, row in dfAdverse2025.iterrows():
    ax.text(
        row[xAxis] + 0.01,
        row[yAxis] + 0.02,
        str(row["campus"]), # column containing labels
        fontsize=9,
        ha="left",
        va="center"
    )

ax.set_title("2025")
ax.set(xlabel="% enrolled from LCFF+ schools / % students in county from LCFF+ schools",
       ylabel="Admission rate LCFF+/non-LCFF+",
       xlim=(0.2, 1.2),
       ylim=(0.7, 1.92))

# Compute a weighted regression separately and then plot the best fit curve
X = dfAdverse2025[xAxis]
y = dfAdverse2025[yAxis]

# *** LINE ***
X_const = sm.add_constant(X) # Add constant for intercept

# Weighted least squares regression
modelWLSq = sm.WLS(y, X_const)
results = modelWLSq.fit()

```

```

# Get fitted line
# x_vals = np.linspace(X.min(), X.max(), 100)
x_vals = np.linspace(X.min(), 1.15, 100)
y_vals = results.predict(sm.add_constant(x_vals))
r2 = results.rsquared

# Add the regression line to the plot
sns.lineplot(x=x_vals, y=y_vals, color="blue", ax=ax)

# *** QUADRATIC ***.
# X_poly = pd.DataFrame({
#     "const": 1,
#     "x": X,
#     "x2": X**2
# })

# modelWLSq = sm.WLS(y, X_poly)
# results = modelWLSq.fit()

# x_vals = np.linspace(X.min(), X.max(), 100)
# X_pred = pd.DataFrame({
#     "const": 1,
#     "x": x_vals,
#     "x2": x_vals**2
# })
# y_vals2 = results.predict(X_pred)

# *** EXPONENTIAL *** ensures monotonicity
# y_log = np.log(y)
# weights = y
# X_const = sm.add_constant(X)

# modelWLSq = sm.WLS(y_log, X_const, weights=weights)
# results = modelWLSq.fit()

# x_vals = np.linspace(X.min(), X.max(), 100)
# X_pred = sm.add_constant(x_vals)

# y_vals3 = np.exp(results.predict(X_pred)) # predict log(y), then exponentiate

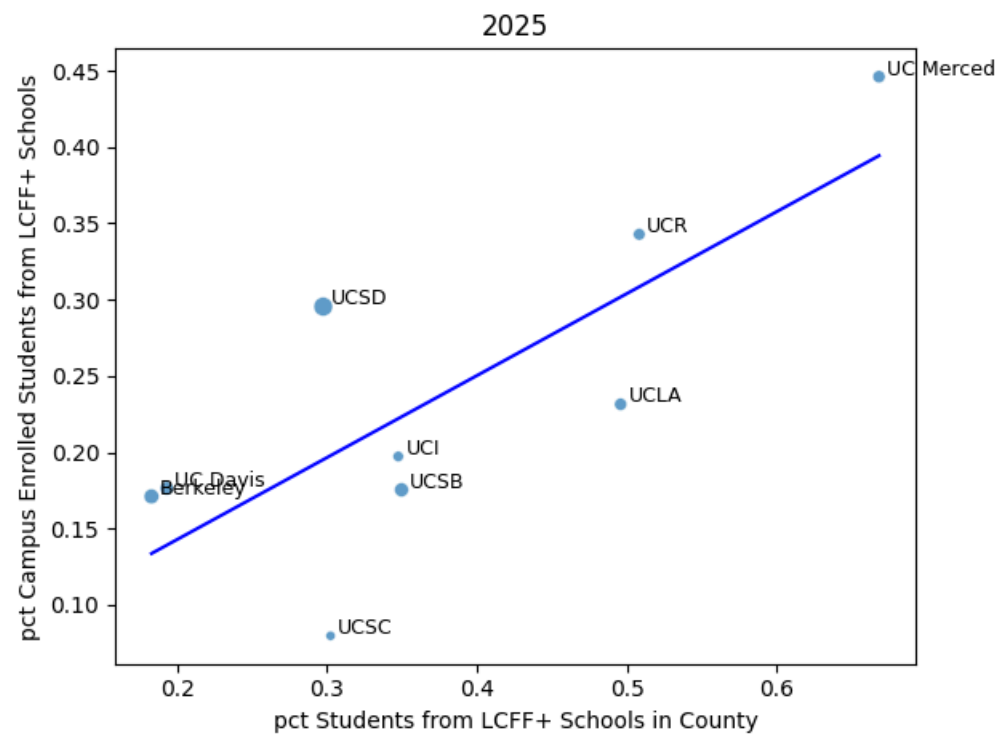
# # Add the regression line to the plot
# sns.lineplot(x=x_vals, y=y_vals3, color="green", ax=ax)

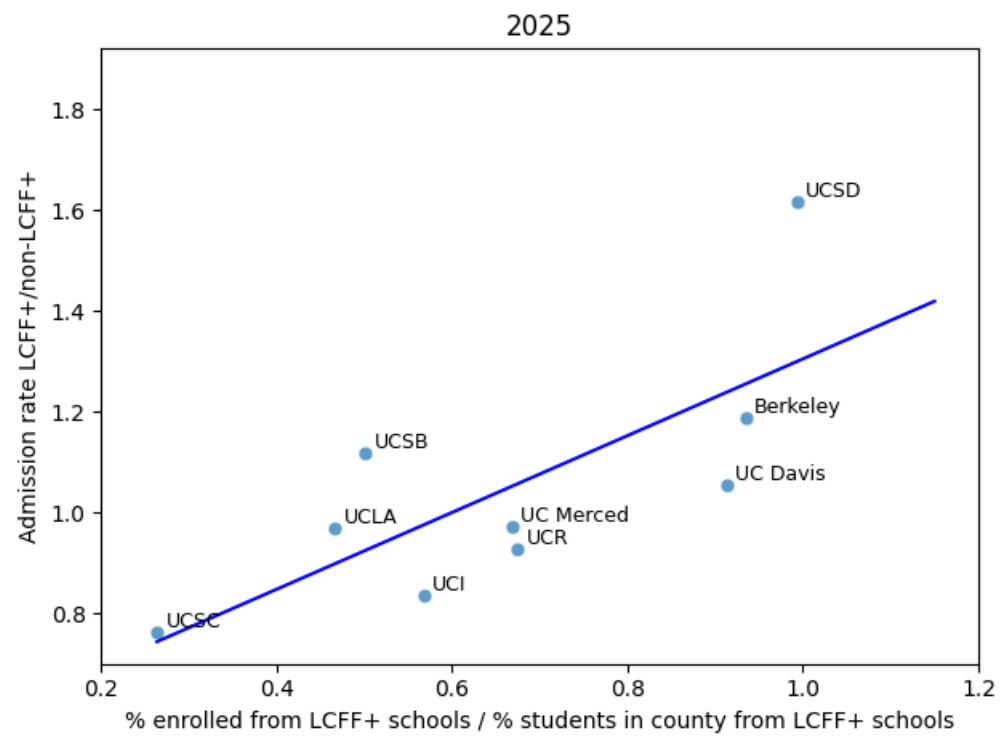
plt.tight_layout()
plt.show()

print(f"R squared for linear fit: {r2:.3f}")

```

County % students LCFF+					
Berkeley	Alameda	15061	2757	18%	
UC Davis	Yolo	2044	395	19%	
UCI	Orange	36050	12531	35%	
UCLA	Los Angeles	98071	48626	50%	
UC Merced	Merced	4227	2824	67%	
UCR	Riverside	32020	16274	51%	
UCSB	Santa Barbara	4886	1709	35%	
UCSC	Santa Cruz	2678	810	30%	
UCSD	San Diego	34933	10395	30%	
Campus admAll admLCFF+ enrAll enrLCFF+ %_enr_LCFF+					
Berkeley	7669	1821	3625	620	17%
UC Davis	21593	4636	4236	747	18%
UCI	16611	4089	4000	789	20%
UCLA	6672	1671	3634	841	23%
UC Merced	37088	12586	1170	522	45%
UCR	47670	15044	5516	1891	34%
UCSB	20375	4640	2720	477	18%
UCSC	31825	5465	3225	257	8%
UCSD	18975	6258	4549	1344	30%
Admission rate LCFF+ / non-LCFF+					
Berkeley	1.2				
UC Davis	1.1				
UCI	0.84				
UCLA	0.97				
UC Merced	0.97				
UCR	0.93				
UCSB	1.1				
UCSC	0.76				
UCSD	1.6				
Campus % enrolled LCFF+ / County % students LCFF+					
Berkeley	0.93				
UC Davis	0.91				
UCI	0.57				
UCLA	0.47				
UC Merced	0.67				
UCR	0.67				
UCSB	0.5				
UCSC	0.26				
UCSD	0.99				
Enrolment rates					
Campus	EnrNL	EnrNR	EnrYL	EnrYR	
Berkeley	0.6416	0.5462	0.6047	0.4409	
UC Davis	0.3837	0.1845	0.6087	0.2307	
UCI	0.3379	0.2627	0.5113	0.2281	
UCLA	0.6167	0.5649	0.6371	0.6	
UC Merced	0.3602	0.01975	0.3995	0.0181	
UCR	0.2526	0.09989	0.2642	0.09505	
UCSB	0.2632	0.1437	0.2258	0.168	
UCSC	0.1538	0.1228	0.1429	0.1478	
UCSD	0.3818	0.2615	0.5346	0.2099	





R squared for linear fit: 0.555

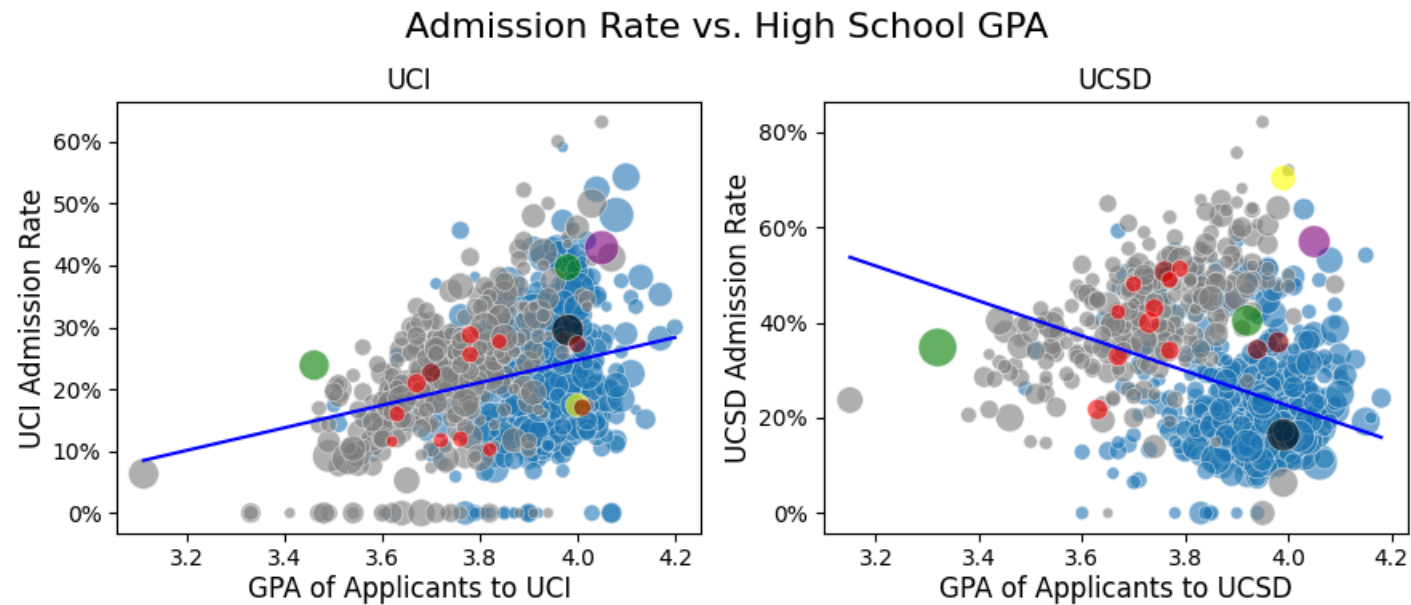
In [97]: `print(dfRight)`

County Code	District Code	District Name	School Code	School Name	Type	ID	Test	ID	Total	Students	Enro					
lled Total	Students	Tested	Total	Students	Tested	with Scores	Mean	Scale	Score	Count	Standard	Exceeded	Count	Standard	Met	\
11469	01		61119	Alameda Unified	0130229	Alameda High	7	1								
467		440		440	2647.1	181	157									
11569	01		61119	Alameda Unified	0130229	Alameda High	7	2								
466		438		438	2623.7	131	109									
11791	01		61119	Alameda Community Learning Center	0130609	Alameda Community Learning Center	9	1								
53		51		51	2556.0	8	10									
11999	01		61119	Alameda Community Learning Center	0130609	Alameda Community Learning Center	9	2								
53		50		50	2515.0	3	10									
12220	01		61119	Alameda Unified	0132142	Encinal Junior/Senior High	7	1								
239		218		218	2627.4	69	77									
...	...		...		...		...						...	...	...	...
...		...		...	...		...						...	...		
4039195	58		72736	Marysville Joint Unified	5830096	South Lindhurst Continuation High	7	2								
140		128		128	2439.4	0	1									
4039649	58		72736	Marysville Joint Unified	5835202	Marysville High	7	1								
231		223		223	2542.9	29	56									
4039751	58		72736	Marysville Joint Unified	5835202	Marysville High	7	2								
231		227		227	2501.7	5	17									
4050422	58		72769	Wheatland Union High	5838305	Wheatland Union High	7	1								
240		237		237	2582.1	48	80									
4050524	58		72769	Wheatland Union High	5838305	Wheatland Union High	7	2								
240		237		237	2545.9	14	41									

Count	Standard	Met	and Above	Count	Standard	Nearly	Met	Count	Standard	Not	Met	PassRate	CAASPP	County	Name	CAASPP	County	Campus	LCFF	Plus	School
l t	Category	LA	Area																		
11469				338		52		50	0.768182					Alameda				Berkeley			
N		N																			
11569				240		72		126	0.547945					Alameda				Berkeley			
N		N																			
11791				18		20		13	0.352941					Alameda				Berkeley			
N		N																			
11999				13		8		29	0.260000					Alameda				Berkeley			
N		N																			
12220				146		39		33	0.669725					Alameda				Berkeley			
N		N																			
...				...		...		...	...												
...	...	...																			
4039195				1		10		117	0.007812					Yuba				UC Davis			
Y	LCFF+	N																			
4039649				85		65		73	0.381166					Yuba				UC Davis			
N		N																			
4039751				22		58		147	0.096916					Yuba				UC Davis			
N		N																			
4050422				128		59		50	0.540084					Yuba				UC Davis			
Y	LCFF+	N																			
4050524				55		62		120	0.232068					Yuba				UC Davis			
Y	LCFF+	N																			

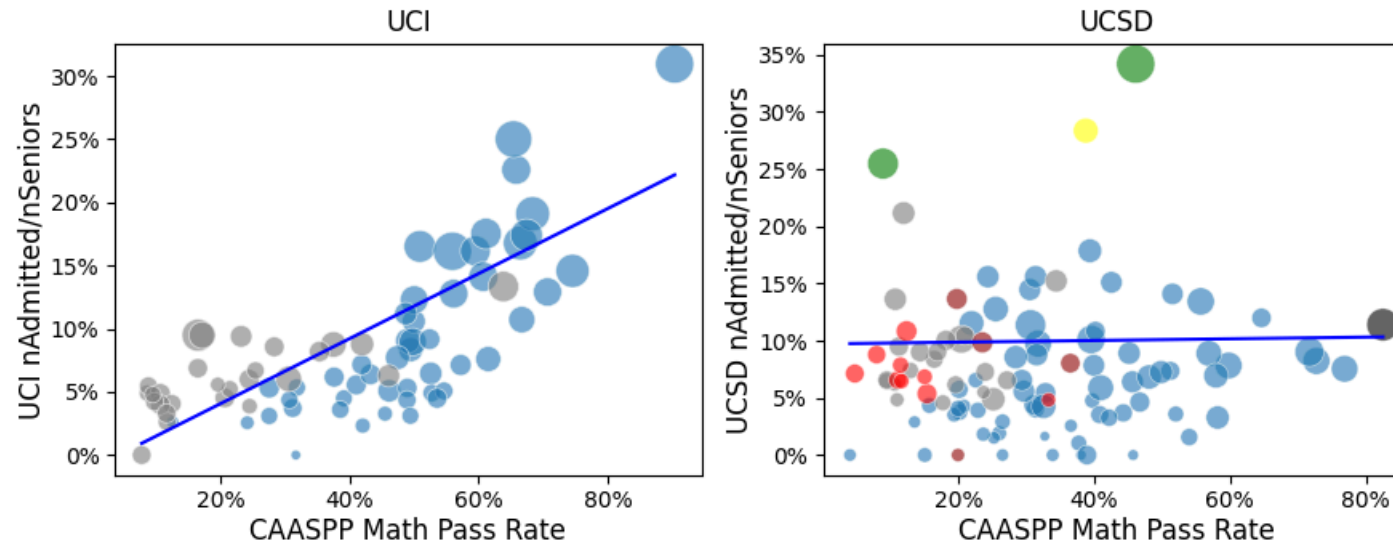
[3105 rows x 22 columns]

```
In [98]: ##### Conclusion plots
# createPlots2(dfLeft, dfRight, "ALL",
createPlots2(dfLeft, dfRight, "All",    xAxis="App GPA",
height=4, width=13.49, campusTitles="Y", title="Admission Rate vs CAASPP M
height=4, width=9,    campusTitles="Y", title=" Admission Rate vs. High
```



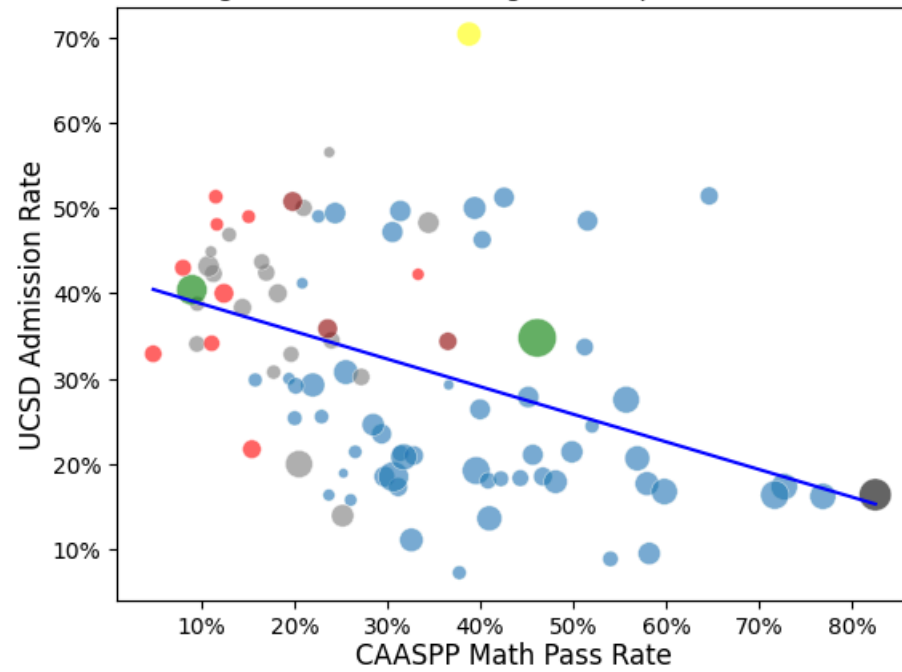
```
In [99]: createPlots2(dfLeft, dfRight, "Local", yAxis="SeniorAdmRate", selectMinApps=0, height=4, width=9,    campusTitles="Y", title=" % of All Seniors Admi
```

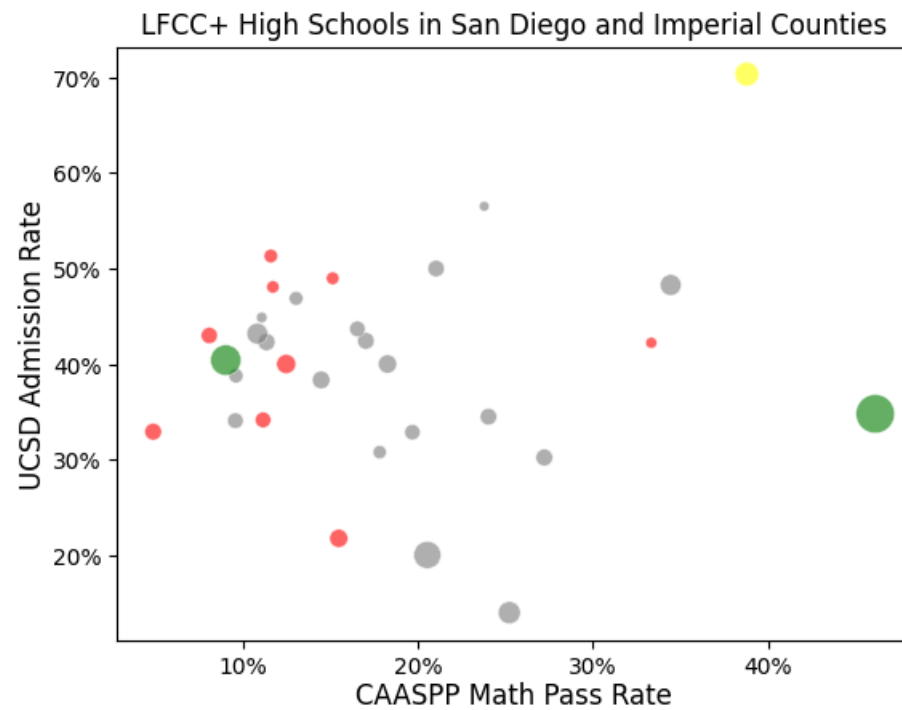

% of All Seniors Admitted vs. CAASPP Math (Local High Schools)



```
In [100... fig, ax = plt.subplots() # Create a Matplotlib Axes object
createPlot(ax, dfLeft, dfRight, "UCSD", "Local", title="High Schools in San Diego and Imperial Coun
plt.show()
fig, ax = plt.subplots() # Create a Matplotlib Axes object
createPlot(ax, dfLeft, dfRight, "UCSD", "Local", filterMetrics=[["LCFF Plus School t", "=", "Y"]], title="LCFF+ High Schools in San Diego and Imperia
plt.show()
```

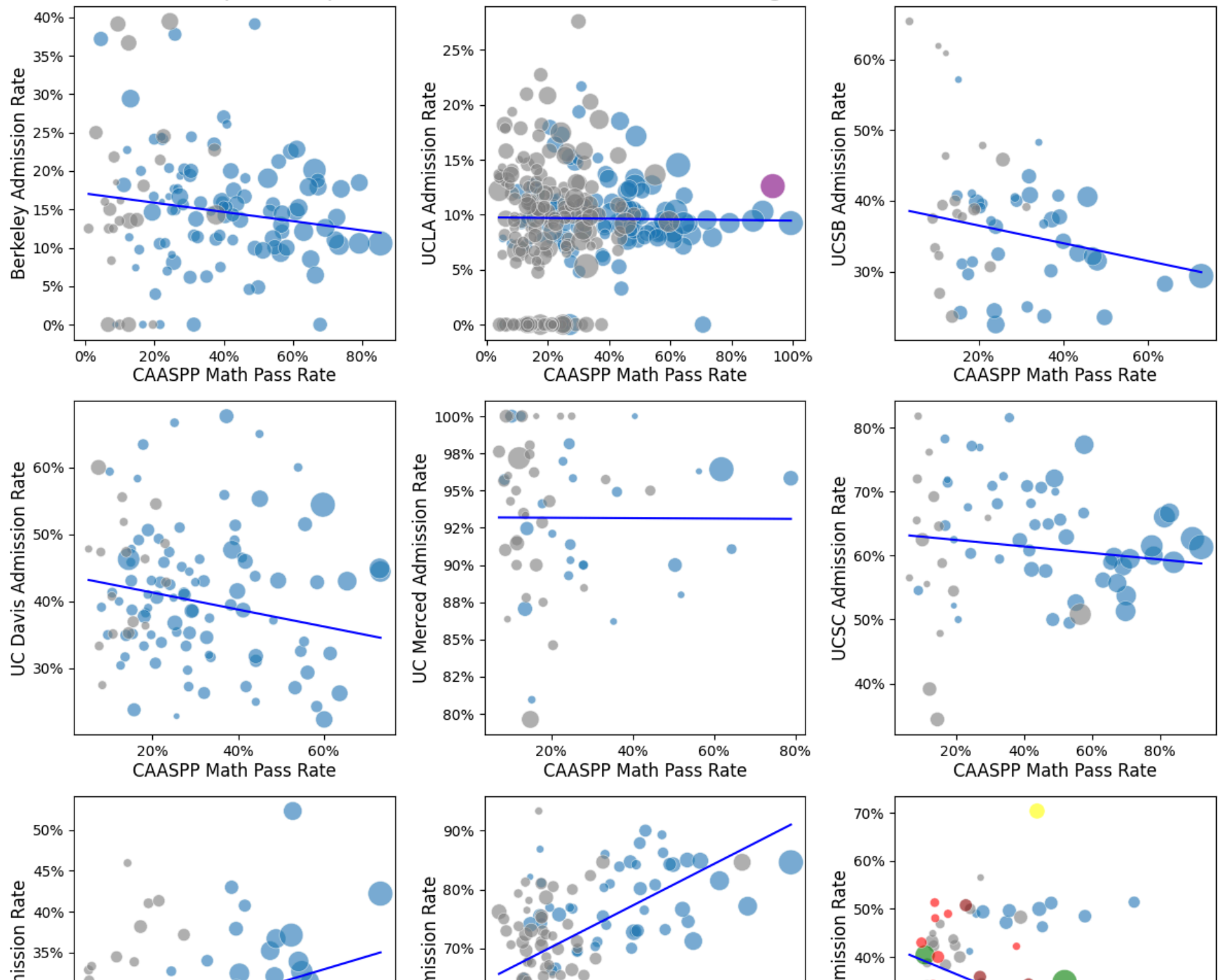
High Schools in San Diego and Imperial Counties

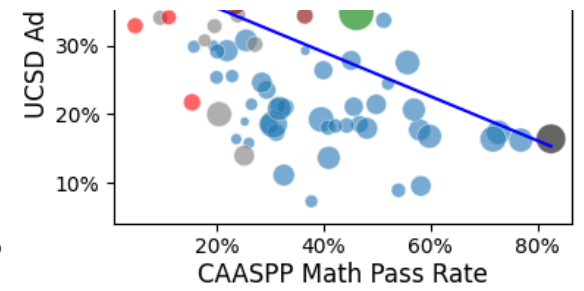
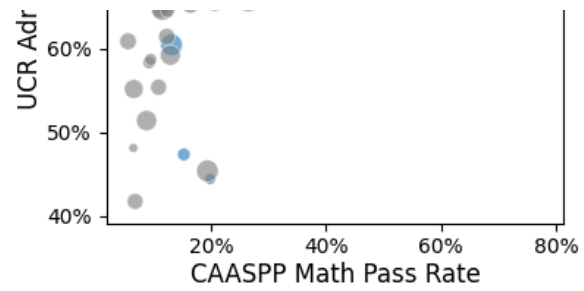
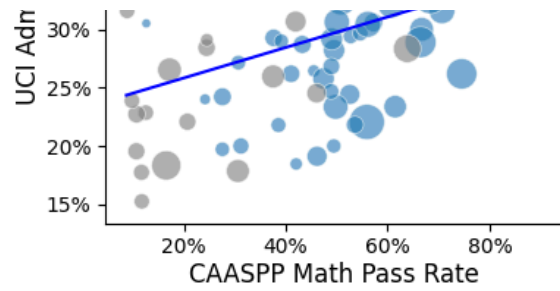




In [101... `createPlots(dfLeft, dfRight, "Local", title="UC Campus Acceptance vs CAASPP Math Pass Rates, High Schools in Local Counties")`

UC Campus Acceptance vs CAASPP Math Pass Rates, High Schools in Local Counties

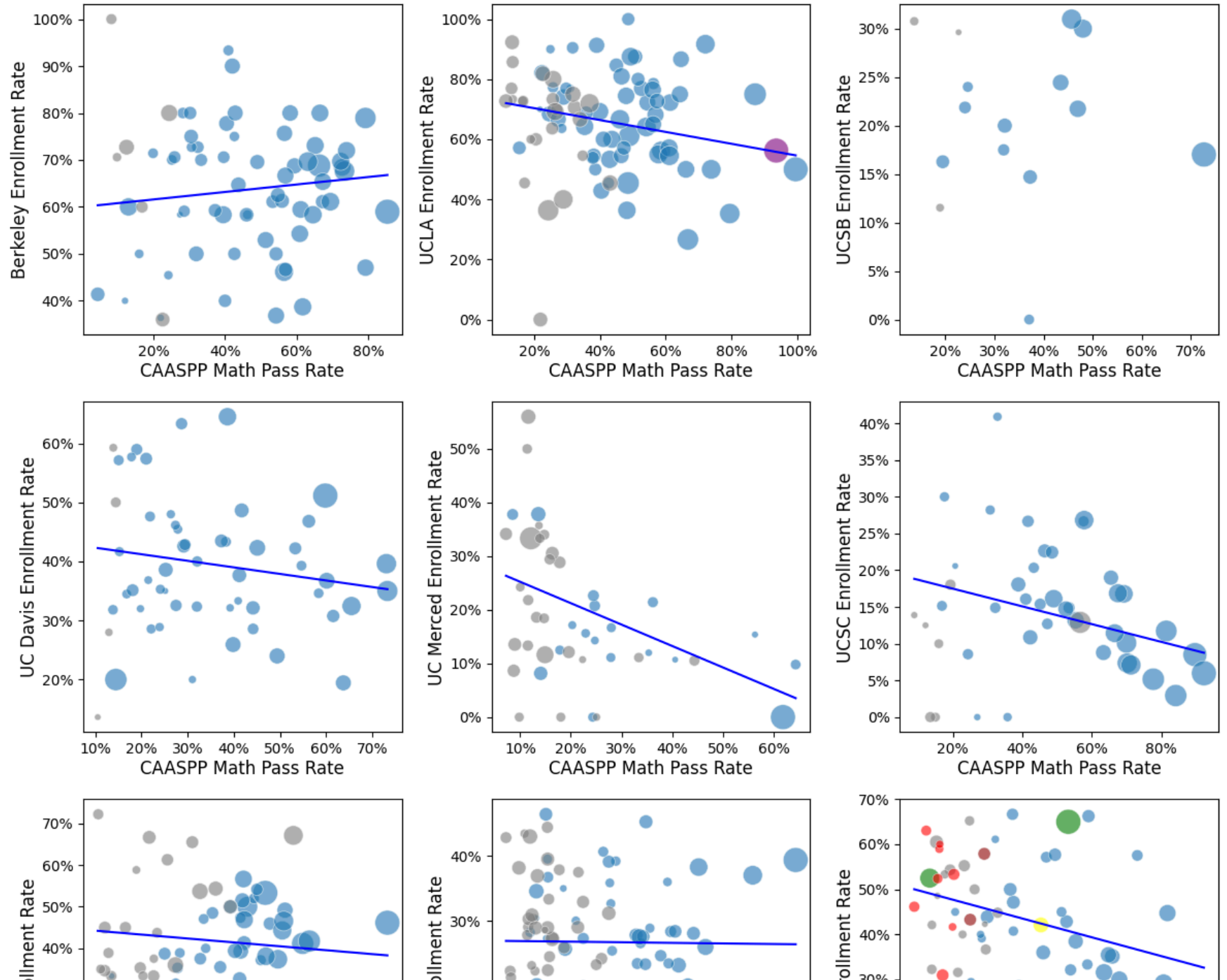


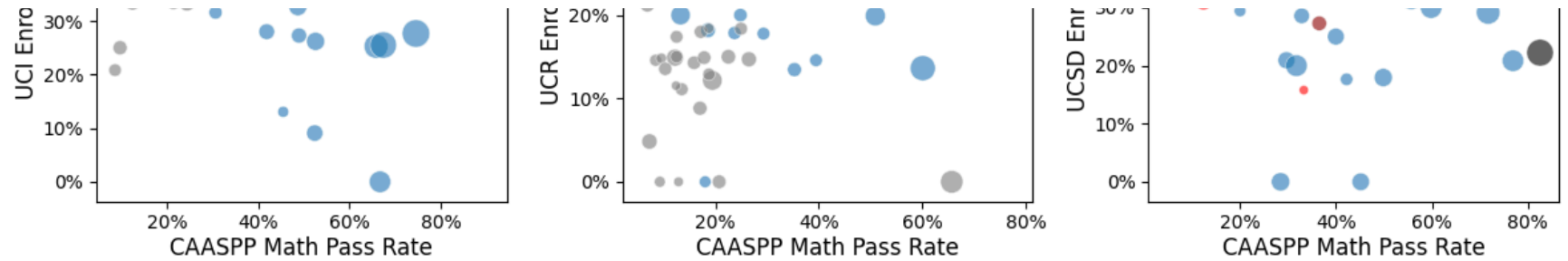


In [102...

```
createPlots(dfLeft, dfRight, "Local", yAxis="EnrRate", title="UC Campus Enrollment vs CAASPP Math Pass Rates, High Schools in Local Counties")
```

UC Campus Enrollment vs CAASPP Math Pass Rates, High Schools in Local Counties

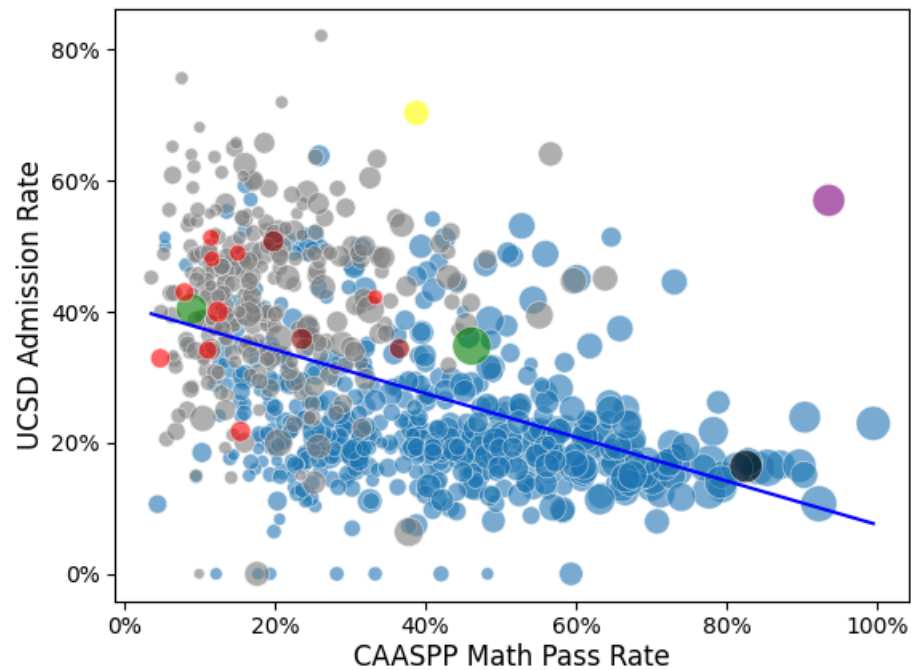




```
In [103... fig, ax = plt.subplots() # Create a Matplotlib Axes object
createPlot(ax, dfLeft, dfRight, "UCSD", "All", title="")

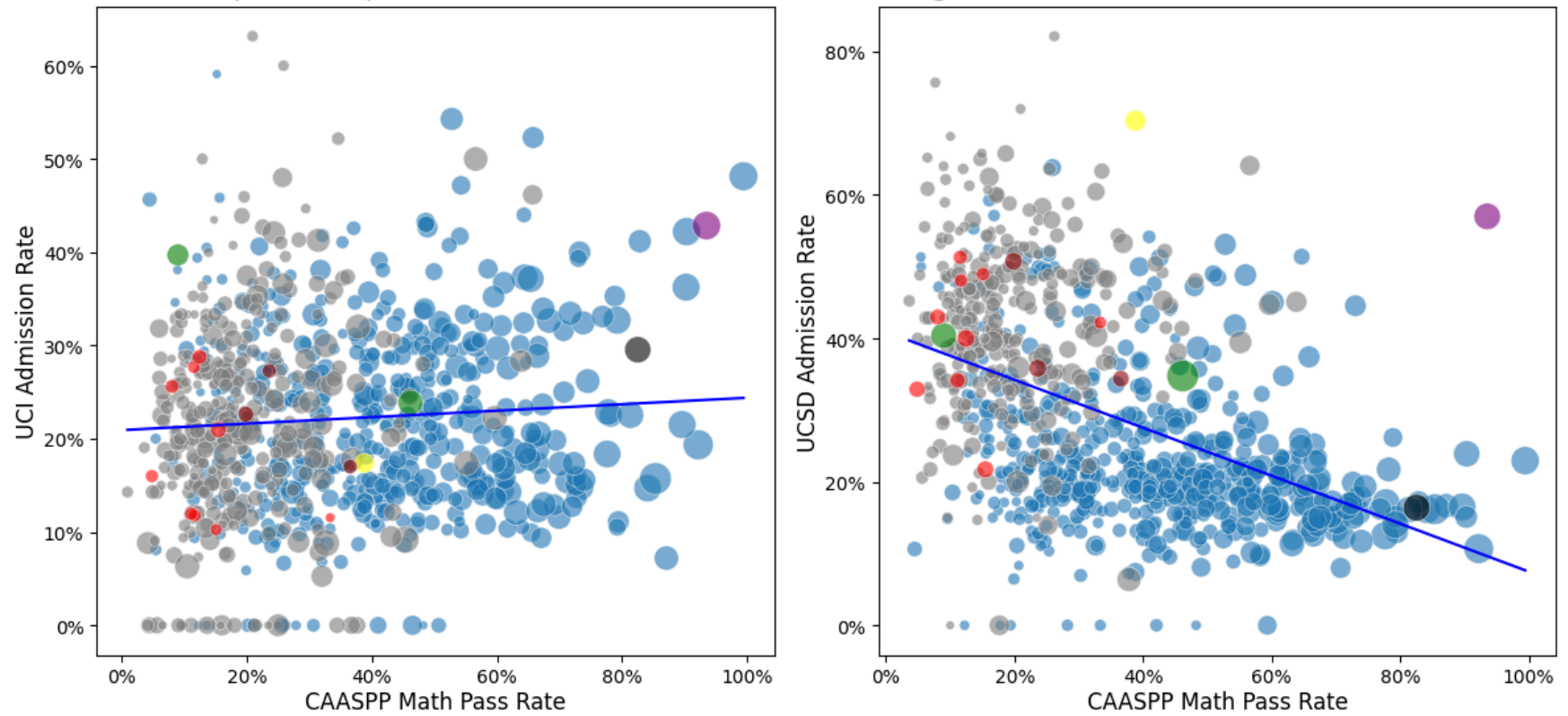
# plt.legend([],[], frameon=False) # Remove Seaborn's automatic legend
# legendElements = [] # Build custom legend handles
# for category in categoryTextNarrow:
#     legendElements.append(Line2D([0], [0], marker='o', color='w', label=categoryTextNarrow[category], markerfacecolor=customPalette[category], marke
# plt.legend(handles=legendElements, bbox_to_anchor=(1.0, 1), loc="upper left") # Add custom legend

plt.show()
```



```
In [104... createPlots2(dfLeft, dfRight, "All", title="UC Campus Acceptance vs CAASPP Math Pass Rates, High Schools in All California Counties")
```

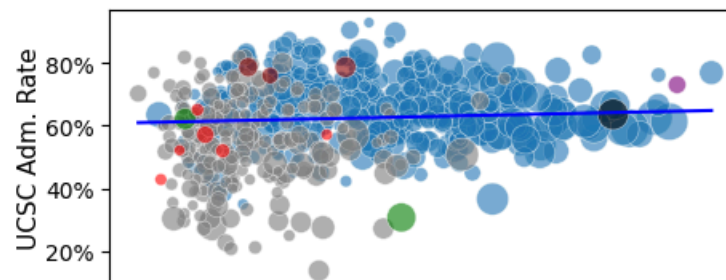
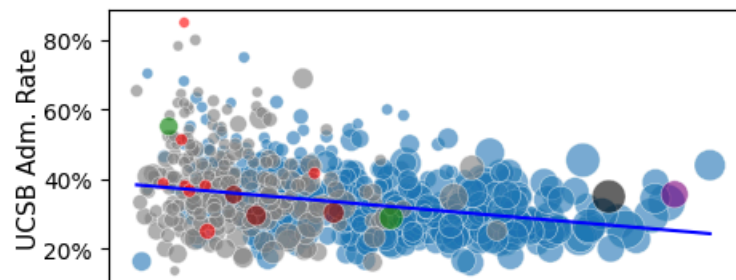
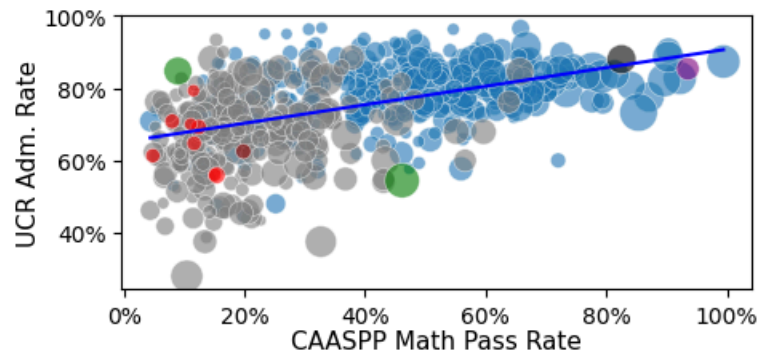
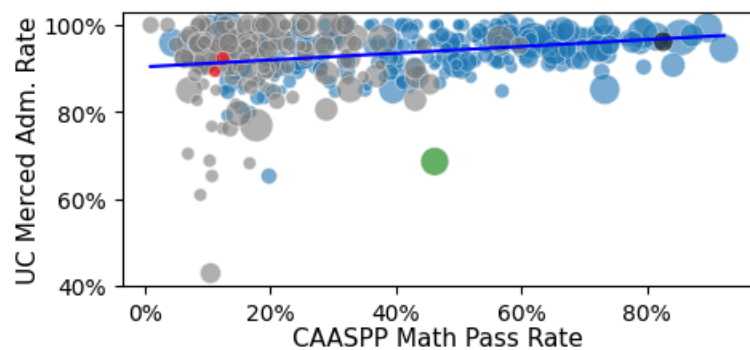
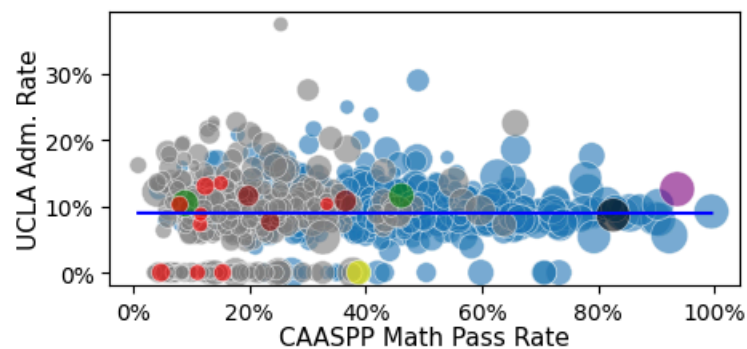
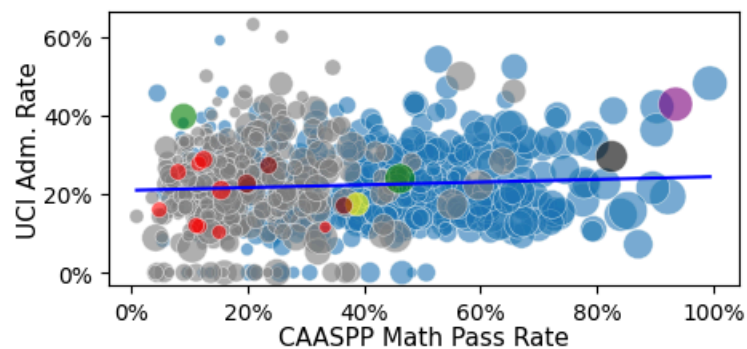
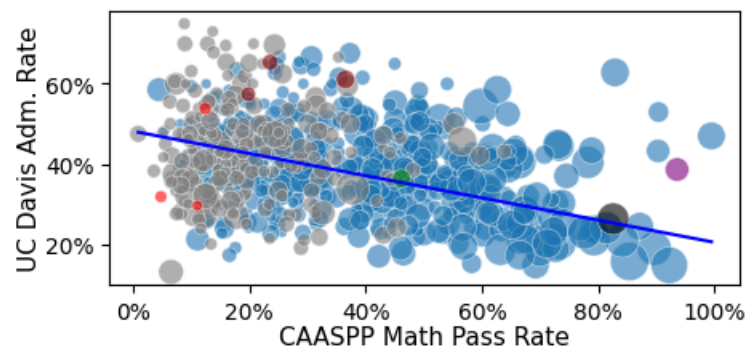
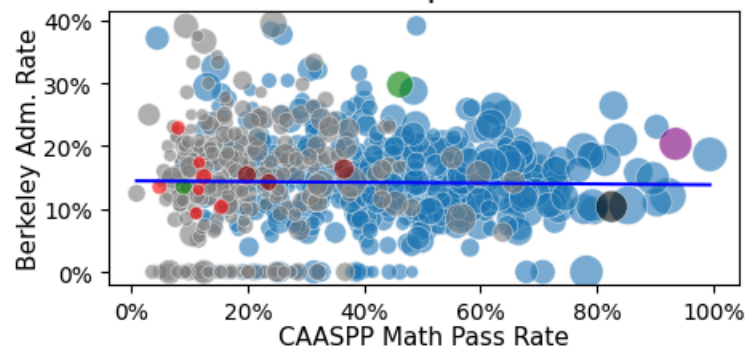
UC Campus Acceptance vs CAASPP Math Pass Rates, High Schools in All California Counties

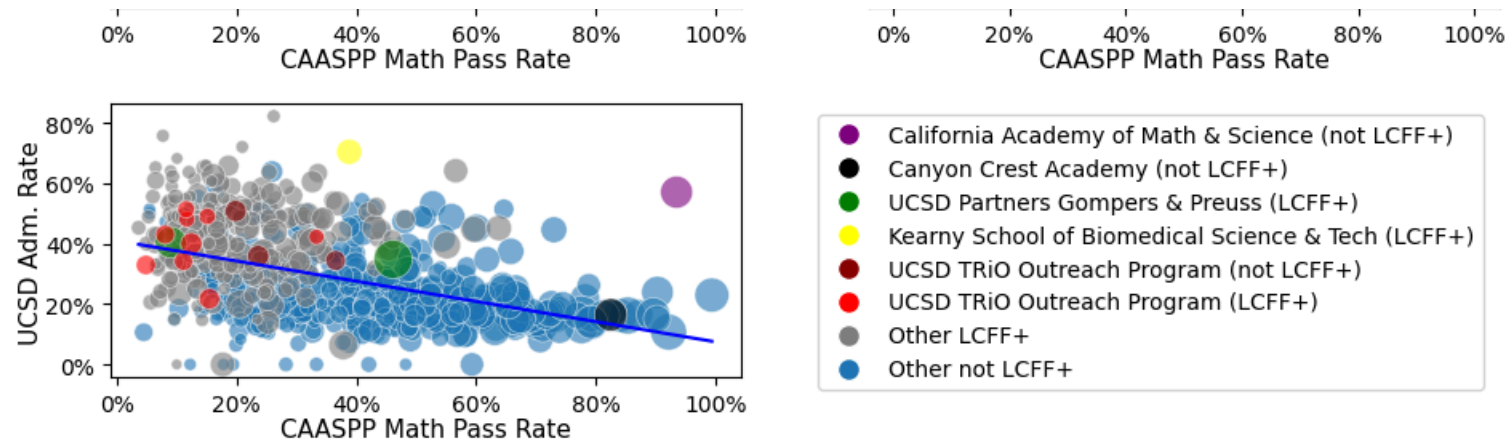


In [105...

```
# 2x5
createPlots10(dfLeft, dfRight, "All", title="UC Campus Admission Rate vs. CAASPP Math Pass Rate")
```


UC Campus Admission Rate vs. CAASPP Math Pass Rate

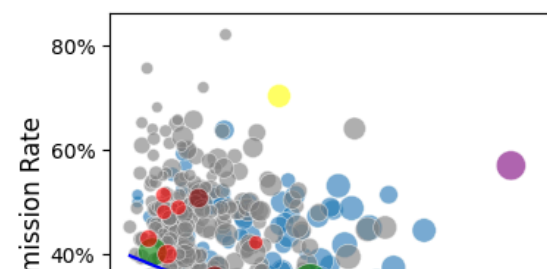
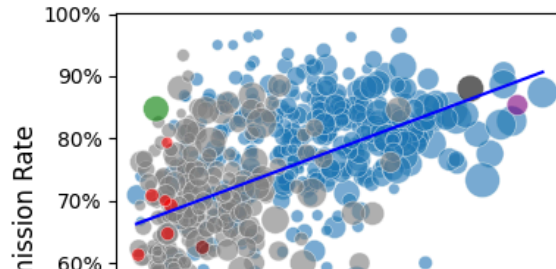
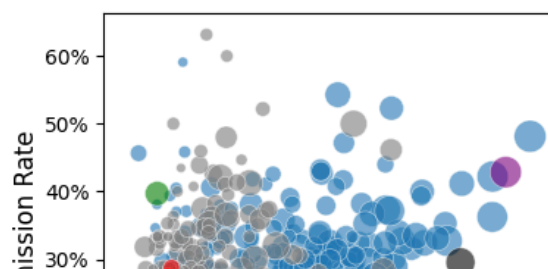
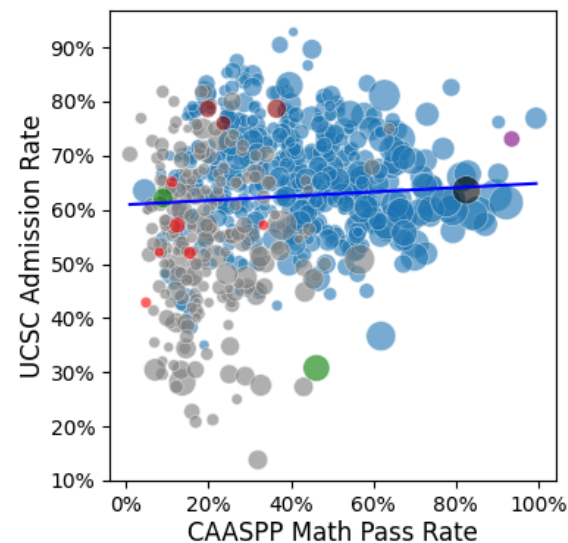
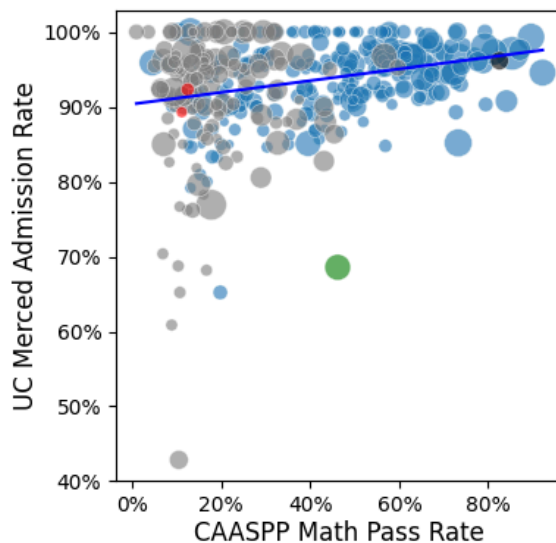
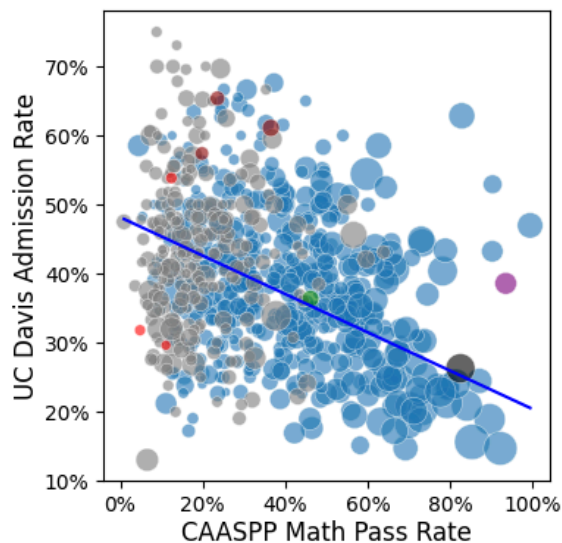
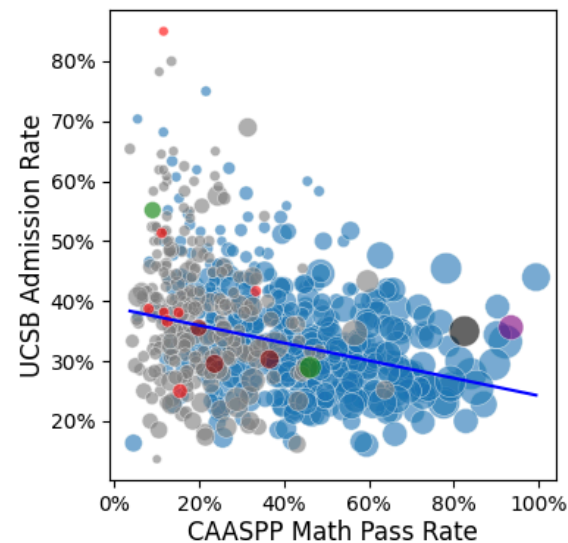
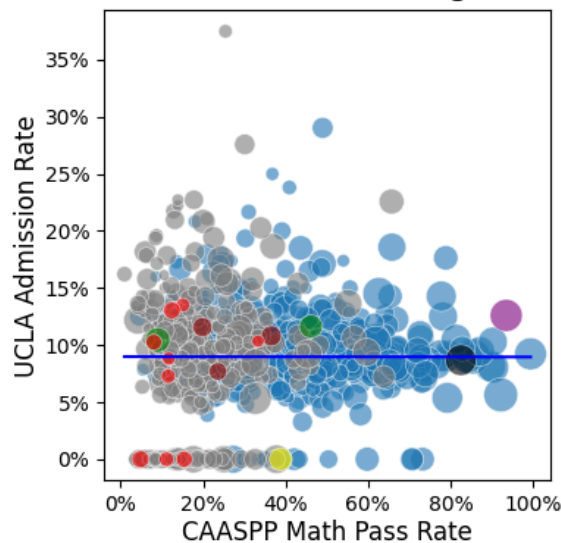
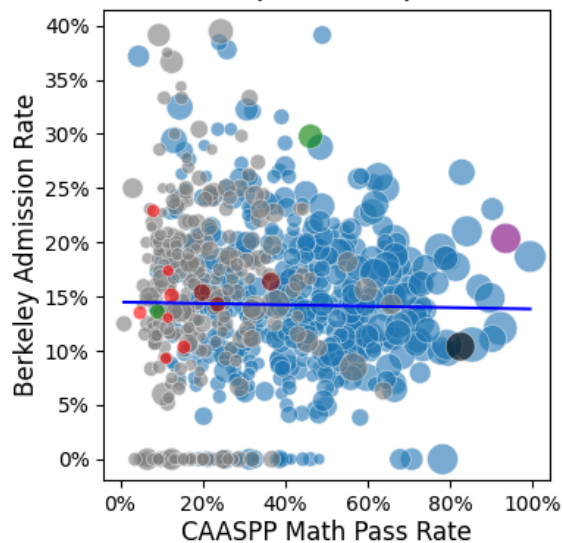


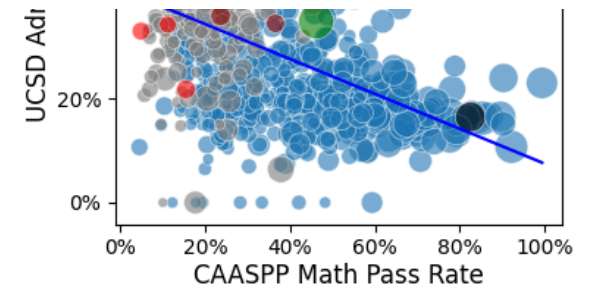
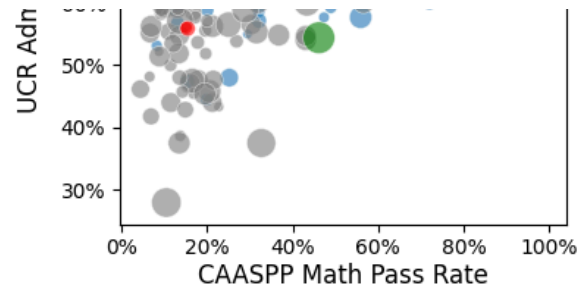
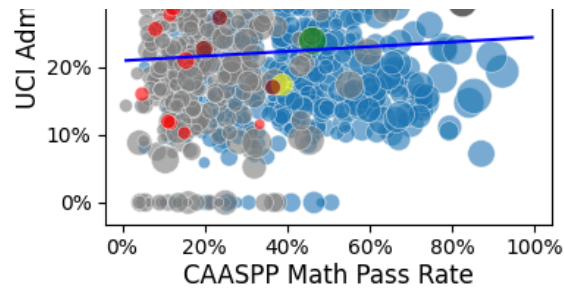


In [106...

```
# 3x3
createPlots(dfLeft, dfRight, "All", title="UC Campus Acceptance vs CAASPP Math Pass Rates, High Sch
```

UC Campus Acceptance vs CAASPP Math Pass Rates, High Schools in All California Counties

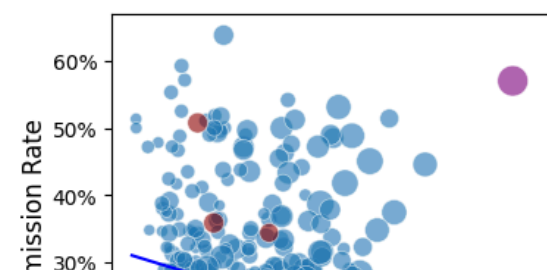
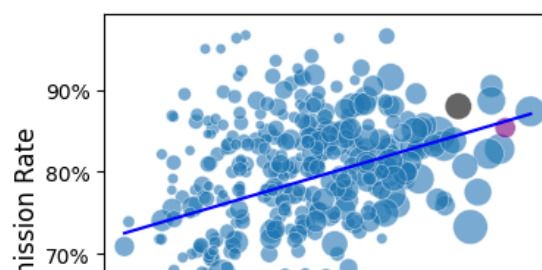
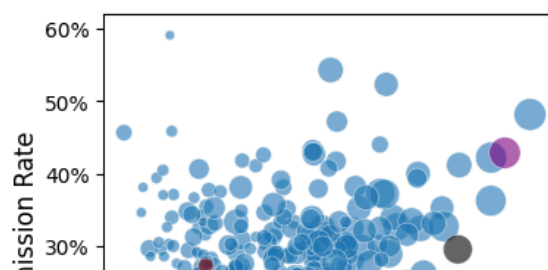
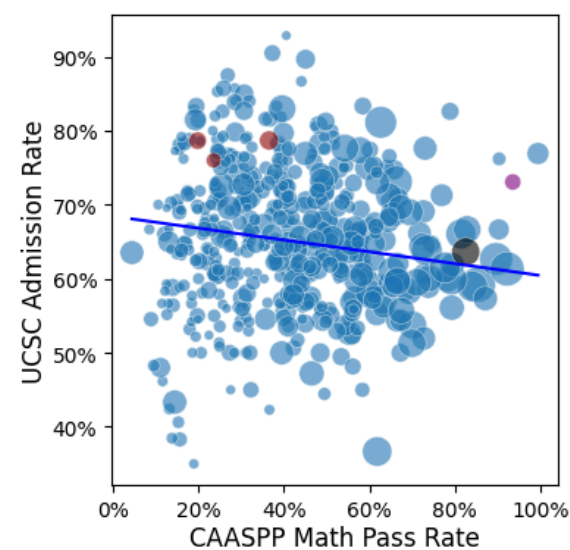
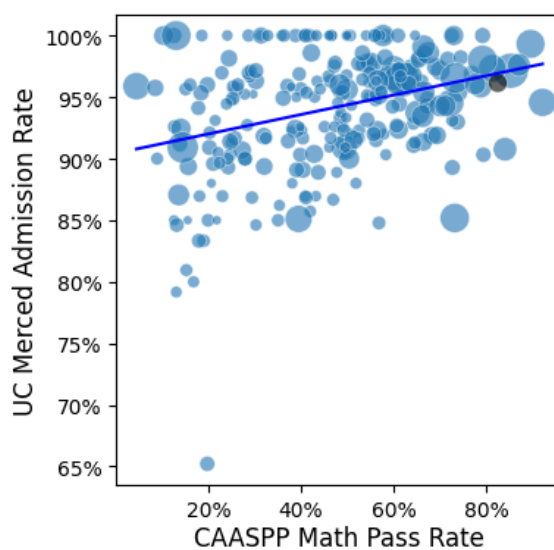
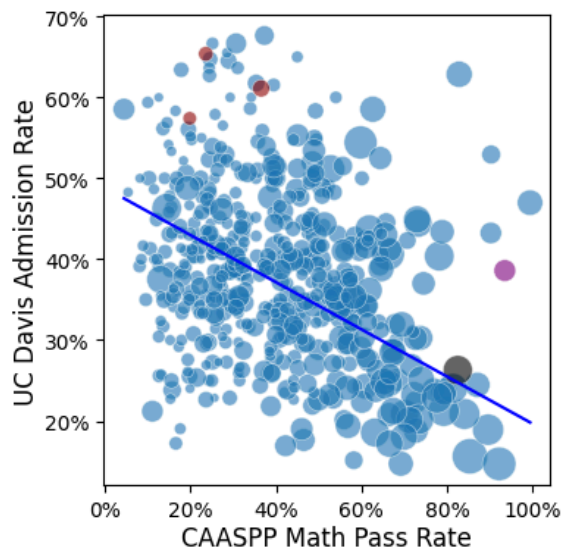
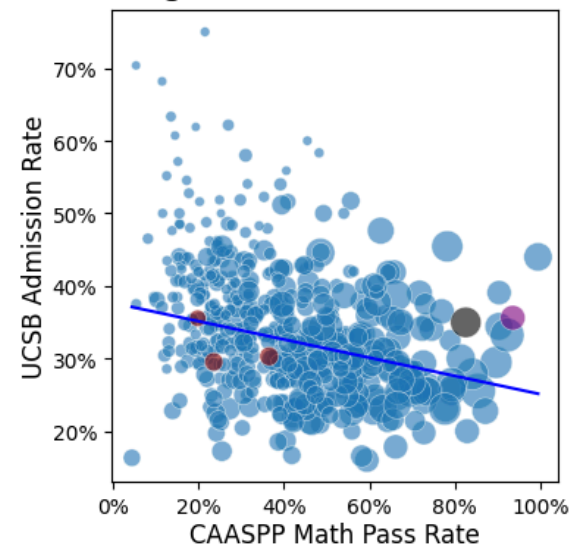
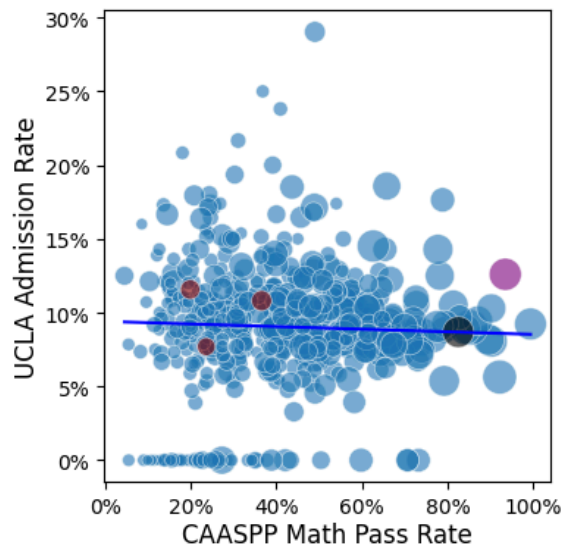
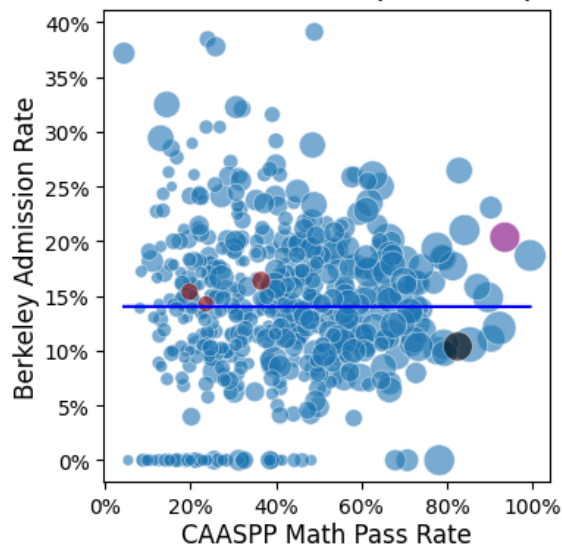


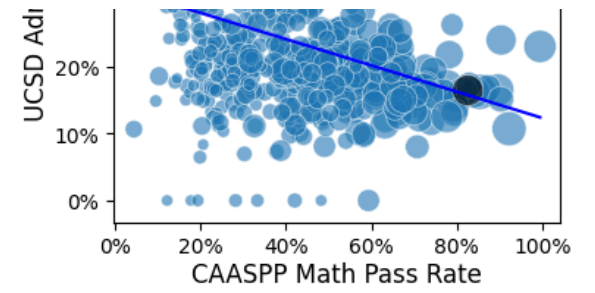
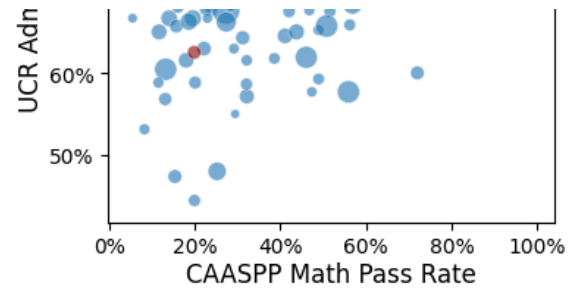
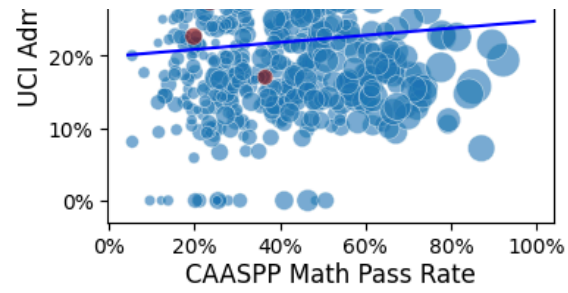


In [107...

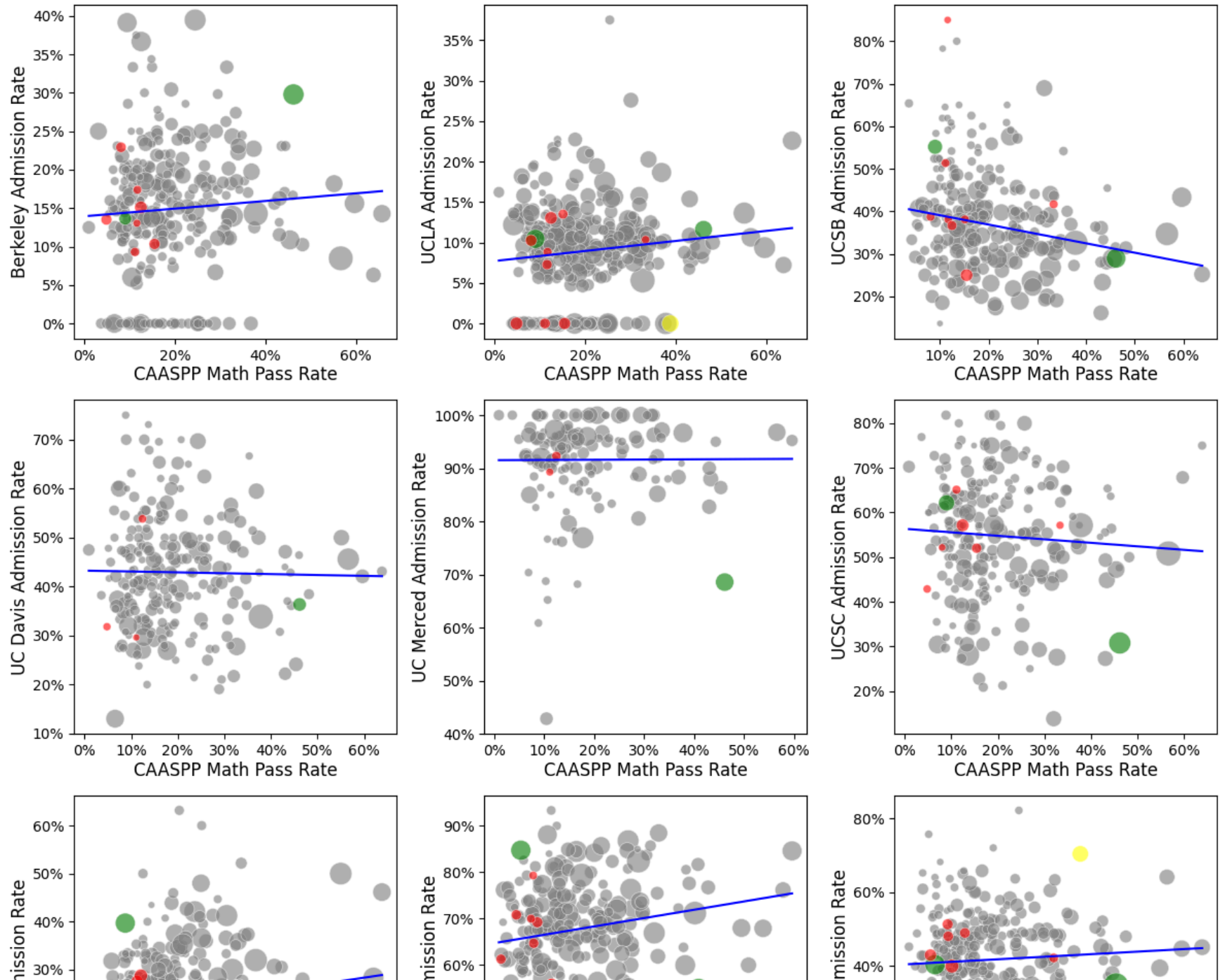
```
# Look separately at LCFF+ schools and only non-LCFF+ schools
createPlots(dfLeft, dfRight, "All", filterMetrics=[["LCFF Plus School t", "==", "N"]], title="UC Campus Acceptance vs CAASPP Math Pass Rates, No LCFF+
createPlots(dfLeft, dfRight, "All", filterMetrics=[["LCFF Plus School t", "==", "Y"]], title="UC Campus Acceptance vs CAASPP Math Pass Rates, Only LCF
```

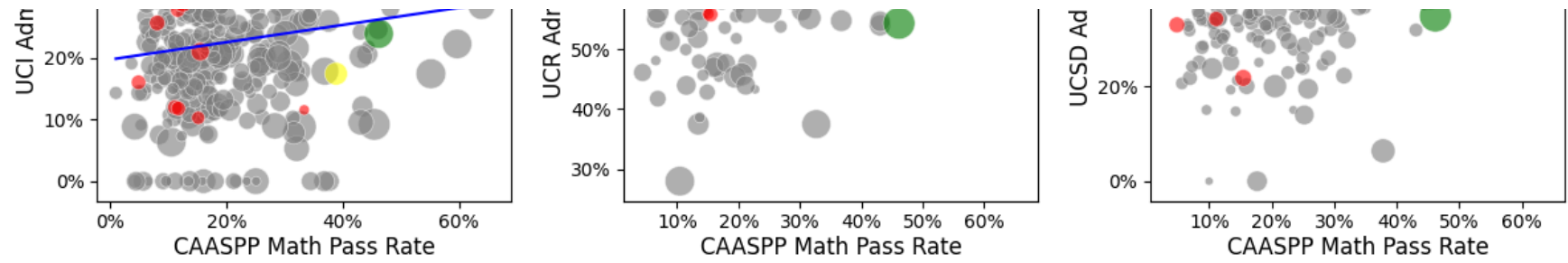

UC Campus Acceptance vs CAASPP Math Pass Rates, No LCFF+ High Schools





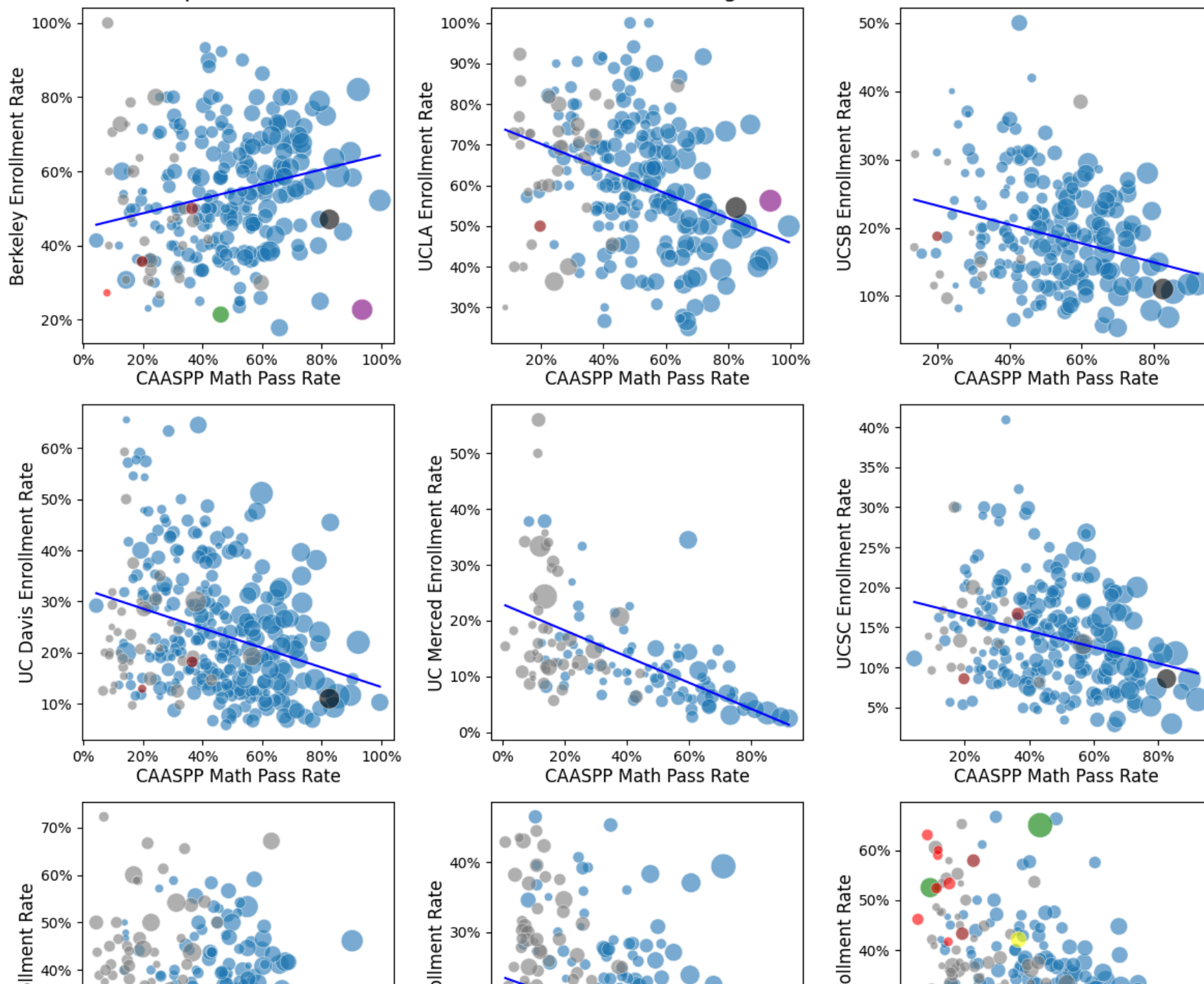
UC Campus Acceptance vs CAASPP Math Pass Rates, Only LCFF+ High Schools

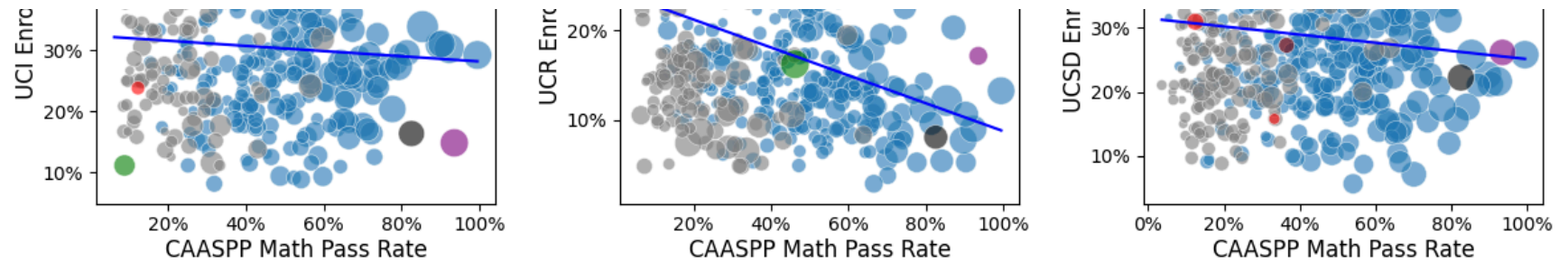




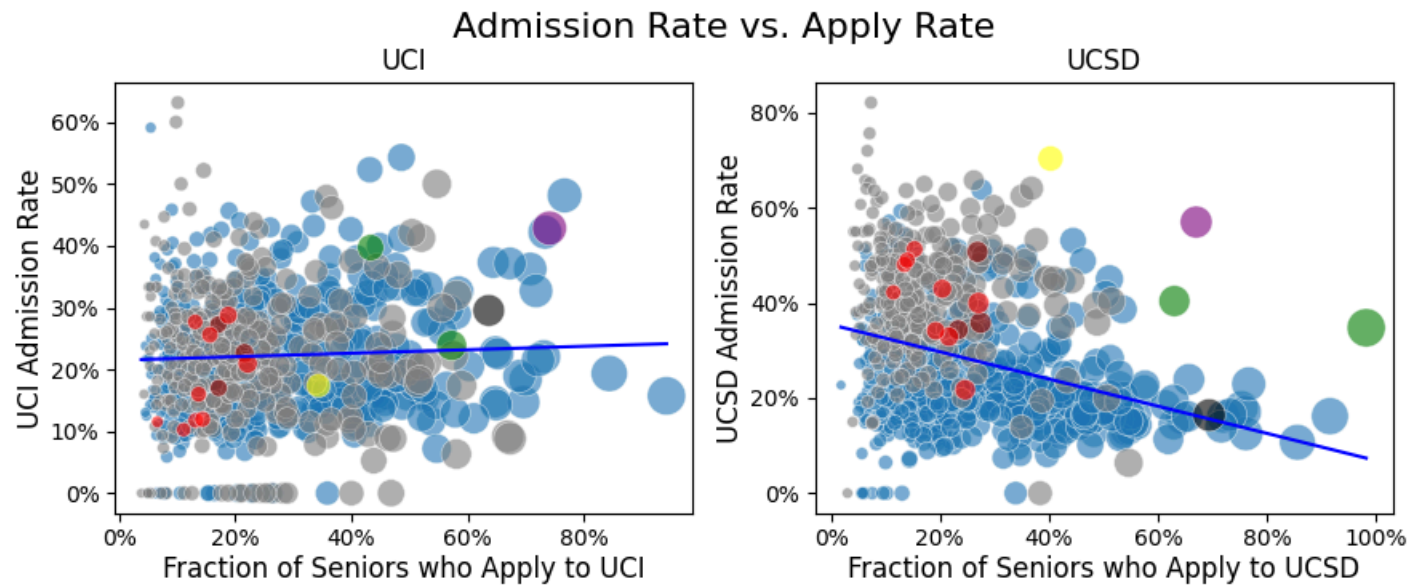
In [108... `createPlots(dfLeft, dfRight, "All", yAxis="EnrRate", filterMetrics=[["Enr", ">=", 3]], title="UC Campus Enrollment vs CAASPP Math Pass Rates, High Sch`

UC Campus Enrollment vs CAASPP Math Pass Rates, High Schools in All California Counties



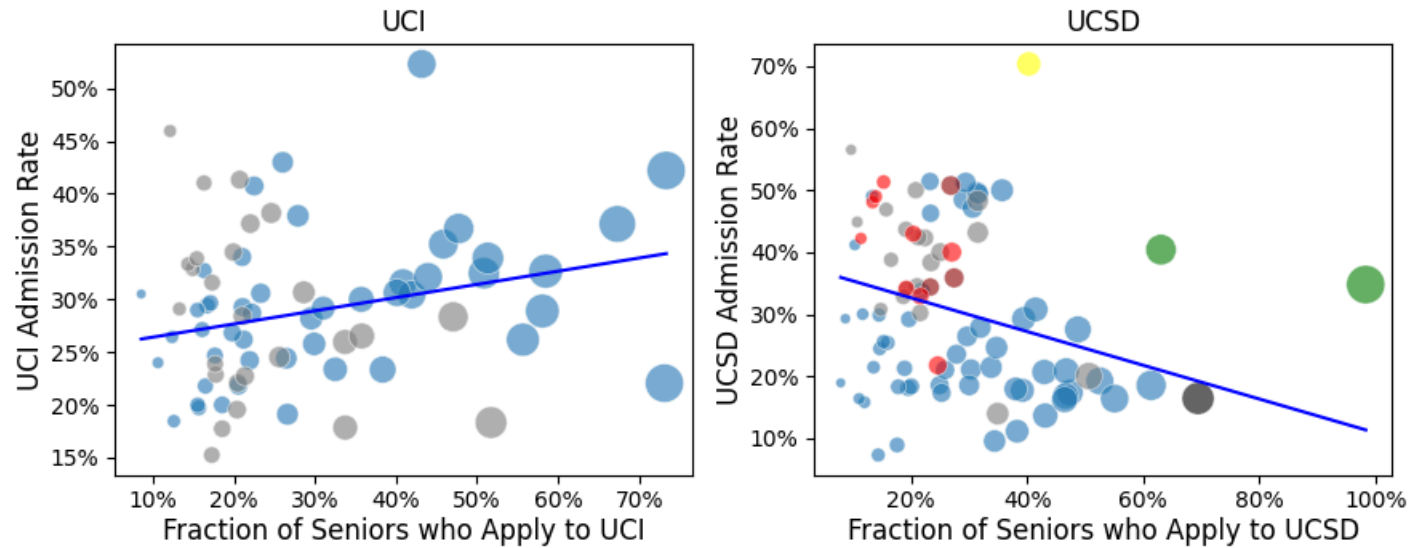


```
In [109... # Acceptance rate vs fraction who apply
createPlots2(dfLeft, dfRight, "All", xAxis="SeniorAppRate", height=4, width=9, campusTitles="Y", title=" Admission Rate vs. Apply Rate", titleGap=0.9
```



```
In [110... createPlots2(dfLeft, dfRight, "Local", xAxis="SeniorAppRate", height=4, width=9, campusTitles="Y", title="UC Campus Admission vs. Apply Rate, High Sch
```

UC Campus Admission vs. Apply Rate, High Schools in All California Counties

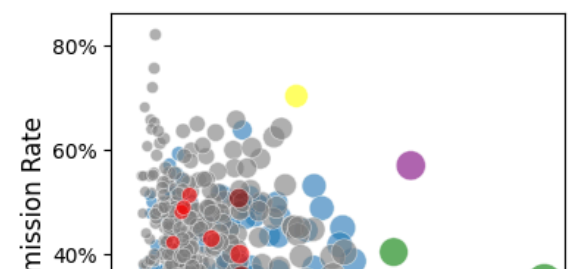
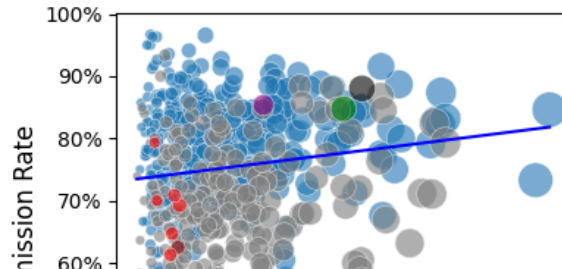
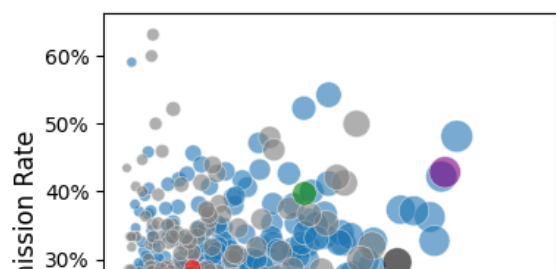
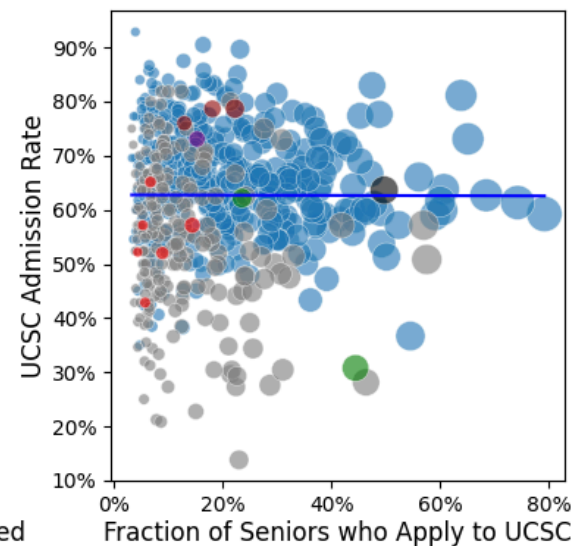
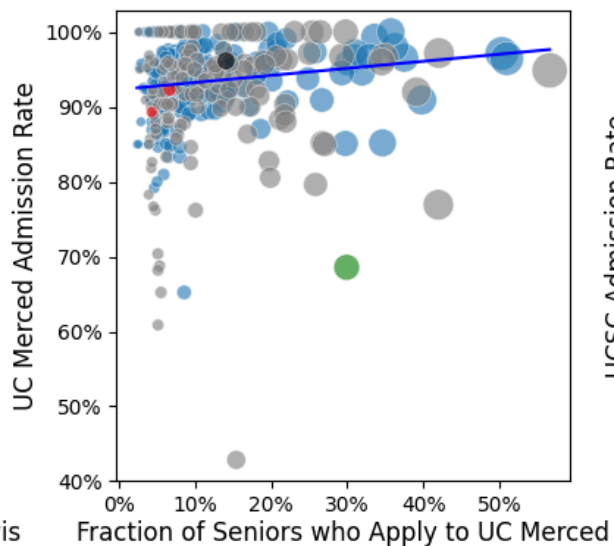
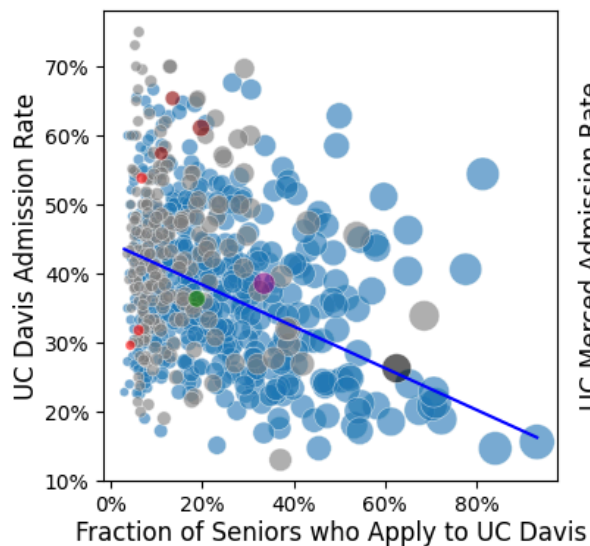
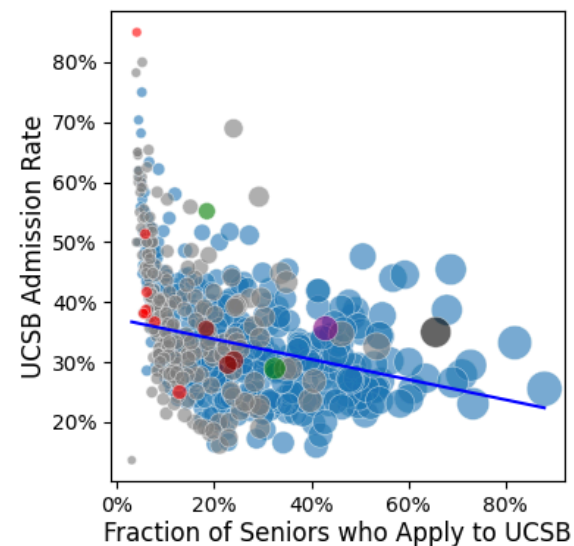
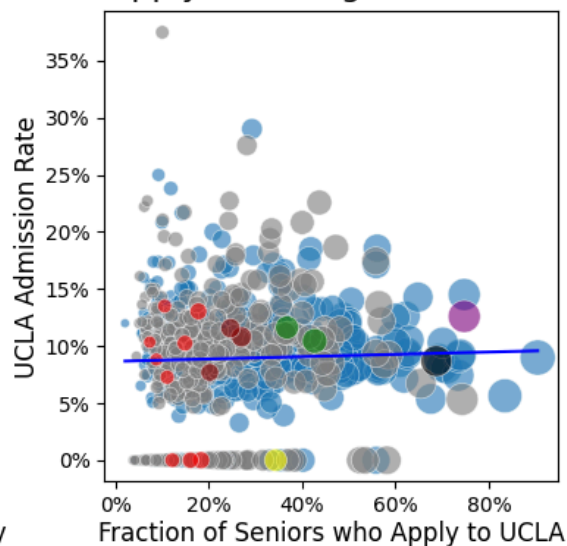
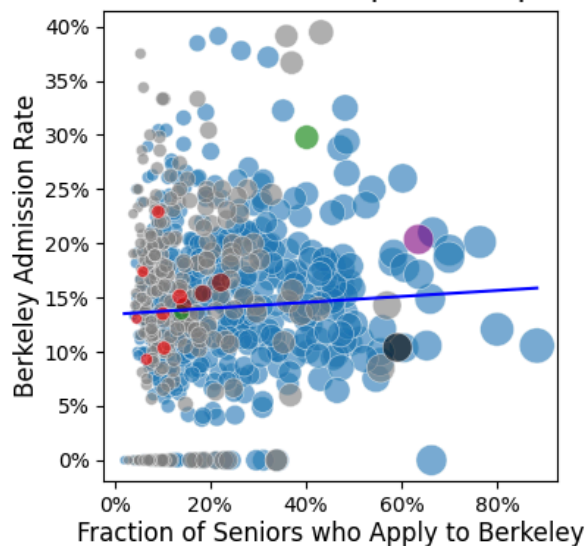


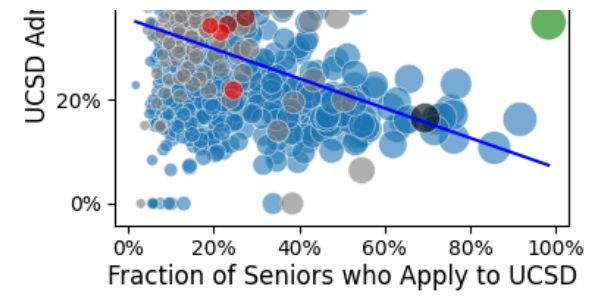
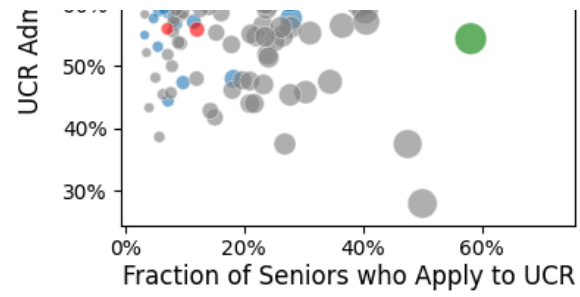
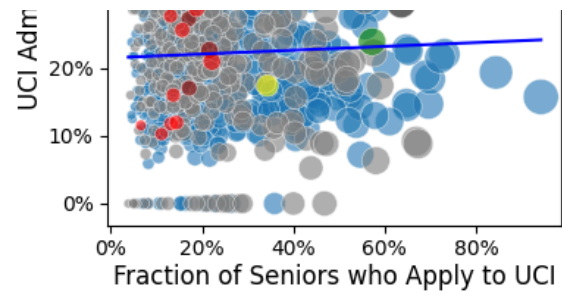
In [111...

```
# Acceptance rate vs fraction who apply
createPlots(dfLeft, dfRight, "All", xAxis="SeniorAppRate", title="UC Campus Acceptance vs Apply Rat

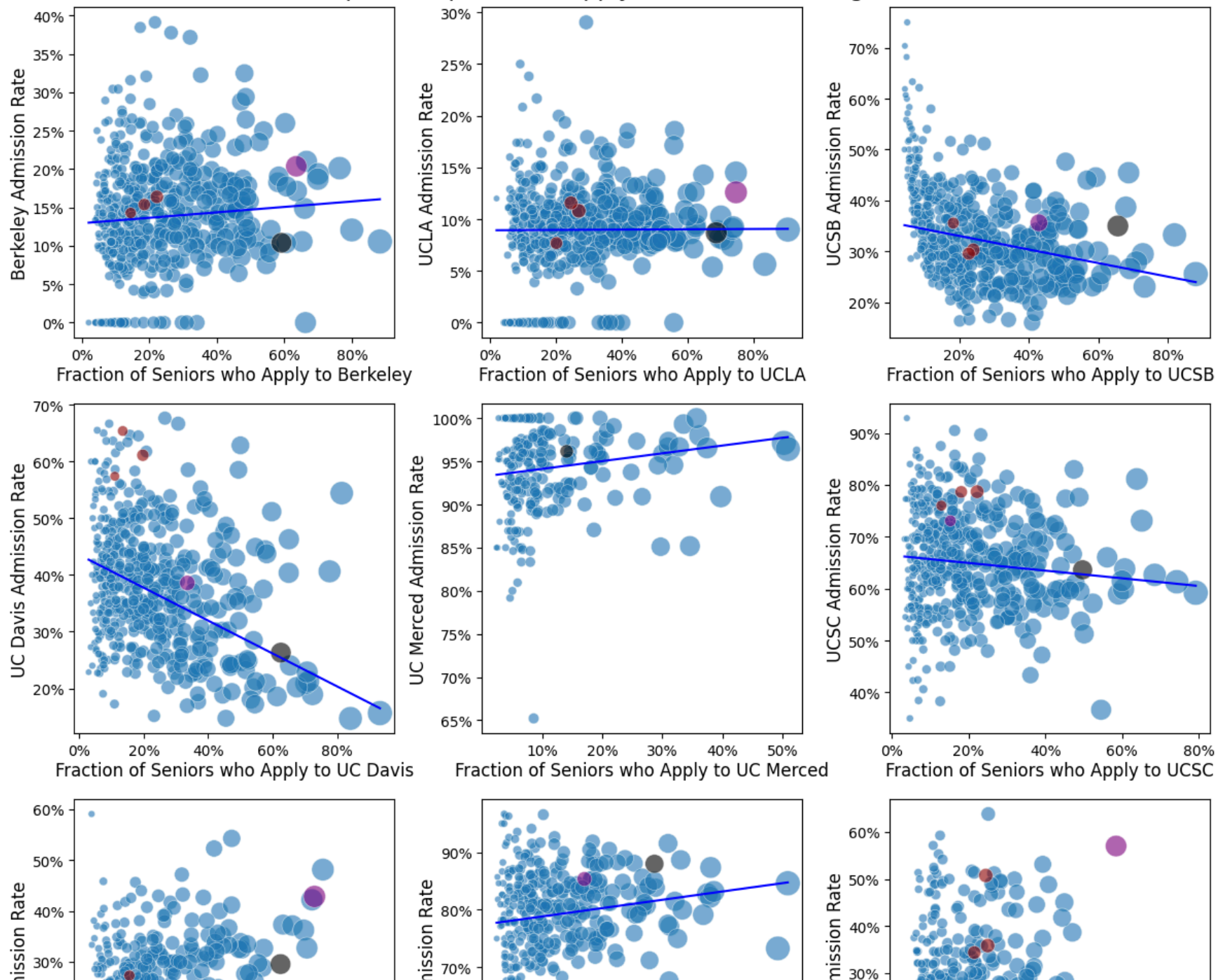
createPlots(dfLeft, dfRight, "All", xAxis="SeniorAppRate", filterMetrics=[["LCFF Plus School t", "=", "N"]], title="UC Campus Acceptance vs Apply Rat
createPlots(dfLeft, dfRight, "All", xAxis="SeniorAppRate", filterMetrics=[["LCFF Plus School t", "=", "Y"]], title="UC Campus Acceptance vs Apply Rat
```

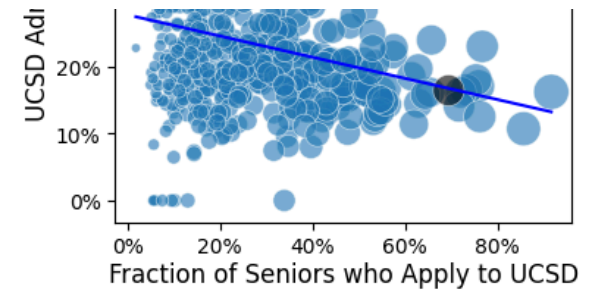
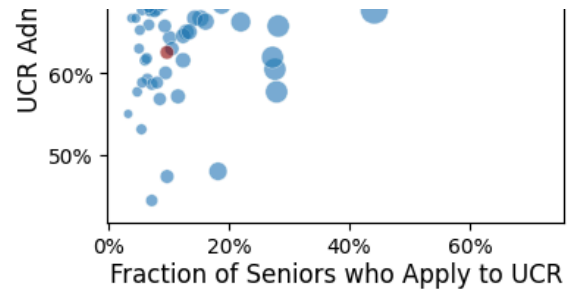
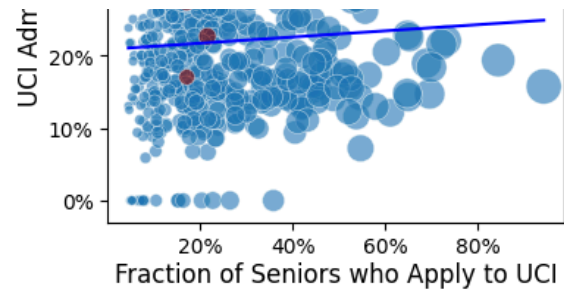
UC Campus Acceptance vs Apply Rate, High Schools in All California Counties



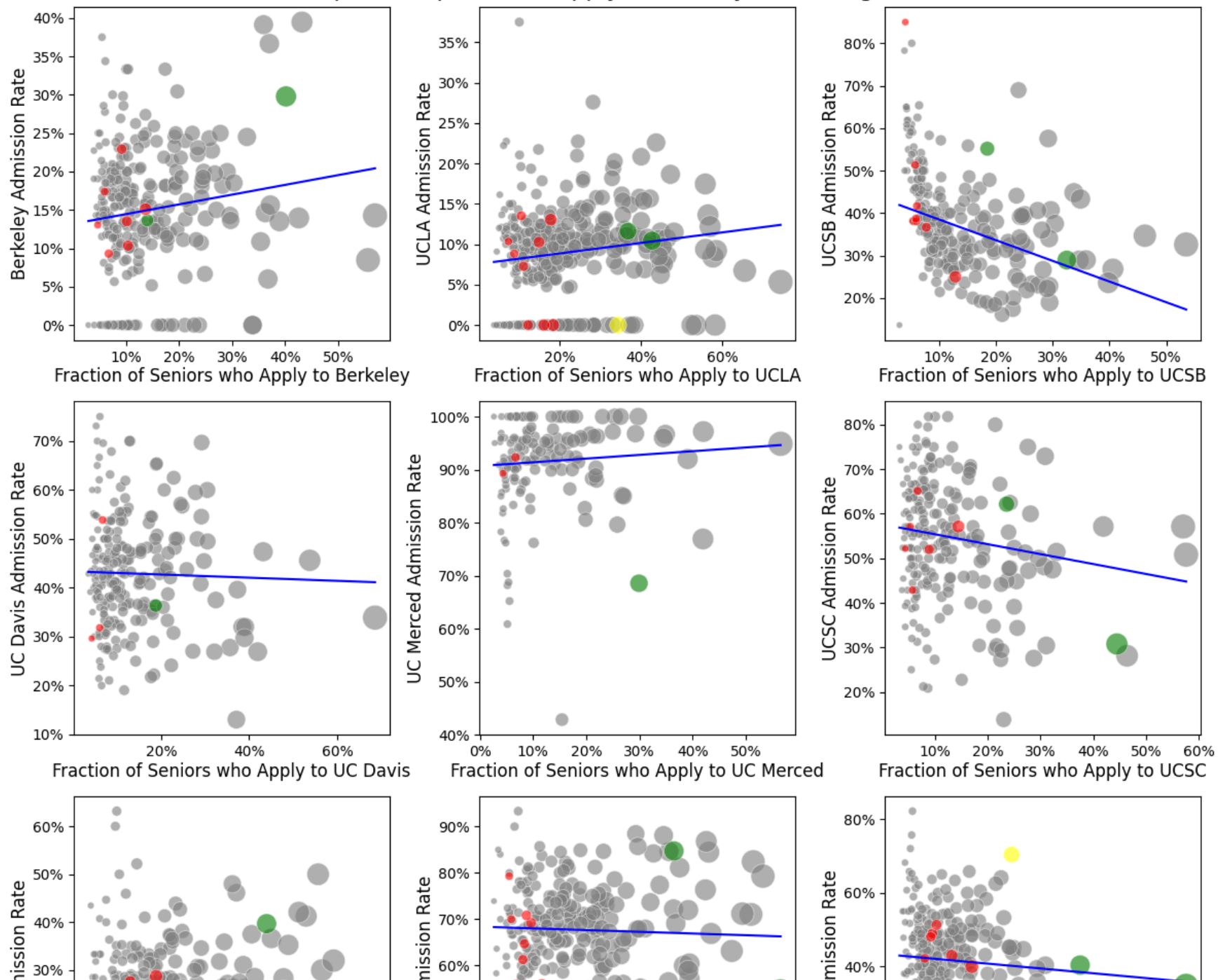


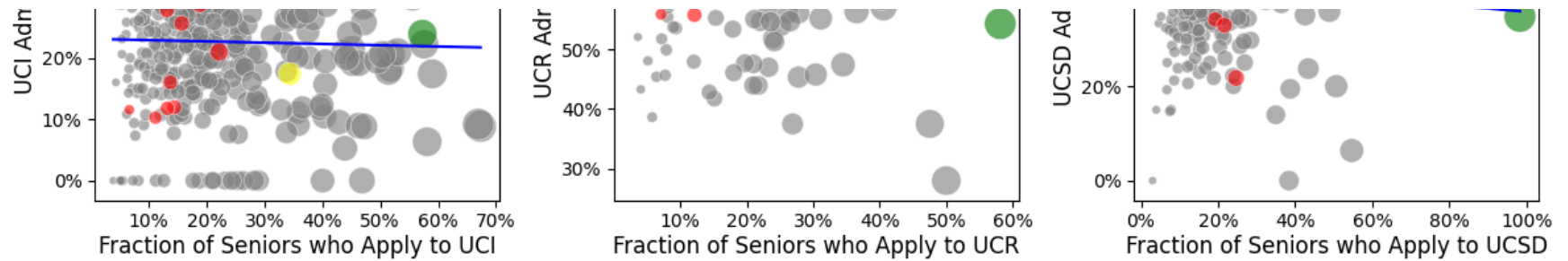
UC Campus Acceptance vs Apply Rate, No LCFF+ High Schools



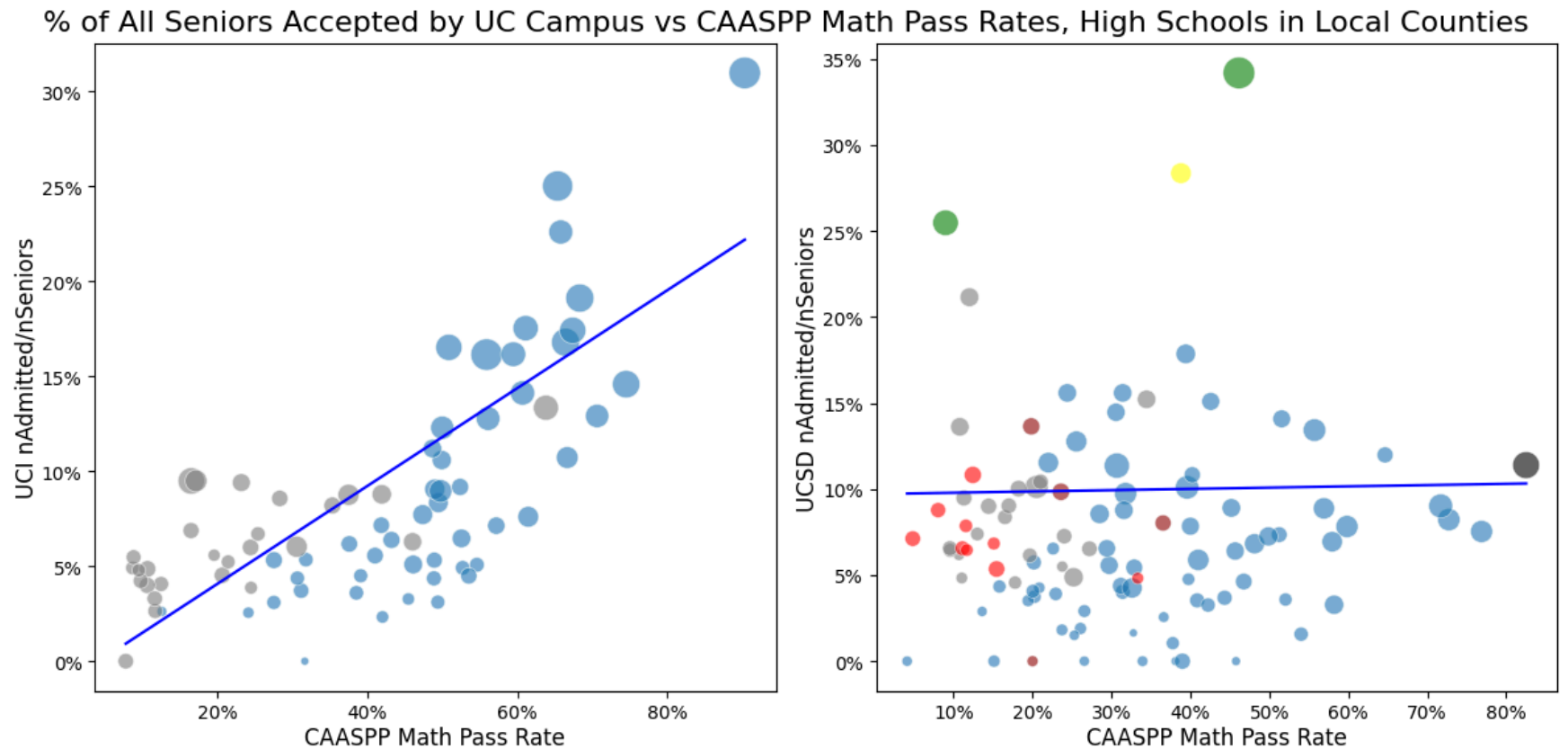


UC Campus Acceptance vs Apply Rate, Only LCFF+ High Schools



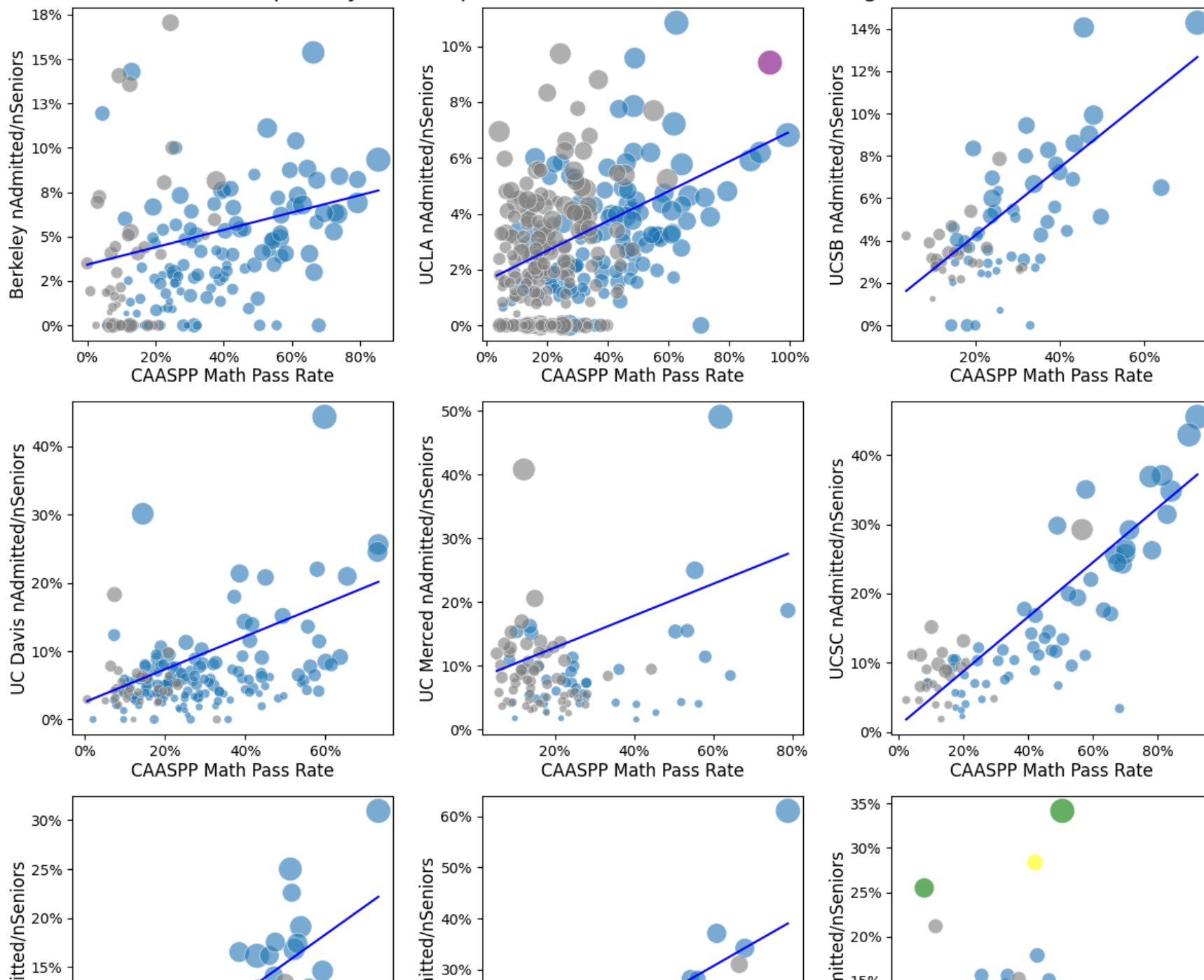


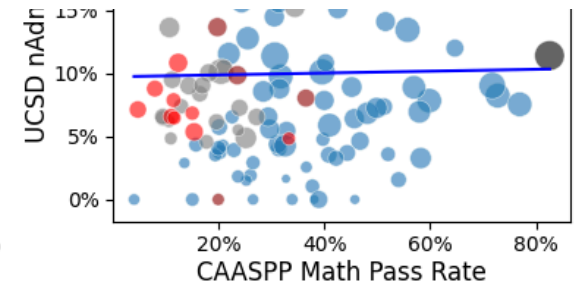
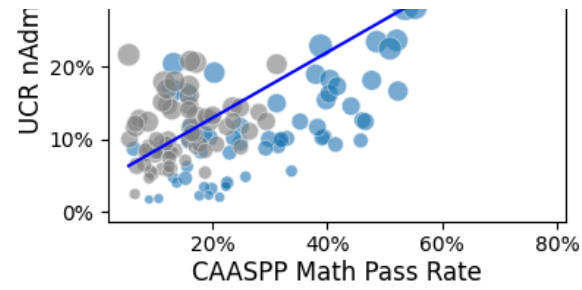
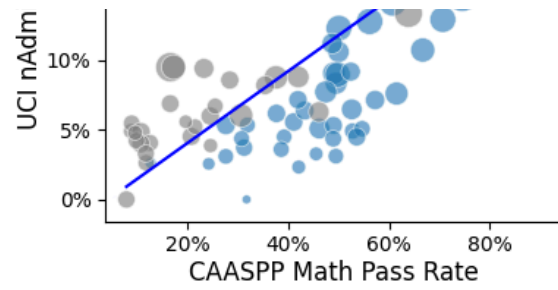
```
In [112... createPlots2(dfLeft, dfRight, "Local", yAxis="SeniorAdmRate", selectMinApps=0, title="% of All Seniors Accepted by UC Campus vs CAASPP Math Pass Rates
```



```
In [113... createPlots(dfLeft, dfRight, "Local", yAxis="SeniorAdmRate", selectMinApps=0, title="% of All Seniors Accepted by UC Campus vs CAASPP Math Pass Rates,
```

% of All Seniors Accepted by UC Campus vs CAASPP Math Pass Rates, High Schools in Local Counties

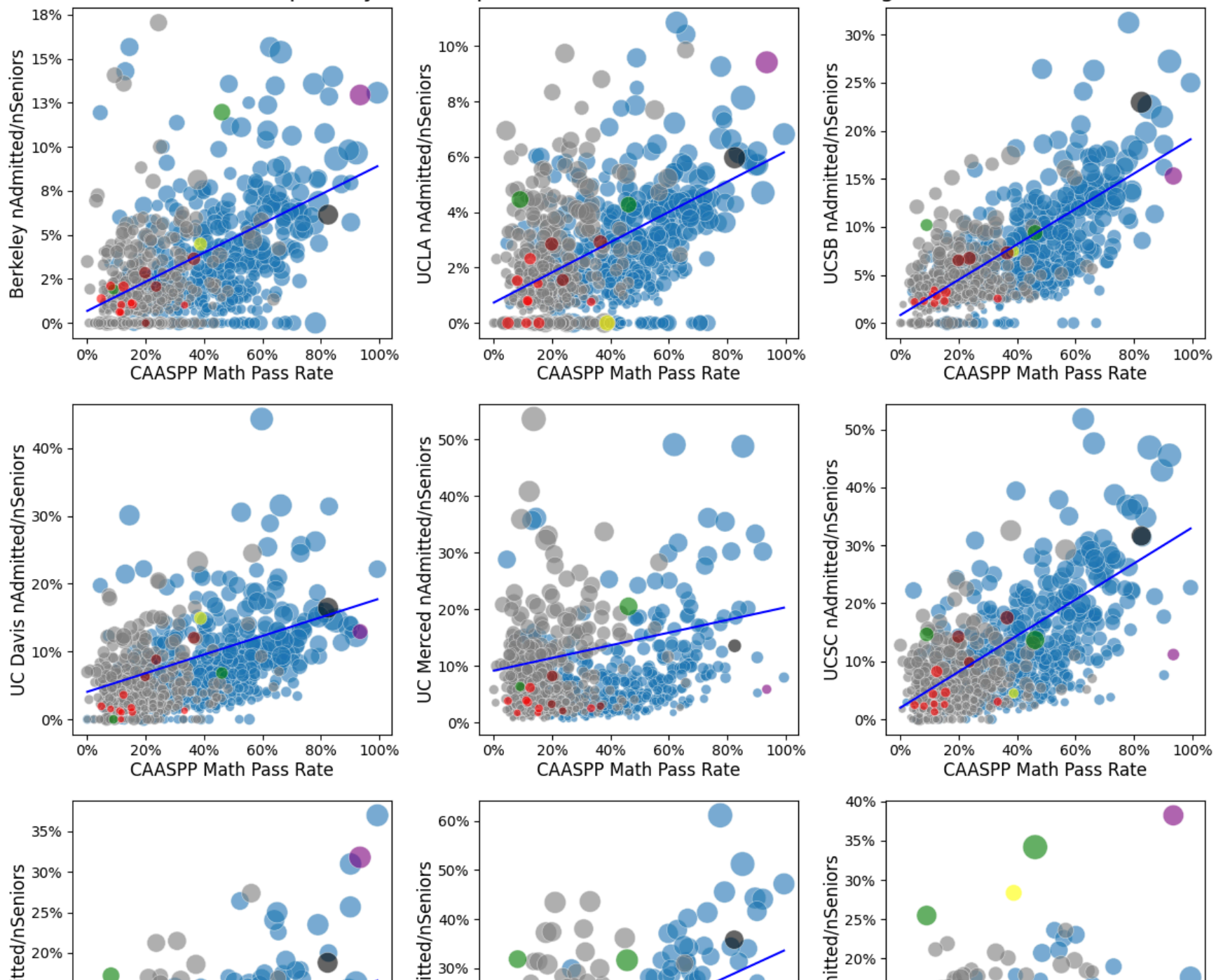


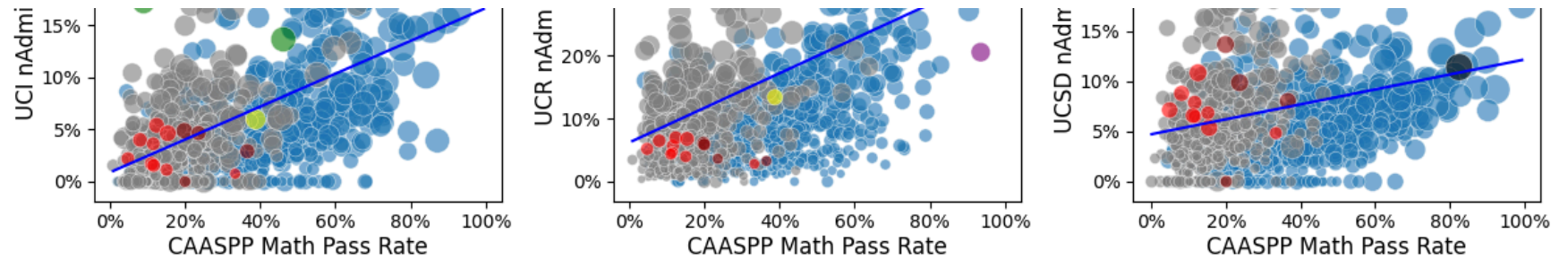


In [114...

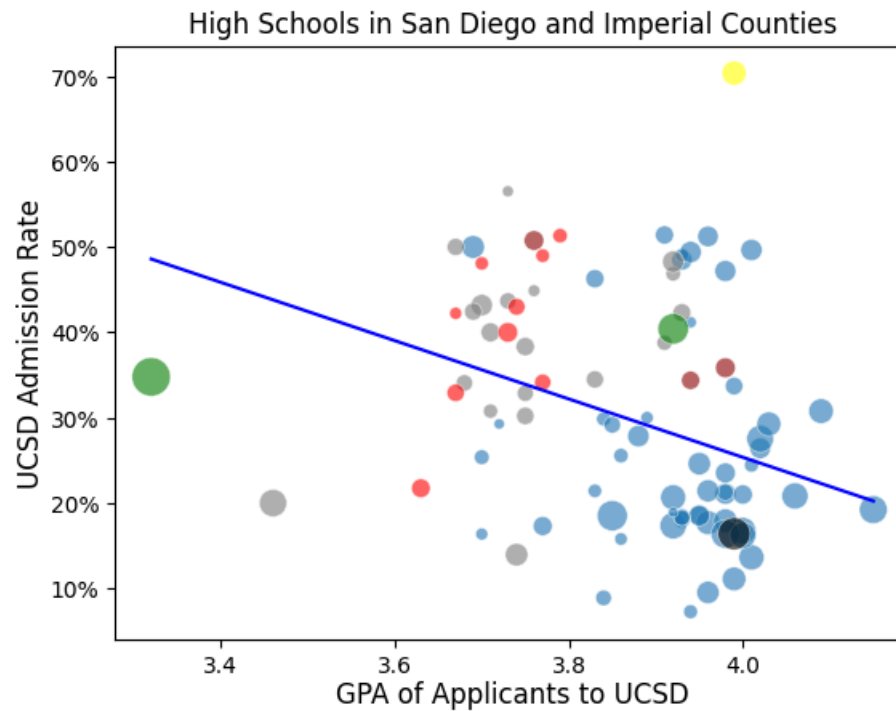
```
createPlots(dfLeft, dfRight, "All", yAxis="SeniorAdmRate", selectMinApps=0, title="% of All Seniors Accepted by UC Campus vs CAASPP Math Pass Rates, H
```

% of All Seniors Accepted by UC Campus vs CAASPP Math Pass Rates, High Schools in All Counties



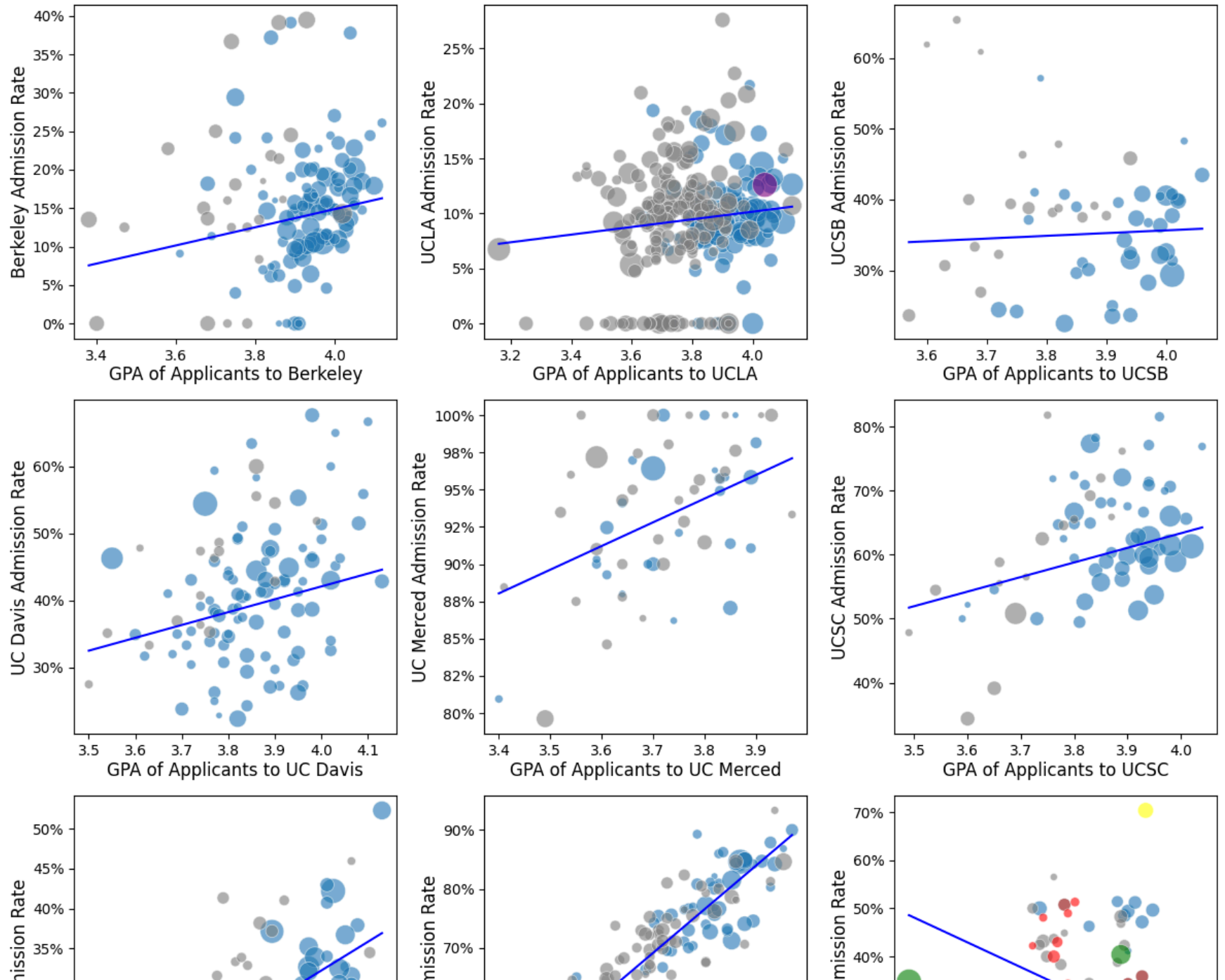


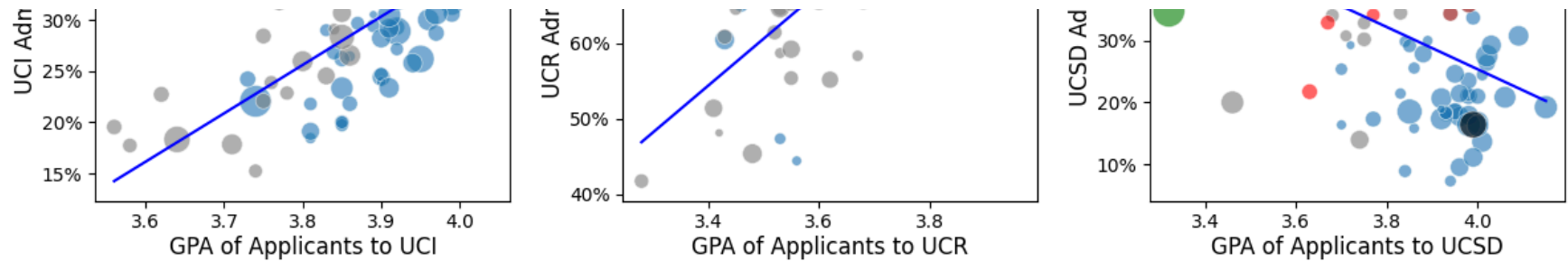
```
In [115... fig, ax = plt.subplots() # Create a Matplotlib Axes object
createPlot(ax, dfLeft, dfRight, "UCSD", "Local", xAxis="App GPA", title="High Schools in San Diego and Imperial Counties")
plt.show()
```



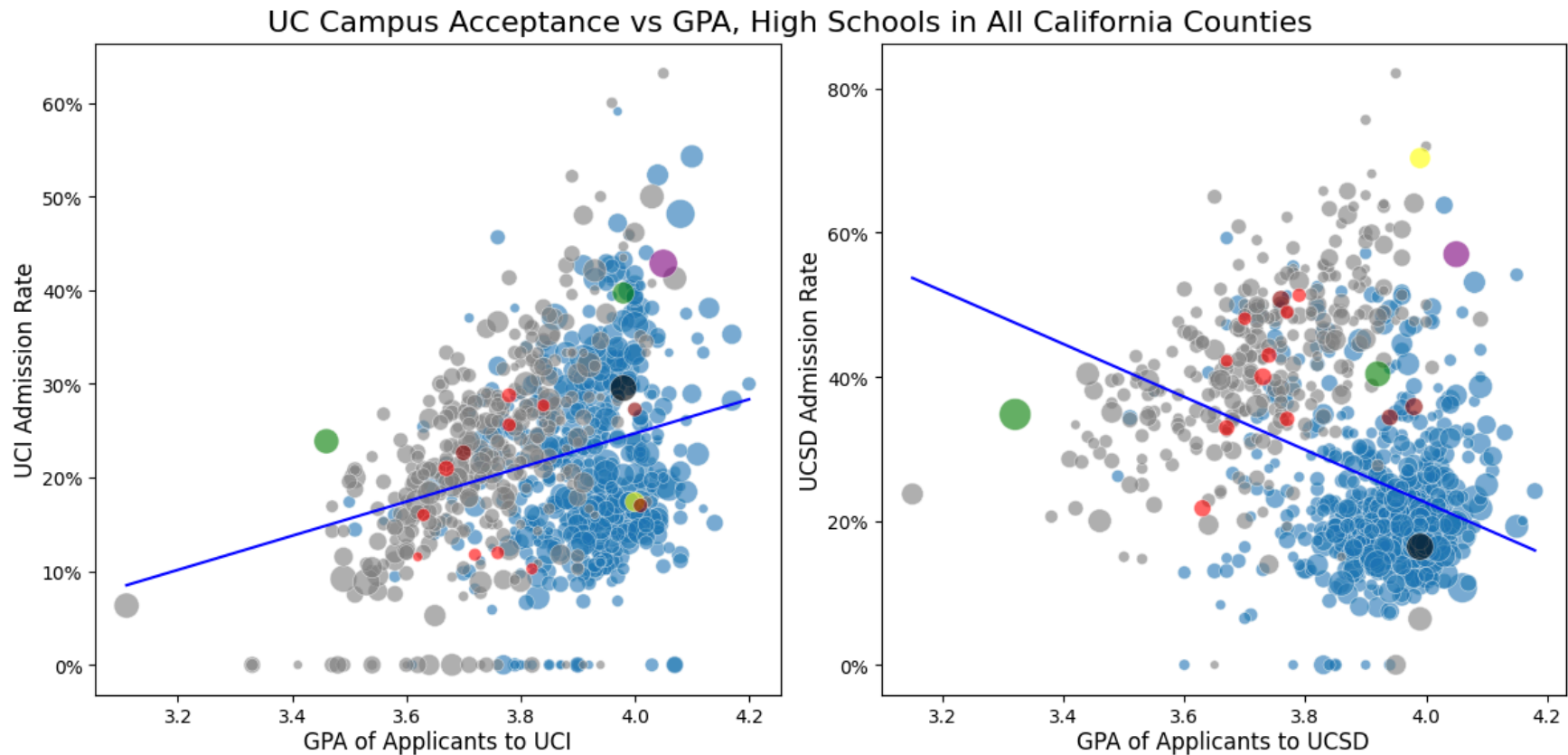
```
In [116... createPlots(dfLeft, dfRight, "Local", xAxis="App GPA", title="UC Campus Acceptance vs GPA, High Schools in Local Counties")
```

UC Campus Acceptance vs GPA, High Schools in Local Counties





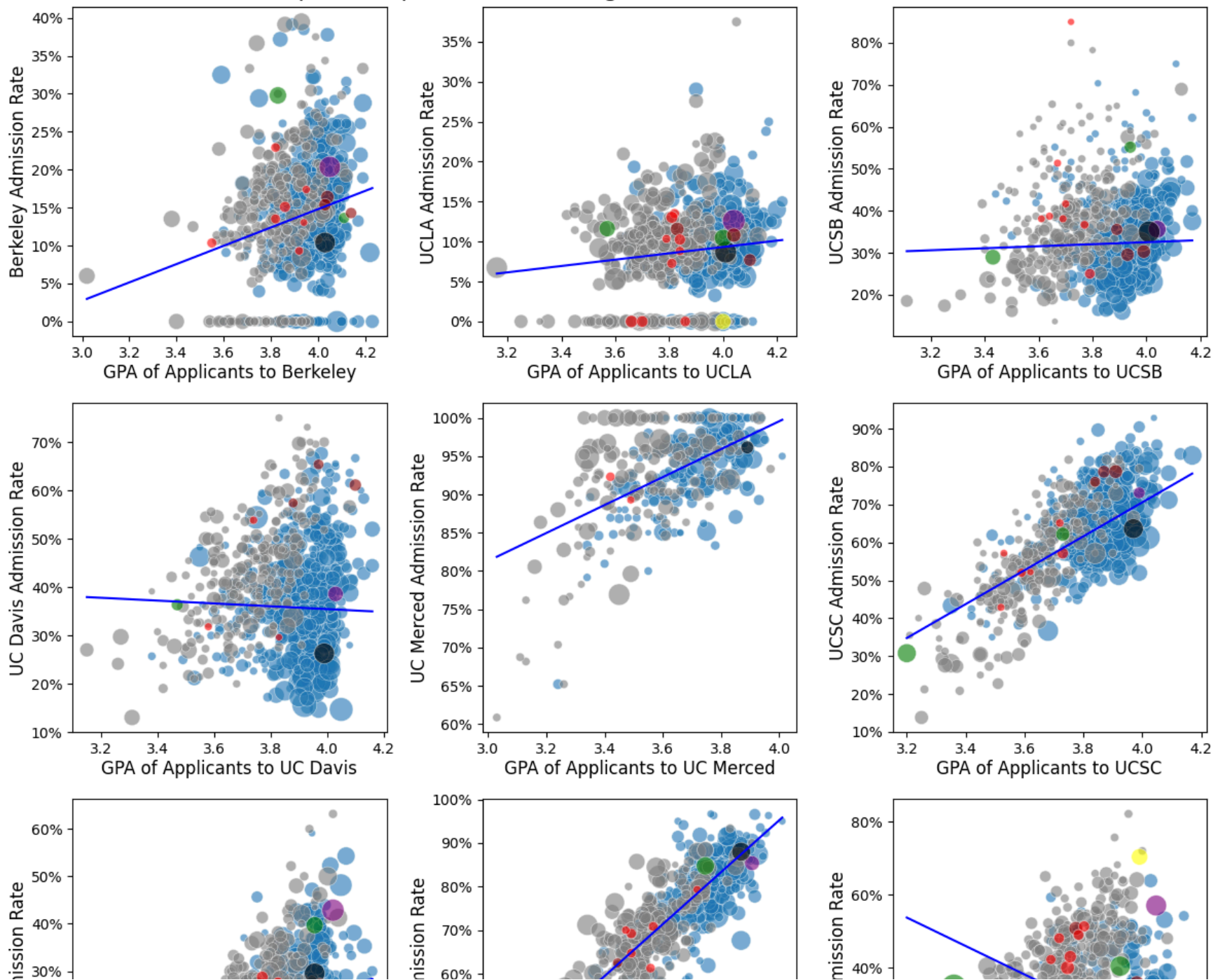
```
In [117... createPlots2(dfLeft, dfRight, "All", xAxis="App GPA", title="UC Campus Acceptance vs GPA, High Schools in All California Counties")
```

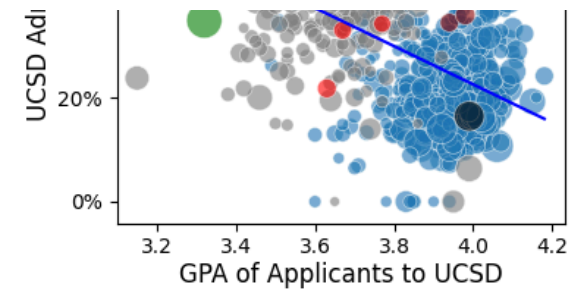
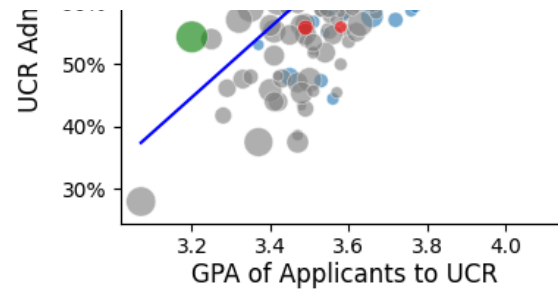
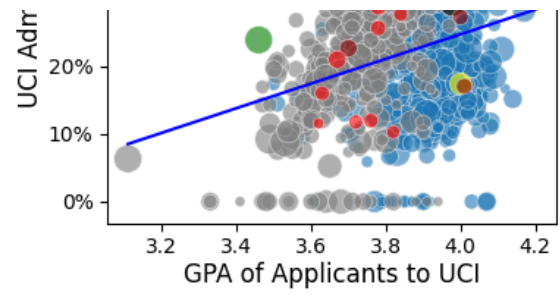


```
In [118... createPlots(dfLeft, dfRight, "All", xAxis="App GPA", title="UC Campus Acceptance vs GPA, High School

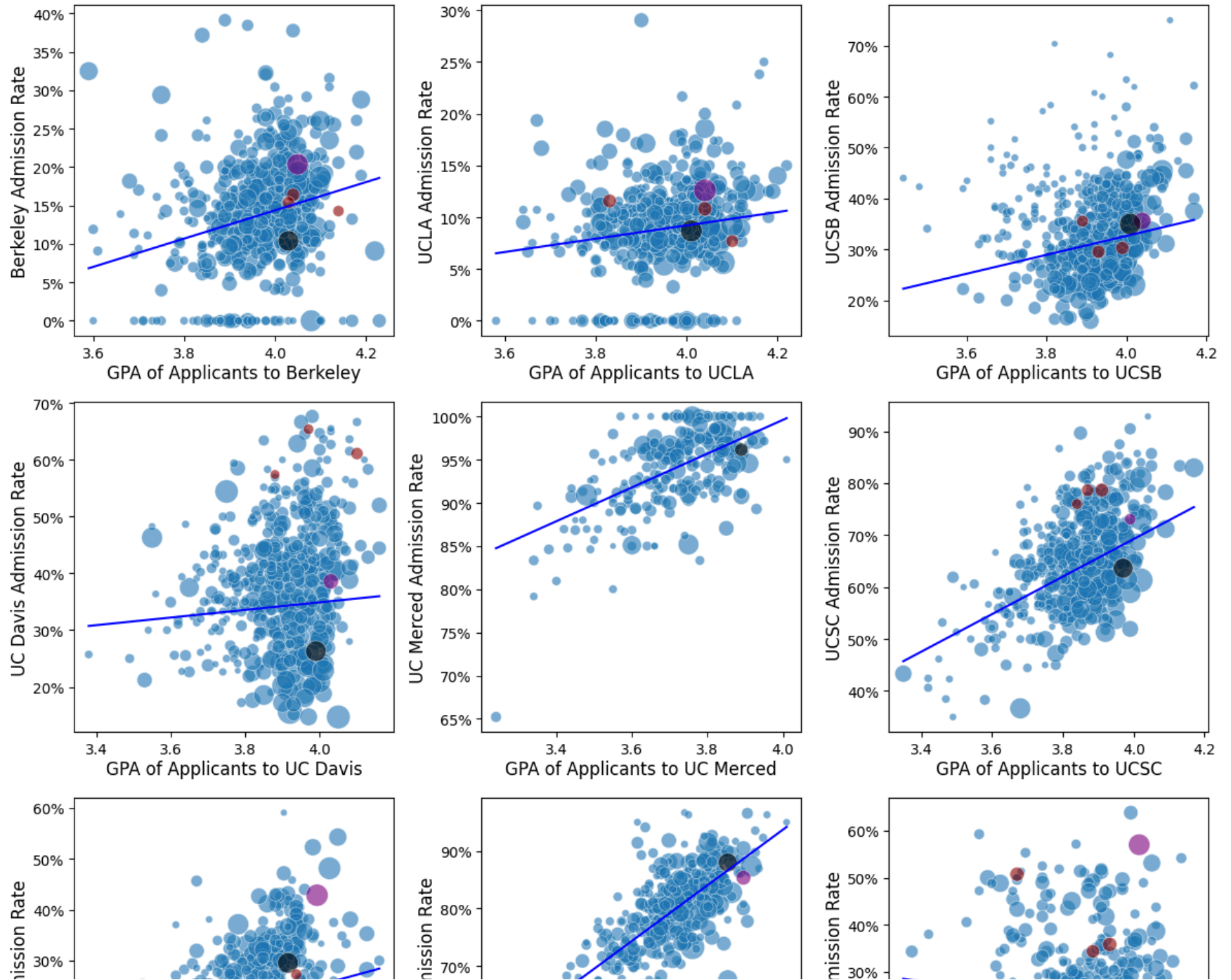
# Look separately at LCFF+ schools and non-LCFF+ schools
createPlots(dfLeft, dfRight, "All", xAxis="App GPA", filterMetrics=[["LCFF Plus School t", "=", "N"]], title="UC Campus Acceptance vs GPA, No LCFF+ H
createPlots(dfLeft, dfRight, "All", xAxis="App GPA", filterMetrics=[["LCFF Plus School t", "=", "Y"]], title="UC Campus Acceptance vs GPA, Only LCFF+
```

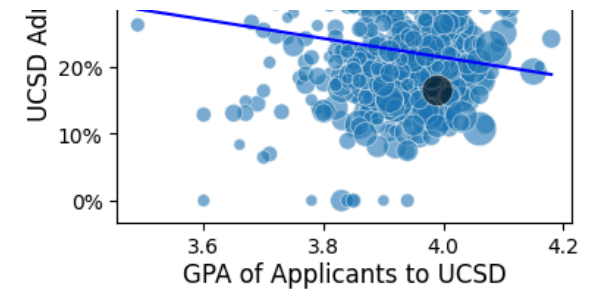
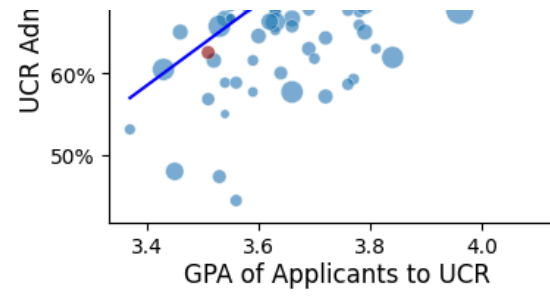
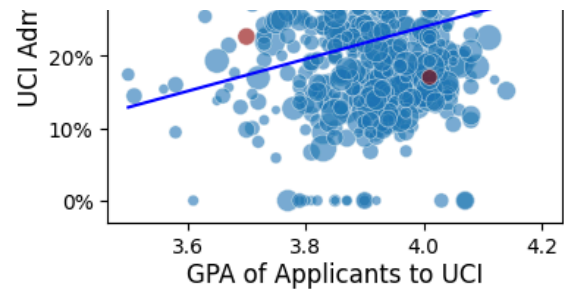
UC Campus Acceptance vs GPA, High Schools in All California Counties



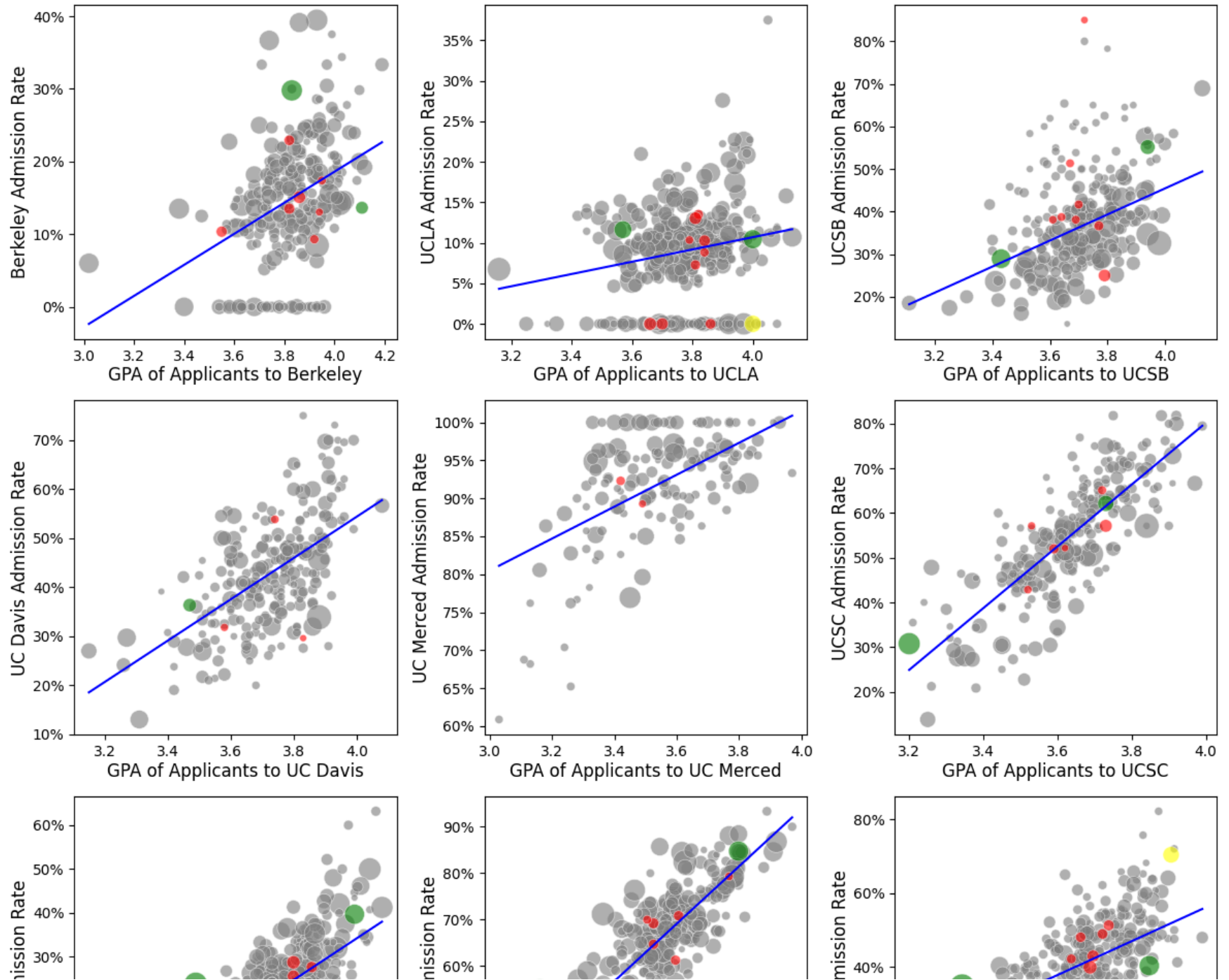


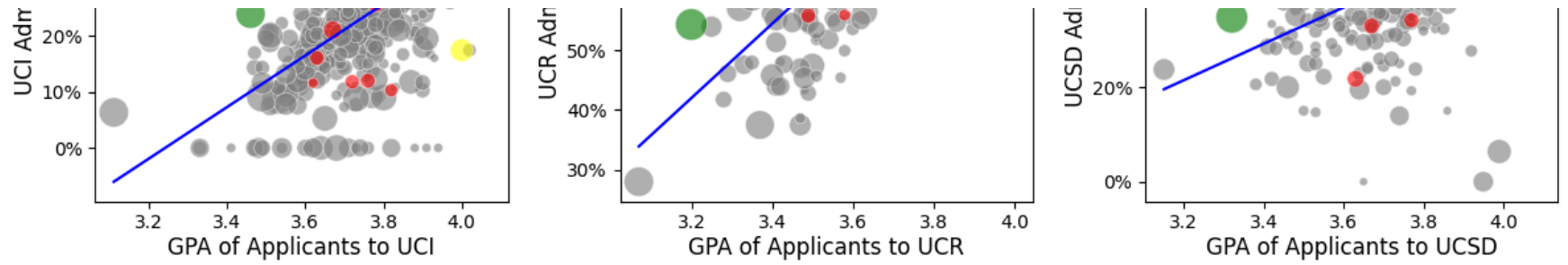
UC Campus Acceptance vs GPA, No LCFF+ High Schools





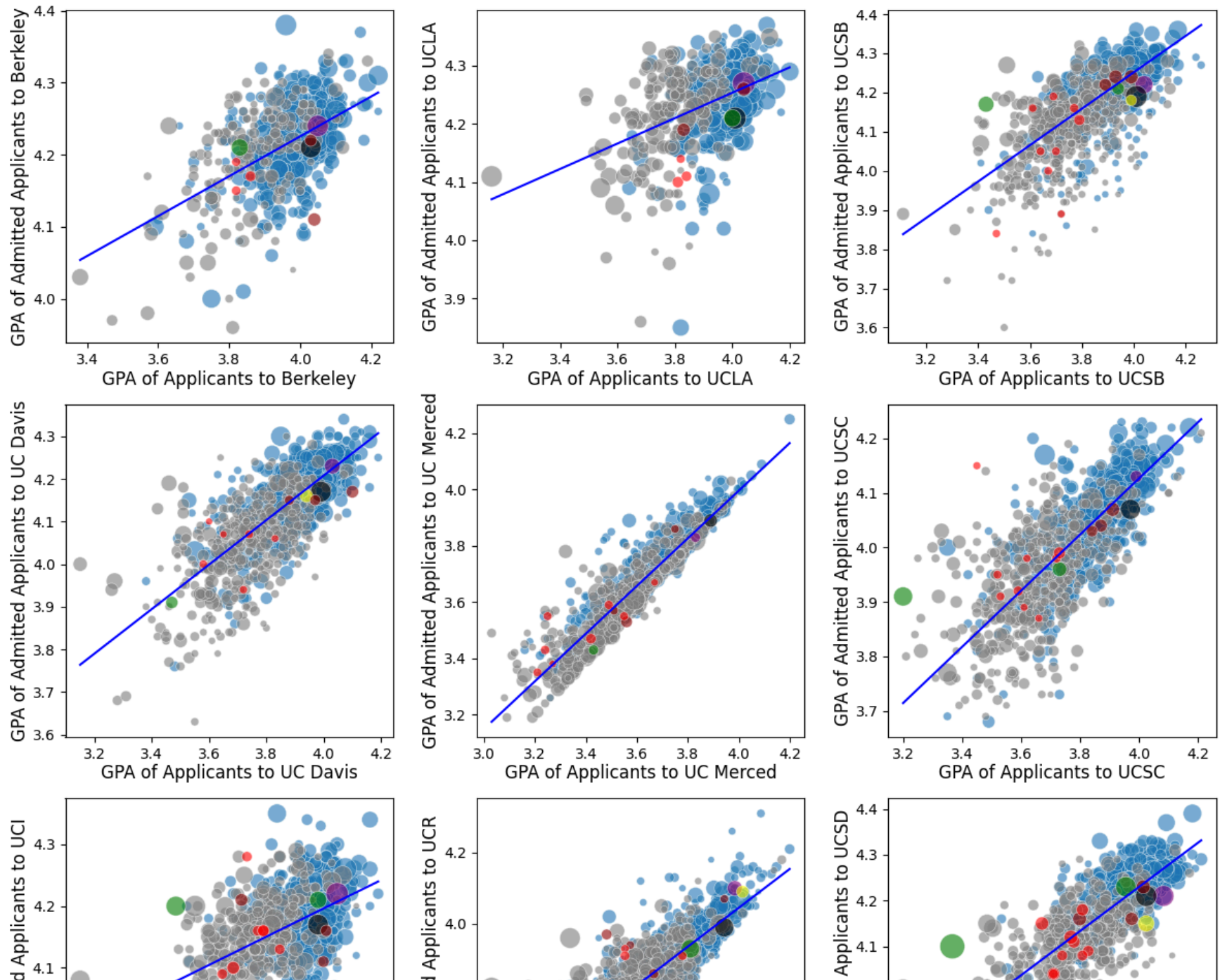
UC Campus Acceptance vs GPA, Only LCFF+ High Schools

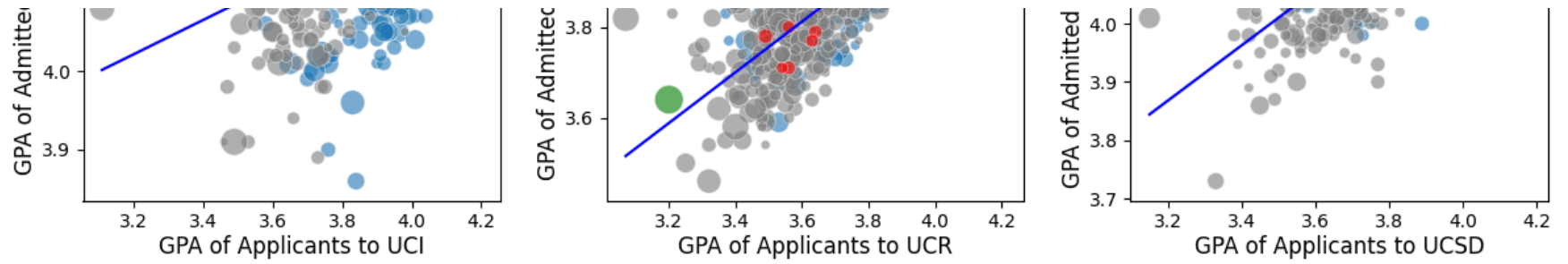




In [119... `createPlots(dfLeft, dfRight, "All", xAxis="App GPA", yAxis="Adm GPA", selectMinApps=0, title="GPA of Accepted Applicants vs All Applicants, High School`

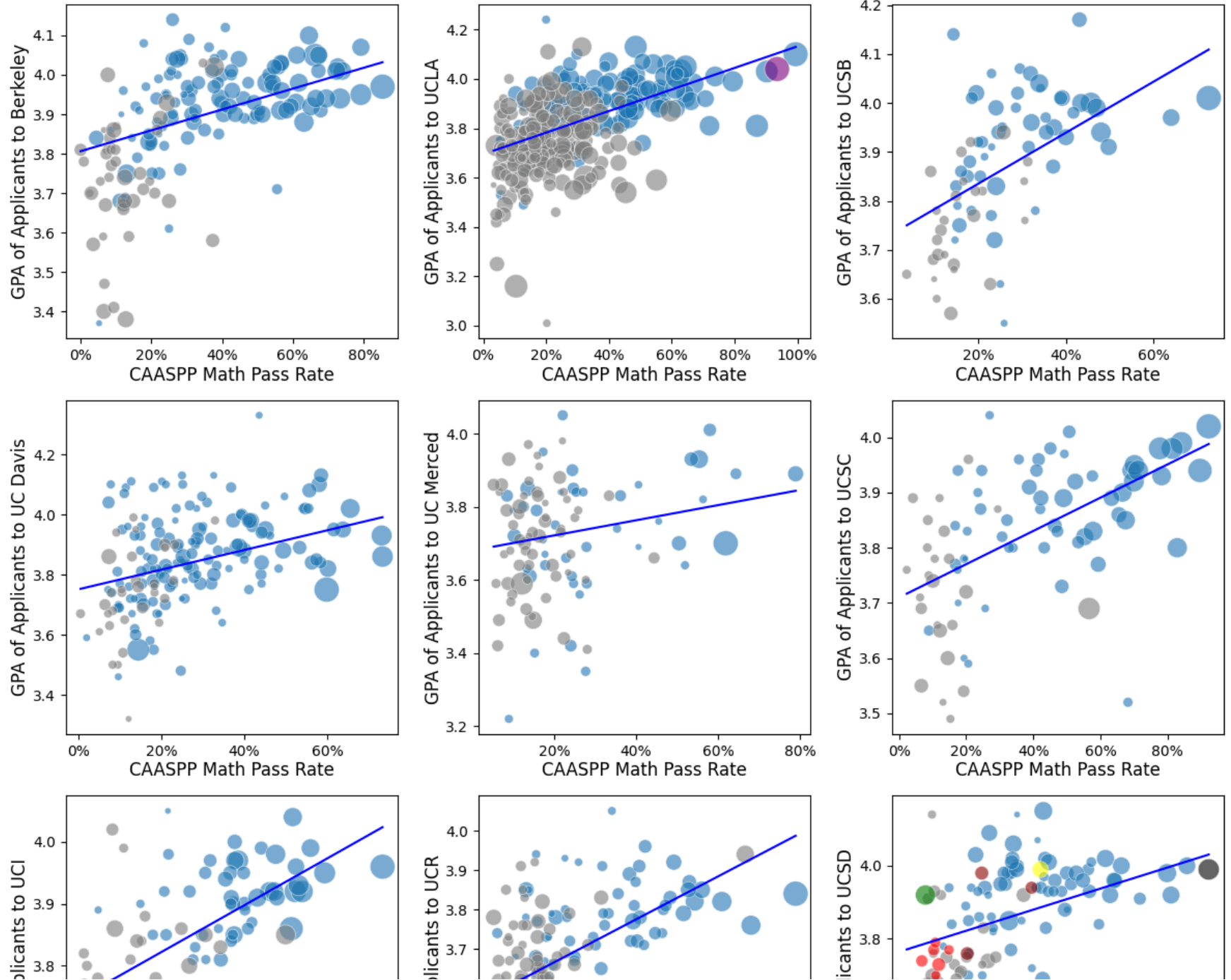
GPA of Accepted Applicants vs All Applicants, High Schools in All California Counties

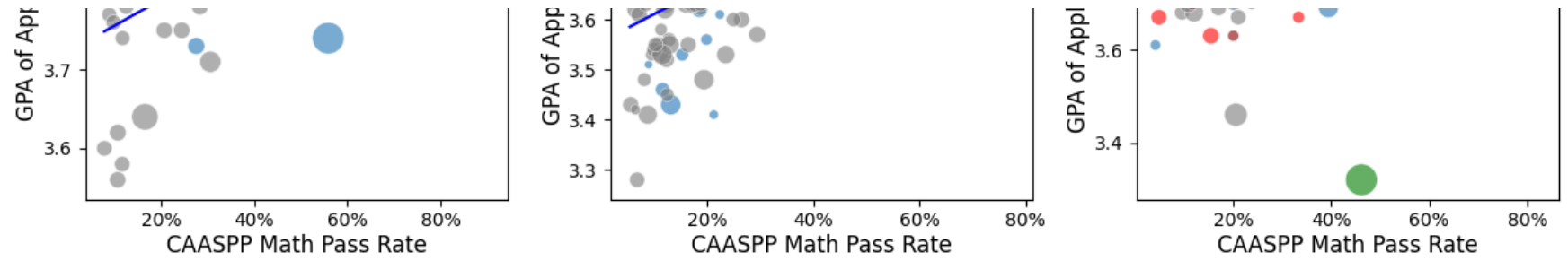




In [120... `createPlots(dfLeft, dfRight, "Local", yAxis="App GPA", selectMinApps=0, title="GPA vs CAASPP Math Pass Rates, High Schools in Local Counties")`

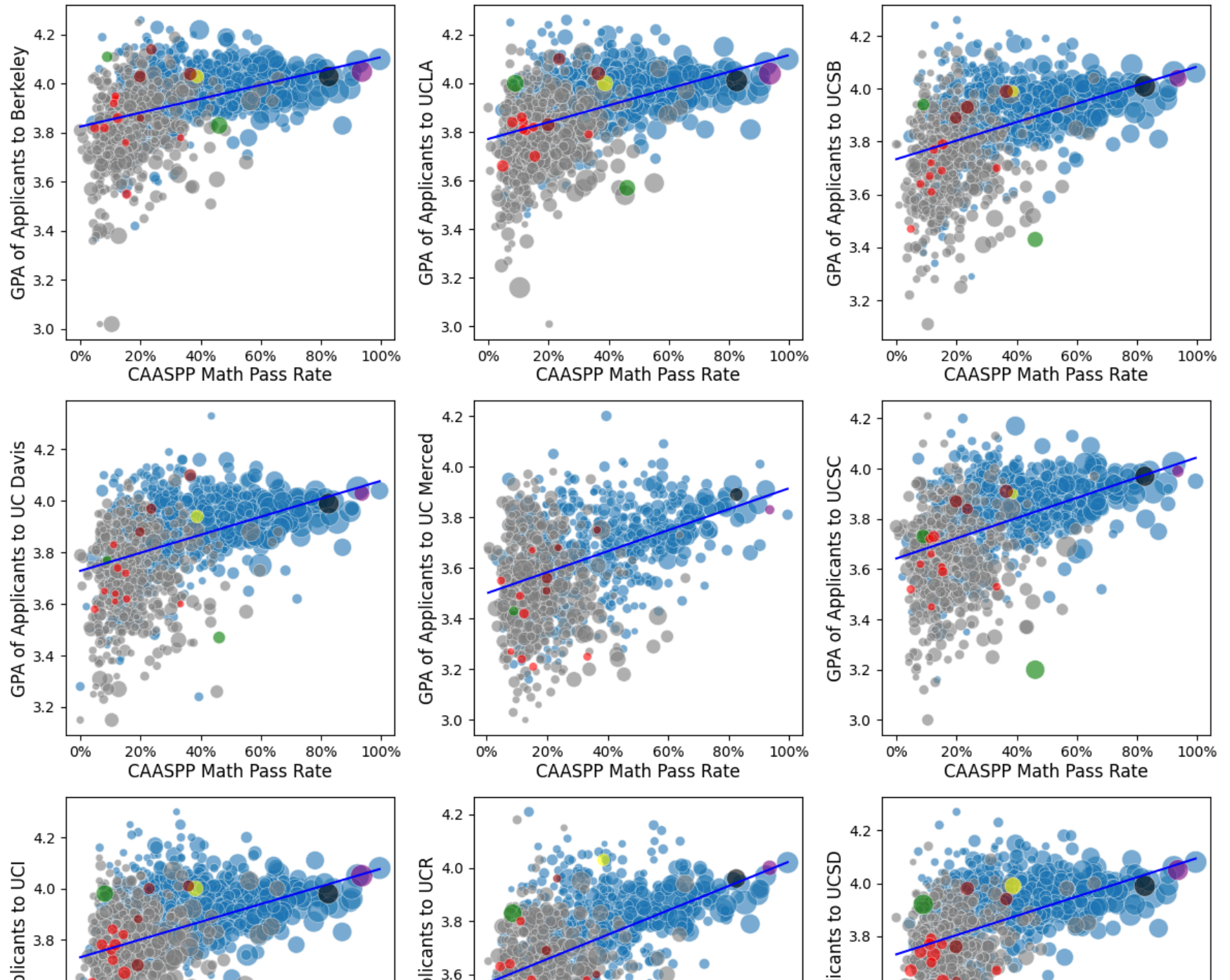
GPA vs CAASPP Math Pass Rates, High Schools in Local Counties

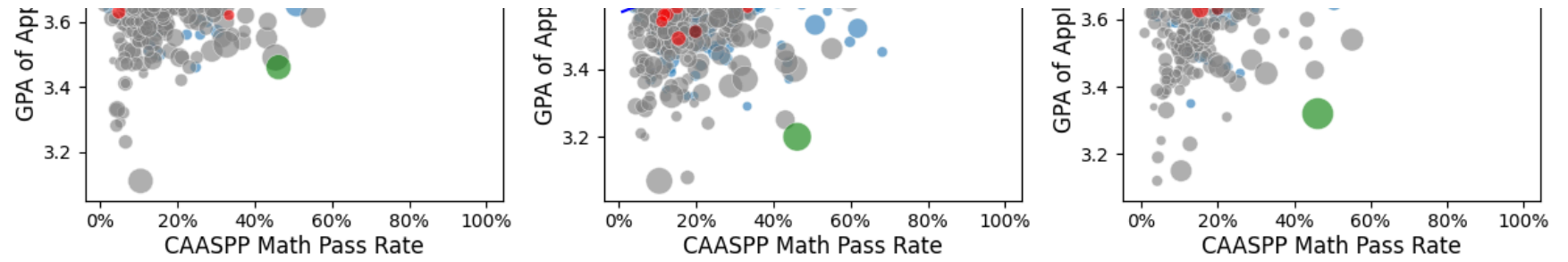




In [121... `createPlots(dfLeft, dfRight, "All", yAxis="App GPA", selectMinApps=0, title="GPA vs CAASPP Math Pass Rates, High Schools in All California Counties")`

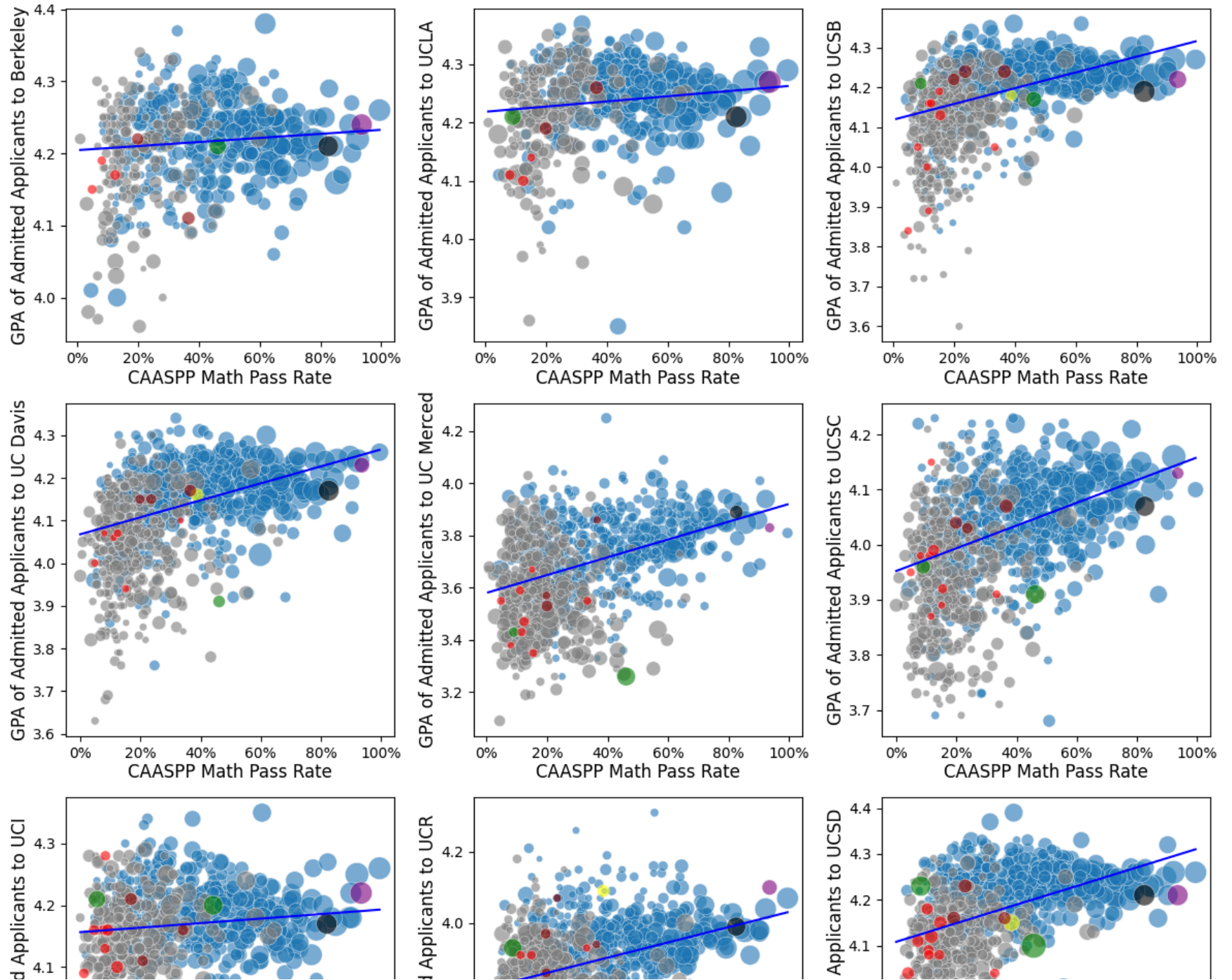
GPA vs CAASPP Math Pass Rates, High Schools in All California Counties

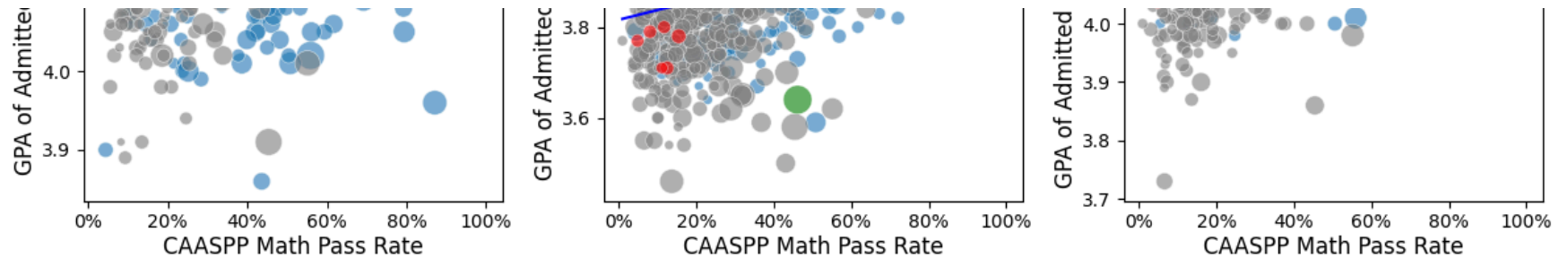




In [122... `createPlots(dfLeft, dfRight, "All", yAxis="Adm GPA", selectMinApps=0, title="GPA of Accepted Applicants vs CAASPP Math Pass Rates, High Schools in All`

GPA of Accepted Applicants vs CAASPP Math Pass Rates, High Schools in All California Counties





```
In [123... # Find high schools where UCSD accepts a greater number of students than pass the CAASPP
joined = pd.merge(dfLeft, dfRight, left_on="CAASPP_ID", right_on="School Code", how="inner")
joined = joined[(joined["Campus"] == "UCSD") & (joined["Test ID"] == "2")]
joined = joined[joined["Adm"].astype(int) > joined["Count Standard Met and Above"].astype(int)]
for index, row in joined.iterrows():
    nAdm = row["Adm"]
    nMet = int(row["Count Standard Met and Above"])
    ratio = float(nAdm / nMet)
    print(f'{row["School Name"]:50} {row["County"]:15} {nMet:2} {nAdm:2} {ratio:.1f} {row["AdmRate"]:.0%}')
```

Ambassador-Global Leadership	Los Angeles	4	7	1.8	47%
Animo Jackie Robinson High	Los Angeles	22	30	1.4	62%
Aspire Golden State College Preparatory Academy	Alameda	4	5	1.2	33%
Aspire Pacific Academy	Los Angeles	3	11	3.7	65%
Crawford High	San Diego	29	38	1.3	43%
Jordan High	Los Angeles	11	12	1.1	43%
Ednovate - East College Prep	Los Angeles	12	13	1.1	57%
East Los Angeles Renaissance Academy at Esteban E.	Los Angeles	7	8	1.1	73%
Fremont High	Alameda	2	6	3.0	38%
Gompers Preparatory Academy	San Diego	14	40	2.9	40%
Diego Rivera Learning Complex Green Design STEAM A	Los Angeles	8	14	1.8	61%
Hoover High	San Diego	40	46	1.1	43%
Humanitas Academy of Art and Technology at Esteban	Los Angeles	6	9	1.5	69%
King-Chavez Community High	San Diego	6	11	1.8	65%
Lincoln High	San Diego	16	26	1.6	33%
McClymonds High	Alameda	2	4	2.0	40%
MetWest High	Alameda	2	6	3.0	50%
Mira Monte High	Kern	14	19	1.4	45%
Rancho Dominguez Preparatory	Los Angeles	13	15	1.2	47%
Marshall (Thurgood) High	San Francisco	2	3	1.5	50%
University Preparatory Value High	Los Angeles	17	19	1.1	50%
West Shores High	Riverside	4	6	1.5	40%
Williams Junior/Senior High	Colusa	6	7	1.2	64%

```
In [124... # Explore: Local schools with UCSD admission rate >= 0.5
joined = pd.merge(dfLeft, dfRight, left_on="CAASPP_ID", right_on="School Code", how="inner")
joined = joined[(joined["Campus"] == "UCSD") & (joined["countyCampus"] == "UCSD") & (joined["Test ID"] == "2")]
joined["SeniorAdmRate"] = joined["Adm"].astype(int) / joined["Total Students Enrolled"].astype(int) # Fraction of seniors admitted
print(joined[joined["AdmRate"] >= 0.5])
```

UC_ID	School	City	County	Campus	county	Campus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr	GPA	AdmRate
EnrRate	County	Code	District	Code	District	Name	School	Code			School	Name	Type	ID	Test	ID	\
16450	50430		CALIPATRIA HIGH SCHOOL		CALIPATRIA	Imperial	UCSD		UCSD	8	5	0	1332501	4.14	4.26	0.00	0.625000
0.000000		13	63107		Calipatria Unified		1332501						Calipatria High		7	2	
16520	50810		CENTRAL UNION HIGH SCHOOL		EL CENTRO	Imperial	UCSD		UCSD	46	26	11	1333004	3.73	4.04	3.90	0.565217
0.423077		13	63115		Central Union High		1333004						Central Union High		7	2	
16562	50580		CHULA VISTA SENIOR HIGH SCHOOL		CHULA VISTA	San Diego	UCSD		UCSD	76	39	23	3731064	3.79	4.09	4.10	0.513158
0.589744		37	68411		Sweetwater Union High		3731064						Chula Vista Senior High		7	2	
16766	50579		EASTLAKE HIGH SCHOOL		CHULA VISTA	San Diego	UCSD		UCSD	203	104	60	3730843	3.96	4.22	4.18	0.512315
0.576923		37	68411		Sweetwater Union High		3730843						Eastlake High		7	2	
16842	50880		ESCONDIDO HIGH SCHOOL		ESCONDIDO	San Diego	UCSD		UCSD	88	44	22	3732062	3.67	4.12	4.12	0.500000
0.500000		37	68106		Escondido Union High		3732062						Escondido High		7	2	
17040	51350		HELIX HIGH SCHOOL		LA MESA	San Diego	UCSD		UCSD	142	73	42	3732732	3.91	4.19	4.16	0.514085
0.575342		37	68130		Helix High		3732732						Helix High		9	2	
17220	53900		KEARNY SCH BIOMED SCIENCE TECH		SAN DIEGO	San Diego	UCSD		UCSD	27	19	8	0107086	3.99	4.15	4.14	0.703704
0.421053		37	68338		San Diego Unified		0107086		Kearny School of Biomedical Science and Technology						7	2	
17226	54507		KING CHAVEZ COMMUNITY HIGH SCH		SAN DIEGO	San Diego	UCSD		UCSD	17	11	6	0118851	3.68	4.00	3.97	0.647059
0.545455		37	68338		King-Chavez Community High		0118851						King-Chavez Community High		9	2	
17683	54629		OFARRELL CHARTER SCHOOL		SAN DIEGO	San Diego	UCSD		UCSD	50	25	9	6061964	3.69	4.17	4.23	0.500000
0.360000		37	68338		The O'Farrell Charter		6061964						The O'Farrell Charter		9	2	
17972	53903		SAN DIEGO HIGH SCHOOL		SAN DIEGO	San Diego	UCSD		UCSD	132	67	29	0107219	3.76	4.16	4.17	0.507576
0.432836		37	68338		San Diego Unified		0107219						San Diego High		7	2	

Total Students Enrolled	Total Students Tested	Total Students Tested with Scores	Mean Scale Score	Count	Standard Exceeded	Count	Standard Met	Count							
Standard Met and Above	Count	Standard Nearly Met	Count	Standard Not Met	PassRate	CAASPP	County	Name	CAASPP	County	Campus	LCFF	Plus	School	t \
16450	81	75			75			2525.4				0			8
8	26	41	0.106667		Imperial			UCSD			Y				
16520	473	467			467			2539.7				24			87
111	120	236	0.237687		Imperial			UCSD			Y				
16562	496	476			476			2497.4				11			44
55	107	314	0.115546		San Diego			UCSD			Y				
16766	689	674			674			2601.4				108			179
287	190	197	0.425816		San Diego			UCSD			N				
16842	422	418			414			2529.7				34			53
87	90	237	0.210145		San Diego			UCSD			Y				
17040	609	587			583			2666.5				190			187
377	137	69	0.646655		San Diego			UCSD			N				
17220	67	67			67			2578.1				8			18
26	13	28	0.388060		San Diego			UCSD			Y				
17226	52	50			50			2508.7				3			3
6	13	31	0.120000		San Diego			UCSD			Y				
17683	140	137			137			2585.9				13			41
54	37	46	0.394161		San Diego			UCSD			N				
17972	491	361			358			2517.5				18			53
71	71	216	0.198324		San Diego			UCSD			N				

	Category	LA Area	SeniorAdmRate
16450	LCFF+	N	0.061728
16520	LCFF+	N	0.054968
16562	TRiOy	N	0.078629
16766		N	0.150943
16842	LCFF+	N	0.104265

17040		N	0.119869
17220	KEARNY	N	0.283582
17226	LCFF+	N	0.211538
17683		N	0.178571
17972	TRiOn	N	0.136456

```
In [125... print(joined[joined["PassRate"] >= 0.75])
```

UC_ID	School	City	County	Campus	county	Campus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enrl	GPA	AdmRate	EnrRate
County Code	District Code	District Name	School Code	School Name	Type	ID	Test ID	Total	Students	Enrolled	Total	Students	Tested	\				
16460	50893	CANYON CREST ACADEMY	SAN DIEGO	San Diego	UCSD	UCSD	384	63	14	0106328	3.99	4.21	4.14	0.164062	0.222222			
37	68346	San Dieguito Union High	0106328	Canyon Crest Academy	7	2		553		544								
16688	54351	DEL NORTE HIGH SCHOOL	SAN DIEGO	San Diego	UCSD	UCSD	296	48	10	0118935	4.00	4.25	4.16	0.162162	0.208333			
37	68296	Poway Unified	0118935	Del Norte High	7	2		637		593								

Total Students Tested with Scores Mean Scale Score Count Standard Exceeded Count Standard Met Count Standard Met and Above Count Standard Nearly	Met Count Standard Not Met	PassRate	CAASPP	County Name	CAASPP	County Campus	LCFF Plus	School t	Category	LA Area	SeniorAdmRate
16460		544	2748.3		346		103			449	
63	32	0.825368	San Diego	UCSD	N	CCA	N		0.113924		
16688		593	2711.9		310		146		456		
69	68	0.768971	San Diego	UCSD	N		N		0.075353		

```
In [126... print(joined[joined["SeniorAdmRate"] >= 0.2])
```

UC_ID	School	City	County	Campus	county	Campus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enrl	GPA	AdmRate
EnrRate	County Code	District Code	District Name	School Code	School Name	Type	ID	Test ID	\								
16974	54353	GOMPERS PREPARATORY ACADEMY	SAN DIEGO	San Diego	UCSD	UCSD	99	40	21	0119610	3.92	4.23	4.20	0.404040			
0.525000	37	68338	Gompers Preparatory Academy	0119610						Gompers Preparatory Academy	9	2					
17220	53900	KEARNY SCH BIOMED SCIENCE TECH	SAN DIEGO	San Diego	UCSD	UCSD	27	19	8	0107086	3.99	4.15	4.14	0.703704			
0.421053	37	68338	San Diego Unified	0107086	Kearny School of Biomedical Science and Technology	7	2										
17226	54507	KING CHAVEZ COMMUNITY HIGH SCH	SAN DIEGO	San Diego	UCSD	UCSD	17	11	6	0118851	3.68	4.00	3.97	0.647059			
0.545455	37	68338	King-Chavez Community High	0118851						King-Chavez Community High	9	2					
17821	51479	PREUSS SCHOOL UCSD	LA JOLLA	San Diego	UCSD	UCSD	115	40	26	3731189	3.32	4.10	4.03	0.347826			
0.650000	37	68338	Preuss School UCSD	3731189						Preuss School UCSD	9	2					

Total Students Enrolled Total Students Tested Total Students Tested with Scores Mean Scale Score Count Standard Exceeded Count Standard Met Count	Standard Met and Above Count Standard Nearly	Met Count Standard Not Met	PassRate	CAASPP	County Name	CAASPP	County Campus	LCFF Plus	School t	\
16974	157	156		156		2502.8		3		11
14	41	101	0.089744	San Diego	UCSD		Y			
17220	67	67		67		2578.1		8		18
26	13	28	0.388060	San Diego	UCSD		Y			
17226	52	50		50		2508.7		3		3
6	13	31	0.120000	San Diego	UCSD		Y			
17821	117	117		117		2617.7		26		28
54	29	34	0.461538	San Diego	UCSD		Y			

Category	LA Area	SeniorAdmRate	
16974	Partner	N	0.254777
17220	KEARNY	N	0.283582
17226	LCFF+	N	0.211538
17821	Partner	N	0.341880

```
In [127... print(joined[joined["AdmRate"] >= 0.5])
```

UC_ID	School	City	County	Campus	county	Campus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr	GPA	AdmRate
EnrRate	County	Code	District	Code	District	Name	School	Code			School	Name	Type	ID	Test	ID	\
16450	50430		CALIPATRIA HIGH SCHOOL		CALIPATRIA	Imperial	UCSD		UCSD	8	5	0	1332501	4.14	4.26	0.00	0.625000
0.000000		13	63107		Calipatria Unified		1332501					Calipatria High		7	2		
16520	50810		CENTRAL UNION HIGH SCHOOL		EL CENTRO	Imperial	UCSD		UCSD	46	26	11	1333004	3.73	4.04	3.90	0.565217
0.423077		13	63115		Central Union High		1333004					Central Union High		7	2		
16562	50580		CHULA VISTA SENIOR HIGH SCHOOL		CHULA VISTA	San Diego	UCSD		UCSD	76	39	23	3731064	3.79	4.09	4.10	0.513158
0.589744		37	68411		Sweetwater Union High		3731064					Chula Vista Senior High		7	2		
16766	50579		EASTLAKE HIGH SCHOOL		CHULA VISTA	San Diego	UCSD		UCSD	203	104	60	3730843	3.96	4.22	4.18	0.512315
0.576923		37	68411		Sweetwater Union High		3730843					Eastlake High		7	2		
16842	50880		ESCONDIDO HIGH SCHOOL		ESCONDIDO	San Diego	UCSD		UCSD	88	44	22	3732062	3.67	4.12	4.12	0.500000
0.500000		37	68106		Escondido Union High		3732062					Escondido High		7	2		
17040	51350		HELIX HIGH SCHOOL		LA MESA	San Diego	UCSD		UCSD	142	73	42	3732732	3.91	4.19	4.16	0.514085
0.575342		37	68130		Helix High		3732732					Helix High		9	2		
17220	53900		KEARNY SCH BIOMED SCIENCE TECH		SAN DIEGO	San Diego	UCSD		UCSD	27	19	8	0107086	3.99	4.15	4.14	0.703704
0.421053		37	68338		San Diego Unified		0107086		Kearny School of Biomedical Science and Technology					7	2		
17226	54507		KING CHAVEZ COMMUNITY HIGH SCH		SAN DIEGO	San Diego	UCSD		UCSD	17	11	6	0118851	3.68	4.00	3.97	0.647059
0.545455		37	68338		King-Chavez Community High		0118851					King-Chavez Community High		9	2		
17683	54629		OFARRELL CHARTER SCHOOL		SAN DIEGO	San Diego	UCSD		UCSD	50	25	9	6061964	3.69	4.17	4.23	0.500000
0.360000		37	68338		The O'Farrell Charter		6061964					The O'Farrell Charter		9	2		
17972	53903		SAN DIEGO HIGH SCHOOL		SAN DIEGO	San Diego	UCSD		UCSD	132	67	29	0107219	3.76	4.16	4.17	0.507576
0.432836		37	68338		San Diego Unified		0107219					San Diego High		7	2		

Total Students Enrolled	Total Students Tested	Total Students Tested with Scores	Mean Scale Score	Count	Standard Exceeded	Count	Standard Met	Count							
Standard Met and Above	Count	Standard Nearly Met	Count	Standard Not Met	PassRate	CAASPP	County	Name	CAASPP	County	Campus	LCFF	Plus	School	t \
16450	81	75			75			2525.4				0			8
8	26	41	0.106667		Imperial			UCSD			Y				
16520	473	467			467			2539.7				24			87
111	120	236	0.237687		Imperial			UCSD			Y				
16562	496	476			476			2497.4				11			44
55	107	314	0.115546		San Diego			UCSD			Y				
16766	689	674			674			2601.4				108			179
287	190	197	0.425816		San Diego			UCSD			N				
16842	422	418			414			2529.7				34			53
87	90	237	0.210145		San Diego			UCSD			Y				
17040	609	587			583			2666.5				190			187
377	137	69	0.646655		San Diego			UCSD			N				
17220	67	67			67			2578.1				8			18
26	13	28	0.388060		San Diego			UCSD			Y				
17226	52	50			50			2508.7				3			3
6	13	31	0.120000		San Diego			UCSD			Y				
17683	140	137			137			2585.9				13			41
54	37	46	0.394161		San Diego			UCSD			N				
17972	491	361			358			2517.5				18			53
71	71	216	0.198324		San Diego			UCSD			N				

	Category	LA Area	SeniorAdmRate
16450	LCFF+	N	0.061728
16520	LCFF+	N	0.054968
16562	TRiOy	N	0.078629
16766		N	0.150943
16842	LCFF+	N	0.104265

17040		N	0.119869
17220	KEARNY	N	0.283582
17226	LCFF+	N	0.211538
17683		N	0.178571
17972	TRiOn	N	0.136456

```
In [128... print(joined[joined["PassRate"] >= 0.7])
```

UC_ID	School	City	County	Campus	county	Campus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enrl	GPA	AdmRate	EnrRate	
County	Code	District	Code	District	Name	School	Code	School	Name	Type	ID	Test	ID	Total	Students	Enrolled	Total	Students	Tested \
16460	50893	CANYON CREST ACADEMY	SAN DIEGO	San Diego	UCSD	UCSD	384	63	14	0106328	3.99	4.21	4.14	0.164062	0.222222				
37	68346	San Dieguito Union High	0106328	Canyon Crest Academy	7	2					553				544				
16688	54351	DEL NORTE HIGH SCHOOL	SAN DIEGO	San Diego	UCSD	UCSD	296	48	10	0118935	4.00	4.25	4.16	0.162162	0.208333				
37	68296	Poway Unified	0118935	Del Norte High	7	2					637				593				
18084	52834	SCRIPPS RANCH HIGH	SAN DIEGO	San Diego	UCSD	UCSD	219	38	17	3730884	3.92	4.24	4.22	0.173516	0.447368				
37	68338	San Diego Unified	3730884	Scripps Ranch High	7	2					462				455				
18430	52986	WESTVIEW HIGH SCHOOL	SAN DIEGO	San Diego	UCSD	UCSD	293	48	14	3731528	3.98	4.23	4.20	0.163823	0.291667				
37	68296	Poway Unified	3731528	Westview High	7	2					532				495				

Total	Students	Tested	with	Scores	Mean	Scale	Score	Count	Standard	Exceeded	Count	Standard	Met	Count	Standard	Met	and	Above	Count	Standard	Nearly
Met	Count	Standard	Not	Met	PassRate	CAASPP	County	Name	CAASPP	County	Campus	LCFF	Plus	School	t	Category	LA	Area	Senior	AdmRate	
16460						544		2748.3			346				103					449	
63			32	0.825368			San Diego				UCSD			N	CCA		N		0.113924		
16688						593		2711.9			310				146					456	
69			68	0.768971			San Diego				UCSD			N			N		0.075353		
18084						455		2692.1			189				142					331	
73			51	0.727473			San Diego				UCSD			N			N		0.082251		
18430						495		2702.1			246				109					355	
86			54	0.717172			San Diego				UCSD			N			N		0.090226		

```
In [129... ##### Explore impact of ethnic composition on admission rate
result = (
    dfUC.groupby("uad_uc_ethn_7_cat", dropna=False)["Pivot Field Values"].sum().reset_index()
)
print(result)
```

uad_uc_ethn_7_cat	Pivot Field Values
0 African American	36864
1 All	987228
2 American Indian	407
3 Asian	342631
4 Domestic Unknown	21456
5 Hispanic/ Latinx	350460
6 Int'l	5511
7 Pacific Islander	39
8 White	151107

```
In [130... # Add empty columns for App, Adm, and Enr to dfLeft for the following ethnic groups with higher counts:
ethnicities = {
    "African American": "AfrAm",
    "Asian": "Asian",
    "Hispanic/ Latinx": "Latinx",
    "White": "White",
```

```

}
for ethnicity in ethnicities.values():
    for countType in ("App", "Adm", "Enr"):
        key = " ".join([ethnicity, countType])
        dfLeft[key] = 0

```

```
In [131... dfExplore = dfUC[dfUC["Calculation1"].str.contains(r"53575", na=False)]
```

```
In [132... print(dfExplore)
```

alues	Campus	Calculation1	School	City	County/State/ Territory	Count	uad_uc_ethn_7_cat	Pivot Field V
12528	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Enr	Hispanic/ Latinx	
3	all							
12529	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	Hispanic/ Latinx	
6	all							
12530	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	Hispanic/ Latinx	
7	all							
12531	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Enr	All	
4	all							
12532	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	All	
8	all							
12533	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	All	
11	all							
35179	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	All	
5	UCI							
42137	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	All	
6	UCLA							
48127	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	Hispanic/ Latinx	
3	UC Merced							
48128	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	All	
3	UC Merced							
54960	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	Hispanic/ Latinx	
3	UCR							
54961	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	All	
3	UCR							
54962	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	All	
6	UCR							
62900	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	Hispanic/ Latinx	
5	UCSD							
62901	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	App	All	
7	UCSD							
70647	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Enr	Hispanic/ Latinx	
3	UCSB							
70648	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	Hispanic/ Latinx	
3	UCSB							
70649	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Enr	All	
3	UCSB							
70650	TWENTYNINE PALMS HIGH SCHOOL	53575	TWENTYNINE PALMS HIGH SCHOOL	TWENTYNINE PALMS	San Bernardino	Adm	All	
3	UCSB							

```
In [133... print(dfLeft[dfLeft["School"] == "AZUSA HIGH SCHOOL"])
```

	UC_ID	School	City	County	Campus	county	Campus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr	GPA	AdmRate	EnrRate	Af			
rAm	App	AfrAm	Adm	AfrAm	Enr	Asian	App	Asian	Adm	Asian	Enr	Latinx	App	Latinx	Adm	Latinx	Enr	White	App	White	Adm	White	Enr
121	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	all		UCLA	54	45	19	1930528	3.86	3.96	4.02	0.833333	0.422222				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
1521	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	Berkeley		UCLA	27	0	0	1930528	3.95	0.00	0.00	0.000000		NaN			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
2768	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UC Davis		UCLA	17	12	0	1930528	3.74	3.97	0.00	0.705882	0.000000				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
4015	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UCI		UCLA	43	17	7	1930528	3.89	4.12	4.16	0.395349	0.411765				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
5266	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UCLA		UCLA	46	3	0	1930528	3.87	0.00	0.00	0.065217	0.000000				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
7672	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UCR		UCLA	31	24	0	1930528	3.83	3.96	0.00	0.774194	0.000000				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
8796	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UCSD		UCLA	28	23	8	1930528	3.95	4.04	3.97	0.821429	0.347826				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
10045	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UCSB		UCLA	15	8	0	1930528	3.79	4.13	0.00	0.533333	0.000000				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
11262	50188	AZUSA	HIGH	SCHOOL	AZUSA	Los	Angeles	UCSC		UCLA	11	7	0	1930528	3.75	4.05	0.00	0.636364	0.000000				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

```
In [134... # Read dfUC and populate the ethnic count columns
for index, row in dfUC.iterrows():
    schoolID = row["Calculation1"]
    ucID = re.sub(r"^\.?(\d{5})$", r"\1", schoolID) # Five digit code, and the first digit is always 5. Already checked that this always succeeds above
    if ucID not in ucID2TestID:
        continue # Don't have CAASPP data for this school
    testID = ucID2TestID[ucID]
    ethn = row["uad_uc_ethn_7_cat"]
    if ethn not in ethnicities:
        continue
    ethnicity = ethnicities[ethn]
    countType = row["Count"]
    key = " ".join([ethnicity, countType])

    school = row["School"]
    city = row["City"]
    county = row["County/State/ Territory"]
    countVal = row["Pivot Field Values"]
    if countVal == "0":
        continue
    campus = row["Campus"]

    matchingIndex = dfLeft.query(f'UC_ID == "{ucID}" and Campus == "{campus}"').index # Find the dfLeft record to modify
    if len(matchingIndex) == 0:
        # dfLeft record does not exist due to UC censoring.
        # If fewer than 5 students applied, nApp = 0 in the UC data.
        # If fewer than 3 students were admitted, nAdm = 0 in the UC data.
        # It happens that nApp = 0 and nAdm = 3 or 4.
        # Examples: "NEEDLES HIGH SCHOOL52110" for "ALL", "AZUSA HIGH SCHOOL50188" for UC Merced, "TWENTYNINE PALMS HIGH SCHOOL53575" for UCSB.
        if (countType not in ("Adm", "Enr")) or (countVal not in (3, 4)):
            print("NOT FOUND", ucID, school, campus, ethnicity, countType, countVal)
        continue
```

```
iLeftRow = matchingIndex[0]
if dfLeft.at[iLeftRow, key] != 0:
    print("DUPLICATE:", ucID, school, campus, ethnicity, countType, countVal, dfLeft.at[iLeftRow, key])
dfLeft.at[iLeftRow, key] += countVal
```

```
In [135... # Compute the admission rate and enrolment rate
dfLeft["AfrAm App Frac"] = dfLeft["AfrAm App"] / dfLeft["App"]
dfLeft["Asian App Frac"] = dfLeft["Asian App"] / dfLeft["App"]
dfLeft["Latinx App Frac"] = dfLeft["Latinx App"] / dfLeft["App"]
dfLeft["White App Frac"] = dfLeft["White App"] / dfLeft["App"]
```

```
In [136... # Compute IsLocal column indicating whether the high school is local to the UC campus (1) or not (0).
dfLeft["IsLocal"] = (dfLeft["countyCampus"] == dfLeft["Campus"]).astype(int)
```

```
In [137... print(dfLeft[dfLeft["IsLocal"] == 1])
```

UC_ID		School			City		County		Campus		countyCampus	App	Adm	Enr	CAASPP_ID	App GPA	Adm GPA	Enrl GPA
AdmRate	EnrRate	AfrAm App	AfrAm Adm	AfrAm Enr	Asian App	Asian Adm	Asian Enr	Latinx App	Latinx Adm	Latinx Enr	White App	White Adm	White Enr	CAASPP_ID	App GPA	Adm GPA	Enrl GPA	
1416	52910	ABRAHAM LINCOLN HIGH SCHOOL	SAN FRANCISCO	San Francisco	Berkeley	Berkeley	190	27	21	3833241	3.90	4.20	4.19					
0.142105	0.777778	0	0	0	154	24	19	22	3	0	6	0						
1425	54439	ACADEMY-SAN FRAN @ MCATEER	SAN FRANCISCO	San Francisco	Berkeley	Berkeley	12	3	0	0119958	3.85	0.00	0.00					
0.250000	0.000000	0	0	0	0	0	0	6	0	0	0	0						
1426	51315	ACALANES HIGH SCHOOL	LAFAYETTE	Contra Costa	Berkeley	Berkeley	96	18	11	0730283	4.01	4.09	4.01					
0.187500	0.611111	0	3	0	26	3	0	9	0	0	53	10						
1433	54176	AIMS COLLEGE PREP HIGH SCHOOL	OAKLAND	Alameda	Berkeley	Berkeley	49	7	6	0111856	4.02	4.26	4.26					
0.142857	0.857143	16	3	0	25	4	4	7	0	0	0	0						
1435	50011	ALAMEDA COMMUNITY LEARNING CTR	ALAMEDA	Alameda	Berkeley	Berkeley	13	3	0	0130609	4.14	0.00	0.00					
0.230769	0.000000	0	0	0	7	0	0	0	0	0	0	0						
...	...	...	...	...	...	...	...	...	...	...	...	...	...					
12345	53107	W C OVERFELT HIGH SCHOOL	SAN JOSE	Santa Clara	UCSC	UCSC	44	24	0	4335428	3.65	4.06	0.00					
0.545455	0.000000	0	0	0	12	6	0	32	18	0	0	0						
12353	53722	WATSONVILLE HIGH SCHOOL	WATSONVILLE	Santa Cruz	UCSC	UCSC	44	36	5	4437901	3.75	3.94	3.82					
0.818182	0.138889	0	0	0	0	0	0	41	33	5	0	3						
12372	53103	WESTMONT HIGH SCHOOL	CAMPBELL	Santa Clara	UCSC	UCSC	98	49	11	4338505	3.73	4.05	3.90					
0.500000	0.224490	0	0	0	58	27	5	7	5	3	22	13						
12386	53105	WILLOW GLEN HIGH SCHOOL	SAN JOSE	Santa Clara	UCSC	UCSC	69	47	7	4338950	3.85	4.02	3.74					
0.681159	0.148936	0	0	0	21	15	3	18	11	0	25	18						
12402	53113	YERBA BUENA HIGH SCHOOL	SAN JOSE	Santa Clara	UCSC	UCSC	39	30	0	4330015	4.04	4.13	0.00					
0.769231	0.000000	0	0	0	32	28	0	7	0	0	0	0						

White Enr	AfrAm App Frac	Asian App Frac	Latinx App Frac	White App Frac	IsLocal
1416	0	0.000000	0.810526	0.115789	0.031579
1425	0	0.000000	0.000000	0.500000	0.000000
1426	6	0.000000	0.270833	0.093750	0.552083
1433	0	0.326531	0.510204	0.142857	0.000000
1435	0	0.000000	0.538462	0.000000	0.000000
...	...	...	...	...	...
12345	0	0.000000	0.272727	0.727273	0.000000
12353	0	0.000000	0.000000	0.931818	0.000000
12372	3	0.000000	0.591837	0.071429	0.224490
12386	0	0.000000	0.304348	0.260870	0.362319
12402	0	0.000000	0.820513	0.179487	0.000000

[1316 rows x 32 columns]

```
In [138... # Add columns to compute
dfLeft["IsLCFF+"] = 0
dfLeft["PassRateELA"] = 0.0
dfLeft["PassRateMath"] = 0.0
dfLeft["SeniorAppRate"] = 0.0
dfLeft["Total Students Enrolled"] = 0
```

```
In [139... # Populate the values. We use dfTests rather than dfRight because schools with fewer than 50 Juniors were filtered out of the latter.
iRow = -1
nOne = 0
nMore = 0
nNone = 0
```

```

for index, rowLeft in dfLeft.iterrows():
    iRow += 1
    schoolCode = rowLeft["CAASPP_ID"]
    school      = rowLeft["School"]
    city        = rowLeft["City"]
    county      = rowLeft["County"]

    # Find the row for ELA (1)
    rowTests = dfTests.loc[(dfTests["School Code"] == schoolCode) & (dfTests["Test ID"] == "1")].iloc[0]
    if rowTests["Grade"] != "11":
        print("WEIRD GRADE", rowTests["Grade"])
    if rowTests["Count Standard Met and Above"] == "*":
        continue
    if rowTests["Total Students Enrolled"] == "*":
        print(rowTests)
        break
    nStudents = int(rowTests["Total Students Enrolled"])
    nStudentsMet = int(rowTests["Count Standard Met and Above"])
    nStudentsNotMet = int(rowTests["Count Standard Nearly Met"]) + int(rowTests["Count Standard Not Met"])
    passRateELA = nStudentsMet / nStudents
    dfLeft.at[index, "Total Students Enrolled"] = nStudents
    dfLeft.at[index, "PassRateELA"] = passRateELA
    dfLeft.at[index, "SeniorAppRate"] = rowLeft["App"] / nStudents # Fraction of seniors who applied

    # Find out whether the school is LCFF+
    cdsCode = "".join([rowTests["County Code"], rowTests["District Code"], rowTests["School Code"]])
    dfResult = dfSchoolsTable[dfSchoolsTable["CDS Code"] == cdsCode]
    if len(dfResult) == 1: # Exactly one match
        nOne += 1
        if dfResult.iloc[0]["LCFF Plus School t"] == "Y":
            dfLeft.at[index, "IsLCFF+"] = 1
    elif len(dfResult) > 1: # More than one match
        nMore += 1
    else: # No school/district match
        print("NO MATCH:", school, row["Test ID"])
        nNone += 1

    # Find the row for Math (2)
    rowTests = dfTests.loc[(dfTests["School Code"] == schoolCode) & (dfTests["Test ID"] == "2")].iloc[0]
    if rowTests["Grade"] != "11":
        print("WEIRD GRADE", rowTests["Grade"])
    if rowTests["Count Standard Met and Above"] == "*":
        continue
    if rowTests["Total Students Enrolled"] == "*":
        print(rowTests)
        break
    nStudents = int(rowTests["Total Students Enrolled"])
    nStudentsMet = int(rowTests["Count Standard Met and Above"])
    nStudentsNotMet = int(rowTests["Count Standard Nearly Met"]) + int(rowTests["Count Standard Not Met"])
    passRateMath = nStudentsMet / nStudents
    dfLeft.at[index, "PassRateMath"] = passRateMath

```

```
print(nNone, nOne, nMore)
```

0 12134 0

In [140...

```
print(dfLeft)
```

Rate	UC_ID	School	City	County	Campus	countyCampus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr	GPA	Adm
0	EnrRate	AfrAm App	AfrAm Adm	AfrAm Enr	Asian App	Asian Adm	Asian Enr	Latinx App	Latinx Adm	Latinx Enr	White App	White Adm	White Enr	White App	White Adm	White Enr	White App
7160	0.402985	9	7	0	0	0	0	66	56	24	3630555	3.75	3.83	3.96	0.82	0.82	0.82
1	54651	ABLE CHARTER	STOCKTON	San Joaquin	all	UC Davis	14	10	3	0126755	3.77	3.79	0.00	0.71	0.71	0.71	0.71
4286	0.300000	0	0	0	0	0	0	11	8	3	1935121	3.69	3.88	3.94	0.74	0.74	0.74
2	51520	ABRAHAM LINCOLN HIGH SCHOOL	LOS ANGELES	Los Angeles	all	UCLA	74	55	18	1935121	3.69	3.88	3.94	0.74	0.74	0.74	0.74
3243	0.327273	0	0	0	27	24	10	45	31	8	3833241	3.82	3.95	4.03	0.66	0.66	0.66
3	52910	ABRAHAM LINCOLN HIGH SCHOOL	SAN FRANCISCO	San Francisco	all	Berkeley	249	166	98	3833241	3.82	3.95	4.03	0.66	0.66	0.66	0.66
6667	0.590361	7	4	0	195	131	83	32	22	9	4333795	3.94	4.04	4.16	0.70	0.70	0.70
4	53075	ABRAHAM LINCOLN HIGH SCHOOL	SAN JOSE	Santa Clara	all	UCSC	104	73	37	4333795	3.94	4.04	4.16	0.70	0.70	0.70	0.70
1923	0.506849	6	0	0	17	15	13	61	43	15	19	19	13	13	13	13	13
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
12403	50663	YGNACIO VALLEY HIGH SCHOOL	CONCORD	Contra Costa	UCSC	Berkeley	24	11	0	0737809	3.52	3.87	0.00	0.45	0.45	0.45	0.45
8333	0.000000	0	0	0	0	0	0	19	8	0	0	0	0	0	0	0	0
12404	54321	YORBA LINDA HIGH SCHOOL	YORBA LINDA	Orange	UCSC	UCI	54	26	0	0119404	3.80	4.02	0.00	0.48	0.48	0.48	0.48
1481	0.000000	0	0	0	32	18	0	6	0	0	10	4	4	4	4	4	4
12405	52176	YOSEMITE HIGH SCHOOL	OAKHURST	Madera	UCSC	UC Merced	8	7	0	2030013	3.99	4.07	0.00	0.87	0.87	0.87	0.87
5000	0.000000	0	0	0	0	0	0	0	0	0	5	4	4	4	4	4	4
12407	53815	YUBA CITY HIGH SCHOOL	YUBA CITY	Sutter	UCSC	UC Davis	40	26	3	5139001	3.71	4.04	0.00	0.65	0.65	0.65	0.65
0000	0.115385	0	0	0	13	10	0	21	14	0	0	0	0	0	0	0	0
12408	53820	YUCAIPA SENIOR HIGH SCHOOL	YUCAIPA	San Bernardino	UCSC	UCR	23	16	0	3638509	3.86	4.02	0.00	0.69	0.69	0.69	0.69
5652	0.000000	0	0	0	0	0	0	11	6	0	8	7	7	7	7	7	7

White Enr	AfrAm App	Asian App	Latinx App	White App	IsLocal	IsLCFF+	PassRateELA	PassRateMath	SeniorAppRate	Total St
udents Enrolled	Frac	Frac	Frac	Frac						
0	0.111111	0.000000	0.814815	0.000000	0	1	0.568233	0.234375	0.181208	
447										
1	0.000000	0.000000	0.785714	0.000000	0	1	0.616438	0.205479	0.191781	
73										
2	0.000000	0.364865	0.608108	0.000000	0	1	0.647059	0.475113	0.334842	
221										
3	0.028112	0.783133	0.128514	0.044177	0	0	0.615385	0.366864	0.491124	
507										
4	0.057692	0.163462	0.586538	0.182692	0	0	0.505181	0.162304	0.269430	
386										
...	...	...	...	...	...	...	...	...	...	...
...										
12403	0.000000	0.000000	0.791667	0.000000	0	1	0.369919	0.077551	0.097561	
246										
12404	0.000000	0.592593	0.111111	0.185185	0	0	0.665653	0.544073	0.164134	
329										
12405	0.000000	0.000000	0.000000	0.625000	0	0	0.292517	0.102041	0.054422	
147										
12407	0.000000	0.325000	0.525000	0.000000	0	0	0.517766	0.157360	0.101523	
394										
12408	0.000000	0.000000	0.478261	0.347826	0	0	0.554795	0.300855	0.039384	
584										

[12153 rows x 37 columns]


```
In [141... # Stepwise variable selection (forward selection)
# Should be able to call it like:
#     selected = forwardSelection(dfFiltered, "AdmRate", candidateVars) # Fails!!!
```

```
def forwardSelection(data, target, candidate_vars):
    remaining = list(candidate_vars)
    selected = []
    current_score, best_new_score = float("inf"), float("inf")

    while remaining:
        scores_with_candidates = []
        for candidate in remaining:
            formula = "{} ~ {}".format(
                target,
                " + ".join(selected + [candidate])
            )
            score = sm.OLS.from_formula(formula, data).fit().aic
            scores_with_candidates.append((score, candidate))

        scores_with_candidates.sort()
        best_new_score, best_candidate = scores_with_candidates[0]

        if best_new_score < current_score:
            remaining.remove(best_candidate)
            selected.append(best_candidate)
            current_score = best_new_score
        else:
            break

    return selected
```

```
In [142... # candidateVars = ["App GPA", "AfrAm App Frac", "Asian App Frac", "Latinx App Frac", "White App Frac", "IsLocal", "IsLCFF+", "PassRateELA", "PassRateM",
candidateVars = ["App GPA", "AfrAm App Frac", "Asian App Frac", "Latinx App Frac", "White App Frac", "IsLocal", "PassRateELA", "PassRateMath", "Senior
# candidateVars = ["App GPA", "AfrAm App Frac", "Asian App Frac", "Latinx App Frac", "White App Frac", "IsLocal", "PassRateELA", "PassRateMath"]
# candidateVars = ["App GPA", "AfrAm App Frac", "Asian App Frac", "Latinx App Frac", "White App Frac", "IsLocal", "PassRateELA", "SeniorAppRate"]
# candidateVars = ["App GPA", "AfrAm App Frac", "Asian App Frac", "Latinx App Frac", "White App Frac", "IsLocal", "PassRateMath", "SeniorAppRate"]
# candidateVars = ["App GPA", "IsLocal", "PassRateELA", "PassRateMath", "SeniorAppRate"]
# candidateVars = ["App GPA", "AfrAm App Frac", "Asian App Frac", "Latinx App Frac", "White App Frac", "IsLocal"]
campus = "UCSD"
dfFiltered = dfLeft[(dfLeft["Total Students Enrolled"] >= 100) & (dfLeft["App"] >= 40) & (dfLeft["Campus"] == campus) & (dfLeft["IsLCFF+"] == 0)]
# dfFiltered = dfLeft[(dfLeft["Total Students Enrolled"] >= 100) & (dfLeft["App"] >= 40) & (dfLeft["Campus"] == campus)]

# Use sklearn + mlxtend for stepwise selection
X = dfFiltered[candidateVars]
y = dfFiltered["AdmRate"]
sfs = SFS(
    LinearRegression(),
    k_features="best", # Number of features to select or "best"
    forward=True,
    floating=False, # Additional exclusion step to remove features once they were included?
    scoring="r2", # Metric for evaluation
    cv=5, # Cross-validation folds
    verbose=0, # 1, 2 provide status updates
```

```

)
sfs = sfs.fit(X, y)
# print(sfs.subsets_, "\n")
print(sfs.k_feature_names_, "\n")
selected = list(sfs.k_feature_names_)

# Evaluate the model
X = sm.add_constant(dfFiltered[selected])
y = dfFiltered["AdmRate"]

finalModel = sm.OLS(y, X).fit()
print(finalModel.summary())

sfs.subsets_

```

('App GPA', 'AfrAm App Frac', 'Latinx App Frac', 'IsLocal', 'PassRateELA', 'PassRateMath', 'SeniorAppRate')

```

=====
                        OLS Regression Results
=====
Dep. Variable:          AdmRate      R-squared:                0.224
Model:                  OLS          Adj. R-squared:           0.211
Method:                 Least Squares   F-statistic:             16.94
Date:                  Sun, 12 Apr 2026   Prob (F-statistic):      1.09e-19
Time:                  23:16:53         Log-Likelihood:          409.87
No. Observations:      419             AIC:                    -803.7
Df Residuals:          411             BIC:                    -771.4
Df Model:              7
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	-0.2598	0.226	-1.148	0.252	-0.705	0.185
App GPA	0.1036	0.057	1.808	0.071	-0.009	0.216
AfrAm App Frac	0.2453	0.077	3.169	0.002	0.093	0.398
Latinx App Frac	0.1532	0.031	4.916	0.000	0.092	0.214
IsLocal	0.0232	0.014	1.621	0.106	-0.005	0.051
PassRateELA	0.1816	0.070	2.609	0.009	0.045	0.318
PassRateMath	-0.3068	0.076	-4.053	0.000	-0.456	-0.158
SeniorAppRate	0.1424	0.050	2.876	0.004	0.045	0.240

```

=====
Omnibus:                 67.392   Durbin-Watson:           1.914
Prob(Omnibus):           0.000   Jarque-Bera (JB):        114.847
Skew:                    0.951   Prob(JB):                 1.15e-25
Kurtosis:                 4.721   Cond. No.                  217.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

Out[142...  {1: {'feature_idx': (3,),
      'cv_scores': array([ 0.23614821,  0.02362941, -0.00789839,  0.2393312 ,  0.16500866]),
      'avg_score': np.float64(0.13124381681944108),
      'feature_names': ('Latinx App Frac',)},
      2: {'feature_idx': (1, 3),
      'cv_scores': array([0.20062426, 0.07238799, 0.00446301, 0.24066189, 0.20037478]),
      'avg_score': np.float64(0.14370238492597787),
      'feature_names': ('AfrAm App Frac', 'Latinx App Frac'))},
      3: {'feature_idx': (1, 3, 7),
      'cv_scores': array([0.17575117, 0.07989074, 0.02355198, 0.25641159, 0.20796049]),
      'avg_score': np.float64(0.1487131949790847),
      'feature_names': ('AfrAm App Frac', 'Latinx App Frac', 'PassRateMath'))},
      4: {'feature_idx': (1, 3, 7, 8),
      'cv_scores': array([0.20727784, 0.04855589, 0.02393905, 0.28347078, 0.21640314]),
      'avg_score': np.float64(0.1559293410597002),
      'feature_names': ('AfrAm App Frac',
        'Latinx App Frac',
        'PassRateMath',
        'SeniorAppRate'))},
      5: {'feature_idx': (1, 3, 6, 7, 8),
      'cv_scores': array([0.19061077, 0.04646404, 0.08244891, 0.28692752, 0.23879858]),
      'avg_score': np.float64(0.1690499622615199),
      'feature_names': ('AfrAm App Frac',
        'Latinx App Frac',
        'PassRateELA',
        'PassRateMath',
        'SeniorAppRate'))},
      6: {'feature_idx': (1, 3, 5, 6, 7, 8),
      'cv_scores': array([0.2063424 , 0.03138949, 0.1016438 , 0.30178054, 0.20930914]),
      'avg_score': np.float64(0.170093072490396),
      'feature_names': ('AfrAm App Frac',
        'Latinx App Frac',
        'IsLocal',
        'PassRateELA',
        'PassRateMath',
        'SeniorAppRate'))},
      7: {'feature_idx': (0, 1, 3, 5, 6, 7, 8),
      'cv_scores': array([0.22149818, 0.04087486, 0.12584551, 0.27836847, 0.1845827 ]),
      'avg_score': np.float64(0.1702339431189003),
      'feature_names': ('App GPA',
        'AfrAm App Frac',
        'Latinx App Frac',
        'IsLocal',
        'PassRateELA',
        'PassRateMath',
        'SeniorAppRate'))},
      8: {'feature_idx': (0, 1, 3, 4, 5, 6, 7, 8),
      'cv_scores': array([0.2138059 , 0.04591146, 0.13068712, 0.28624802, 0.15897896]),
      'avg_score': np.float64(0.16712629301687254),
      'feature_names': ('App GPA',
        'AfrAm App Frac',
        'Latinx App Frac',

```

```

'White App Frac',
'IsLocal',
'PassRateELA',
'PassRateMath',
'SeniorAppRate'})),
9: {'feature_idx': (0, 1, 2, 3, 4, 5, 6, 7, 8),
'cv_scores': array([0.2171198 , 0.03600494, 0.13933852, 0.28457384, 0.164258  ]),
'avg_score': np.float64(0.16825902001924983),
'feature_names': ('App GPA',
'AfrAm App Frac',
'Asian App Frac',
'Latinx App Frac',
'White App Frac',
'IsLocal',
'PassRateELA',
'PassRateMath',
'SeniorAppRate')}}

```

In [143... `print(dfTests[dfTests["School Code"].str.contains(r"0106401", na=False)])`

ID	Grade	Total Students	Enrolled	Total Students	Tested	Total Students	Tested with Scores	Mean Scale Score	Percentage Standard Exceeded	\
9563		01	61119	Alameda Unified	0106401	Alameda Science and Technology Institute	7	2024	B	1
1	11	47			46		46	2708.3	63.04	
9641		01	61119	Alameda Unified	0106401	Alameda Science and Technology Institute	7	2024	B	2
1	11	47			46		46	2698.2	47.83	

Count	Standard Exceeded	Percentage Standard Met	Count	Standard Met	Percentage Standard Met and Above	Count	Standard Met and Above	Percentage Stand
9563	29	36.96	17		100.00		46	
0.00	0	0.00	0		46			
9641	22	28.26	13		76.09		35	
10.87	5	13.04	6		46			

In [144... `print(dfRight[dfRight["School Name"].str.contains(r"Alameda", na=False)])`

County Code	District Code	District Name	School Code	School Name	Type	ID	Test ID	Total Students	Enroll
11469	01	61119	Alameda Unified	0130229	Alameda High	7	1		4
67		440	440	2647.1	181	157			
11569	01	61119	Alameda Unified	0130229	Alameda High	7	2		4
66		438	438	2623.7	131	109			
11791	01	61119	Alameda Community Learning Center	0130609	Alameda Community Learning Center	9	1		
53		51	51	2556.0	8	10			
11999	01	61119	Alameda Community Learning Center	0130609	Alameda Community Learning Center	9	2		
53		50	50	2515.0	3	10			

Count Standard Met and Above	Count Standard Nearly Met	Count Standard Not Met	PassRate	CAASPP	County Name	CAASPP	County Campus	LCFF	Plus	School
11469	338	52	50	0.768182	Alameda		Berkeley			
N	N									
11569	240	72	126	0.547945	Alameda		Berkeley			
N	N									
11791	18	20	13	0.352941	Alameda		Berkeley			
N	N									
11999	13	8	29	0.260000	Alameda		Berkeley			
N	N									

In [145...

```
dfExamine = dfLeft[dfLeft["Campus"].isin({"UCI", "UCSD"})]
dfExamine = dfExamine[dfExamine["School"].isin({"AZUSA HIGH SCHOOL", "CANYON CREST ACADEMY", "DEL NORTE HIGH SCHOOL", "TORREY PINES HIGH SCHOOL"})]
print(dfExamine)
```

	UC_ID	School		City	County	Campus	countyCampus	App	Adm	Enr	CAASPP_ID	App	GPA	Adm	GPA	Enr	GPA	AdmRate	EnrR						
ate	AfrAm	App	AfrAm	Adm	AfrAm	Enr	Asian	App	Asian	Adm	Asian	Enr	Latinx	App	Latinx	Adm	Latinx	Enr	White	App	White	Adm	White	Enr	\
4015	50188		AZUSA	HIGH	SCHOOL		AZUSA		Los Angeles		UCI		UCLA	43	17	7	1930528	3.89	4.12	4.16	0.395349	0.411			
765		0		0	0	0		3	3			39		14	4		0	0	0						
4082	50893		CANYON	CREST	ACADEMY		SAN DIEGO		San Diego		UCI		UCSD	352	104	17	0106328	3.98	4.17	4.02	0.295455	0.163			
462		0		0	0		196		59	10		34		9	0		79	19	3						
4198	54351		DEL NORTE	HIGH	SCHOOL		SAN DIEGO		San Diego		UCI		UCSD	293	97	26	0118935	4.00	4.20	4.12	0.331058	0.268			
041		10		3	0		184		62	14		20		4	0		55	20	8						
5029	50466		TORREY PINES	HIGH	SCHOOL		SAN DIEGO		San Diego		UCI		UCSD	247	43	4	3730033	3.99	4.21	0.00	0.174089	0.093			
023		0		0	0		111		22	0		24		0	0		79	8	0						
8796	50188		AZUSA	HIGH	SCHOOL		AZUSA		Los Angeles		UCSD		UCLA	28	23	8	1930528	3.95	4.04	3.97	0.821429	0.347			
826		0		0	0		0		3	0		25		20	7		0	0	0						
8866	50893		CANYON	CREST	ACADEMY		SAN DIEGO		San Diego		UCSD		UCSD	384	63	14	0106328	3.99	4.21	4.14	0.164062	0.222			
222		0		0	0		202		42	8		39		8	0		95	5	0						
8984	54351		DEL NORTE	HIGH	SCHOOL		SAN DIEGO		San Diego		UCSD		UCSD	296	48	10	0118935	4.00	4.25	4.16	0.162162	0.208			
333		8		0	0		186		34	6		19		0	0		64	5	0						
9816	50466		TORREY PINES	HIGH	SCHOOL		SAN DIEGO		San Diego		UCSD		UCSD	298	50	15	3730033	4.00	4.25	4.22	0.167785	0.300			
000		6		0	0		117		25	5		36		4	0		108	12	5						
led	AfrAm	App	Frac	Asian	App	Frac	Latinx	App	Frac	White	App	Frac	IsLocal	IsLCFF+	PassRateELA	PassRateMath	SeniorAppRate	Total	Students	Enrol					
4015		0.000000		0.000000		0.906977		0.000000		0		1		0.494764	0.246073	0.112565									
382																									
4082		0.000000		0.556818		0.096591		0.224432		0		0		0.905967	0.811935	0.636528									
553																									
4198		0.034130		0.627986		0.068259		0.187713		0		0		0.778650	0.715856	0.459969									
637																									
5029		0.000000		0.449393		0.097166		0.319838		0		0		0.792188	0.580595	0.385937									
640																									
8796		0.000000		0.000000		0.892857		0.000000		0		1		0.494764	0.246073	0.073298									
382																									
8866		0.000000		0.526042		0.101562		0.247396		1		0		0.905967	0.811935	0.694394									
553																									
8984		0.027027		0.628378		0.064189		0.216216		1		0		0.778650	0.715856	0.464678									
637																									
9816		0.020134		0.392617		0.120805		0.362416		1		0		0.792188	0.580595	0.465625									
640																									

In []: